

Ecological Macroeconomics: The path towards a just low-carbon transition for countries in the European Union

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Ph.D. Thesis

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Abstract

The contemporary capitalist socioeconomic system has led to essential and existential socioeconomic and ecological challenges. Addressing these issues requires a fundamental system transition that centers around socioeconomic and ecological considerations. Within this context, this thesis aims to address the question of how countries in the European Union (EU) can achieve a low-carbon transition of the socioeconomic system that is socially just, i.e. a simultaneous substantial reduction in emissions and inequality. To this end, the individual Chapters focus on specific implications regarding: 1) potential policy mixes, 2) the specific role of the energy transition, and 3) decoupling and the role of trade. This work is conducted within the framework of Ecological Macroeconomics, which aims to enhance methods and insights towards analysing the socioeconomic transition and thereby addresses key shortcomings of main Integrated Assessment Models. Specifically, the Eurogreen model is revised and applied to Italy, as example for EU countries. Moreover, its national simulation-based approach is complemented by an empirical consumption-based analysis utilizing environmentally extended multiregional input-output analysis. The results illustrate how the intricate interactions between economic activity, inequality and emissions render a just transition challenging to achieve. In particular, the trade-off between emission and inequality reductions poses a meaningful but not surmountable impediment. The findings underline the importance of an integrated analysis of the emission and inequality goals and the need for a range of coherent policy measures to address them in combination. These include specifically genuine redistribution from [higher to lower incomes](#) and a rapid expansion of the energy transition, while generic economic growth does not appear particularly conducive towards a just low-carbon transition. Focusing explicitly on the energy transition, the findings of this work show that it can overcome the emission-inequality trade-off, if featuring the appropriate characteristics, particularly a rebate of carbon costs to households. Extensive redistribution alone increases economic activity and emissions. It can also shift the process of innovation towards increased labour productivity and away from intermediate input efficient production, additionally increasing emissions. Thus, a structural reduction of consumption and production is potentially necessary to combine low emission and low inequality goals. The need to not solely rely on national energy transition efforts to achieve low-emission targets in particular in the context of substantial redistribution is further highlighted by the results of the empirical consumption-based embedded fossil fuel use of the EU. While the regional energy transition historically only led to a modest reduction in fossil fuel use, trade patterns have

counteracted these developments. The findings underline the importance, from an EU perspective, to extend the scope of energy transition efforts towards the global supply chain, but also to consider demand side measures to achieve emission reductions in line with environmental targets. Thus, overall this thesis suggests that the most promising path towards a just low-carbon transition is a post-growth one, featuring extensive redistribution measures in combination with rapid global energy transition efforts and a structural reduction in certain types of consumption and production. Yet, a substantial amount of research on this fundamental transition of the socioeconomic system is still needed for a comprehensive assessment of associated implications and its feasibility.

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Contents

1	Introduction	1
1.1	General context	1
1.2	Theoretical framework	6
1.2.1	Heterodox Economics	6
1.2.2	Ecological Economics	8
1.2.2.1	Post-growth	10
1.2.3	Ecological Macroeconomics	11
1.2.3.1	Integrated Assessment Models and the issues with orthodox approaches	12
1.2.3.2	Ecological Macroeconomics as part of a new economic approach	14
1.2.4	Post-Keynesian Economics	15
1.2.4.1	Post-Keynesian and Ecological Economics	17
1.3	Methodological framework	19
1.3.1	Methodology in Ecological Macroeconomics	19
1.3.2	Ecological Macroeconomic models to analyse a just low-carbon transition	20
1.3.2.1	Eurogreen	21
1.3.3	Environmentally Extended Multiregional Input-Output Analysis	22
1.4	Structure and contributions of the thesis	24
2	Scenario discovery for a just low-carbon transition	27
2.1	Introduction	28
2.2	Scenarios and policies in ecological macroeconomics	30
2.3	Methods	33
2.3.1	The model	33
2.3.2	Simulation and data analysis approach	36
2.4	Results	40
2.4.1	The emissions-inequality-GDP nexus	40

2.4.2	Main policy parameters	42
2.4.3	Low emissions, low inequality pathways	46
2.5	Concluding remarks	47
2.A	Additional results	50
2.A.1	Emissions, inequality and GDP	50
2.A.2	Partial dependence plots	51
2.A.2.1	Policy objective: emissions	51
2.A.2.2	Policy objective: inequality	52
2.A.2.3	Policy objectives: emissions and inequality	52
2.A.2.4	Policy objectives: emissions, inequality and GDP	53
2.A.3	Kernel densities	53
2.B	Robustness checks	53
2.C	Simulation parameters	59
2.6	Supplementary Material	63
2.6.1	The Eurogreen model	63
2.6.1.1	Population	63
2.6.1.2	Input-output and GDP	63
2.6.1.3	International trade	67
2.6.1.4	Innovation	68
2.6.1.5	Investment and profits	73
2.6.1.6	Finance and wealth distribution	79
2.6.1.7	Labour	83
2.6.1.8	Government	87
2.6.1.9	Prices	96
2.6.1.10	Consumption	97
2.6.1.11	Energy	104
2.6.1.12	Climate damage	111
2.6.1.13	Time use	113
2.6.2	Stock-flow matrices	116
2.6.3	List of variables and parameters	118
2.6.3.1	Subscripts and Superscripts	118
2.6.3.2	Variables and Parameters	119
2.6.3.3	Calibration Parameters	130

3 Emissions, inequality and employment – The role of the energy transition in a just low-carbon transition	131
3.1 Introduction	133
3.2 Methods	135
3.2.1 The main structure of the Eurogreen model	135
3.2.2 Novel energy-economy links in Eurogreen	136
3.2.2.1 Energy and production structure	136
3.2.2.2 Electricity module	138
3.2.3 Scenarios	140
3.2.3.1 The energy transition	141
3.2.3.2 Sensitivity to stochastic processes	141
3.3 Results	142
3.3.1 Main dynamics	142
3.3.2 Employment structure	145
3.4 Discussion	146
3.5 Conclusion	149
3.A Additional details on scenarios	151
3.A.1 Socioeconomic paths	151
3.A.2 Quantitative electrification scenario	152
3.B Aggregate industries	153
3.C Additional figures	153
3.C.1 Dynamics of totals	153
3.C.2 Final period structures	155
3.6 Supplementary Material	156
3.6.1 Demography	156
3.6.1.1 Migration	156
3.6.2 Input-Output	156
3.6.2.1 Sector classification	156
3.6.2.2 Output constraints	157
3.6.3 Innovation	158
3.6.3.1 Capital weighted technological development	158
3.6.4 Investment	159
3.6.4.1 Residential investment	159
3.6.4.2 Investment matrix	159
3.6.5 Labour	159

3.6.5.1	Skill-shares in labour demand	159
3.6.5.2	Job Transition	160
3.6.6	Government	160
3.6.7	Households	161
3.6.7.1	Households and consumption	161
3.6.7.2	Transformation of purchaser prices to basic prices	162
3.6.8	Prices	163
3.6.8.1	Input-output price model	163
3.6.8.2	Consumer Price Index and inflation	163
3.6.9	Carbon costs	164
3.6.9.1	ETS	164
3.6.9.2	Carbon costs and prices	164
3.6.10	Energy	166
3.6.10.1	Energy input-output structure	166
3.6.10.2	Energy module scenario implementation	174
3.6.10.3	Emission factors	179
3.6.11	Electricity	179
3.6.11.1	Exogenous pathway	180
3.6.11.2	Capacity investment	181
3.6.11.3	Dispatch and production structure	182
3.6.11.4	Integration in the Eurogreen structure	183
3.6.12	Climate damage	185
4	Trade and decoupling of fossil fuel use embedded in EU consumption	187
4.1	Introduction	188
4.2	Material and Methods	189
4.2.1	Footprints	190
4.2.2	Subsystem Approach	191
4.2.3	Simulation	192
4.2.4	Structural Decomposition Analysis	192
4.2.5	Data	193
4.3	Results	193
4.3.1	Fossil Fuel Footprints	193
4.3.2	Drivers	194
4.3.2.1	Trade	196

4.3.3	The Carbon Border Adjustment Mechanism	197
4.4	Discussion	199
4.4.1	Implications for Decoupling	200
4.4.2	Policy Implications	201
4.4.3	Methods and Limitations	202
4.5	Conclusion	203
4.A	Additional Graphs	206
4.A.1	Fossil fuel intensities	206
4.A.2	Structural decomposition analysis results by good	207
4.A.3	Input structure	209
4.A.4	Emissions	210
4.6	Supplementary Material	212
4.6.1	Footprint analysis	212
4.6.2	Structural decomposition analysis	212
4.6.2.1	Components	212
4.6.2.2	Full decomposition	213
4.6.2.3	Chaining	221
5	Discussion & conclusion	223
5.1	The role of economic growth in a just transition	224
5.2	Inequality and emissions	226
5.3	Policy implications	229
5.3.1	Individual policies	229
5.3.2	Policy combinations and transition pathways	230
5.4	Limitations and future research	231
5.5	Concluding remarks	234
	Bibliography	237

List of Figures

1.1	Conditions of global biophysical and social thresholds in 1992 and 2015. .	2
1.2	Top 10% income shares for selected countries and regions between 1980 and 2016.	3
1.3	The economy as embedded within the social and ecological sphere.	8
1.4	The economic schools dominating different types of IAMs.	15
1.5	Comparative GDP developments for equilibrium and non-equilibrium mod- elling approaches with respect to a baseline following a policy-induced en- ergy transition.	17
2.1	Summary of literature review.	32
2.2	Model Overview.	34
2.3	Simulation results.	40
2.4	Main policy parameters.	43
2.5	Decomposition of emissions and inequality.	47
2.A.1	Inequality, emissions and GDP in the final simulation year.	50
2.A.2	ICEs and PDPs of the 10 main policy parameters (objective: GHG)	51
2.A.3	ICEs and PDPs of the 10 main policy parameters (objective: Gini)	52
2.A.4	ICEs and PDPs of the 10 main policy parameters (objective: GHG-Gini) .	52
2.A.5	ICEs and PDPs of the 10 main policy parameters (objective: GHG-Gini- GDP)	53
2.A.6	Kernel density distribution of the 10 main continuous policy parameters .	54
2.B.1	Permutation feature importance of the 10 main policy parameters under different resampling methods	58
2.6.1	Conceptual framework of the population module	64
2.6.2	Conceptual framework of the input-output module	66
2.6.3	Possible cases of technological development.	71
2.6.4	Conceptual framework of the innovation module	73
2.6.5	Conceptual framework of the Investments and profits module	77

2.6.6	Conceptual framework of the labour module	87
2.6.7	Conceptual framework of the government module	94
2.6.8	Conceptual framework of the consumption module	102
2.6.9	Conceptual framework of the energy module	109
2.6.10	Probability distribution of Λ at different temperature levels	112
3.1	Illustration of the electrification process.	137
3.2	Simplified structure of the electricity module.	138
3.3	Main macroeconomic dynamics.	142
3.4	Main ecological and inequality dynamics.	144
3.5	Employment in the main energy supplying sectors.	146
3.6	Employment share by industry and skill.	147
3.C.1	Dynamics of total final demand and output.	153
3.C.2	Dynamics of labour compensation.	154
3.C.3	Main macroeconomic, ecological and inequality dynamics with and without climate income.	154
3.C.4	Main macroeconomic, ecological and inequality dynamics with different energy transition parameters.	155
3.C.5	Final demand and output structure.	155
3.6.1	Illustration of energy input-output structure.	167
3.6.2	Illustration of the electrification process.	174
3.6.3	Simplified structure of the electricity module.	180
4.1	Flowchart linking the methodologies used and the results obtained.	190
4.2	The evolution of the EU and global fossil fuel footprint between 2000 and 2014 in TJ.	194
4.3	The relative cumulative impact of drivers for the fossil fuel footprint and its components.	195
4.4	The contribution to intermediate and final production of the EU fossil fuel footprint by region of origin.	196
4.5	The relative cumulative impact of within and outside EU developments of energy efficiency and energy mix.	197
4.6	The EU fossil fuel footprint disaggregated according to origin and element in the production chain.	198
4.A.1	The dynamics of fossil fuel intensities from a consumption-based accounting framework per chain linked volume of final consumption.	206

List of Figures

4.A.2	The relative cumulative impact of drivers for sector specific fossil fuel footprints and their components.	208
4.A.3	Absolute levels of fossil fuels embedded in final and intermediate production of the EU fossil fuel footprint by EU and non-EU origin.	209
4.A.4	Fossil fuels embedded in direct and indirect imports into the EU, structured by region and sector of origin respectively.	209
4.A.5	EU 27 GDP, consumption and fossil CO2 emission footprint.	210
4.A.6	The evolution of the EU emission footprint decomposed into fossil fuels embedded in intermediate and final production.	210
4.A.7	The EU emission footprint disaggregated according to origin and element in the production chain.	211
4.6.1	The EU fossil fuel footprint based on current year price data compared to results using previous year price data.	222

List of Tables

1.1	Main characteristics of orthodox and heterodox economic schools.	6
2.B.1	Training sample sizes under different resampling methods	55
2.B.2	Prediction accuracy of the random forest models under different resampling methods	57
2.C.1	List of sensitivity parameters	60
2.6.1	List of sectors	65
2.6.2	Tax schedule for taxable annual income	89
2.6.3	General government expenditure on social protection	92
2.6.4	List of consumption categories	100
2.6.5	Total supply and use energy flows by sectors and households.	107
2.6.6	Aggregate Balance sheet	116
2.6.7	Transaction Flow Matrix	117
2.6.8	Subscripts and Superscripts	118
2.6.9	Variables and Parameters	119
2.6.10	Calibration Parameters	130
3.1	Policies, parameters and type of implementation for the socioeconomic paths. 140	
3.2	The six scenario combinations.	141
3.B.1	Aggregate Industry Structure.	153
3.6.1	Sector classification.	157
3.6.2	Energy product aggregation.	172
4.6.1	Sectoral aggregation and their NACE Rev. 2 equivalent.	212
4.6.2	Energy product aggregation and their equivalents.	212

Abbreviations and Acronyms

ABM	Agent-based model
CB IAM	Cost-Benefit Integrated Assessment Model
CBAM	Carbon Border Adjustment Mechanism
CCS	Carbon capture and storage
CGE	Computational General Equilibrium
CPI	Consumer Price Index
DP	Fossil fuels used in direct final production
DPB IAM	Detailed Process Based Integrated Assessment Model
DSGE	Dynamic Stochastic General Equilibrium
EE	Ecological Economics
EEA	European Environment Agency
EEMRIO	Environmentally Extended Multiregional Input-Output
EM	Ecological Macroeconomics
ETS	Emissions Trading System
EU	European Union
FP	Embedded fossil fuel footprint
GDP	Gross Domestic Product
GFCF	Gross Fixed Capital Formation
GHG	Greenhouse gas
GIC	Gross Inland Consumption
HBS	Household Budget Survey
IAM	Integrated Assessment Model
IO	Input-Output
IPCC	Intergovernmental Panel on Climate Change
MRIO	Multiregional Input-Output
NIC	Net Inland Consumption

OECD	Organisation for Economic Co-operation and Development
PEFA	Physical Energy Flow Accounts
PKE	Post-Keynesian Economics
RES	Renewable Energy Sources
RPC	Representative Concentration Pathway
SC	Fossil fuels embedded in intermediate inputs throughout the supply chain
SD	System Dynamics
SDA	Structural Decomposition Analysis
SFC	Stock-Flow Consistent
SHIW	Survey on Household Income and Wealth
SSP	Shared Socioeconomic Pathway
tCO ₂ eq	Tonnes of Carbon Dioxide equivalents
TJ	Terajoule
VAT	Value Added Tax
WIOD	World Input-Output Database

Chapter 1

Introduction

1.1 General context

The contemporary dominant capitalist socioeconomic system has produced a multitude of fundamental and existential challenges societies are faced with. On the one hand, its priority for profit and growth over ecological concerns has led to an era of multifaceted ecological crises. The most salient of these is human-induced climate change, with anthropogenic warming reaching 1.31°C in 2023 (Forster et al., 2024) and causing widespread harm and damages (IPCC, 2023).

While most prominent in the public discourse, climate change is neither an isolated nor the only ecological challenge humanity faces. Planetary boundaries, which reflect the stability of the Earth system, thereby defining a safe operating space, feature nonlinear interactions and impacts on the whole Earth system (Richardson et al., 2023; Rockström et al., 2009; Steffen et al., 2015). Six out of nine planetary boundaries are currently transgressed, underlining that humanity is far exceeding its safe operating space (Richardson et al., 2023).

Evidently, its oblivious strive for profit and growth has led to societies eroding the very foundation of their existence. Still, growth hailed as the panacea for economic and social matters (Schmelzer, 2015) has seemingly not delivered: Despite the overexploitation of the resources, humanity at a global level is not able to provide a decent social foundation for all to thrive (Fig. 1.1). Nor does any individual country exhibit a sustainable rate of resource use while meeting the basic needs for its people (Fanning et al., 2022; O'Neill et al., 2018).

A major factor in this simultaneous existence of high levels of material wealth and basic deprivation is clearly inequality, with highly unequal income and wealth distributions (Chancel et al., 2022). Moreover, the issue of inequality is tightly linked with ecological ones, in particular climate change: Not only are the contributions to energy use and emissions strikingly unequal



Figure 1.1: Conditions of global biophysical and social thresholds in a) 1992 and b) 2015. The blue wedges indicate performance of social indicators and the green ones resource use, both relative to their respective thresholds. Red wedges represent ecological overshoot or social shortfall. Source: Fanning et al., 2022.

(Bruckner et al., 2022; Chancel, 2022; Oswald et al., 2020), with consumption and affluence playing a dominant role in driving environmental impacts (Wiedmann et al., 2020), but also the impact of climate change falls disproportionately upon vulnerable groups who bear the least responsibility for it (IPCC, 2023).

Consequently, addressing inequality will have direct implications for ecological concerns and vice versa, which is why these issues should be addressed together and failing to do so can have unintended consequences. For instance, climate policies without redistribution or compensation mechanisms can contribute to exacerbating social issues like poverty (Soergel et al., 2021) or inequality (D'Alessandro et al., 2020). Conversely, addressing inequality might have important implications for emissions. A mechanism commonly discussed in the literature (e.g. Rao and Min, 2018, Sommer and Kratena, 2017) is that the carbon intensity of income decreases with income, i.e. that an additional Euro in the hands of the wealthy leads to less emissions than a Euro in the hands of the less affluent. Thus, redistribution towards the bottom of the income distribution would increase emissions.

While the ecological challenges are extremely urgent – retaining a 50% likelihood of keeping global warming below 1.5°C only allows for 7 more years of 2023 emissions from 2024 on (Friedlingstein et al., 2023) –, the current socioeconomic system appears to be arguably unable or unwilling to adequately address them. Implemented policies and nationally determined contributions (NDCs) are not in line with climate goals of 1.5 or even 2°C (IPCC, 2023), while CO₂

emissions have rebounded from the COVID-19 drop and continue to increase (Friedlingstein et al., 2023).

A similar argument can be made for the issue of inequality. It remains a political decision with policy options to address inequality ranging from wealth, corporate and progressive income taxations to tackling tax evasion and public investment (Alvaredo et al., 2018b; Chancel et al., 2022). Still, while direct redistribution policies (i.e. taxes and transfers) reduce inequality throughout, their impact is limited with some tax systems even featuring regressive characteristics (Chancel et al., 2022). After a range of liberalization and deregulation measures, inequality has been generally increasing almost everywhere in recent decades (Chancel et al., 2022). In particular, the concentration of wealth and income follow increasing trends in most regions (Alvaredo et al., 2018a; Piketty and Saez, 2014), as illustrated for income in Figure 1.2.

Taken together, current trends point towards a future worsening of the overshoot of ecological

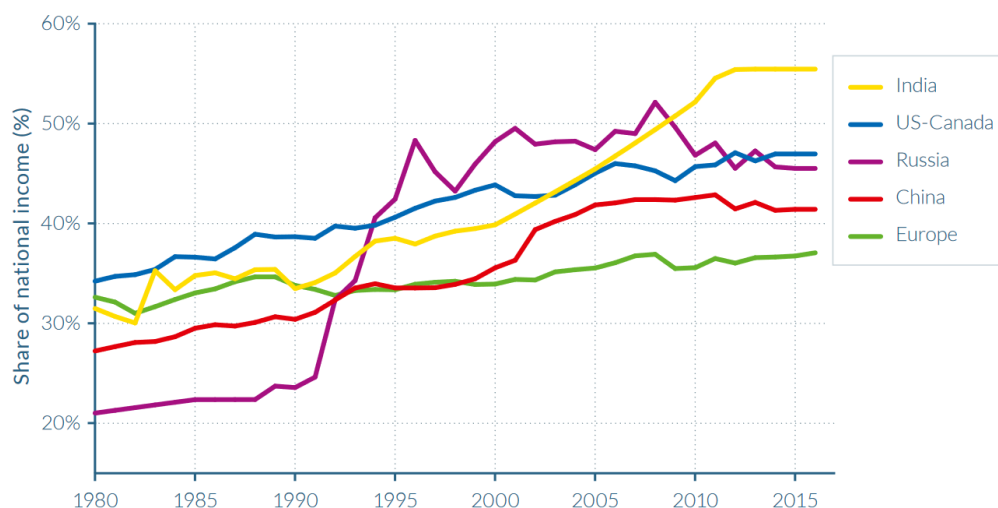


Figure 1.2: Top 10% income shares for selected countries and regions between 1980 and 2016. Source: Alvaredo et al. (2018b), based on wid.WORLD (2017).

boundaries without closing all basic needs gaps (Fanning et al., 2022), pointing towards the inefficiency of translating resources into satisfying basic needs as well as a general trade-off between social and ecological thresholds in the context of the current system.

Thus, tackling social issues like inequality and ecological ones together requires a rapid and extensive transition of the current socioeconomic system, as also called for by the IPCC, 2023, p. 28:

“Rapid and far-reaching transitions across all sectors and systems are necessary to achieve deep and sustained emissions reductions and secure a liveable and sustainable future for all.”

As argued above, inequality should play a central and integrated role within this transition. Notably, focusing on inequality is also important from a practical perspective, since it can help address perceived losses of transition measures, which is crucial for their political acceptance (Groh and Möllendorff, 2020). Thus, the concept of a just transition is essential. The IPCC defines a just transition as “A set of principles, processes and practices that aim to ensure that no people, workers, places, sectors, countries or regions are left behind in the transition from a high-carbon to a low-carbon economy” (IPCC, 2022a, p.1806). It is often framed in relation to the energy transition (García-García et al., 2020). However, in this thesis, it is conceptualised more holistically as referring to the whole transition of the socioeconomic system and instrumentalised as a simultaneous reduction of emissions and inequality.

While the need for a substantial transition is generally accepted, there is an ongoing debate about its fundamental approach and scope. It ranges from retaining the broad growth-based structure of the current socioeconomic system and banking on technological solutions to reduce environmental pressures in the form of green growth (Jackson and Victor, 2019b; Schmelzer, 2015) to an extensive overhaul of the contemporary system with a focus on wellbeing, equality and ecological impact in the form of degrowth (Kallis et al., 2018). As illustrated by the respective names, the role of economic growth for effective environmental mitigation plays a particularly contentious role. While green growth proponents are typically optimistic towards the continuation of economic growth, degrowth tends to be more skeptical towards its desirability and compatibility with effective environmental mitigation.

So far the debate has not yet addressed the substantial uncertainties about the ecological and socioeconomic impacts of different transition pathways¹ comprehensively. While simulation models provide a suitable tool for this, key issues and gaps remain in the current state of the respective modelling literature (see Sections 1.2 and 1.3). Yet, society and policy makers require credible analyses regarding the feasibility and desirability of possible policy measures, transitions and futures to make informed decisions.

To address these shortcomings in the prevalent literature, this thesis mainly operates within the field Ecological Macroeconomics, which is part of a new approach to Integrated Assessment Models, addressing several of their prevalent issues and biases (see section 1.2); while also adding an empirical consumption-based perspective and the issue of trade to the analysis. This work aims to contribute to the debate about potential transition pathways, by analysing the ecological and in particular the socioeconomic implications of different policy measures and tran-

¹Throughout the thesis, the concepts of scenarios and pathways are mentioned, which refer to the following: While scenarios describe different potential futures grounded in a comprehensive range of assumptions, pathways refer to the dynamic developments of systems approaching a future state (IPCC, 2022a).

sition pathways [in the context of European Union \(EU\) countries](#), focusing on climate change via energy and emissions on the one hand and inequality, in particular income inequality, on the other. Thus, it is often framed around the concept of a just transition. In line with Ecological Macroeconomics (see section [1.2.3](#)), the goal of this work is to contribute novel insights and perspectives to inform the debate around the implications of different shapes of policy measures and transition pathways, further adding to the understanding of trade-offs and synergies between ecological and socioeconomic policy measures.

The specific overall research question of this work is: How can countries in the EU achieve a low-carbon transition of the socioeconomic system that is socially just? To address specific aspects of this overarching research question, the specific research questions analysed by the individual chapters are:

1. What are potential policy mixes towards a just low-carbon transition?
2. Which role does the energy transition play in a just low-carbon transition?
3. What role does trade play in the EU energy transition from a consumption-based perspective and what are the implications for decoupling?

This Chapter proceeds as follows: Sections [1.2](#) and [1.3](#) provide an overview of the theoretical and methodological frameworks that form the basis of this thesis. Subsequently, section [1.4](#) introduces the remaining Chapters of the thesis and outlines the contributions of this work to the major topics within the main related fields of research.

1.2 Theoretical framework

This thesis falls within the field of heterodox economics. Following Lavoie (2014) economics is defined here not as an allocation of scarce resources, but as “*the study of the process by which society brings its available resources into production, and the distribution of that production among its members.*” (Weeks, 2012). As illustrated in the following section, the allocation of scarce resources is at the core of the orthodox economic approach, whereas the definition presented here is more in line with a heterodox view (see also Lavoie, 2014).

This subsection outlines and contrasts the main characteristics of heterodox and orthodox economics, before presenting several important heterodox schools and zooming in on the economic school this thesis can be allocated to more narrowly, namely ecological economics and in particular ecological macroeconomics.

1.2.1 Heterodox Economics

There are fundamental distinctions in the overall approach between orthodox and heterodox economic schools in general. Here, they are presented mainly following Lavoie (2014). The main respective assumptions and approaches of heterodox and orthodox economic approaches are summarized in Table 1.1.

In terms of epistemology, orthodox schools follow Instrumentalism, considering not whether

Table 1.1: Main characteristics of orthodox and heterodox economic schools. Source: Lavoie (2014).

Presupposition	Orthodox schools	Heterodox schools
Epistemology/Ontology	Instrumentalism	Realism
Rationality	Hyper model-consistent rationality, optimizing agent	Environment-consistent rationality, satisficing agent
Method	Individualism, atomicism	Holism, organicism
Economic core	Exchange, allocation, scarcity	Production, growth, abundance
Political core	Unfettered markets	Regulated markets

a theoretical statement is true but whether it allows precise predictions. In contrast, heterodox schools’ approach is more in line with Realism, i.e. for them, theory must be based on realistic hypotheses.

For orthodox economics its epistemological approach tends to translate into unbounded rationality, where agents have full information, can predict the future and optimize their behaviour accordingly. By contrast heterodox economics recognizes that economic agents are commonly

faced with a lack or an overload of information and hence resort to heuristics and satisficing behaviour.

In addition, for orthodox economics all analysis has to be micro-founded and based on an isolated individual agent not influenced by other agents. By contrast heterodox economics argues that agents are interdependent and that the economy cannot be understood solely on the basis of individual preferences. Examples for this include in the emergence of macroeconomic phenomena in agent-based models or macroeconomic paradoxes, for instance of costs or debt in the post-Keynesian literature.

The main core of the economy in orthodox economics is determined by scarcity, thus the focus is on the proper allocation of consequently scarce resources. The assumptions of full capacity utilization and employment for instance guarantee that the respective resources are scarce. By contrast, heterodox economics and in particular post-Keynesian economics do not see resources as scarce (e.g. capital and labour are not fully deployed), and hence focus on increasing production and growth. This approach is however counter to heterodox schools like ecological economics which underlines the scarcity of natural resources and limits to growth (see subsection 1.2.2).

Finally, these distinct overall approaches of orthodox and heterodox economics lead to fundamentally different political implications. Whereas orthodox economics tend to believe in the merit of uninhibited markets and see little role for state intervention, heterodox economics generally view unregulated capitalism critical and call for state intervention.

Whereas orthodox economics is mainly associated with Neoclassical Economics, heterodox economics spans a range of different schools of economics with many commonalities and differences. Important heterodox schools and their characteristics include:

- **Marxist political economy** focuses its analysis on social structures and class and observes the systematic exploitation of workers in capitalism (Fine and Saad-Filho, 2017).
- **Feminist economics** underlines the structural gender dimension of the economic system. As such it, recognizes that the economy relies on activities outside its realm and focuses on the analysis of unpaid and care work (Himmelweit, 2017).
- Classical **institutional economics** highlights that individuals' preferences and behaviour are not fixed but context dependent and influenced by institutions (Hodgson, 2017; Vatn, 2017).
- **Complexity economics** sees the economy as a constantly evolving system driven by interaction between agents which leads to aggregate emerging phenomena (Kirman, 2017).

- **Post-Keynesian economics** analyses the economy as a demand-driven monetary production economy, where investment determines savings (Lavoie, 2014). This school of economics is described in more detail in section 1.2.4.

Notable, Feminist Economics recognizes the incompleteness of the system, as analysed by dominant economics (Himmelweit, 2017), and Marxist Economics observes the destructive character of capitalism (Fine and Saad-Filho, 2017). Still, Spash and Asara (2017) argue that what distinguishes **Ecological Economics** from other heterodox approaches is its explicit conceptualisation of the embeddedness of the economy in the ecological system and a consequently more critical attitude towards economic growth. Within heterodox economics, this thesis falls into this approach of Ecological Economics, the main characteristics of which are subsequently outlined.

1.2.2 Ecological Economics

As defined by Costanza (1989) “*Ecological Economics addresses the relationship between ecosystems and economic systems in the broadest sense.*” As such it conceptualizes the economy as being fundamentally embedded in the ecological and social system (Figure 1.3).

Overall, Ecological Economics (EE) is characterized by pluralism and transdisciplinarity

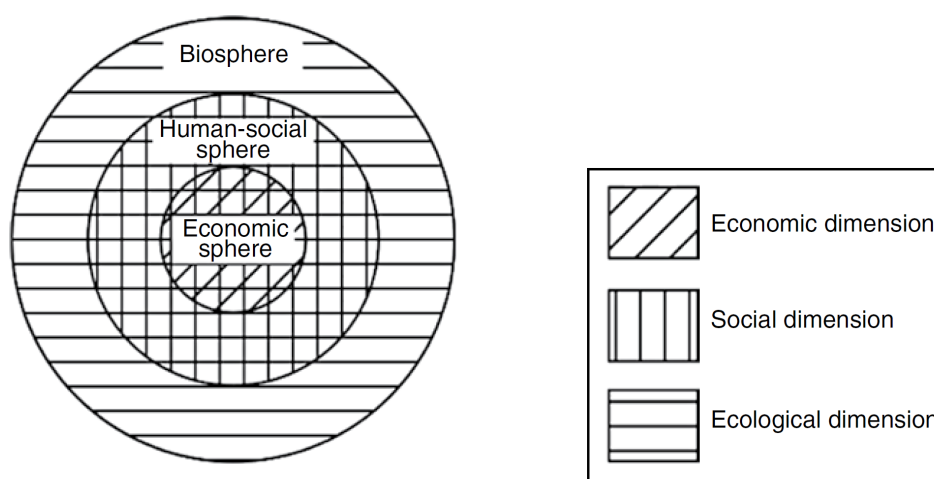


Figure 1.3: The economy as embedded within the social and ecological sphere. Source: Martinez-Alier and Muradian (2015).

(Røpke, 2005) and tends to underline the incommensurability of values to avoid reductionism to monetary or physical terms (Martinez-Alier et al., 1998). In contrast to environmental economics, EE generally views externalities as cost-shifting, where economic agents shift costs to

others for their own benefit (Kapp, 1978). Other contrasts to environmental economics – which is dominated by Neoclassical theory and assumptions – include: the tendency towards the principle of strong sustainability, a more sceptical attitude towards price mechanisms, the focus on the scale of the economy and the precautionary principle instead of efficiency and a more doubtful perspective towards the desirability and feasibility of continuous economic growth (Røpke, 2005; van den Bergh, 2001).

Following Georgescu-Roegen (1975) Ecological Economics applies the laws of thermodynamics to the economic process. The first law, stating that energy is always conserved, in combination with the Law of Conservation of Mass underlines the biophysical base of the economy. The economic process cannot create nor destroy matter or energy and hence relies on the environment for resource extraction and waste dumping. Consequently, the economy is an open system. In addition, the second law stipulates that entropy² in an isolated system only increases and hence underlines the quality of energy and the irreversibility of the economic process. As an open system the economy can receive low entropy inputs from outside (i.e. the environment), but the economic process always must increase entropy. Hence, energy transformation losses can be minimized but never eliminated. Consequently, energy cannot be recycled and there are ultimate limits to growth if economic activity relies on energy. Here, for instance Ayres and Warr (2005) show that energy services (i.e. useful work) are a crucial factor of production and underline the importance of the conversion efficiency³.

The consideration of the laws of thermodynamics subsequently leads to the logical step to quantify the material and energy throughput of the society. This study refers to the social metabolism, which encompasses all energy and material flows that are mobilized to (re)produce the stocks⁴ or biophysical structure of a social system (Fischer-Kowalski and Haberl, 2015; Krausmann, 2017). For a thorough understanding of the social metabolism, energy and material metabolism as well as consumption and production perspectives are required (Fischer-Kowalski and Haberl, 2015). For instance, it has long been argued that the energy available for use is crucial for the development of societies (Fischer-Kowalski and Haberl, 2015). Finally, the understanding of the social metabolism can provide insights for the study of ecologically unequal exchange (e.g. Hornborg, 1998) as well as ecological distribution conflicts (Martinez-Alier, 2021; Martinez-Alier et al., 2010).

Many of the insights generated from the EE perspective and approach to economic analysis underline the aforementioned cautious attitude towards the scale of the economy and the goal

²Entropy can be understood as the quantity of unavailable energy (Georgescu-Roegen, 1975).

³The conversion efficiency describes the ratio of useful work to raw exergy inputs. Exergy is the amount of energy that can be used for useful work (Ayres et al., 2007; Ayres and Warr, 2005).

⁴Stocks typically include durable infrastructure, humans and livestock (Fischer-Kowalski and Haberl, 2015).

of continuous economic growth. Out of these concerns, a post-growth scholarship developed which is presented below.

1.2.2.1 Post-growth

Ecological economists tend to be skeptical towards continuous economic growth for several key reasons. On the one hand, there is a clear deficiency of historic decoupling of economic growth and environmental impacts like resource use and emissions with respect to environmental targets (Haberl et al., 2020; Hickel and Kallis, 2020), as well as a generally doubtful view on sufficient developments in the future due to phenomena like the rebound effect (Brockway et al., 2021). On the other hand, there is a lack of evidence of a long term benefit of economic growth for well-being (Easterlin et al., 2010). These arguments are particularly relevant for rich nations, where economic growth is likely already uneconomic growth, i.e. the costs of economic growth exceed the benefits society derives (Daly, 2005). In addition, even if it were desirable, some ecological economists are doubtful about the feasibility of continuous economic growth (e.g. Jackson, 2019).

As also outlined above, the contemporary growth-based socioeconomic system is evidently unable to provide a good life for all within planetary boundaries and is highly unequal. Given these problematic implications of striving for continuous economic growth, the reliance of the current socioeconomic system on continuous expansion – with periods of lack of economic growth (e.g. recessions) leading to socially undesirable outcomes – is generally perceived as problematic by ecological economists. Consequently, a stream of literature critiquing and providing alternatives to the growth-based system has emerged, namely post-growth.

Here, post-growth is applied as an umbrella term for approaches that counter the current growth-based narrative and often follow all or several of the three goals of 1) addressing throughput and ecological impact, 2) reducing inequality and 3) ensuring well-being. In particular degrowth is defined “*as an equitable downscaling of throughput, with a concomitant securing of wellbeing*” (Kallis et al., 2018, p. 297). *Notably, a potential decline in GDP is not seen as a goal but as a probable outcome of such a transition (Kallis et al., 2018).*

Other approaches typically included under the term post-growth are the steady state economy – which aims to stabilize physical stocks as well as economic throughput and activity at sustainable levels (Czech, 2017; Daly, 1974), albeit it is still heavily based on neoclassical foundations (Pirgmaier, 2017) – and the economics of prosperity beyond growth – which addresses macroeconomic stability and is tightly linked to Ecological Macroeconomics and post-Keynesian economics (Durand et al., 2023). Within the field of post-growth, the concept of doughnut economics (Raworth, 2017), navigating society within a social foundation and ecological ceiling

(see also Figure 1.1), is also prevalent. These different concepts are not applied in isolation, but for instance degrowth in rich countries can be conceptualised as a transition towards a steady state economy that is more equitable on a global scale (Kerschner, 2010).

Prominent policy proposals in this literature include resource caps and green taxes, more progressive taxation and maximum incomes, universal basic income, a job guarantee and working time reduction (Fitzpatrick et al., 2022; Kallis et al., 2018). These policies aim to steer the economy towards a more ecologically sustainable structure on the one hand; on the other, they are designed to reduce inequality, guarantee the fulfillment of basic needs for everyone and inhibit unemployment, while freeing up time (Fitzpatrick et al., 2022; Kallis et al., 2018). Moreover, these policies would render individuals' livelihoods less dependent on for instance wage labour and hence economic activity (Vogel et al., 2024).

Thus, a post-growth future aims to provide high quality of social outcomes (e.g. universal access to aspects like healthcare, education, housing) that does not rely on growth, which, in particular in the degrowth context, allows for the reduction in economic activities that provide little social benefits while exhibiting high ecological costs (Hickel et al., 2021; Kallis et al., 2018). Whether a post-growth society is compatible with a capitalist socioeconomic system is often viewed skeptically (e.g. Blauwhof, 2012; Cahen-Fourot and Lavoie, 2016).

An integrated quantitative analysis of pathways towards a post-growth future however has been neglected in the dominant literature so far (Hickel et al., 2021; Keyßer and Lenzen, 2021; O'Neill et al., 2020) with an emerging literature within the field of Ecological Macroeconomics gradually addressing this gap.

1.2.3 Ecological Macroeconomics

Ecological Macroeconomics is a relatively recent field of research that acknowledges drawbacks in current dominant approaches towards analysing the socioeconomic transition, which it aims to improve. As Hardt and O'Neill (2017, p. 200) phrase it:

“the emerging literature on ecological macroeconomics is driven by a need to develop better analytical frameworks to understand economy-environment interactions on a macro-scale and to provide tools to manage the transition towards a sustainable economy.”

As aforementioned, the massive uncertainties arising from a fundamental transition of the socioeconomic system and its complexity and interaction with the environmental system, require integrated quantitative scrutinization of potential alternative transition pathways for informed decision making. While there already exists an established literature of Integrated Assessment

Models aiming to address this issue, dominant approaches therein suffer from a range of significant shortcomings.

1.2.3.1 Integrated Assessment Models and the issues with orthodox approaches

Integrated Assessment Models (IAMs) constitute a single modelling framework that combines insights from several domains (IPCC, 2022a). Here particularly relevant are the links between socioeconomic and ecological system, notably including the energy system.

There are generally two types of IAMs: detailed process based (DPB) and cost-benefit (CB) ones: While CB IAMs feature a higher level of aggregation in the representation of the economy, they integrate costs from climate change. These types of models are typically applied to determine optimal policies (where marginal costs of climate policies equate marginal benefits) and the social cost of carbon, reflecting the marginal costs of one additional ton of carbon emissions. DPB IAMs on the other hand exhibit a more disaggregated economic structure and are mainly used to study socioeconomic-technology-climate interactions and the effects of climate policies, i.e. mitigation analysis (IPCC, 2022a; Weyant, 2017).

For an integrated analysis, IAMs are often framed within shared socioeconomic pathways (SSPs; Kriegler et al., 2012; O'Neill et al., 2014) and representative concentration pathways (RCPs; van Vuuren et al., 2011), which together provide distinct futures of developments for society and climate (O'Neill et al., 2020; van Vuuren et al., 2014).

Thus, there is already a range of established IAMs that are capable of and have been applied to analysing transition pathways. However, this literature is dominated by general equilibrium models (Computational General Equilibrium (CGE) and Dynamic Stochastic General Equilibrium (DSGE)). Here, these types of IAMs are referred to as orthodox IAMs (see Figure 1.4). They mainly rely on neoclassical frameworks and assumptions like perfect markets and market clearing, optimising behaviour, rationality, representative agents, and complete information (Farmer et al., 2015; Hafner et al., 2020; Köberle et al., 2021). Even though DSGE models allow for the inclusion of some more realistic elements (like imperfect information), they still feature optimising and equilibrium behaviour (Hafner et al., 2020). The structure and assumptions of these orthodox economic models and IAMs render them ill equipped to analyse the socioeconomic implications of a fundamental system transition (e.g. unemployment and inequality) and lead them to provide problematic policy recommendations. These issues are subsequently outlined in more detail:

Beginning with problematic microfoundations that do not capture behaviour adequately (Stiglitz, 2018), the atomistic, rational optimising and fully informed agent reacts overly strong to price changes as other determinants of behaviour are ignored, creating a bias towards the effectiveness

of price policies (Hafner et al., 2020). Part of this behavioural issue is also driven by the use of the representative agent, wrongly assuming away institutional contexts (Asefi-Najafabady et al., 2021) and that a collection of agents might exhibit different behaviour towards policies (Farmer et al., 2015). As Mercure et al. (2019, p. 1030) puts it:

“Furthermore, a danger exists in interpreting the results of optimization models – inherently normative – in a descriptive philosophy. For example, cost-optimization and pure representative agent equilibrium models offer the attractive but potentially misleading suggestion that pricing policies are the best way to correct market failures such as climate change, since agents, when assumed to behave optimally, always find the best possible use of resources according to prices.”

In addition, Stern et al. (2022) argue that the optimisation framework in IAMs cannot adequately integrate deep uncertainty and under such conditions agents can tend towards more precautionary behaviour.

Regarding inequality, the aggregate representative agent framework prevalent in orthodox IAMs renders these models to neglect heterogeneity and distribution (Emmerling and Tavoni, 2021; Rao et al., 2017), with important policy implications. In particular, (Stern et al., 2022) note that ignoring distribution in the context of climate damage can reduce the derived climate policy ambition.

Another highly policy relevant issue is the cost of the transition, which orthodox IAMs tend to overestimate for several reasons. Firstly, marginal abatement costs increase with abatement (Stern, 2022), as shown by Kaya et al. (2017) for the CES (constant elasticity of substitution) production function, which by construction tends toward factor share preservation (Kaya et al., 2017). This is in contrast to actual developments in technological progress, which are characterized by economies of scale and learning effects, exhibiting logistic dynamics, not by diminishing returns (Kaya et al., 2017; Stern, 2022). Secondly, these models assume the economy to be at the efficiency frontier in the absence of interventions, which leads to any policy intervention incurring costs and negates their potential positive effect by addressing other occurring distortions or creating co-benefits (Köberle et al., 2021; Riahi et al., 2022; Stern et al., 2022), which in turn would call for a multiplicity of different types of measures (Stern et al., 2022). In addition, in the IPCC mitigation framework, climate damage is omitted in the reference scenario, preventing the inclusion of benefits of mitigation by reducing climate impact (Köberle et al., 2021; Riahi et al., 2022). Finally, the underlying neoclassical economic theory might also imply a more pessimistic attitude towards the economic impacts of such a transition (see subsection 1.2.4).

These issues illustrate that these models exhibit a bias towards little climate action and a carbon

price as the primary policy instrument (Stern et al., 2022). In addition, they feature characteristics that render them unsuitable to analyse fundamental and disruptive transitions:

The equilibrium framework dominant in IAMs is inadequate to capture fundamental features of extensive changes in complex systems (Stern, 2022), involving non-linear feedbacks and out-of-equilibrium dynamics (Solé et al., 2020). Adjustments are always smooth (Rezai et al., 2013) and amplification and persistence of shocks are not captured among others due to the omission of banks and the financial system (Stiglitz, 2018).

In addition, cleared markets and a supply side approach imply that all resources are fully utilized. This precludes involuntary unemployment, which is certainly a central aspect to recognize in the context of such a transition (Rezai and Stiglitz, 2016).

Other problematic elements of many orthodox IAMs include but are not limited to the role and handling of the discount rate (Pindyck, 2013; Stern, 2022), climate damage functions (Keen, 2021), the role of energy for production (Ayres et al., 2007; Keen et al., 2019) as well as aggregation and the role of structural change in the economy (Stern, 2022).

Overall, their reliance on neoclassical economic theory renders orthodox IAMs ill suited to analyse certain central aspects of the socioeconomic transition and their results inherently biased. Consequently, there is a need for different types of economic approaches and models.

1.2.3.2 Ecological Macroeconomics as part of a new economic approach

Acknowledging the fundamental issues of neoclassical economics, there has been a call for new economic approaches that should be able to capture among others path-dependency, non-linearity and feedback-loops, heterogeneity and interactions of agents, the financial and environmental system and multi-dimensional goals (Hafner et al., 2020). Ecological Macroeconomics is part of this new approach, focusing on the linkage to the environmental system, but ecological macroeconomic models are also able to address several other crucial points called for depending on their exact configuration (Hafner et al., 2020). Thus, ecological macroeconomic models can be conceptualised as a different form of IAMs, in distinction to orthodox IAMs. While both types are IAMs as defined in section 1.2.3.1, i.e. frameworks spanning several domains, they feature fundamentally different economic foundations, as illustrated in Figure 1.4.

While applying a range of different modelling frameworks (see section 1.3.1), the model closure in Ecological Macroeconomics tends to be dominated by post-Keynesian economics theory (Hardt and O'Neill, 2017). Accordingly, the following subsection will provide an overview of the key characteristics and implications of post-Keynesian Economics.

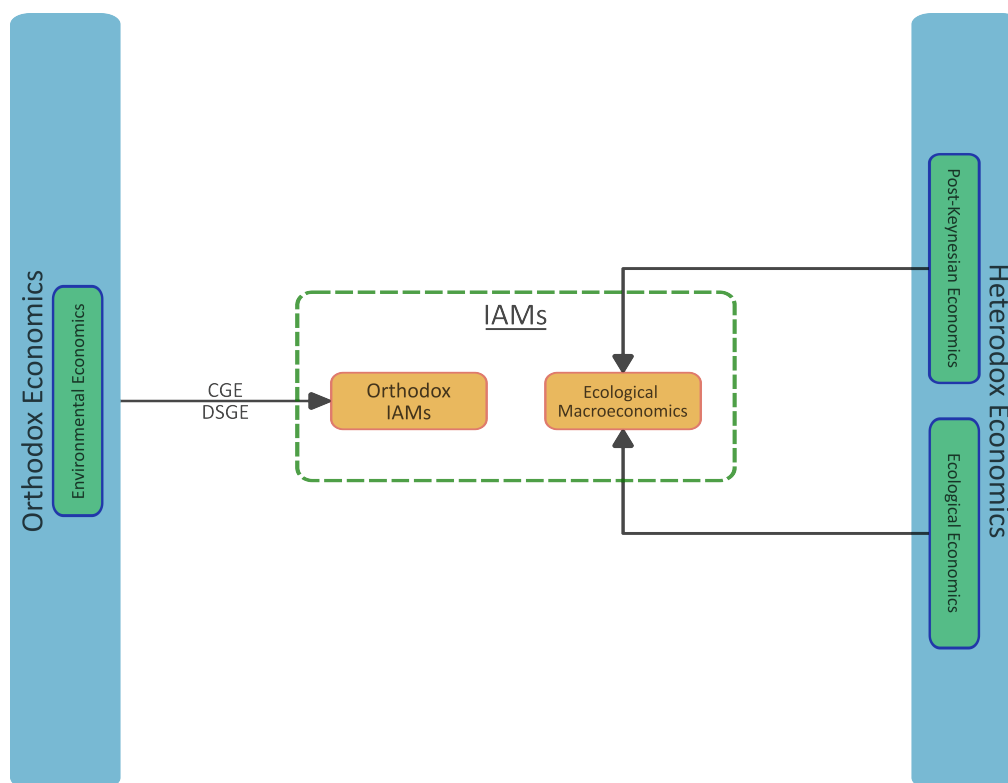


Figure 1.4: The economic schools dominating different types of IAMs. Whereas most Orthodox IAMs are based on Computational General Equilibrium (CGE) or Dynamic Stochastic General Equilibrium (DSGE) models with their neoclassical economic assumptions, the emerging field of IAMs based on Ecological Macroeconomics is often based on a combination of Ecological Economics and post-Keynesian Economics.

1.2.4 Post-Keynesian Economics

Already briefly mention in section 1.2.1, post-Keynesian Economics is an economic school in the heterodox tradition. This section outlines its main features while focusing on several main aspects that have crucial implications for modelling assumptions and mechanisms as well as for resulting dynamics and policy implications. Where not explicitly citing other sources, the following paragraphs are based on Lavoie (2014).

First and foremost post-Keynesian Economics (PKE) understands the economy as a monetary production economy. Money is not neutral but essential for and tied to production. Crucially, the money supply is endogenous and not controlled by the central bank. This means that banks create loans *ex nihilo* if the customer is deemed credit worthy and credits create deposits, not vice versa as in neoclassical theory. Thus, money is supply led and credit rationing can occur, in particular in economic downturns due to the pro-cyclicality of banking as Nadal (2015) notes. The central bank is a lender of last resort and acts to control the interest rate. Thus, it generally

determines the price but not the quantity of money. Accordingly, an increase in the demand for loans does not necessarily imply an increase in the interest rate as opposed to neoclassical theory where the interest rate aims to reconcile supply and demand for loans.

Inflation is not seen as a monetary phenomenon driven by excessive money supply but generally as arising from the supply side. Here, cost-push inflation is typically the result of a conflict regarding wages and markups, and hence the income distribution between workers and firms.

Another central aspect of PKE is the principle of effective demand. Say's law is rejected and the economy is demand driven in the short as well as the long run. Notably demand and hence economic activity are influenced by the distribution of income. For instance, in some post-Keynesian long run growth models (so called Kalecki-Steingl models), growth regimes are differentiated between wage and profit led growth depending on whether an increase in the profit share, via its impact on consumption and investment, decreases or increases demand, utilisation and growth (Hein, 2023).

Employment is not determined by supply and demand in a labour market, but by the effective demand for and consequently production of goods. A consequence of PKE assumptions is that investment determines savings and not the other way around as in neoclassical theory, which has fundamental implications for model outcomes for instance with regard to the energy transition as Mercure et al. (2019) argue: In neoclassical economic theory there is a limited pool of funds (savings) that can be used for investment, which is why policy induced investments in the energy transition crowd out potentially more productive other investment implying costs and reduced growth. On the contrary, in the post-Keynesian framework policy induced investment can boost economic activity. Thus, Mercure et al. (2019) find that whether this crowding out effect occurs determines fundamental differences between models as depicted in Figure 1.5.

Furthermore PKE captures fundamental uncertainty in agents' behaviour. They typically follow procedural rationality, i.e. they follow simple rules that allow them to make decisions under incomplete information. Even though this behaviour is generally non-optimising and satisficing, it is nevertheless rational as it renders agents capable of action. In addition, fundamental uncertainty prevents the full utilization of resources in the economy (Mercure et al., 2019), leading among others to unemployment. This, as well as path-dependence and irreversible time (see also Fontana and Sawyer, 2013; Berr, 2017) represent other features from PKE that are crucial with regards to the analysis of the socioeconomic transition.

Instead of identical preferences reflected in the representative agent approach in neoclassical economics, PKE argues for similar income groups exhibiting comparable consumption structures. Markets are characterized by imperfect competition leading to cost plus pricing. Thus, prices do not reflect scarcity but costs and market power. In addition, firms typically operate

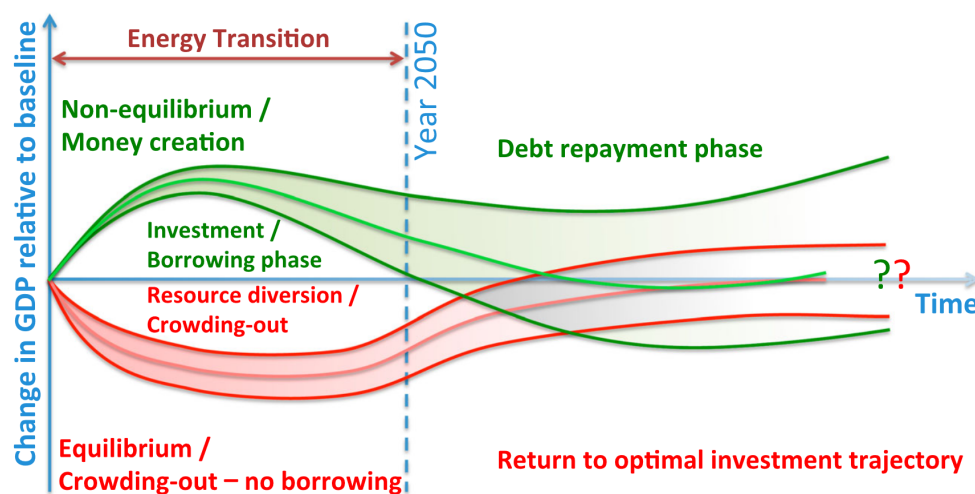


Figure 1.5: Comparative GDP developments for equilibrium and non-equilibrium modelling approaches with respect to a baseline following a policy-induced energy transition. Note that the equilibrium approach here is in line with neo-classical/orthodox economics and the non-equilibrium approach includes post-Keynesian economics. Source: Mercure et al. (2019).

below full capacity and face approximately constant marginal costs, leading to falling average costs⁵. Profits are crucial for firms to obtain external financing and hence to grow.

Thus, PKE provides a fundamentally different analytical approach to economics than the orthodox perspective. While there is a range of heterodox economic schools featuring distinct approaches to economic analysis (see section 1.2.1), the focus of Ecological Macroeconomics on PKE can be explained by several considerations as outlined subsequently.

1.2.4.1 Post-Keynesian and Ecological Economics

At first glance PKE might appear to be an unsuitable fit for EE given its focus on the demand side largely neglecting issues of scarcity and its overall positive connotation towards growth to reach full employment (Kronenberg, 2010; Lavoie, 2014). Yet there are many commonalities. Production theories between the two schools are in line, focusing on complementarity of inputs and being sceptical towards substitutability, as are the dynamic theories, underlining irreversibility and path-dependence (Kronenberg, 2010). In addition, the two schools' consumer theories are aligned, with PKE allowing for instance for limited substitutability, satiable needs as well as relative and conspicuous consumption (Lavoie, 2014). Moreover, Rezai et al. (2013) note that a non-utility maximising approach can allow for analyses considering various goals, on line with

⁵These characteristics might not hold for natural resources or production that takes a long time to adjust like agriculture, where supply and demand as well as the marginal costs are more akin to the behaviour and role they have in neoclassical theory (Lavoie, 2014).

value pluralism.

Regarding the apparent contradictions between the schools, firstly, as PKE mainly strives for growth to ensure full employment, other policies directed at this issue (e.g. a working time reduction) can potentially achieve the goal without the need for continuous growth. On the demand driven nature of PKE, aims to describe the current economic system and this is the system that has to be steered before supply constraints due to scarcity of natural resources are reached. In particular, the substantial overshoot of several planetary boundaries illustrates that the economy can cause dangerous environmental damage before being reined in by absolute scarcity constraints. This is not to say that scarcity constraints are irrelevant, but that contemporarily they do not dominate economic behaviour. There are indeed efforts to introduce environmental supply constraints like energy into ecological macroeconomic models with post-Keynesian elements that are generally demand driven, e.g. Dafermos et al. (2017) and Dafermos et al. (2018) or MEDEAS (Capellán-Pérez et al., 2020; Nieto et al., 2020) and pymedeas (Solé et al., 2020).⁶ Thus, while at first glance appearing contradictory to EE, PKE shares many fundamental concepts. Thus, while EE lacks its own macroeconomic theory, PKE provides a suitable framework for Ecological Macroeconomic models.

After having clarified the theoretical foundation of this thesis and Ecological Macroeconomics, the following section introduces specific methodologies applied.

⁶For a discussion on the role of energy constraints, also in relation to the energy transition, see for instance (Brockway et al., 2019).

1.3 Methodological framework

This section first introduces general modelling approaches used in Ecological Macroeconomics, before discussing ability of specific models to address the research question of this thesis and presenting the model deemed most suitable for this task, namely Eurogreen. Finally, the section presents the concrete methodology applied in the complementary consumption-based analysis.

1.3.1 Methodology in Ecological Macroeconomics

Overall, numerical ecological macroeconomic simulation models are based on one or more of the following frameworks and tools: input-output (IO) production, a stock flow consistent (SFC) framework, system dynamics, econometric and agent-based modelling (Hafner et al., 2020; Hardt and O'Neill, 2017), which are subsequently outlined:

Essentially, an IO model is a system of linear equations allowing to trace industries' output through the economy. Thus, it illustrates the interlinkages between industries, conceptualizing them as producers on the one hand, but also as consumers and allows to capture direct and indirect effects (Miller and Blair, 2009). Inputs are conceptualised as complements, rendering this production approach in line with PKE and EE (see section 1.2.4.1).

Stock-flow consistent principles call for the complete representation of all stocks on a balance sheet and all monetary flows between them for all sectors ensuring the representation of the economy to be coherent (Caverzasi and Godin, 2015; Hafner et al., 2020). The vital benefit this accounting framework adds, is the portrayal of financial aspects and their interdependence with elements of the real economy (Caverzasi and Godin, 2015; Godley and Lavoie, 2006; Hafner et al., 2020).

While IO and SFC are crucial elements within ecological macroeconomic models, they do not determine the overall modelling approach, but specific elements of the framework (i.e. the production structure and the representation and link of financial stocks and flows). As such, they can be combined with the three overarching approaches of system dynamic, agent-based and econometric modelling.

System dynamics (SD) centres around stocks, flows and feedback mechanisms (Sterman, 2000). The system essentially consists of nonlinear ordinary differential equations and has to be solved by way of numerical integration (Sterman, 2000). A system dynamics approach is aligned with the call for a new economics as it allows for irreversibility and path-dependence as well as complexity analysis and multiple equilibria (Hafner et al., 2020).

In contrast to the more aggregate framework in SD, agent-based models (ABMs) are characterized by a bottom-up approach. The interaction of autonomous adaptive agents and institutions

drives the evolution of the model and ultimately determines macroeconomic outcomes like GDP (Farmer and Foley, 2009; Hafner et al., 2020; Tesfatsion, 2003). Thus, they are tightly linked to the school of Complexity Economics (Section 1.2.1). As for system dynamics, ABMs represent many aspects of the call for a new economics, in particular complexity and the heterogeneity and interaction of agents (Hafner et al., 2020).

Lastly, the behaviour of agents in macro-econometric models is based the application of econometric methods to historic data (Dwesar et al., 2022). While allowing for elements like multiple equilibria they tend to be more limited with regards to aspects like complexity and non-ergodicity as well as heterogeneity, due to the prevalence of linear and log-linear relationships and their reliance on historic data (Hafner et al., 2020).

Thus, while all of the approaches have merit to them, their ability to analyse the specific object of this thesis depends the concrete model they are applied to.

1.3.2 Ecological Macroeconomic models to analyse a just low-carbon transition

As illustrated in subsection 1.2.3, orthodox IAMs are inadequate to analyse the ramifications of a fundamental transition of the socioeconomic system and accordingly to provide reliable policy recommendations. On the other hand, there is an emerging range of ecological macroeconomic models, with distinct strengths and weaknesses (see Hafner et al. (2020) for a recent overview). The research questions outlined in this thesis require an ecological macroeconomic model that allows for the analyses of the low-carbon transition, which is why this section is based on the models presented in Hafner et al. (2020). In particular, their ability to analyse inequality as well as their alignment with principles of EE are subsequently discussed.

While in theory well suited to reflect heterogeneity among agents, disaggregation and inequality, most ABMs in Hafner et al. (2020) are small, stylized (i.e. more abstract) models, the exception being the EURACE model (e.g. Teglio et al., 2019). However, the EURACE model, while allowing for heterogeneity among households, is aggregated on the production side with only one homogeneous consumption and investment good (Teglio et al., 2019), rendering it unable to capture sectoral dynamics and inter-industrial links. In addition, consumption goods are produced with a Cobb-Douglas production function, featuring substitutability between its two only inputs capital and labour. This type of production function is clearly not in line with the production theory of EE (see section 1.2.4.1).

On the other hand, the macro-econometric models E3ME (Dwesar et al., 2022) and PANTA RHEI (e.g. Lehr et al., 2012) are restricted in their representation of the income distribution.

Given data limitations, E3ME relies on a representative household (Dwesar et al., 2022). While it allows for a supplementary analysis of inequality, this does not feed back into the model and features a range of other limitations (Dwesar et al., 2022). Moreover, PANTA RHEI is restricted to a functional income distribution (Hafner et al., 2020).

Lastly, in terms system dynamics models, iSDG (Millenium Institute, 2021) features Cobb-Douglas production functions and is the only supply-led model in Hafner et al. (2020), which renders it prone to many aforementioned issues (see for instance section 1.2.3.1). Other system dynamics models are LowGrow (Jackson and Victor, 2019a) and MEDEAS (Capellán-Pérez et al., 2020), both featuring a demand-driven frameworks. While the relative strength of LowGrow is its implementation of financial and stock-flow consistent elements, MEDEAS features an elaborate conceptualisation of the energy system and biophysical constraints. Yet, LowGrow features no disaggregation among households or business sector firms and little differentiation among types of goods. On the other hand, MEDEAS does distinguish between sectors and goods, applying an IO production structure, but does not differentiate between types of households. Moreover, demand in MEDEAS is driven by an exogenous functional income distribution and exogenous expected economic growth, which can however be endogenously constrained by available energy (Nieto et al., 2020). Thus, while both of these models have their respective advantages, their high level of aggregation and exogenous distribution and growth features, render them unsuitable for an integrated analysis of inequality.

Finally, the Eurogreen model (D'Alessandro et al., 2020) exhibits a high level of disaggregation on the household as well as the sectoral side which, in combination with its national focus and detailed integration of the government and policies, allows for a thorough and integrated analysis of socioeconomic policies and inequality. In addition, its modelling of technological development allows for the inclusion of dynamics that are important for inequality, for instance labour productivity and its implications for employment. Thus, given the focus of the thesis on socioeconomic policies and implications, in particular inequality, the Eurogreen model is particularly well suited for this work and subsequently presented in more detail.

1.3.2.1 Eurogreen

Eurogreen is an established ecological macroeconomic dynamic simulation model with past applications to France (Cieplinski et al., 2021b; D'Alessandro et al., 2020) and Italy (Cieplinski et al., 2021a; Cieplinski et al., 2023; Distefano and D'Alessandro, 2023). Here, only an overview about the general features of the model is presented to avoid repetition and confusion between the different versions. The respective model and relevant features are outlined in more detail in

Chapter 2 and 3 and their respective supplementary materials.

Eurogreen is a national system dynamics model applying post-Keynesian economic theory and following stock-flow consistent principles, while integrating economy-environment interactions. Thus, it is generally demand driven, features cost-plus pricing, spare capacity and endogenous money. Fundamental uncertainty is reflected in non-optimising agent behaviour like investment and household consumption, which is based on income specific marginal propensities to consume and consumption structures. Production inputs (i.e. capital, labour and intermediate products) are complements and production follows an IO structure.

Eurogreen is calibrated using generally publicly available data and its reproducibility of historical data is evaluated. However, note that basing validation of Eurogreen on real world data is challenging, for instance since the system can change over time or through unmodelled developments as noted for energy economy optimisation models by DeCarolis et al. (2012). This also has to be taken into account when interpreting model results, where in particular a post-growth transition could fundamentally change the structure of the socioeconomic system. Thus, outcomes should not be interpreted as being predictive but rather as scenarios providing insights into potential narratives and futures in a complex system.

1.3.3 Environmentally Extended Multiregional Input-Output Analysis

In addition to the Eurogreen model, this work expands its national simulation based approach by conducting an empirical consumption-based analysis applying an environmentally extended multiregional input-output (EEMRIO) framework.

A consumption-based accounting framework assigns accountability of environmental impacts throughout the supply chain to final consumption, thus addressing leakage (Davis and Caldeira, 2010; Peters and Hertwich, 2008b). Here, bottom-up process-based life cycle assessments provide a higher level of sectoral or product detail than IO approaches, but they do not provide a coherent representation of the whole economic system, which is why they are prone to issues of system boundaries and exclusion (Castellani et al., 2019; Lenzen and Dey, 2000; Steubing et al., 2022). In this work, the IO approach (Leontief, 1936) is applied.

For global supply chains multi-regional input-output (MRIO) frameworks are applicable, which combine domestic matrices with trade data enabling to trace interactions between all included regions (Wiebe, 2016; Wiedmann et al., 2007), allowing to allocate the environmental impact along the global supply chain to the final demand in a country (Pablo-Romero and Sánchez-Braza, 2017).

This complementary perspective is particularly crucial against the backdrop of substantial and

increasing proportions of environmental impacts being embodied in trade, including energy use and GHG emissions (Wiedmann and Lenzen, 2018). Consumption-based accounting is particularly relevant when analysing rich societies which tend to feature net imports of environmental impacts (Wiedmann and Lenzen, 2018).

Thus, this approach contributes additional insights by allowing to more credibly assess the state and impacts of the energy transition and thereby the feasibility of (energy) transition scenarios. In addition, it adds aspects of international inequality and facilitates considering as well as addressing outsourcing and leakage.

1.4 Structure and contributions of the thesis

Following the Introduction, the thesis consists of three separate analytical Chapters and a final overarching Discussion & Conclusion Chapter. The subsequent Chapters are briefly introduced in this section and their contributions to major topics within the main related fields of research outlined. More specific contributions of each Chapter are presented in the Chapters themselves.

Chapter 2: Scenario discovery for a just low-carbon transition

Chapter 2 widens the scope for and underlines important potential interventions towards a just low-carbon transition. It explores a wide range of leverage points by applying a scenario discovery approach to the Eurogreen model. The Chapter highlights that the intertwined relations between inequality, economic activity and emissions imply that a just transition requires a number of coordinated policy measures. In particular the role of the energy transition, redistribution policies and economic growth are presented. The findings underline that studying policies and policy goals in isolation neglects trade-offs and synergies and hence can lead to distorted results.

Contributions:

Chapter 2 contributes to the field of Ecological Macroeconomics in several methodological aspects: It revises the Eurogreen model and produces a detailed documentation of the revised framework. Moreover, this work is the first to conduct a scenario discovery approach on an ecological macroeconomic model. This Chapter also adds important results to the literature. By assessing leverage points as well as synergies and trade-offs of interventions, it substantially broadens the perspective of and provides specific insights into relevant debates around the role of economic growth and post-growth, the emission-inequality trade-off, a just transition, as well as corresponding proposals of policies and policy packages.

Chapter 3: Emissions, inequality and employment – The role of the energy transition in a just low-carbon transition

Chapter 3 extends the analysis by improving the integration of the economic feedback effects from the energy transition into the Eurogreen model, allowing for a more comprehensive analysis of a just transition. It subsequently utilizes the revised Eurogreen framework to analyse the role of the energy transition within three socioeconomic paths (reference, redistribution and post-growth) characterized by distinct socioeconomic policies. The results highlight and compare the implications of socioeconomic as well as energy transition measures for a just transition.

Contributions:

Methodological contributions of Chapter 3 primarily consist of the improved integration of el-

ements of the energy transition into the Eurogreen framework, facilitating an enhanced assessment of implications of the energy transition. In addition, by analysing specific policy packages and pathways within the revised framework, it provides relevant insights into debates surrounding specific prominent policy proposals (like basic income or working time reduction) and socioeconomic transition pathways. Thus, it contributes to debates around the emission-inequality trade-off and the just transition as well as post-growth, also by adding to the analysis of notably underrepresented post-growth scenarios (Hickel et al., 2021; Keyßer and Lenzen, 2021; O'Neill et al., 2020).

Chapter 4: Trade and decoupling of fossil fuel use embedded in EU consumption

Chapter 4 complements and broadens the framework and perspective of the thesis by taking an empirical consumption-based approach to study the embedded fossil fuel footprint of EU consumption. In particular, it analyses decoupling dynamics and the impact of the energy transition, as well as underlines the role of trade patterns. In this context, the Chapter also assesses the potential role of a Carbon Border Adjustment Mechanism.

Contributions:

While Chapter 4 also contributes methodological novelties to the field of EEMRIO analysis, as outlined in the Chapter, it particularly highlights empirical results. By analysing the historical impact of the energy transition as well as the role of trade patterns, it adds important elements to the empirical assessment of transition policies, scenarios and pathways. Specifically, the results provide contributions to debates around empirical decoupling, as well as outsourcing and leakage and corresponding measures.

Chapter 5: Discussion & conclusion

Chapter 5 synthesises the findings of the previous Chapters to address the overall research question of how the EU can achieve a low-carbon transition that is socially just. In order to approach the answer to this question, it discusses the role of economic growth on the path towards a just transition as well as the relation between emissions and inequality, and presents specific policies and policy combinations that this work finds to be conducive towards a just low-carbon transition. The Chapter also outlines limitations and opportunities for future research.

Contributions:

While not directly contributing novel findings, this Chapter combines and structures previously gained insights around the main research questions. It thus contributes to all aforementioned topics, but in particular relates to debates around a just low-carbon transition in the EU and in this context refers to issues like the role of economic growth and post-growth, the emission-

inequality trade-off, and specific policies and policy combinations. Overall, the contributions of this thesis are mainly located in the field of Ecological Macroeconomics (Figure 1.4).

Chapter 2

Scenario discovery for a just low-carbon transition

This chapter corresponds to: Campigotto, N., Catola, M., Cieplinski, A., D'Alessandro, S., Distefano, T., Guarnieri, P., and Heydenreich, T. (2024). Scenario discovery for a just low-carbon transition. Discussion Papers, Dipartimento di Economia e Management – University of Pisa, no. 304. (*Currently in Review in Ecological Economics*)*

Abstract

There is currently no consensus among scholars on how to achieve a just low-carbon transition. To address this issue, this paper subjects a macrosimulation model to an extensive sensitivity analysis and then fits random forests to the simulation results to identify policy combinations that can reduce carbon emissions while promoting income equality. Our findings indicate that interventions aimed at supporting low-income groups result in an increase in these groups' energy demand and emissions, and that this negative effect should be offset through a faster deployment of renewable energy sources and measures that redistribute income away from top earners. Our analysis also confirms the importance of well-known policies such as carbon taxation and working time reduction. On the other hand, we do not find energy efficiency to be crucial in achieving inequality and emission reduction goals.

*Please note that the order of authors here does not reflect individual contribution but is alphabetical. Given the extensive and complex character of the Eurogreen model, this work is a highly collaborative effort. My main contributions in this work relate to the model revision, development and application, as well as the writing of the paper. More details on individual contributions are provided in the authorship statement of this Chapter.

2.1 Introduction

Seldom before have concerns about the environment and inequality been so closely related. Evidence from surveys conducted around the world indicates that inequality and climate change are frequently identified as among the most pressing issues of our day (see e.g. Bowles and Carlin, 2020; European Parliament, 2019), and public and scholarly debates abound on the kinds of measures necessary to secure people's livelihoods in the transition away from fossil fuels. These challenges are often framed within the concept of a 'just transition', which calls for action at the international, national and local levels to ensure a fair and equitable transition for all individuals (Fanning et al., 2020; McCauley and Heffron, 2018; Millward-Hopkins and Oswald, 2021; O'Neill et al., 2018). A related principle has become central to the way some policymakers conceive climate interventions. In late 2019, European Commission's President Ursula von der Leyen presented the European Green Deal as 'a strategy for growth that gives more back than it takes away', stressing that 'we have to be sure that no one is left behind [...] this transition will either be working for all and be just or it will not work at all' (European Commission, 2019).

The need for tools to understand and guide the transition to sustainability has given impetus to the emerging field of ecological macroeconomics (henceforth EM; Rezai and Stigl, 2016). The models developed in this literature typically feature a range of economy-environment interactions, energy use, and disaggregated production and consumption (Barker et al., 2016; Dafermos et al., 2017). Moreover, consistent with empirical evidence (Haberl et al., 2020), they share a general skepticism about the possibility of rapidly achieving an absolute decoupling of economic activity from environmental pressures.

It is perhaps surprising, however, that despite burgeoning advocacy for just transition efforts, EM has paid relatively little attention to how to jointly meet low-carbon and distributive goals. Research in this area tends to focus on the energy-environment-growth nexus, while inequality and other aspects of well-being receive considerably less attention (Hardt and O'Neill, 2017). Often the income distribution is considered only in functional terms (that is in terms of profit and wage shares, e.g. Dafermos et al., 2017; Jackson and Victor, 2016), which makes it difficult to thoroughly assess inequality. Inevitably this modelling stance influences the kinds of policies EM seeks to examine: as detailed in Section 2.2, most studies are concerned with direct climate mitigation measures, whereas those investigating the impact of socioeconomic policies are considerably fewer in number.

A different but related point is that the standard approach in EM is to consider pre-conceived scenarios featuring a small number of policies. This approach consists in choosing one or more

(usually two) policy measures that are deemed relevant to the context being studied, translating them into suitable input parameters, and simulating the resulting dynamics. As a consequence, on one hand, research ends up being guided by *a priori* prescriptions on how to address social and environmental challenges; on the other, a large portion of the policy space is left unexamined.

This paper proceeds in the opposite direction, taking a scenario discovery perspective (Groves and Lempert, 2007; Lempert et al., 2006) to explore interventions for a low-carbon, low-inequality transition. We introduce a revised version of the Eurogreen macrosimulation model (D'Alessandro et al., 2020) and use data from about 16,000 simulation runs to identify policy combinations that address distributive and climate challenges. Our findings and recommendations are the result of an ex-post assessment. First, the model is repeatedly run within the parameter space. In each simulation, more than one hundred parameters are randomly drawn from a wide range of possible values. Second, random forests are used on the database of simulation results to understand which parameter combinations can simultaneously improve social, economic and environmental indicators. Finally, successful combinations are interpreted either as policy prescriptions or modelling choices. The paper addresses questions such as: what policy combinations, if any, make decarbonisation and inequality reduction compatible? Does the pursuit of growth affect the set of policies available for achieving a just low-carbon transition? Has the literature overlooked relevant policy alternatives or overemphasized others?

The results indicate that the achievement of low-carbon-and-inequality goals can benefit from interventions targeted at both ends of the income distribution, which however have different effects on growth and emissions. Policies supporting bottom income groups do certainly play a role in addressing inequality, but the consequent increase in these groups' demand for goods and energy calls for stronger environmental policies to cut emissions. Conversely, measures that redistribute income away from high earners can make the income distribution more equal while balancing changes in energy consumption and emissions across income groups. Our findings further suggest that although a just low-carbon transition neither prevents nor requires sizeable GDP growth, the primacy of economic growth as a policy objective makes the path towards a just transition narrower, that is it reduces the range of feasible policy combinations.

The remainder of the paper is organised as follows. Section 2.2 presents some stylised facts about the use of scenarios in EM. Section 2.3 describes the main features and novelties of the Eurogreen model, and then introduces our scenario discovery approach. Section 2.4 discusses the main results and their policy implications. Section 2.5 concludes and suggests avenues for further research.

2.2 Scenarios and policies in ecological macroeconomics

To place our work in perspective, we conducted a systematic review of the use of scenarios in the EM literature. We limited the analysis to this specific research area to improve comparability with the model presented in this paper. The term ‘scenario’ denotes a consistent, model-based description of how the future may evolve under a certain set of input assumptions. Different scenarios result from alternative assumptions, which in turn reflect different policies or hypotheses about socio-economic and environmental conditions (Moss et al., 2010). The full list of articles and scenarios is available for download at Zenodo (see Code availability section below).

All publications dated 2010 or later and retrieved by posing the query ‘ecological macroeconomics’ in Scopus (34 results) and Web of Science (33 results) were initially considered for analysis. Additionally, we considered the 44 publications included in the literature review by Hafner et al. (2020, Table 2). After discarding duplicates from the three sources, we were left with 87 publications. The sample was then restricted to articles featuring scenarios and published in peer-reviewed journals. This reduced the number of publications to 25.

The next step was to determine how many scenarios were simulated in each article and what policies comprised each scenario. To do so, we went through all articles and identified all input parameters that varied across simulations. Often these exogenous parameter changes are framed as hypotheses rather than policy measures. This is the case, for instance, for variations in the pace of technological progress, changes in the wage and profit shares of income, and higher or lower projections of emissions and temperature increases. We made no explicit distinction between policies and hypotheses, as we understand that both are equally important in allowing articles to make their points. Moreover, whether a model can incorporate well-defined, real-world policies rather than general hypotheses depends on its level of abstraction and geographical coverage. For brevity, hereafter we use the term ‘policies’ to refer to both actual policies and hypotheses.

The review led to the identification of 199 scenarios and 105 policies. The latter were finally grouped into 12 environmental and socio-economic categories (see Figures 2.1a and 2.1b for the complete list of policy groups and for how groups were classified as environmental or socio-economic). The classification process was largely based on the frequency with which similar policies appeared in the literature. Some groups, such as *Carbon price* and *Working time reduction*, are narrow in scope, as the policies that belong to them were observed frequently and implemented in similar ways for simulation; the former comprises carbon taxes, cap-and-trade schemes (such as the EU-ETS) and border carbon adjustments.

Other groups consist of several related policies. We briefly describe them in turn. The *Di-*

rect RES investments and incentives group mainly includes variations in the share of renewable energy sources (RES) in total energy use; feed-in tariffs for wind and solar energy were also grouped in this category. The *Environmental taxes and regulations* group comprises all kinds of environmental taxes and regulations, excluding those included in the Carbon price and Direct RES investments groups. Examples include regulations to prevent the construction of new coal plants, material input taxes, taxes on the consumption of carbon-intensive goods, and subsidies for green capital. The *Technological progress* group is composed of changes in energy and fuel efficiency, input-output technical coefficients, labour productivity, and R&D investments.

The *Income distribution* group includes changes in the functional income distribution, basic income, job guarantee, and rebates to households from carbon tax revenues. *Aggregate demand* policies encompass direct variations in aggregate demand components, such as pro- or counter-cyclical government spending. Finally, the *Behavioural change* group comprises all changes in agents' behaviour, including voluntary reductions in private consumption and network and snob effects in agent-based models; by definition these are not actual policies, although they may depend indirectly on policy measures such as information campaigns on climate change or energy efficiency improvements in household appliances.

Policies in the remaining categories — *Population growth*, *Climate damage*, *Green finance* and *Financial Stability* — were observed sporadically, but were considered too different from other policy measures to be grouped together with them. The *Climate damage* group consists of alternative hypotheses about the functional form of the climate damage function (e.g. quadratic rather than linear) and the likelihood of extreme climate events. The *Green finance* group includes reductions in interest rates and various forms of credit rationing influencing investment in green capital. The *Financial stability* group is comprised of bailout measures and other policies to sustain the financial system in face of increased climate risks.

Figure 2.1a shows the frequency of the 12 policy categories. About 70 percent of policies consisted of environmental measures. Of these, 21 percent belonged to the Environmental taxes and regulations group, 15.2 percent to the Technological progress group, 12.4 percent to the Carbon price group, and 10.5 percent to the Direct RES investments group. Socio-economic policies add up to the remaining 30 percent of the total, with Income distribution, Aggregate demand, and Behavioural change being the most common groups (8.6, 6.7 and 5.7 percent, respectively).

Figure 2.1b was constructed by considering pairwise combinations of policy types. Each unique policy type combination featured in a given scenario adds 1 to the corresponding cell in the matrix. For example, a scenario consisting of a single Carbon price policy adds 1 to the (Single, CP) cell; a scenario with one Carbon price and one Income distribution policies adds 1

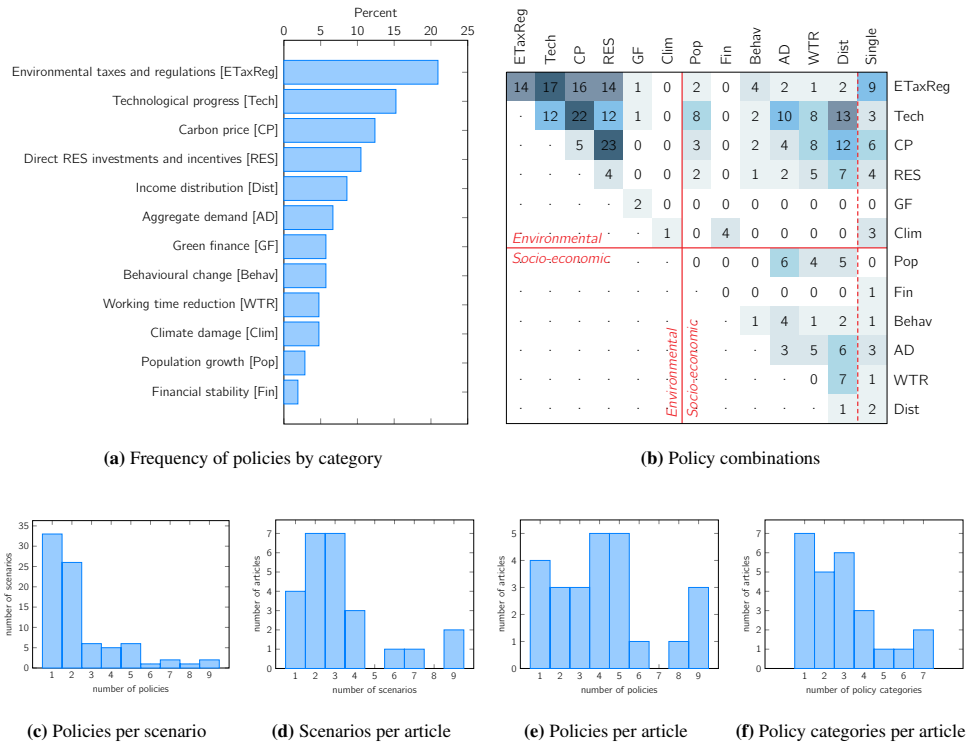


Figure 2.1: Summary of literature review

to the (Dist, CP) cell; a three-policy scenario featuring two different Carbon price policies and one RES policy adds 1 to the (CP, CP) cell and 2 to the (CP, RES) cell. The figure shows that most policy mixes combined policy instruments from the Environmental taxes and regulations, Technological change, Carbon price, and Direct RES investments groups. Socio-economic interventions were most often observed together with Carbon price and Technological progress policies; this is so particularly in the case of Aggregate demand, Working time reduction, and Income distribution.

The four bottom-row panels of Figure 2.1 provide summary information about the number of policies per scenario and the numbers of scenarios, policies and policy categories per article. About 60 percent of scenarios featured either 1 or 2 policies (Figure 2.1c), with the mean number of policies per scenario being 2.43. On average, articles included 3.28 scenarios and examined 4.20 policies from 2.88 policy groups. Eighteen out of 25 articles considered 3 or less scenarios (Figure 2.1d) and 3 or less policy categories (Figure 2.1f), and all except five articles considered 5 or less policies (Figure 2.1e). Overall, these findings seem to indicate that a gap exists between the positive and policy aspects of ecological macroeconomics. On one hand, research in this field seeks to investigate the complex relation between social, economic and environ-

mental sustainability. On the other hand, however, scholars tend to rely on scenarios featuring a small number of policies, which are hardly suitable for the analysis of multidimensional policy objectives. This point is further discussed in Sections 2.3.2 and 2.5.

2.3 Methods

2.3.1 The model

The Eurogreen model is based on Post-Keynesian economic theory and combines system dynamics and stock-flow consistent methods. It is formulated at the country level and has previously been applied to France (Cieplinski et al., 2021b; D'Alessandro et al., 2020) and Italy (Cieplinski et al., 2021a). The simulations presented in this study were obtained for Italy and run from 2010 to 2050, while calibration is based on data for the period 2010-2020. Exogenous shocks to private consumption, investment, exports and imports were included to account for the Covid-19 pandemic. This section outlines the model structure, with emphasis on the new features introduced in this version. For a comprehensive discussion, see the supplementary material.

Aggregate demand drives production and consists of exports and government spending (which are mainly governed by exogenous trends), consumer spending, and gross fixed capital formation.

Households' consumption depends on disposable income, income-dependent marginal propensities to consume, and prices. Consumption is allocated among 16 different goods¹ as a function of relative price changes, with elasticities ranging from 0 to 1.5. Disposable income is determined by government transfers (such as unemployment benefits and pensions), labour and financial income, social security contributions, and income taxes. These variables differ by skill, gender and employment status (employed, unemployed, out of labour force, and retired, with the top 1 percent of individuals designated as capitalists or rentiers who only earn financial income). The new version of the model incorporates gender differences among agents, leading to a total of 25 different population groups and allowing for a thorough analysis of distributional issues. Moreover, since consumers' behaviour depends on income and prices, the model captures the feedback effects that arise from distributional and price changes, which in turn may result from such causes as technological progress, wage increases, or the introduction of a carbon tax.

Employment varies by skill and gender, and is determined as a function of labour productivity at the industry level, previous-period output, and weekly work hours. The skill composition

¹Categorised according to the Classification of Individual Consumption According to Purpose (COICOP) scheme.

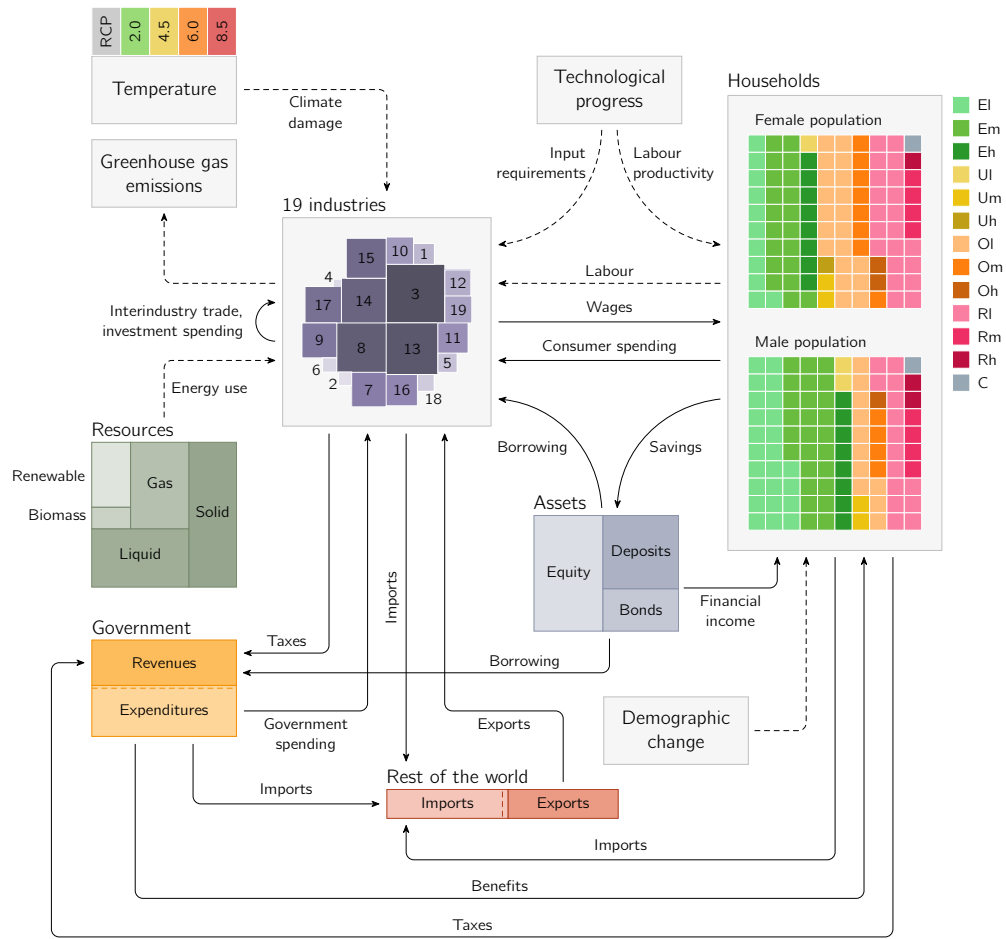


Figure 2.2: Model overview. The solid and dashed arrows represent monetary and non-monetary flows, respectively. The Households, Industries, Resources, Assets, Government, and Rest of the world boxes summarily represent first-period simulation results. The dashed lines in the Government and Rest of the world boxes are drawn for reference and cut the area of the rectangles in half. Abbreviations in the Households box describe the following groups: E = employed; U = unemployed; O = out of labour force; R = retired; l = low-skilled; m = middle-skilled; h = high-skilled; C = capitalists. List of industries: 1 = Agriculture, forestry and fishing; 2 = Mining and quarrying; 3 = Manufacturing; 4 = Coke and refined petroleum products; 5 = Electricity, gas and steam; 6 = Water supply; 7 = Construction; 8 = Wholesale and retail trade; 9 = Transportation and storage; 10 = Accommodation and food service activities; 11 = Information and communication; 12 = Financial and insurance activities; 13 = Real estate activities; 14 = Professional, scientific, technical, administrative and support service activities; 15 = Public administration and defence; 16 = Education; 17 = Human health and social work activities; 18 = Arts, entertainment and recreation; 19 = Other.

of labour demand reflects industry-specific historical trends, whereas the gender composition depends on the difference between female and male unemployment rates within each skill group. Pensions and unemployment benefits are paid in proportion to wages, which in turn are affected by labour productivity, inflation, and group-specific employment rates. Financial income is made up of dividends on equity and interests on government bonds.

Industries adjust their desired investment on the basis of the difference between actual and normal capacity utilisation, striving to produce at the normal rate of capacity utilisation. However, investment spending is constrained by profits after debt repayment and taxes, which determine the maximum investment that each industry is able to finance. Another novel feature of this version of the model is that financing conditions are negatively affected by industry leverage.

Output is obtained by multiplying domestic final demand by the Leontief inverse matrix, and is constrained by fixed capital and capital productivity. Input-output technical coefficients change endogenously over time with technological progress. The innovation process can be summarised as follows. In each period, one or more new technologies can be discovered with a certain probability. Innovations may be labour-saving, intermediate-input-saving, or both. The probability that a new technology is discovered depends on labour and intermediate input costs; for example, if in a certain industry the unit cost of intermediate inputs increases relative to the unit cost of labour, then firms become less likely to develop new labour-saving technologies than new intermediate-input-saving technologies. Once a technology is discovered, the extent of technological progress in each industry is randomly determined from normal distributions calibrated to historical data. Finally, in each industry a choice is made (based on cost minimisation criteria) about whether to adopt a new technology and, if so, which one. This version of the model allows for improvements in production efficiency even in the absence of new innovations, due to the progressive diffusion of the newest available technology.

Technological progress also affects energy demand by increasing energy efficiency. Energy flows are linked to real domestic monetary flows, with energy demand to output coefficients specific to every industry-to-industry cell of the input-output matrix, and industry-specific coefficients for household consumption. Energy demand is met by five energy sources (solid, liquid, gas, biomass and renewables) according to industry- and household-specific shares, which change in time according to the projections of MISE-MATTM-MIT (2019). Greenhouse gas emissions are then determined, once again, using industry- and household-specific energy-source-to-emissions conversion coefficients. This approach allows energy demand and emissions to respond to changes in both the amount and composition of inputs required for production.

This new version also includes a climate damage function, defined as the fraction by which production varies relative to what it would be in the absence of global warming. Temperature projections are exogenous and based on Representative Concentration Pathways (RCPs). The functional form of climate damage draws on Desmet and Rossi-Hansberg (2015), with modifications to account for extreme climate events. In each period, industry-specific damages are drawn from a beta distribution and deducted from output, which is equivalent to an increase in

intermediate input requirements.

The Government collects social security contributions, value added taxes, carbon taxes (if levied), and taxes on labour, financial and corporate income. It also makes transfers to households and purchases goods and services. Prices are determined as a markup over unit production costs. The population dynamic is exogenous and depends on demographic projections.

Although this version of the model addresses several shortcomings of the original version (such as the inclusion of a damage function, price elasticities of consumption demand, the introduction of leverage as a determinant of financing capacity, and a direct correspondence between inter-industry trade and energy demand), other limitations remain. The share of energy demand met by renewable sources is driven by an exogenous trends based on the growth rates that achieve national targets (MISE-MATTM-MIT, 2019). Total energy demand depends directly on output and, therefore, on the level overall level of economic activity and on changes in technical coefficients; however, there is no direct feedback from the expansion of renewable sources to other socio-economic variables in the model.

The results are also influenced by the level of aggregation of industries and individuals. This is most relevant for income and carbon taxes. The lack of within-group variability in income results in a very limited number of individuals whose income falls in the highest and second-highest tax rate brackets. The homogeneity of emissions within industries also reduces the capacity of a carbon tax capacity to incentive renewable energy adoption among the most polluting plants in an industry. Moreover, homogeneous groups of income earners tend to reduce inequality due to a lower dispersion of income on the top end of the distribution. Finally, the model does not include non-energetic resources (such as water, land and raw materials) and does not consider how the use of natural resources might impact the local ecological processes (e.g. biodiversity loss, water pollution).

2.3.2 Simulation and data analysis approach

The use of scenarios in macrosimulation models provides an intuitive and appealing way to envision plausible futures and evaluate policy alternatives in contexts of deep uncertainty. As shown in Section 2.2, a common practice in the literature is to construct ex-ante a limited number of alternative scenarios, each featuring a different combination of input parameters, and frame them as possible states of the world. However, this approach is complicated by issues concerning the choice of how scenarios are constructed, which strongly relies on the modellers' prior knowledge about the phenomena being examined and the main causal relations among the model variables.

An alternative strategy, which we adopt in this study, is to identify relevant scenarios that

emerge spontaneously from a database of simulations (Gerst et al., 2013; Groves and Lempert, 2007; Guivarch et al., 2016; Lempert et al., 2006). This approach requires running the model a large number of times, each time with a new set of inputs randomly drawn from the parameter space. The database of simulation outcomes is then partitioned into regions according to a pre-defined criterion, and statistical learning methods are used to determine which parameters best predict the simulations' position in the partitioned outcome space.

The data used in our analysis were obtained by running the Eurogreen model 50,000 times. In each simulation run, the value of 107 different parameters was selected at random from a given distribution. The list of parameters and their respective distributions are given in Appendix 2.C, while a detailed discussion of the role of each parameter is provided in the supplementary material. Some parameters, named 'structural parameters', can take only integer values corresponding to different economic or environmental assumptions. For instance, the *Warming scenarios* parameter can take values ranging from 1 to 4, each corresponding to a different Representative Concentration Pathway and temperature projection. The *Carbon tax* parameter takes value 1 if emissions are taxed (starting in 2022), and 0 otherwise. The *Output constraint* parameter indicates whether production capacity is constrained by fixed capital (value 0) or not (value 1). The *Investment constraint* parameter takes a value of 0 if no restriction is placed on firms' investment, a value of 1 if firms must internally finance a fixed proportion of their investment expenditures, and a value of 2 if this proportion depends on leverage.

Other parameters, named 'non-calibration parameters', can vary over a wide range of values, which generally spans from -50 to $+50$ percent of the parameter's initial value. This group includes, among others, the depreciation rate of fixed capital, the number of working hours per year, the pension-to-wage and unemployment benefits-to-wage ratios, and several tax rates. Each non-calibration parameter follows a linear trend, starting from a given initial value in 2022 and reaching the randomly selected value in the final simulation year (2050). For example, if a value of 0.25 is drawn for the *Value added tax (VAT) rate* parameter, then the VAT rate will remain constant from 2010 to 2022 and then progressively increase in a linear fashion, ultimately exceeding its initial level by 25 percent by the end of the simulation run.

Finally, the 'calibration parameters' group mainly comprises the parameters for which the model was calibrated. Since calibration was performed to fit observed data, these parameters usually span a smaller range of values than non-calibration parameters. Moreover, their value is randomly drawn in the first period and then remains constant throughout the simulation run. Examples include the sensitivity of wages and consumption to price changes, and the percentage of profits paid out as dividends.

Although most parameters can easily be understood in policy terms, there are also some

parameters which are beyond the direct control of policies. The latter usually reflect different, alternative modelling assumptions. For example, changes in the *output constraint* parameter can help understand the consequences of letting output be constrained by the capital stock. Thus, our work speaks to both the policymaking and EM modelling literatures.

Out of the 50,000 simulations, only 16,023 were deemed suitable for analysis. We stress that this is not a cause for concern. Since the output of each run depends on the joint realisation of many random variables, it can well be the case that a combination of several high- or low-valued parameters is drawn as input data, leading to economically implausible results. This point is substantiated by the kernel density estimates presented in Appendix 2.A, which show that the share of simulations suitable for analysis and which featured extreme-valued parameters is small. The most common reasons for dropping simulations were a negative or very high (300 percent or more) debt-to-GDP ratio (about 24 and 9 percent of dropped simulations, respectively) and a negative unemployment rate of male and female low, middle, or high-skilled workers (37 percent, 7 percent and 22 percent, respectively).²

The simulations were grouped into mutually exclusive classes according to four different criteria, each reflecting a policy objective: (1) reducing greenhouse gas emissions; (2) reducing income inequality (measured by the Gini coefficient of net income); (3) reducing emissions and inequality; (4) reducing emissions and inequality while increasing GDP. We refer to these as the GHG, Gini, GHG-Gini, and GHG-Gini-GDP objectives, respectively. In the GHG case, simulations were grouped into a ‘high GHG’ or ‘low GHG’ class depending on whether the level of emissions in the final simulation period was above or below the last-period median value. Similarly, in the Gini case, simulations were grouped into a ‘high Gini’ or ‘low Gini’ class. In the GHG-Gini case, the classification was based on two different thresholds resulting in four classes: ‘high GHG, high Gini’, ‘high GHG, low Gini’, ‘low GHG, high Gini’, and ‘low GHG, low Gini’. Finally, in the GHG-Gini-GDP case, simulations were grouped into eight different classes: ‘high GHG, high Gini, high GDP’, ‘high GHG, high Gini, low GDP’, . . . , ‘low GHG, low Gini, low GDP’.

For each policy objective, we trained a random forest that predicts the outcome of simulations. To do so, we first split the data into a training set (70 percent of simulations) and a test set (30 percent) using stratified sampling. In the GHG-Gini and GHG-Gini-GDP cases, due to moderate imbalances in the class distribution of simulations, we also used a combination of

²This results from an incompatibility between the randomly drawn input values determining the gender and skill distribution of labour demand (e.g. *Investment sensitivity*, *Technologies probability sensitivity*; see Appendix 2.C), and those contributing to variations in the labour force participation rate (*Labour force participation sensitivity*), the substitution between male and female workers of the same skill level (*Gender employment substitution sensitivity*), and skill transitions (*Skill transition sensitivity*).

majority class undersampling and synthetic minority class oversampling of the training data to improve the sensitivity of under-represented classes. Appendix 2.B provides a description of the resampling procedure and shows that our results are robust to alternative resampling methods.

Random forests combine the results of many classification trees, each based on a bootstrapped subset of the training data. A classification tree is grown by partitioning the space of input parameters using a recursive binary splitting algorithm, so as to predict which outcome class each simulation belongs to (Breiman et al., 1984). Initially, at the top of the tree, all simulations belong to a single region. The algorithm then repeatedly splits the data into increasingly homogeneous regions. Each split is made according to a specific cut-point value of a selected input parameter. Simulations in a region are subdivided in two regions based on whether or not their value of the selected parameter exceeds the parameter's cut-point value. The parameter and its cut-point value are chosen to minimise the classification error rate, i.e. the fraction of observations in a region that do not belong to the region's most common outcome class. At each step of the splitting procedure, a different random subset of input parameters is drawn as split candidates. Following standard practice, the number of split candidates was chosen to approximate the square root of the total number of parameters (James et al., 2021); that is, at each new split in the tree a random sample of 10 parameters was drawn as split candidates from the full set of 107 parameters.

The prediction of each random forest was obtained by averaging the predictions of 500 classification trees. In the GHG and Gini cases, the random forest correctly predicted the outcome of more than 83 and 81 percent of simulations, respectively, whereas the GHG-Gini model had a 65 percent accuracy rate. In the GHG-Gini-GDP case, prediction accuracy decreased to about 53 percent, which is modest but in line with other studies on multi-objective scenario discovery (e.g. Gerst et al., 2013).

The predictive value of input parameters was evaluated based on their permutation feature importance, which is a measure of the decrease in the performance of the random forest when the values of a parameter are randomly shuffled (Breiman, 2001; Strobl et al., 2008). The shuffling procedure breaks the relation between the parameter and the outcome variable, and the resulting variation in model performance is indicative of the extent to which predictions depend on that parameter. A high permutation feature importance denotes an important predictor of the simulation outcome.

2.4 Results

2.4.1 The emissions-inequality-GDP nexus

Before considering which parameter values are compatible with the simultaneous curtailment of emissions and inequality, it is useful to assess the likelihood of achieving both goals simultaneously. Figure 2.3a plots greenhouse gas emissions and the Gini coefficient of net income for all simulations in the final year 2050. Starting from the same position in 2010, different parameter combinations can drive the economy to a variety of outcomes. The Gini coefficient ranges from 0.13 to 0.30, starting from an initial value of 0.23; greenhouse gas emissions fall between 50 and 85 percent relative to their initial value of more than 500 Mtons CO₂ eq./year; real GDP ranges between 1.25 and 2.25 trillion euros, starting from an initial level of 1.46 trillion, meaning that the GDP growth rate ranges from slightly negative to moderately positive values.³

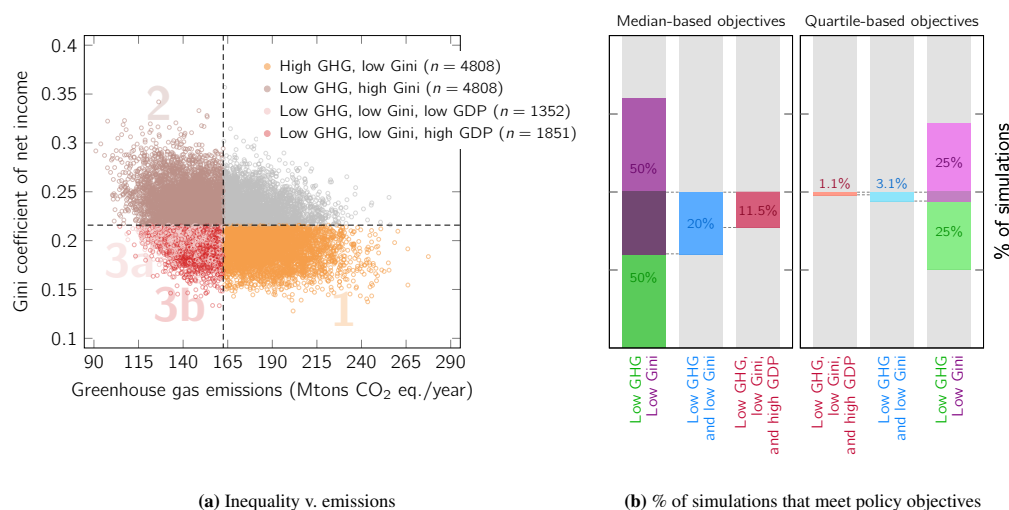


Figure 2.3: Simulation results. The left-hand panel shows the relation between greenhouse gas emissions and inequality in the final simulation year 2050. Each point represents a different simulation. The vertical and horizontal dashed thresholds are the median values calculated from all simulations. The right-hand panel shows the percentages of simulations that meet median-based (columns 1-3) and quartile-based (columns 4-6) policy objectives.

Figure 2.3a shows a negative correlation between inequality and emissions, suggesting that decarbonisation efforts can have an adverse effect on income distribution. Furthermore, efforts to counterbalance this effect through economic growth may jeopardise low-carbon goals, given that GDP is positively correlated with emissions (see Figure 2.A.1).

³To see this note that a yearly growth rate of about 1.08 percent between 2010 and 2050 would result in a real GDP of 2.25 trillion euros at the end of the simulation. Note also that demographic change cannot explain the differences in GDP in the model, because population does not vary across simulations.

Figure 2.3b shows the percentages of simulations that meet one or multiple policy goals. Each grey column represents the whole set of 16,023 simulations. The green and violet rectangles are the subsets of simulations that meet the ‘low GHG’ and ‘low Gini’ objectives, respectively, while the blue rectangles denote the simulations that meet the ‘low GHG, low Gini’ objective and the red rectangles are the simulations that meet the ‘low GHG, low Gini, high GDP’ objective. In the first three columns, policy goals are defined based on final-period median values of emissions, inequality, and GDP; in the last three columns, objectives are instead based on top and bottom quartile thresholds. Thus, for example, the observations that meet the ‘low (below-median) GHG’ goal in the first column of Figure 2.3b (green rectangle) are those in quadrants 2, 3a and 3b of Figure 2.3a, whereas the observations that meet the ‘low Gini’ goal (violet rectangle) are those in quadrants 1, 3a and 3b. The set of observations that meet the ‘low GHG, low Gini’ goal in the second column of Figure 2.3b (blue rectangle) is the intersection of the green and violet rectangles in the first column; these observations are those in quadrants 3a and 3b of Figure 2.3a. Finally, the observations that meet the ‘low GHG, low Gini, high GDP’ goal in the third column of Figure 2.3b (red rectangle) are those in quadrant 3b of Figure 2.3a.

Approximately 20 percent of simulations ($n = 3,203$) exhibit below-median levels of Gini and emissions. When we also consider above-median GDP, this figure drops to approximately 11.5 percent of simulations ($n = 1,842$). Note that if the three indicators were uncorrelated, then 25 percent of simulations would meet the ‘low GHG, low Gini’ objective, while 12.5 percent would meet the ‘low GHG, low Gini, high GDP’ objective. Focusing on simulations with emissions and inequality levels below the bottom quartile would make achieving policy objectives even more challenging. Merely 3.1 percent of simulations ($n = 496$) meet bottom-quartile inequality-and-emissions goals, which is half of what it would be if there was no correlation. If we add the condition of above-top-quartile GDP, then only 1.1 percent of simulations ($n = 175$) remain, in contrast 1.5 percent if the three indicators were orthogonal.

Empirical studies have yielded mixed results on the existence of a trade-off between emissions and inequality, whereas the positive relation between growth and emissions has stronger support in the literature. Ravallion et al. (2000) and Rojas-Vallejos and Lastuka (2020), examining different country samples, find evidence of a trade-off. However, the former study argues that this relation is weak, while the latter suggests that it does not hold for high-income countries. Uddin et al. (2020), using a longer series of data (1870-2014) for G7 countries, provide evidence of a non-linear relation, concluding that there was a trade-off only between the 1950s and the end of the 1990s. Lastly, Jorgenson et al. (2017), examining US state data for the period 1997-2012, find no correlation between the Gini coefficient and emissions but do find a positive correlation between emissions and the income share of the top decile. These findings are con-

sistent with our simulation results, which show that emissions and inequality reductions are not complementary but can nevertheless be achieved jointly, although at a lower frequency than if they were pursued individually.

The results also reveal that achieving a just transition becomes more challenging when pursuing GDP growth. Considering growth as a necessary condition for a low-carbon, low-inequality transition would result in discarding all parameter combinations that achieve emissions and inequality objectives at lower GDP levels. This means that all simulations falling in region 3a of Figure 2.3a would be excluded. These findings suggest that the policies adopted to achieve a certain objective may not be neutral with respect to other objectives, that is, pursuing one goal may negatively impact the achievement of another goal. This highlights the need to understand which policy parameters can serve either or both policy objectives, and what combinations of parameters can balance possibly contrasting forces towards a just transition.

2.4.2 Main policy parameters

The main results of the random forest analysis are presented in the left-hand panel of Figure 2.4, which shows the permutation feature importance of the 10 most important predictors for each policy objective (GHG, Gini, GHG-Gini and GHG-Gini-GDP), plotted on a logarithmic scale. The right-hand panel gives information about the direction and magnitude of the effect of each parameter, and shows the difference between the parameter's mean value calculated from the simulations that meet a certain policy goal ('low GHG', 'low Gini', 'low GHG, low Gini' and 'low GHG, low Gini, high GDP') and its mean value calculated from all 16,023 simulations.

The majority of key predictors can be directly related to policies. Moreover, importantly, our analysis shows that most of the parameters that strongly predict GHG emissions (indicated by green circles) do not play a major role in predicting income inequality (indicated by violet squares), and vice versa. This finding supports the view that no single intervention can be a game changer for achieving multidimensional objectives, and that a combination of measures is needed. This is further evidenced by the results obtained for the GHG-Gini and GHG-Gini-GDP cases (denoted by blue triangles and red pentagons, respectively).

In three out of four cases, the policy parameter with the highest permutation importance is the increase in the share of renewable energy sources in industry energy consumption. The right-hand panel of Figure 2.4 shows that if we aim for inequality reduction and GDP growth along with decarbonisation, then achieving low emission levels becomes more challenging and requires greater deployment of renewables. The increase in the share of renewable energy sources in household energy consumption also plays a role, albeit less significant, mainly due to lower household energy use and emissions compared to industry.

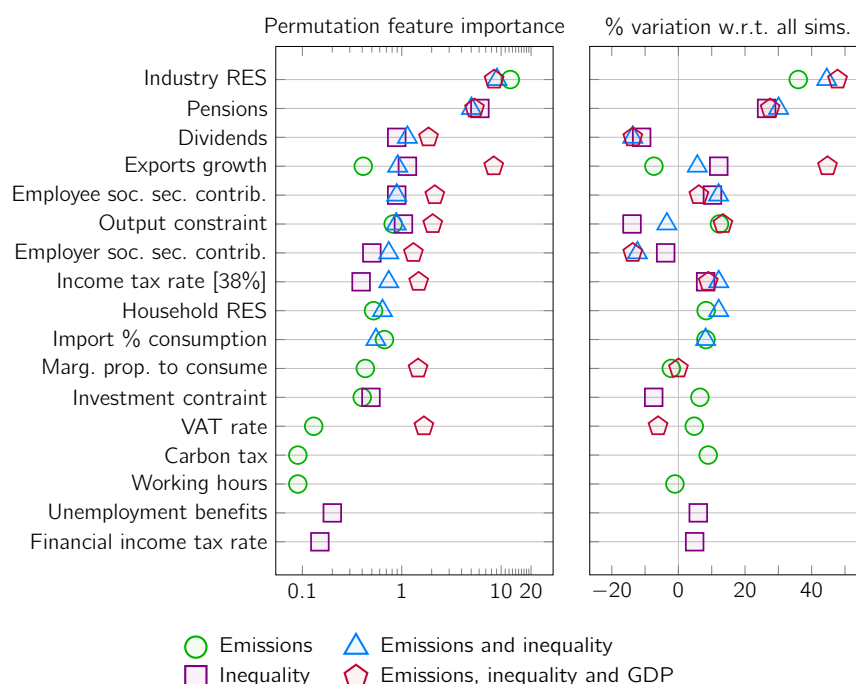


Figure 2.4: Main policy parameters. The two panels provide information on the 10 main parameters associated with each policy goal or combination of goals: emissions (green circles); inequality (violet squares); emissions and inequality (blue triangles); emissions, inequality and GDP (red pentagons). The left-hand panel shows the permutation feature importance of each parameter, plotted on a log scale. Higher values indicate an important predictor of the simulation outcome. The right-hand panel shows the difference between the mean value of each policy parameter calculated from all the simulations that meet a policy objective and the mean value calculated from all simulations.

A critical leverage point towards a just transition is to support the lowest segments of the income distribution. In the model, pensions play a key role in this respect.⁴ Although higher pensions tend to increase aggregate demand and therefore emissions, their positive effect on income distribution is much more significant, because a large proportion of pensioners are low-skilled and concentrated in the lower end of the income distribution. This also applies to unemployment benefits, which however are less effective in reducing inequality as they are directed towards a smaller subset of the population.

Another group of parameters points to the potential of addressing the income distribution from the opposite angle, that is by limiting top incomes. Here it is worth noting that, despite mounting evidence of tax dodging by the very wealthy (Saez and Zucman, 2019) and of their disproportionate environmental impact (IPCC, 2022b; Oswald et al., 2021), policies directed at high-income earners are somewhat neglected in the EM and climate mitigation literatures.

⁴The parameter that we let vary is the pensions-to-wage ratio, which determines gross pensions as a percentage of workers' mean annual wage.

To our knowledge, there is currently no scenario-based study that explores the consequences of redistributing income away from the rich. Our analysis identifies at least three relevant parameters directly related to limiting top incomes, including lower dividends, a higher labor income tax rate in top brackets, and a higher tax rate on financial income, which can yield a double dividend by reducing inequality and limiting consumption and emissions among high earners.⁵ Similarly, higher employee social security contributions reduce inequality because they reduce the net labor income of employed individuals, who tend to earn more than their unemployed, retired, and out-of-the-labor-force counterparts of the same gender and skill level.

A third set of parameters suggests a negative relation between investment and inequality. Reductions in employer social security contributions, value-added tax rates, and the percentage of profits paid out as dividends tend to increase retained profits, enabling industries to finance more investment. The resulting acceleration of fixed capital formation has two main effects: on income distribution through reduced dividends, and on employment and aggregate demand through increased investment.

The potential of addressing inequality by increasing aggregate demand is confirmed by the inclusion of the output and investment constraints in the sets of strongest predictors. While these parameters are not policy measures themselves, their significance sheds light on how limiting output and investment could impact emissions and inequality. The findings in Figure 2.4 suggest that excluding these constraints from a simulation model tends to lead to lower inequality, while their inclusion restricts production and investment spending, resulting in lower emissions. Along a related line, a decrease in individuals' marginal propensity to consume reduces demand and thus emissions.

Exports play a uniquely relevant role as the only variable identified as a strong predictor in all four cases.⁶ Since exports provide a strong stimulus to economic growth, they tend to have below-average values in simulations that meet the 'low GHG' objective, but above-average values in simulations that meet the 'low Gini', 'low GHG, low Gini', and 'low GHG, low Gini, high GDP' objectives. However, we note that relying on net exports to drive the growth and distribution of national income is not a feasible option at the global level. The same holds for the percentage of consumer goods that are imported (which is one of the strongest predictors in the GHG and GHG-Gini cases), since in the model an increase in imports roughly corresponds

⁵In the model, due to the aggregation of individuals into groups, the majority of high income earners consists of high-skill employed males and females falling into the third income tax bracket (28,000 to 55,000 euros per year). A notable exception is that of capitalists, who make up one percent of the total population and are assumed to earn solely from financial income.

⁶The Exports growth parameter determines the magnitude of an exogenous trend in exports. In the model, exports are also influenced by a price elasticity parameter, which however does not rank among the main predictors.

to exporting emissions.

Finally, the main policy parameters include working time reduction and carbon taxation, which are both central to the current debate on achieving a just transition and are well represented in our literature review. Empirical findings suggest that, on average, a small decrease in working hours results in lower emissions (Antal et al., 2020), which is consistent with our results. However, we find no evidence of the double dividend (i.e. the joint improvement of socioeconomic and environmental indicators) that is sometimes associated with working time reduction (Fitzgerald et al., 2018).

The carbon tax is unsurprisingly identified as an effective means to reduce emissions, but its role is somewhat less significant than is generally credited in the environmental economics literature (Hájek et al., 2019; Metcalf, 2019). Although recent evidence suggests a small effect of carbon taxation on low-carbon innovation (van den Bergh and Savin, 2021), in our case the less-than-stellar impact of the tax is likely related to the aggregation level of the model. Production and energy demand are in fact modelled at the industry level, while the burden of a carbon tax can actually vary considerably within industries, with heavy polluters facing higher costs⁷. Moreover, the reason why the carbon tax does not appear among the 10 main policy parameters in the GHG-Gini case is likely due to the fact that no automatic recycling mechanism of carbon tax revenues is implemented in the model. Nonetheless, our analysis does accommodate the fact that the distributional effects of carbon taxation crucially depend on whether the tax is accompanied by compensatory measures (Callan et al., 2009; Fremstad and Paul, 2019). In the model, explicit revenue recycling schemes are proxied by an increase in public benefits which is concomitant with the introduction of the carbon tax.

An interesting finding from our analysis is the absence of technological progress among the strongest predictors, which may come as a surprise given its prevalence in the literature reviewed in Section 2.2. This does not mean that the transition towards renewables is not influenced by new technologies. Rather, it suggests that innovations modelled at the macro level — related to labour productivity, intermediate input requirements, and energy efficiency — do not appear to play a major role. The limited impact of energy efficiency is likely related to economy-wide rebound effects (Brockway et al., 2021), which make efficiency gains lead to increased consumption (through lower prices of energy-intensive goods) and investment (through a decline in production costs and an increase in profit rates).

These findings emphasize that a low-carbon, low-inequality transition calls for a variety of measures spanning several different policy domains, certainly more than is commonly seen in

⁷For instance, in the electricity sector, although the industry as a whole may exhibit a relatively low emission intensity, high polluters like coal plants would typically face high carbon costs.

the literature. They also indicate that targeted measures to address income inequality play an essential role, whereas interventions aimed at raising national income (e.g. through government expenditure or a proportional increase in wages) seem not to be as relevant. Finally, they show that policies which boost aggregate demand, mostly directed at low-income groups, are crucial to achieve a just transition, and that the adverse effects of these policies on emissions must be compensated for by measures that decarbonise the additional demand and reduce the environmental impact of the wealthy.

2.4.3 Low emissions, low inequality pathways

Some of the results discussed earlier can also be seen by considering the drivers of emissions and inequality summarised in Figure 2.5. Panel 2.5a shows the dynamic of greenhouse gas emissions (GHG), which are broken down into GDP , the energy intensity of production (NRG/GDP), and the emissions intensity of energy demand (GHG/NRG); panel 2.5b decomposes the Gini coefficient of net income (N) into the Gini coefficient of gross income before taxes and transfers (G) and the ratio between the former and the latter:

$$GHG = GDP \times \frac{NRG}{GDP} \times \frac{GHG}{NRG}, \quad N = G \times \frac{N}{G}.$$

The figure is based on a new set of 500 simulations, performed by letting the mean and standard deviation of the distributions from which parameters are drawn be those of the simulations that jointly reached below-median emissions and inequality levels in Figure 2.3a. Moreover, in order to highlight the most relevant drivers of environmental and distributional outcomes, we selected the subset of simulations ($n = 55$) that reach bottom quartile emissions ($GHG \leq 146$ Mtons CO₂ eq./year.) and inequality (Gini ≤ 0.197) in the final simulation year⁸.

The dynamics further underline the importance of a fast expansion of renewables in all industries, as suggested in Figure 2.4. Improvements in energy efficiency due to technological progress slow down in 2025 and are less than enough to offset the (modest) growth of GDP. Thus, a substantial curbing of greenhouse gas emissions depends heavily on the expansion of renewable energy sources.

Panel 2.5b illustrates how variations in the Gini coefficient, calculated using disposable income, are affected by the evolution of market income (from labour and capital) and by the capacity of public taxes and transfers (including pensions and unemployment benefits) to redistribute income. The figure confirms the role of overall economic activity in driving inequality through changes in employment, wages and profits. However, it also underlines the importance of a more progressive tax and transfer system. The redistribution of income induced by some

⁸These are the bottom quartile values calculated from all simulations shown in Figure 2.3a.

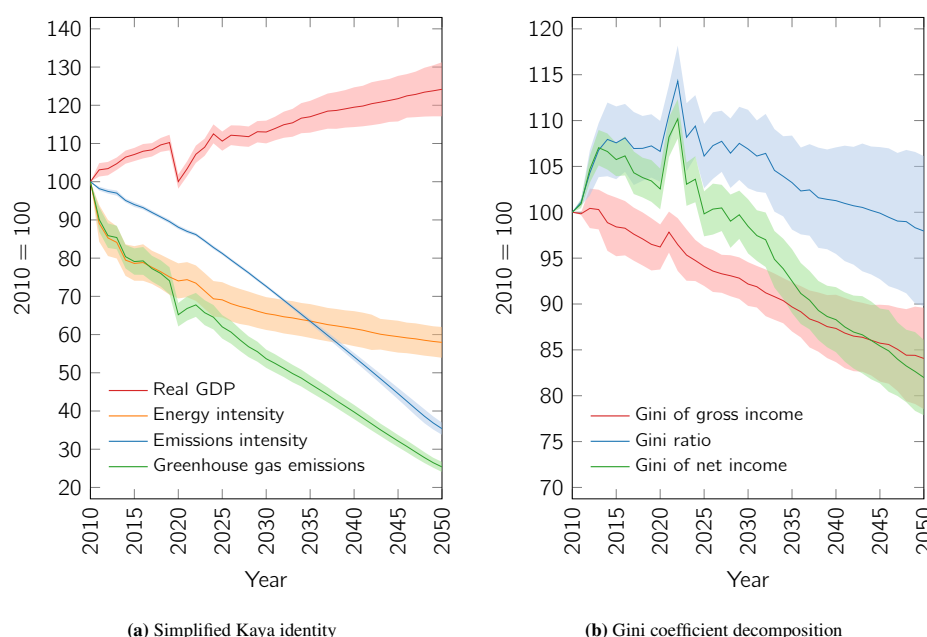


Figure 2.5: Decomposition of emissions and inequality. The left-hand panel decomposes greenhouse gas emissions into real GDP, net inland energy consumption divided by GDP (energy intensity), and total emissions divided by net inland energy consumption (emissions intensity). The right-hand panel decomposes the Gini coefficient of net income into the Gini coefficient of gross income and the net-to-gross Gini ratio. Both panels plot the means and one standard deviation confidence intervals.

of the parameters listed in Figure 2.4 (such as dividends, labour and financial income taxes, pensions and unemployment benefits) seems to be crucial in influencing the Gini coefficient of net income, and contributes to its sharper decline after 2030.

2.5 Concluding remarks

The simulation exercise presented in this study focuses on the simultaneous reduction of emissions and inequality, and identifies combinations of model parameters that promote a just low-carbon transition. The results discussed in Section 2.4.1 also indicate that the pursuit of continuous economic growth narrows the path to achieving low-emissions-and-inequality goals.

The expansion of renewable energy sources is found to be an essential pre-condition for decarbonisation, and can be complemented by carbon taxes, measures that discourage consumption, and working time reduction. Policies that affect the distribution of income are also crucial and should be designed to work in synergy with one another: the increase in demand and emissions caused by transfers and benefits targeted at low-income groups can in fact be partially

offset by policies that limit top incomes, which reduce the emissions of high earners while at the same time improving the income distribution. Increases in aggregate demand components, such as investment and exports, tend to impact positively on growth and income equality but also result in a direct increase in emissions. Finally, a notable absence among the main policy parameters are those directly related to energy efficiency. We interpret these results as suggesting that a just transition will require a progressive tax and benefits system that redistributes income without major increases in aggregate demand, coupled with a considerable increase in renewable generation capacity.

Our simulations encompass the case of a carbon tax accompanied by progressive tax reductions and transfers, the only difference being that in our model there is no earmarking of tax revenues. However, the opposite is not true, that is, the policies discussed in this paper cannot be subsumed into a carbon tax with progressive revenue recycling. Furthermore, although there is convincing evidence that revenue recycling can more than offset the regressive impact of carbon taxation (Budolfson et al., 2021; Williams et al., 2015), it is uncertain whether it can serve as a one-size-fits-all solution that allows for a sufficient reduction in inequality alongside a decrease in emissions.

The sets of policy parameters identified by random forests as the most important to reduce emissions and inequality have only a few elements in common. Thus, in contrast to the majority of studies reviewed in Section 2.2, the joint pursuit of these two goals is likely to require a number of simultaneous measures. A related point is that studies which examine policies in isolation or in very small groups risk under- or overestimating the necessary scale of these measures. For instance, the yearly growth rate of renewable energy used in the baseline of our model replicates the rates predicted by Italy's Integrated National Energy and Climate Plan (MISE-MATTEM-MIT, 2019), according to which renewables will supply 55% of electricity demand, 22% of transport energy demand and 30% of total energy demand by 2030 (starting from 34.1, 5.5 and 18.3% in 2020, respectively). However, the mean growth rate of renewable energy sources in industry calculated from the simulations that meet the 'low GHG, low Gini' objective is some 40% higher than the baseline, mainly due to the need to offset the increase in energy demand caused by redistributive policies directed at low income groups. By 2030 this would result in a share of renewables of about 64% in electricity consumption, 36% in transportation and 35% in total energy consumption. Thus, commitment by governments to jointly achieve distributional and low-carbon goals will make it necessary for them to increase renewables at a significantly faster pace than if climate mitigation were pursued without regard to, or even at the expense of, equity issues.

Data and code availability

The complete results of the literature review, the Vensim code for the model, the database of simulations, and the Stata and R scripts to replicate the results are available for download at Zenodo (DOI: [10.5281/zenodo.5711126](https://doi.org/10.5281/zenodo.5711126)). All files are open access and subject to a Creative Commons Attribution 4.0 International Public License (CC BY 4.0).

Authorship statement

NC: methodology (model development, programming), investigation (model input data), data curation, formal analysis, visualisation, writing (original draft). *MC*: methodology (model development). *AC*: conceptualisation, supervision, methodology (model development, programming, calibration, simulations), investigation (model input data, literature review), data curation, formal analysis, visualisation, writing (original draft). *SD*: methodology (model development), project administration, funding acquisition. *TD*: methodology (model development, programming), investigation (model input data), data curation, writing (original draft). *PG*: investigation (literature review). *TH*: methodology (model development, programming, simulations), investigation (model input data), data curation, formal analysis, writing (original draft).

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Appendix

This appendix presents additional results and details on data processing and analysis. Section 2.A complements Section 2.4 in presenting our findings. Section 2.B provides a series of robustness checks, showing that the random forest results hold under a variety of resampling methods to deal with imbalanced data. The parameter value ranges used to generate the simulation dataset are listed in Section 2.C.

2.A Additional results

2.A.1 Emissions, inequality and GDP

Figure 2.A.1 shows the relation between greenhouse gas emissions, inequality, and GDP in the final simulation year. Each point represents a different simulation. The vertical and horizontal dashed lines represent the median values calculated from all simulations.

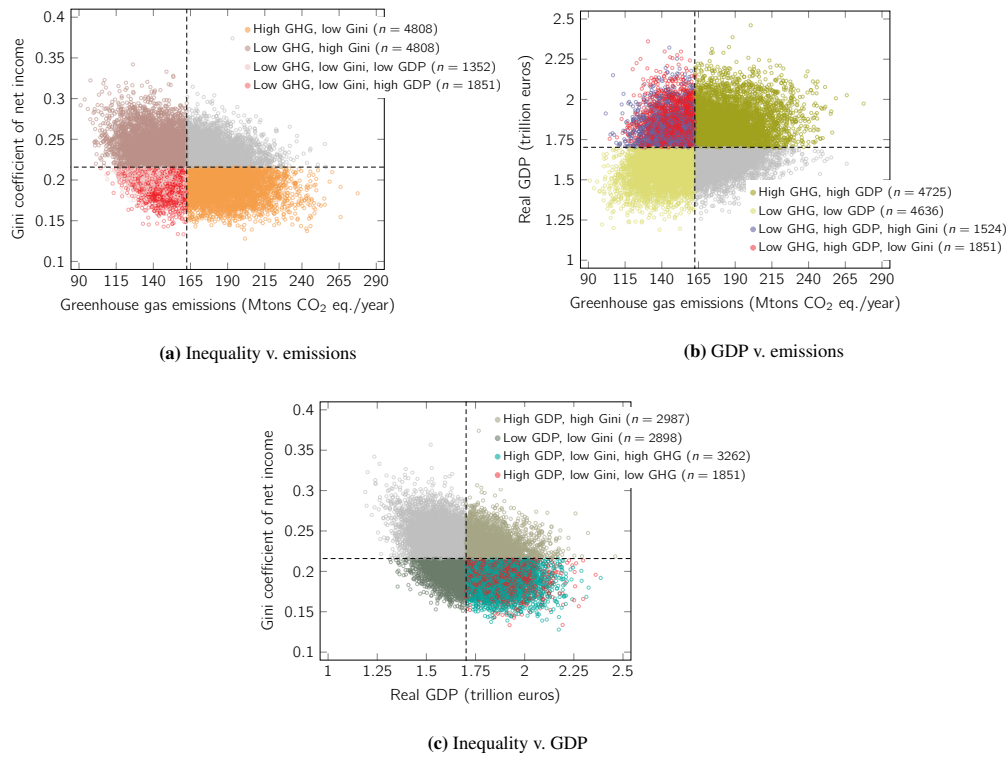


Figure 2.A.1: Inequality, emissions and GDP in the final simulation year

2.A.2 Partial dependence plots

Figures 2.A.2, 2.A.3, 2.A.4 and 2.A.5 show the impact of policy parameters on Prob (low GHG), Prob (low Gini), Prob (low GHG, low Gini) and Prob (low GHG, low Gini, high GDP), respectively. The black curves (one for each observation in the training sample) are the individual conditional expectations describing how the probability of the desired policy outcome changes with the value of a certain parameter of interest, keeping all other parameters constant at their respective last-period level (Goldstein et al., 2015). The green, purple, blue and red curves are the partial dependence plots obtained by averaging over all observations. Each plot is anchored at the lower end of the value range, and shows the difference in the prediction with respect to that point. The blue ticks at the bottom of each panel represent the deciles of the parameter distribution.

2.A.2.1 Policy objective: emissions

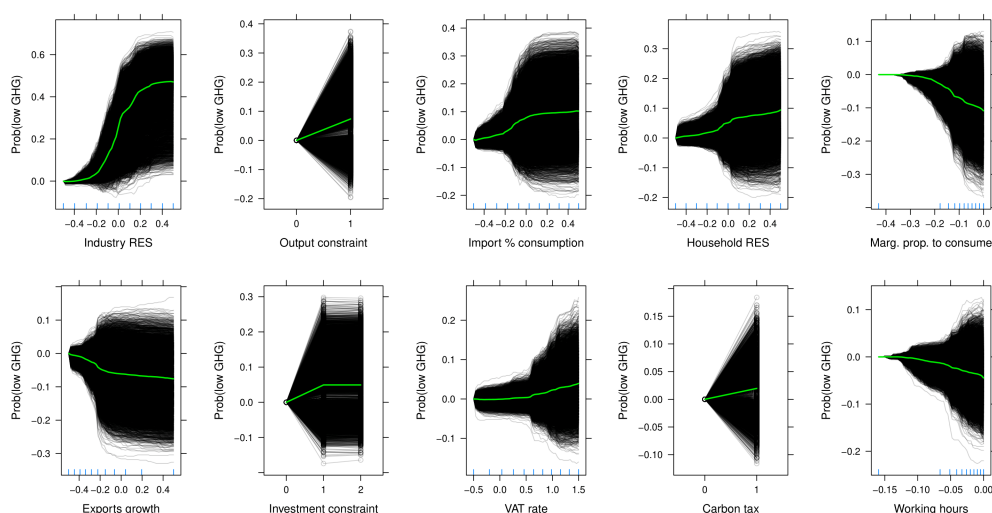


Figure 2.A.2: ICEs and PDPs of the 10 main policy parameters (objective: GHG)

2.A.2.2 Policy objective: inequality

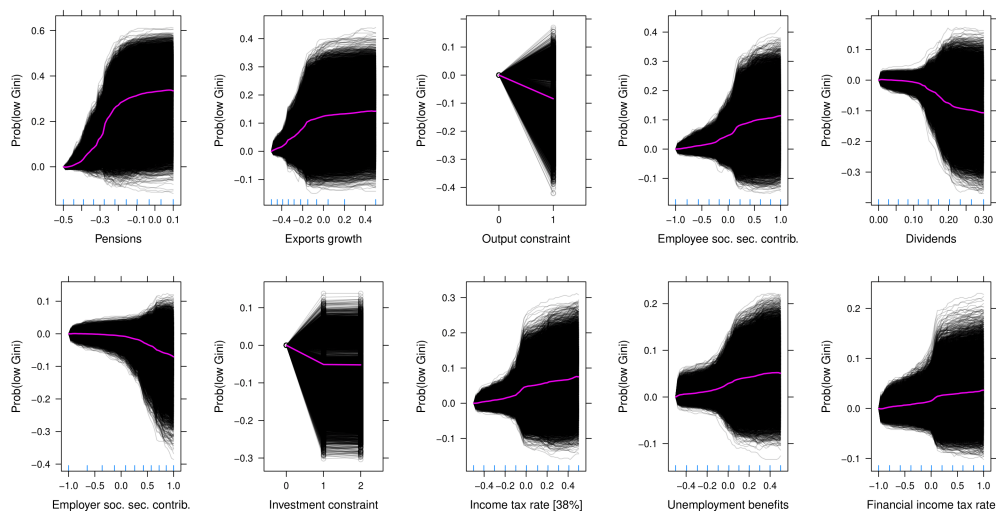


Figure 2.A.3: ICEs and PDPs of the 10 main policy parameters (objective: Gini)

2.A.2.3 Policy objectives: emissions and inequality

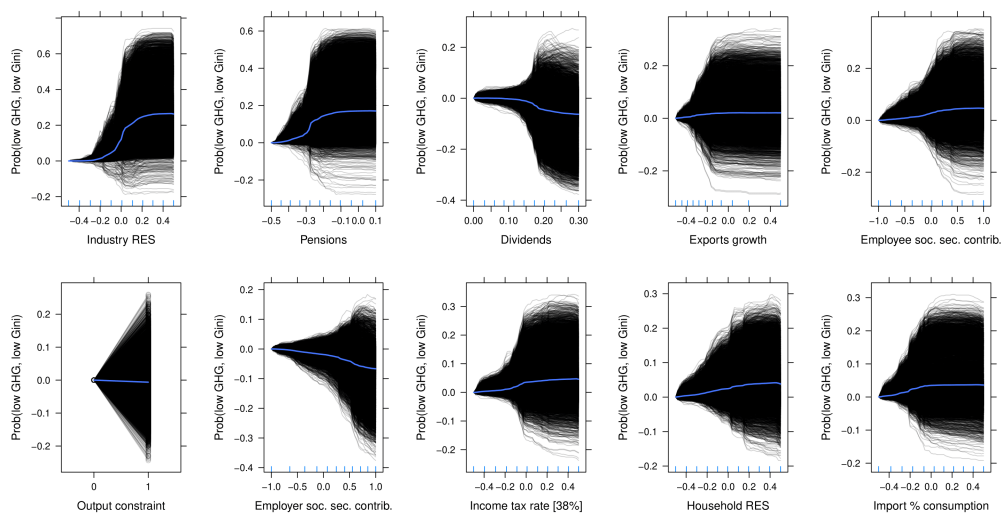


Figure 2.A.4: ICEs and PDPs of the 10 main policy parameters (objective: GHG-Gini)

2.A.2.4 Policy objectives: emissions, inequality and GDP

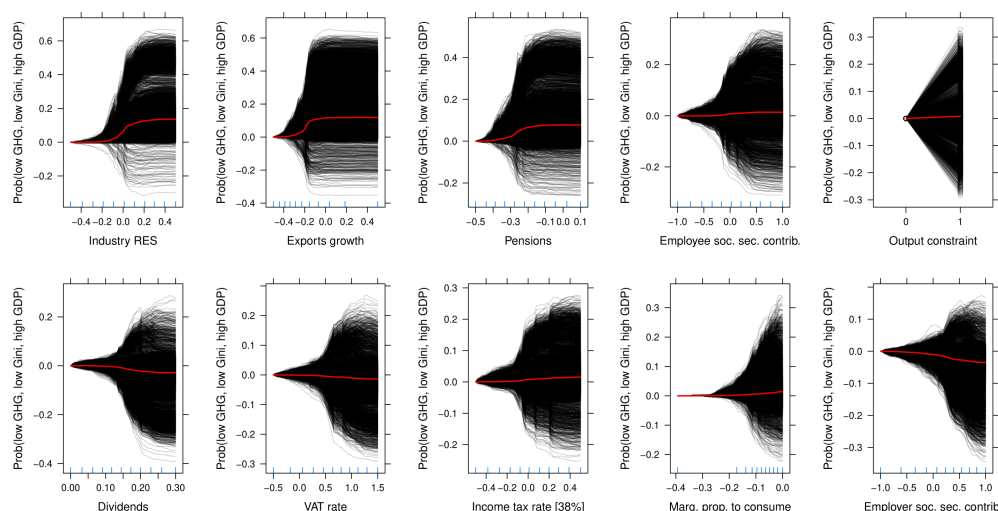


Figure 2.A.5: ICEs and PDPs of the 10 main policy parameters (objective: GHG-Gini-GDP)

2.A.3 Kernel densities

Figure 2.A.6 shows the kernel density distribution of the main continuous policy parameters. The coloured lines describe the distribution of input parameters in the subset of simulations that meet a certain policy goal: low emissions (green); low inequality (purple); low emissions and low inequality (blue); low emissions, low inequality and high GDP (red). The black dashed lines represent the parameters' distribution across all simulations. In all panels the number of observations decreases near the extremes of the range of possible values; this is because the simulations dropped from the dataset (due to economically meaningless or unreasonable results) are typically those featuring extreme values of policy parameters.

2.B Robustness checks

As discussed in Section 2.4, when simultaneously considering emissions and inequality, the four classification categories — ‘high GHG, high Gini’, ‘high GHG, low Gini’, ‘low GHG, high Gini’ and ‘low GHG, low Gini’ — are not equally represented in the data. In particular, the observations in the ‘low GHG, low Gini’ region of Figure 2.3a are relatively fewer in number than those in the ‘low GHG, high Gini’ and ‘high GHG, low Gini’ regions. A consequence of this mild class imbalance is that the random forest model will tend to overlook, and therefore have poor prediction performance on, the ‘low GHG, low Gini’ class, which however is the class

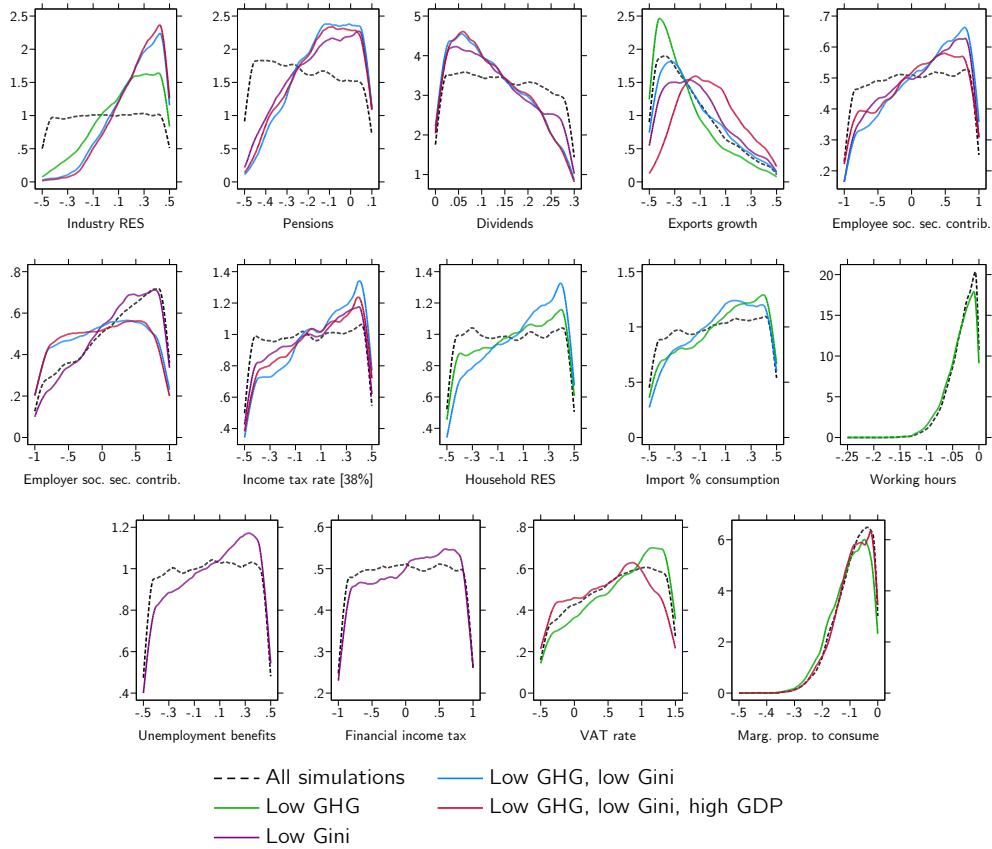


Figure 2.A.6: Kernel density distribution of the 10 main continuous policy parameters

we are most interested in.

A common method to deal with class imbalances is to resample the dataset, either by undersampling the majority classes or by oversampling the minority classes. The undersampling approach involves drawing observations at random from the majority classes and dropping them from the training dataset, so as to balance the class distribution before training the model; conversely, the oversampling approach consists in randomly duplicating observations from the minority classes and adding them to the training dataset. Yet another method is to synthesize new observations from the minority classes using the Synthetic Minority Oversampling Technique (SMOTE). This method consists in randomly drawing a minority class observation, finding its m nearest neighbours in terms of characteristics, and choosing one of these neighbours at random; the synthetic observation is created as a convex combination of the two neighbours, that is at a random point on the line connecting them.

Following common practice in the literature, we combined undersampling and synthetic minority oversampling methods (Chawla et al., 2002), with the number nearest neighbours m equal to 5. For example, in the GHG-Gini case, we downsampled the majority classes — ‘high

GHG, low Gini’ and ‘low GHG, high Gini’ — by a factor of about 0.83 and then synthetically oversampled the minority classes — ‘high GHG, high Gini’ and ‘low GHG, low Gini’ — by a factor of 1.25. This reduced the number of observations in the training dataset from 11,214 to 11,198, i.e. about 2,800 observations per class. Table 2.B.1 gives the numbers of observations in the original training dataset, in the SMOTEd-and-undersampled dataset, and in the datasets resulting from 3 alternative resampling methods: undersampling, oversampling, and SMOTE without undersampling.

Table 2.B.1: Training sample sizes under different resampling methods

	No resampling	Majority undersampling	Minority oversampling	SMOTE	SMOTE & majority undersampling
<i>Objective: GHG</i>					
<i>n</i> high GHG	5,608	-	-	-	-
<i>n</i> low GHG	5,608	-	-	-	-
<i>n</i> (overall)	11,216	-	-	-	-
<i>Objective: Gini</i>					
<i>n</i> high Gini	5,608	-	-	-	-
<i>n</i> low Gini	5,608	-	-	-	-
<i>n</i> (overall)	11,216	-	-	-	-
<i>Objectives: GHG and Gini</i>					
<i>n</i> high GHG, high Gini	2,242	2,242	3,365	3,365	2,800
<i>n</i> high GHG, low Gini	3,365	2,242	3,365	3,365	2,799
<i>n</i> low GHG, high Gini	3,365	2,242	3,365	3,365	2,799
<i>n</i> low GHG, low Gini	2,242	2,242	3,365	3,365	2,800
<i>n</i> (overall)	11,214	8,968	13,460	13,460	11,198
<i>Objectives: GHG, Gini and GDP</i>					
<i>n</i> high GHG, high Gini, high GDP	1,006	952	2,323	2,322	1,700
<i>n</i> high GHG, high Gini, low GDP	1,215	952	2,323	2,323	1,699
<i>n</i> high GHG, low Gini, high GDP	1,069	952	2,323	2,322	1,700
<i>n</i> low GHG, high Gini, high GDP	2,304	952	2,323	2,322	1,699
<i>n</i> high GHG, low Gini, low GDP	2,323	952	2,323	2,322	1,700
<i>n</i> low GHG, high Gini, low GDP	1,066	952	2,323	2,323	1,700
<i>n</i> low GHG, low Gini, high GDP	1,279	952	2,323	2,322	1,699
<i>n</i> low GHG, low Gini, low GDP	952	952	2,323	2,322	1,700
<i>n</i> (overall)	11,214	7,616	18,584	18,578	13,597

Table 2.B.2 compares the prediction accuracy, sensitivity (true positive rate) and specificity (true negative rate) of random forest models trained on the original and resampled datasets. All random forests were based on 500 classification trees built on bootstrapped training samples. At each split in each tree, a random sample of 10 out of 107 parameters was chosen as split

candidates. The results of the GHG-Gini case indicate that all resampling methods improve the sensitivity of the ‘low GHG, low Gini’ class at the cost of some decrease in specificity and overall accuracy. No technique clearly dominates the others; the SMOTE-and-undersampling method was chosen as the preferred option because it represents a reasonable compromise between sensitivity and accuracy. For consistency, this resampling method was also used in the GHG-Gini-GDP case, although in this case the the oversampling approach approach yields slightly better results in terms of both accuracy and sensitivity of the ‘low GHG, low Gini, high GDP’ class.

Figure 2.B.1 shows that the random forest results are robust to whether and how training data are resampled. The four panels report the permutation feature importance of the main policy parameters in the imbalanced case and for each resampling method (the results obtained in the SMOTE-and-undersampling case are shown in the left-hand panel of Figure 2.4). The set of 10 policy parameters with the greatest predictive power remains essentially unchanged (with some minor exceptions in the imbalanced case with no resampling), and their ranking is similar for all resampling methods. This indicates that our arguments do not hinge on specific data processing choices.

Table 2.B.2: Prediction accuracy of the random forest models under different resampling methods

	No resampling	Majority undersampling	Minority oversampling	SMOTE	SMOTE & majority undersampling
<i>Objective: GHG</i>					
Overall accuracy	.830	-	-	-	-
Sensitivity	.826	-	-	-	-
Specificity	.834	-	-	-	-
<i>Objective: Gini</i>					
Overall accuracy	.811	-	-	-	-
Sensitivity	.794	-	-	-	-
Specificity	.826	-	-	-	-
<i>Objectives: GHG and Gini</i>					
Overall accuracy	.672	.647	.671	.658	.652
Sens. high GHG, high Gini	.518	.704	.583	.634	.664
Sens. high GHG, low Gini	.769	.618	.711	.667	.639
Sens. low GHG, high Gini	.833	.559	.704	.642	.599
Sens. low GHG, low Gini	.556	.765	.650	.692	.739
Spec. high GHG, high Gini	.946	.880	.926	.904	.894
Spec. high GHG, low Gini	.821	.885	.847	.866	.876
Spec. low GHG, high Gini	.756	.913	.854	.875	.893
Spec. low GHG, low Gini	.948	.856	.924	.895	.873
<i>Objectives: GHG, Gini and GDP</i>					
Overall accuracy	.552	.520	.563	.545	.532
Sens. high GHG, high Gini, high GDP	.267	.512	.363	.484	.449
Sens. high GHG, high Gini, low GDP	.485	.601	.589	.620	.595
Sens. high GHG, low Gini, high GDP	.807	.442	.673	.484	.541
Sens. low GHG, high Gini, high GDP	.358	.519	.413	.501	.505
Sens. high GHG, low Gini, low GDP	.385	.558	.456	.529	.506
Sens. low GHG, high Gini, low GDP	.800	.446	.700	.513	.570
Sens. low GHG, low Gini, high GDP	.526	.568	.586	.552	.563
Sens. low GHG, low Gini, low GDP	.223	.670	.430	.643	.608
Spec. high GHG, high Gini, high GDP	.989	.938	.978	.946	.956
Spec. high GHG, high Gini, low GDP	.956	.929	.944	.932	.934
Spec. high GHG, low Gini, high GDP	.829	.940	.877	.931	.917
Spec. low GHG, high Gini, high GDP	.973	.932	.959	.935	.944
Spec. high GHG, low Gini, low GDP	.970	.918	.950	.930	.929
Spec. low GHG, high Gini, low GDP	.803	.959	.872	.937	.928
Spec. low GHG, low Gini, high GDP	.956	.933	.941	.927	.931
Spec. low GHG, low Gini, low GDP	.986	.907	.965	.927	.938

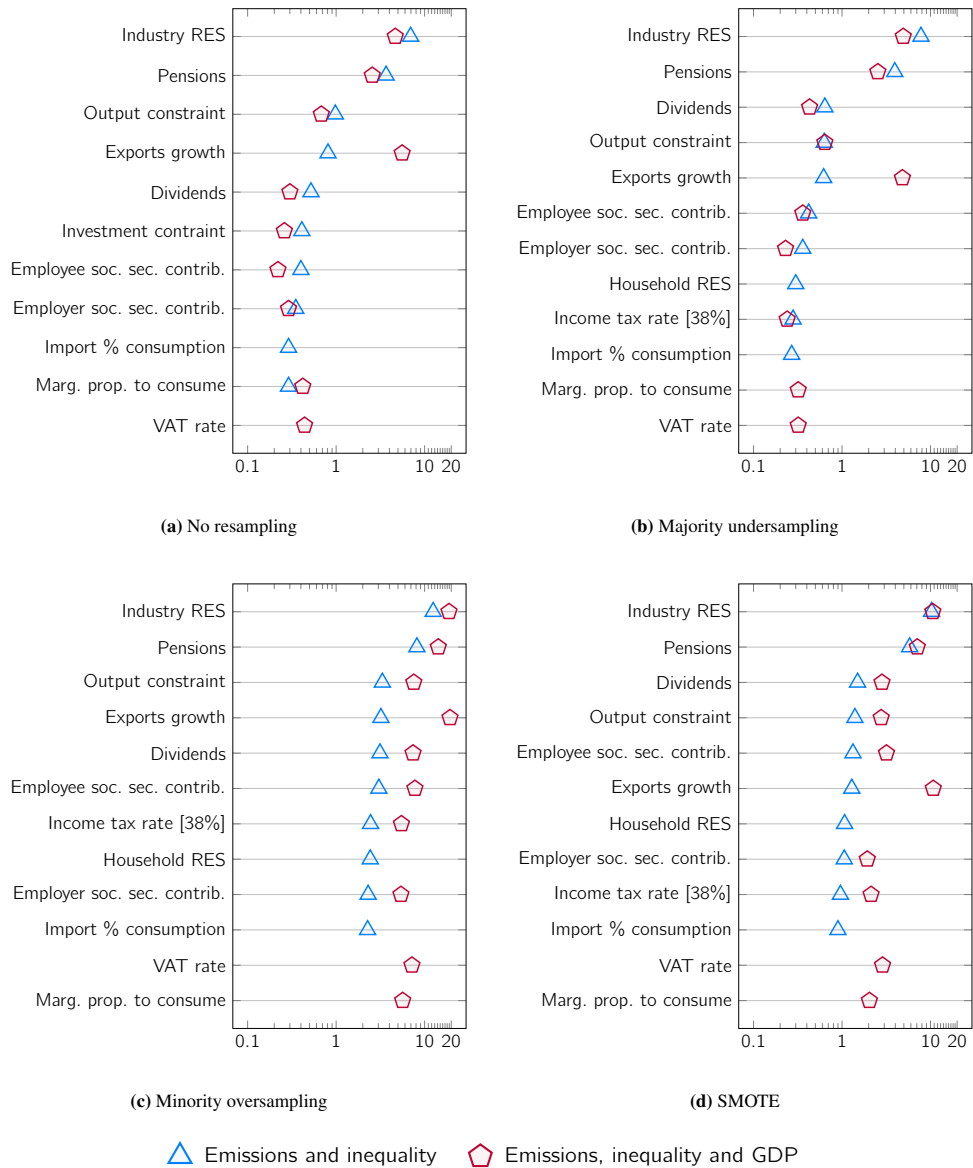


Figure 2.B.1: Permutation feature importance of the 10 main policy parameters under different resampling methods

2.C Simulation parameters

Table C.1 lists all parameters that vary across simulations. Based on their characteristics, they were grouped into Structural parameters, Non-calibration parameters, and Calibration parameters. References to the equations in the model documentation are given in the last column of the table.

Non-calibration parameters follows a linear trend, starting from a fixed value in 2022 and reaching a randomly selected value in the final simulation year. Conversely, Structural and Calibration parameters are randomly drawn in the first period and then remain fixed throughout the simulation run. The value range of Non-calibration parameters generally spans from -50 to $+50$ percent of their initial level, except when a different value range was available from cross-country comparisons. The *Working hours*, *Marginal propensity to consume*, and *Carbon tax rate* parameters were drawn from a half-normal (rather than uniform) distribution in order to focus on moderate, more plausible and politically feasible parameter values. Finally, as discussed in Section 2.3.2, Calibration parameters span a smaller range of values because they are calibrated to fit historical data.

The letter i within parentheses indicates that an independent random draw is made for each of the 19 industries featured in the model. Thus, for example, the sensitivity of investment to capacity utilisation is determined at the industry level by 19 independent draws. Similarly, the letters c and s indicate that a draw is made for each of the 16 consumption categories and 3 skill levels, respectively. The total number of random draws per period is 107.

Table 2.C.1: List of sensitivity parameters

Parameter	Unit	Min	Max	Distribution	Baseline	Equation
<i>Structural parameters</i>						
Investment constraint	{0,1,2}	0	2	vector	2	2.27,2.27'
Output constraint	{0,1}	0	1	vector	1	2.4'
Carbon tax	{0,1}	0	1	vector	0	2.20
Warming scenarios	{1,2,3,4}	0	4	vector	1	2.159
<i>Non-calibration parameters</i>						
Skill supply trends	%	-0.5	+0.5	continuous uniform	0	2.63
Depreciation rates	%	-0.5	+0.5	continuous uniform	0	2.30
Equity-to-liabilities ratio	%	-0.5	+0.5	continuous uniform	0	2.27
Import share of consumption	%	-0.5	+0.5	continuous uniform	0	2.7
Import share of government final demand	%	-0.5	+0.5	continuous uniform	0	2.9
Import share of investment spending	%	-0.5	+0.5	continuous uniform	0	2.8
Exports growth rate	%	-0.5	+0.5	continuous uniform	0	2.11
Households' RES growth rate	%	-0.5	+0.5	continuous uniform	0	2.145,2.148
Industries' RES growth rate	%	-0.5	+0.5	continuous uniform	0	2.145,2.147
Change in technical coefficients	%	-0.5	+0.5	continuous uniform	0	
Change in labour productivity	%	-0.5	+0.5	continuous uniform	0	
Tax rate on financial income ⁹	%	-1	+1	continuous uniform	0	2.85
Employee social security contrib. ¹⁰	%	-1	+1	continuous uniform	0	2.77
Employer social security contrib. ¹¹	%	-1	+1	continuous uniform	0	2.78

⁹<https://tinyurl.com/3k6nfjkd>¹⁰<https://tinyurl.com/v9sxn9kv>¹¹<https://tinyurl.com/334mc44v>

VAT rate ¹²	%	-0.5	+1.5	continuous uniform	0	2.87
Corporate income tax rate ¹³	%	-0.6	+0.6	continuous uniform	0	2.89
Government expenditure trend	%	-0.5	+0.5	continuous uniform	0	2.102
Unemployment benefits to wage ratio ¹⁴	%	-0.5	+0.5	continuous uniform	0	2.93
Pension to wage ratio ¹⁵	%	-0.5	+0.1	continuous uniform	0	2.96
Sickness and disability benefits	%	-0.5	+0.5	continuous uniform	0	2.117 – 2.120
Family and children benefits	%	-0.5	+0.5	continuous uniform	0	2.98
Other benefits	%	-0.5	+0.5	continuous uniform	0	2.120
Income tax rate [0.23]	%	-0.5	+0.5	continuous uniform	0	2.83
Income tax rate [0.27]	%	-0.5	+0.5	continuous uniform	0	2.83
Income tax rate [0.38]	%	-0.5	+0.5	continuous uniform	0	2.83
Income tax rate [0.41]	%	-0.5	+0.5	continuous uniform	0	2.83
Income tax rate [0.43]	%	-0.5	+0.5	continuous uniform	0	2.83
Working hours	%	-0.25	0	$N(0, 0.05)$	0	2.69
Marginal prop. to consume	%	-0.5	0	$N(0, 0.1)$	0	2.122
Carbon tax rate	%	0	10	$N(3, 2.5)$	0	2.20

Calibration parameters

Technologies probability sens.		9	15	continuous uniform	11.93	2.13-14
Initial prob. labour-saving innov.	%	0.35	0.75	continuous uniform	0.67	2.13
Initial prob. intermediate-input-saving innov.	%	0.35	0.75	continuous uniform	0.47	2.14
Skill transition sens. (<i>s</i>)		0.65	0.85	continuous uniform	[0.69,0.75]	2.63-65
Labour force participation sens. (<i>s</i>)	%	0.65	0.85	continuous uniform	0.75	2.68
Gender employment subst. sens. (<i>s</i>)		0	0.1	continuous uniform	[0.03,0.08]	2.72

¹²<https://tinyurl.com/5k9j9cac>

¹³<https://tinyurl.com/ufvztxph>

¹⁴<https://tinyurl.com/ymyu3xu7>

¹⁵<https://tinyurl.com/mez7354w>

Wage sens. to employment		0.35	0.55	continuous uniform	0.45	2.73
Wage sens. to lab. productivity		0.7	1	continuous uniform	0.99	2.73
Wage sens. to price		0.7	1	continuous uniform	1	2.73
Investment sens. (<i>i</i>)		0	0.35	continuous uniform	[0,0.225]	2.22
Dividends rate	%	0	0.3	continuous uniform	0.3	2.40
Interest on loans sens.		0	0.25	continuous uniform	0.12	2.36
Price-elasticity of exports	%	0	-1	continuous uniform	-0.5	2.11
Price-elasticity of consumption (<i>c</i>)	%	0	-1.5	continuous uniform	0	2.131
Mark-up sens. (<i>z</i>)		0	0.05	continuous uniform	[0,0.067]	2.112
Seed		0	5,076	discrete uniform	1	

Structural parameters can take the integer values listed in column 2. Policy parameters vary in percentage according to the outcome of a random draw from a uniform or half-normal distribution. Calibration parameters are drawn from a uniform distribution. The extremes of the support of the distributions are given in columns 3 and 4. The footnotes contain references to the sources used to define plausible value ranges.

2.6 Supplementary Material

2.6.1 The Eurogreen model

2.6.1.1 Population

The population is partitioned into five age groups: 0-14 (group I), 15-24 (group II), 25-44 (group III), 45-64 (group IV), and 65+ (group V). The working-age population comprises groups 2 to 4, whereas the reproductive-age population comprises groups 2 and 3. The model also distinguishes between female (f) and male (m) individuals. Time unfolds discretely and is indexed by $t = 1, 2, \dots$, with one period corresponding to one year. The number of individuals of gender $g \in \{f, m\}$ and age group $n \in \{I, II, \dots, V\}$ in period t is denoted by $Pop_{g,t}^n$. Let RL be the number of reproductive years of a woman's life and κ_t^+ be the fertility rate (births per woman) at time t . Thus, a number $\frac{\kappa_t^+}{RL} (Pop_{f,t}^{II} + Pop_{f,t}^{III})$ of individuals is born in each period, half of which are assumed to be females and half males. Moreover, let $\kappa_{n,g}^-$ be a gender- and age group-specific mortality rate, and for each $n < V$ let ϱ_n be the fraction of group- n individuals who mature to the next age group in every period. The latter is given by the total number of surviving individuals in group n divided by the number of years group n spans (e.g. 15 years for group 1). Changes in the size of each group are therefore described by:

$$\Delta Pop_{g,t}^n = \begin{cases} \frac{\kappa_t^+}{2RL} (Pop_{f,t}^{II} + Pop_{f,t}^{III}) - (\varrho_n + \kappa_{n,g}^-) Pop_{g,t-1}^n & \text{if } n = I \\ \varrho_{n-1} Pop_{g,t-1}^{n-1} - (\varrho_n + \kappa_{n,g}^-) Pop_{g,t-1}^n & \text{if } n \in \{II, III, IV\} \\ \varrho_{n-1} Pop_{g,t-1}^{n-1} - \kappa_{n,g}^- Pop_{g,t-1}^n & \text{if } n = V \end{cases} \quad (2.1)$$

Individuals in age groups II to V are further grouped according to their skill level. As detailed in Section 2.6.1.7, a distinction is made between low-, middle- and high-skilled workers.

Data sources: population

Population data for the first period (by age and gender) and fertility and mortality rates are taken from Eurostat.

2.6.1.2 Input-output and GDP

Economic activity is separated into 19 sectors (or industries), which are listed in Table 2.6.1 together with their NACE counterparts.

Each sector produces goods and consumes goods from other sectors. The monetary value of the transactions from sector i to sector j is denoted by $z_{i,j}$ (the time subscript is omitted for

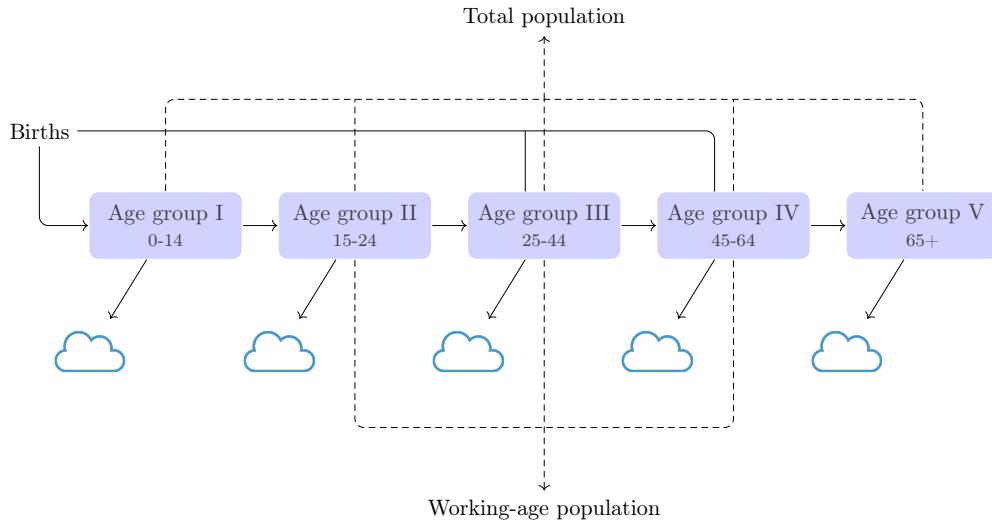


Figure 2.6.1: Conceptual framework of the population module. Half the newborns are females and half males. Reproductive lifespan is 30 years. Clouds represent death.

simplicity). For instance, $z_{1,3}$ is the interindustry flow from sector 1 to sector 3, that is the input of agricultural products bought by manufacturing firms. Domestic consumption expenditures by households (f^C), gross fixed capital formation (i.e. investments, f^I), government expenditures (f^G), and exports (f^E) constitute the final demand for goods and services. The distribution of sector i 's output is then given by:

$$y_i = \sum_{j=1}^{19} z_{i,j} + f_i, \quad (2.2)$$

where $f_i \equiv f_i^C + f_i^I + f_i^G + f_i^E$ is the total final demand for sector i 's goods that are produced domestically. The 19-element vector of final demands, with typical element f_i , is denoted by $\mathbf{f}$.

The relationship between a sector's output and its inputs is given by:

$$a_{i,j} = \frac{z_{i,j}}{y_j}, \quad (2.3)$$

where $a_{i,j}$ is the technical coefficient measuring the euros' worth of sector i 's input per euros' worth of sector j 's output. Technical coefficients change endogenously over time due to innovations (modelled in Section 2.6.1.4) and climate change (Section 2.6.1.12). Let $\mathbf{A}$ and $\mathbf{y}$ be, respectively, the 19×19 matrix with typical element a_{ij} and the 19×1 vector with typical element y_i . Following standard practice in the input-output literature (Miller and Blair, 2009), the vector of sectoral outputs is obtained from the matrix of technical coefficients and the vector of final demands:

$$\begin{aligned} \mathbf{y} &= \mathbf{L}\mathbf{f} \\ &= (\mathbf{I} - \mathbf{A})^{-1} \mathbf{f}, \end{aligned} \quad (2.4)$$

Table 2.6.1: List of sectors

Sector no.	Sector name	Nace Rev. 2 code
1	Agriculture, forestry and fishing	<i>A</i>
2	Mining and quarrying	<i>B</i>
3	Manufacturing	<i>C</i> (excl. <i>C19</i>)
4	Coke and refined petroleum products	<i>C19</i>
5	Electricity, gas, steam and air conditioning supply	<i>D</i>
6	Water supply	<i>E</i>
7	Construction	<i>F</i>
8	Wholesale and retail trade	<i>G</i>
9	Transportation and storage	<i>H</i>
10	Accommodation and food service activities	<i>I</i>
11	Information and communication	<i>J</i>
12	Financial and insurance activities	<i>K</i>
13	Real estate activities	<i>L</i>
14	Professional, scientific, technical, administrative and support service activities	<i>M, N</i>
15	Public administration and defence	<i>O</i>
16	Education	<i>P</i>
17	Human health and social work activities	<i>Q</i>
18	Arts, entertainment and recreation	<i>R</i>
19	Other	<i>S, T, U</i>

where $\mathbf{I}$ is the 19×19 identity matrix and $\mathbf{L} \equiv (\mathbf{I} - \mathbf{A})^{-1}$ is the so-called Leontief inverse matrix. The model contains an option to let output be constrained by capacity utilisation, in which case for each industry $i = 1, \dots, 19$ we have:

$$y_i = \min \left\{ \sum_{j=1}^{19} l_{i,j} f_j, y_i^{FC} \right\}, \quad (2.4')$$

where $l_{i,j}$ is the ij th element of matrix $\mathbf{L}$ and y_j^{FC} is industry j 's full-capacity output (defined in Section 2.6.1.5). The default setting is capacity constraint = ON.

Gross Domestic Product is calculated as

$$GDP = \sum_{i=1}^{19} f_i - \sum_{i=1}^{19} \sum_{j=1}^{19} z_{i,j}^{imp}, \quad (2.5)$$

where $z_{i,j}^{imp}$ represents the imports of sector i goods by domestic firms in sector j (see Section 2.6.1.3).

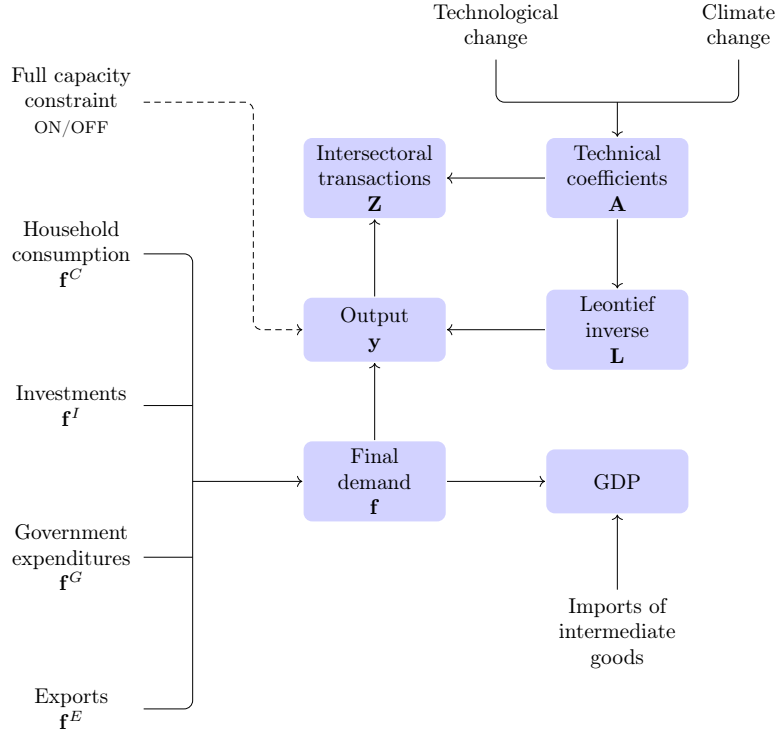


Figure 2.6.2: Conceptual framework of the input-output module

Final demands, sectoral outputs, and intermediate inputs can be deflated by commodity price indices. Let $p_{i,t}$ be the price of a unit of sector i 's product at time t . If we choose any period b to be the reference (or base) period, then the price index for sector i is:

$$\tilde{p}_{i,t} = \frac{p_{i,b}}{p_{i,t}}$$

and the vector of price indices is $\tilde{\mathbf{p}}_t = [\tilde{p}_{1,t}, \tilde{p}_{2,t}, \dots, \tilde{p}_{19,t}]$. The relation between current- and constant-price interindustry inputs, final demands, and outputs is therefore given by:

$$\mathbf{Z}_t^b = \langle \tilde{\mathbf{p}}_t \rangle \mathbf{Z}_t,$$

$$\mathbf{f}_t^b = \langle \tilde{\mathbf{p}}_t \rangle \mathbf{f}_t,$$

and

$$\mathbf{y}_t^b = \langle \tilde{\mathbf{p}}_t \rangle \mathbf{y}_t,$$

where $\mathbf{Z}$ is the 19×19 matrix of interindustry domestic transactions, with typical element $z_{i,j}$, and $\langle \tilde{\mathbf{p}}_t \rangle$ is the diagonal matrix with the elements of $\tilde{\mathbf{p}}_t$ along the main diagonal.

Data sources: input-output and GDP

The input-output data used to initialise the model are taken from Eurostat.

2.6.1.3 International trade

2.6.1.3.1 Imports

Imports are determined using constant import share coefficients, which are calculated for the initial period and assumed not to change over time. Imports of sector i goods by sector j are assumed to be a fraction $\gamma_{i,j}^Z$ of the corresponding domestic demand, that is:

$$\gamma_{i,j}^Z = \frac{z_{i,j,0}^{imp}}{z_{i,j,0}}, \quad (2.6)$$

where the subscript 0 denotes the initial period. In the case of final demand, import shares are instead given by:

$$\gamma_i^C = \frac{f_{i,0}^{C,imp}}{\hat{f}_{i,0}^C}, \quad (2.7)$$

$$\gamma_i^I = \frac{f_{i,0}^{I,imp}}{\hat{f}_{i,0}^I}, \quad (2.8)$$

$$\gamma_i^G = \frac{f_{i,0}^{G,imp}}{\hat{f}_{i,0}^G}, \quad (2.9)$$

where $\hat{f}_{i,t}^x = f_{i,t}^x + f_{i,t}^{x,imp}$ is the overall (domestic plus imported) demand of category $x \in \{C, I, G\}$ for sector i goods. All three import shares are bound within $[0,1]$.¹⁶ Total imports by industry at time t are therefore given by:

$$\begin{aligned} M_{i,t} &= \sum_{j=1}^{19} z_{j,i,t}^{imp} + f_{i,t}^{C,imp} + f_{i,t}^{I,imp} + f_{i,t}^{G,imp} \\ &= \sum_{j=1}^{19} \gamma_{j,i}^Z z_{j,i,t} + \gamma_i^C \hat{f}_{i,t}^C + \gamma_i^I \hat{f}_{i,t}^I + \gamma_i^G \hat{f}_{i,t}^G. \end{aligned} \quad (2.10)$$

All in all, variations in imports depend on the level of national income and its composition, since import shares differ by product and by demand component (household consumption, capital stock formation, government consumption, and interindustry consumption).

¹⁶The shares are calculated differently for intermediate and final demand, as the IO calculations are conducted with a domestic $\mathbf{A}$ matrix, distinguishing between domestic and foreign demand and production. Consequently, the most straightforward approach is to calculate domestic values for intermediate trade, to which imports have to be added. For the final demand categories on the other hand, data availability only allows for the calculation of overall demand of domestic actors, which subsequently has to be divided into demand for the domestic economy and for imports.

2.6.1.3.2 Exports

The complex nature of international trade and the absence in our model of international capital flows make it particularly challenging to model exchange rates. For this reason, and since exports largely depend on how competitive domestic prices are in the global market, we assume that exports (denoted by f^E) are a decreasing function of domestic inflation and change over time according to:

$$f_{i,t,+}^E = (1 + g_i^E) f_{i,t-1}^E (1 + \varepsilon^E r_{i,t}), \quad (2.11)$$

where g_i^E is an exogenous industry-specific growth rate, $\varepsilon^E < 0$ is the elasticity of exports to domestic price variations, and $r_{i,t}$ is the percentage change in the domestic price of sector i goods between period $t - 1$ and period t . Note that productivity gains can have a positive effect on the trade balance, because they result in a decrease in unit labour and intermediate costs (see Section 2.6.1.4).

An exogenous negative shock to the demand for exports, caused by the Covid-19 pandemic, is assumed for the years 2020 and 2021.

Data sources: international trade

The import shares ($\gamma_{i,j}^Z, \gamma_i^C, \gamma_i^G$ and γ_i^I) are calculated using Eurostat's input-output data, as are initial exports ($f_{i,0}^E$) and the industry-specific growth rate of exports (g_i^E , which is based on data for the period 2010-2017).

The covid shock is based on the Bank of Italy's projections for the variations of consumption, investments, exports and imports, available at <https://www.bancaditalia.it/pubblicazioni/proiezioni-macroeconomiche/2020/Proiezioni-Macroeconomiche-Italia-dicembre-2020.pdf.pdf>

Other parameters and values in this section are calibrated to data from 2010-2019/20

2.6.1.4 Innovation

Technological progress is endogenous and determined at the industry level. Innovation can reduce intermediate input requirements or increase labour productivity (denoted by λ) or both. Importantly, a new technology does not necessarily affect all technical coefficients equally: some coefficients may decrease while others may increase or remain unchanged.

Let $\tilde{a}_j = \frac{1}{19} \sum_{i=1}^{19} a_{i,j,t}$ be industry j 's mean intermediate input requirement per unit of output at time t , and let $\Delta \tilde{a}_{j,t} = \tilde{a}_{j,t} - \tilde{a}_{j,t-1}$ and $\Delta \lambda_{j,t} = \lambda_{j,t} - \lambda_{j,t-1}$. The adoption of a new *intermediate-input-saving technology* in period t leads to a variation in mean intermediate input requirements of $\Delta^a \tilde{a}_{j,t} < 0$ and a variation in labour productivity of $\Delta^a \lambda_{j,t} \geq 0$. Similarly,

the adoption of a new *labour-saving technology* leads to a variation in labour productivity of $\Delta^\lambda \lambda_{j,t} > 0$ and a variation in mean input requirements of $\Delta^\lambda \tilde{a}_{j,t} \geq 0$. (As described below, the signs of $\Delta^a \lambda_{j,t}$ and $\Delta^\lambda \tilde{a}_{j,t}$ are based on historical data.) There is assumed to be a general trade-off between improvements in intermediate input requirements and labour productivity, which is captured by the following inequalities:

$$\Delta^a \tilde{a}_{j,t} < \Delta^\lambda \tilde{a}_{j,t} \quad \text{and} \quad \Delta^\lambda \lambda_{j,t} > \Delta^a \lambda_{j,t}.$$

Finally, the adoption of a new *intermediate-input-and-labour-saving technology* (henceforth referred to as a *win-win technology*) leads to an increase in labour productivity of $\Delta^\lambda \lambda_{j,t}$ and a decrease in mean intermediate input requirements of $\Delta^a \tilde{a}_{j,t}$. Thus, on average the win-win technology combines the best features of the intermediate-input-saving and the labour-saving technologies.

For each industry j and period t , one or more of four possible cases can occur, each corresponding to a different technology (labelled $T1$ - $T4$):

$T1$. No new technology is discovered, in which case

$$\Delta \lambda_{j,t} = \Delta \tilde{a}_{j,t} = 0;^{17}$$

$T2$. Discovery of a new labour-saving technology, in which case

$$\Delta \lambda_{j,t} = \Delta^\lambda \lambda_{j,t} \quad \text{and} \quad \Delta \tilde{a}_{j,t} = \Delta^\lambda \tilde{a}_{j,t};$$

$T3$. Discovery of a new intermediate-input-saving technology, in which case

$$\Delta \lambda_{j,t} = \Delta^a \lambda_{j,t} \quad \text{and} \quad \Delta \tilde{a}_{j,t} = \Delta^a \tilde{a}_{j,t};$$

$T4$. Discovery of a new win-win technology, in which case

$$\Delta \lambda_{j,t} = \Delta^\lambda \lambda_{j,t} \quad \text{and} \quad \Delta \tilde{a}_{j,t} = \Delta^a \tilde{a}_{j,t}.$$

The innovation process is modelled in four steps. First, new technologies are discovered. Second, the magnitude of the $\Delta \lambda$ and Δa coefficients (i.e. the extent of the innovations) is determined. Third, a choice is made on whether to adopt one of the new technologies or not. Fourth, the chosen technology is implemented.

¹⁷This is actually a simplification, made to keep the exposition as simple as possible for now. As detailed later in this section, labour productivity and intermediate input requirements can vary over time even when no new technology is discovered.

Step 1: discovery The probability that new technologies are discovered depends on the evolution of labour and intermediate input costs over time, which is captured by:

$$\dot{\theta}_{j,t} = \frac{UIC_{j,t-1}/ULC_{j,t-1}}{UIC_{j,t-2}/ULC_{j,t-2}} - 1, \quad (2.12)$$

where UIC and ULC are the unit cost of labour and the unit cost of intermediate inputs, respectively (see Section 2.6.1.9).¹⁸ If in a certain industry j the unit cost of intermediate inputs increases relative to the unit cost of labour, then $\dot{\theta}_{j,t} > 0$ and firms become relatively more likely to develop $T3$ technologies than $T2$ technologies. Conversely, if $\dot{\theta}_{j,t} < 0$, then the probability that a $T2$ technology is developed increases.

Formally, for each period t and each $z \in \{2, 3, 4\}$, a random number is drawn from a uniform distribution on $[0, 1]$. A new Tz technology is discovered if the draw exceeds a threshold value $1 - \text{Prob}_{j,t}(Tz)$. The probabilities corresponding to $T2$ and $T3$ technologies are

$$\text{Prob}_{j,t}(T2) = \text{Prob}_{j,t-1}(T2) \left[1 - \vartheta \dot{\theta}_{j,t} \right] \quad (2.13)$$

and

$$\text{Prob}_{j,t}(T3) = \text{Prob}_{j,t-1}(T3) \left[1 + \vartheta \dot{\theta}_{j,t} \right], \quad (2.14)$$

respectively, where $\vartheta > 0$ is a positive parameter. Note that both probabilities are bound within $[0.05, 0.95]$. The probability corresponding to a $T4$ -type technology is

$$\text{Prob}_{j,t}(T4) = \text{Prob}_{j,t}(T2 \cap T3) = \text{Prob}_{j,t}(T2) \text{Prob}_{j,t}(T3). \quad (2.15)$$

Finally, the old technology is always available, that is $\text{Prob}_{j,t}(T1) = 1$. Assuming independent random draws, the probabilities corresponding to different combinations of new technologies being available in a certain period are the following (subscripts are omitted for brevity).

$$\begin{aligned} \text{Only } T1 \text{ is available:} & \quad [1 - \text{Prob}(T2)] [1 - \text{Prob}(T3)] [1 - \text{Prob}(T4)]. \\ T1 \text{ and } T2 \text{ are available:} & \quad \text{Prob}(T2) [1 - \text{Prob}(T3)] [1 - \text{Prob}(T4)]. \\ T1 \text{ and } T3 \text{ are available:} & \quad [1 - \text{Prob}(T2)] \text{Prob}(T3) [1 - \text{Prob}(T4)]. \\ T1, T2 \text{ and } T3 \text{ are available:} & \quad \text{Prob}(T2) \text{Prob}(T3) [1 - \text{Prob}(T4)]. \\ T1, T2 \text{ and } T4 \text{ are available:} & \quad \text{Prob}(T2) [1 - \text{Prob}(T3)] \text{Prob}(T4). \\ & \quad \dots \quad \dots \\ T1, T2, T3 \text{ and } T4 \text{ are available:} & \quad \text{Prob}(T2) \text{Prob}(T3) \text{Prob}(T4). \end{aligned}$$

¹⁸Note that here intermediate input costs also include unit carbon costs if the carbon tax is active (see section 2.6.1.9).

Step 2: how much innovation? The extent of technological progress in industry j depends on the values of $\Delta a_{1,j}, \dots, \Delta a_{19,j}$ and $\Delta \lambda_j$ associated with each available new technology. Each of these values is drawn from a different normal distribution, the mean and variance of which are derived from time series data on intermediate trade and labour productivity, respectively.

All distributions are adjusted by a scaling factor calibrated to the historical data. Subsequently, each distribution is shifted to the right or left by adding or subtracting the standard deviation to the mean. In the case of a $T4$ (win-win) technology, the distribution of labour productivity is shifted to the right and the distributions of intermediate input requirements are shifted to the left; in the case of a $T2$ (labour-saving) technology, all distributions are shifted to the right; in the case of a $T3$ (intermediate-input-saving) technology, all distributions are shifted to the left.

A schematic representation of all possible innovations for a given industry pair ij is shown in Figure 2.6.3. The pink circles represent the sets of possible $(\Delta a_{i,j}, \Delta \lambda_j)$ pairs within a certain confidence interval, whereas the blue and the red points represent the average values and the actual realisations of the random variables, respectively.

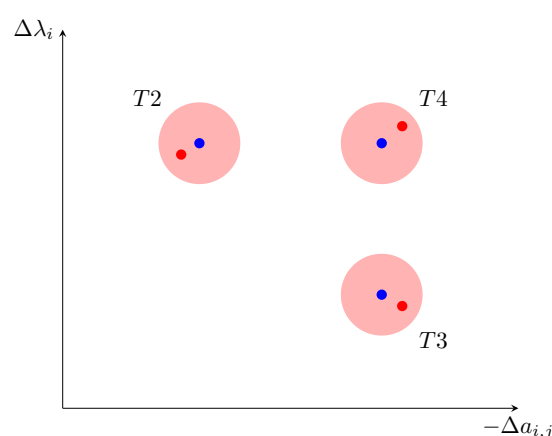


Figure 2.6.3: Possible cases of technological development. The pink circles represent the sets of possible $(\Delta a_{i,j}, \Delta \lambda_j)$ pairs for each technology; the blue points represent the average values; the red points represent the actual realisations of the random variables.

Step 3: choice Firms decide whether to adopt a new technology and, if so, which one. To do this they calculate the unit full cost of production (defined in Section 2.6.1.9) that would result from adopting each of available technologies, and then choose the cost-minimising alternative¹⁹.

¹⁹Again, intermediate input costs also include unit carbon costs if applicable.

Note that technologies $T2$ - $T4$ are not necessarily cost-saving relative to $T1$, and therefore will not always be adopted (although this case is quite unlikely).

Step 4: implementation If a new technology ($T2$, $T3$ or $T4$) is adopted at time t , then each technical coefficient $a_{i,j}$ changes by the amount $\Delta a_{i,j,t}$ determined in Step 2; analogously, labour productivity changes by the amount $\Delta \lambda_{j,t}$.

Technical coefficients and labour productivity can change over time even when no technological innovation occurs. This reflects the idea that production processes can become more efficient even without new discoveries, due to the progressive diffusion of the newest available technology. Suppose that a new technology is adopted in period t and that $T1$ is adopted in the following m periods. In each of these m periods, labour productivity will be given by:

$$\lambda_{j,t+g} = \frac{\lambda_{j,t+g}^{\max} [K_{j,t+g} - (1 - \delta_j) K_{j,t+g-1}] + \lambda_{j,t+g-1} (1 - \delta_j) K_{j,t+g-1}}{K_{j,t+g}}, \quad g = 1, \dots, m, \quad (2.16)$$

where K_j and δ_j are the capital stock and the capital depreciation rate in industry j , respectively, and $\lambda_{j,t+g}^{\max} \equiv \lambda_{j,t+g-1} + \Delta \lambda_{j,t+g-1}$, so that $\Delta \lambda_{j,t+g-1}$ is the greatest possible change in labour productivity between periods $t+g-1$ and $t+g$, which would occur if the whole capital stock at time $t+g$ consisted of new capital goods (that is investments). By construction, this upper bound for the change in productivity decreases period after period, because as time goes on there is less and less room for further implementation of the technology discovered at time t in the economy. As shown in equation (2.16), labour productivity is calculated as a weighted average between $\lambda_{t+g}^{\max}$ and λ_{t+g-1} , where the weights are the proportions of new and old capital at time $t+g$. If $(1 - \delta_j) K_{j,t+g-1} = 0$ (i.e. all capital goods are new), then $\lambda_{j,t+g} = \lambda_{j,t+g}^{\max}$; conversely, if $(1 - \delta_j) K_{j,t+g-1} = K_{j,t+g}$ (i.e. no investment is made), then $\lambda_{j,t+g} = \lambda_{j,t+g-1}$. Similar reasoning applies in the case of technical coefficients, meaning that:

$$a_{i,j,t+g} = \frac{a_{i,j,t+g}^{\max} [K_{j,t+g} - (1 - \delta_i) K_{j,t+g-1}] + a_{i,j,t+g-1} (1 - \delta_i) K_{j,t+g-1}}{K_{j,t+g}}. \quad (2.17)$$

The nature and extent of technological progress at one point in time can influence future innovations, that is technological progress has feedback effects on its own evolution. For example, suppose that technology $T2$ is adopted in industry j at time t . Ceteris paribus, this will result in an increase in $\dot{\theta}_{j,t+1}$, which will reduce $\text{Prob}_{j,t+1}(T2)$ and increase $\text{Prob}_{j,t+1}(T3)$. The opposite result will occur if technology $T3$ is adopted. Taxes and other policy measures can influence $\dot{\theta}_j$ through changes in the relative unit costs of labour and intermediate inputs, thereby affecting innovation. Finally, changes in intermediate input requirements can affect the entire structure of the economy, e.g. by altering capital productivity and energy efficiency (see

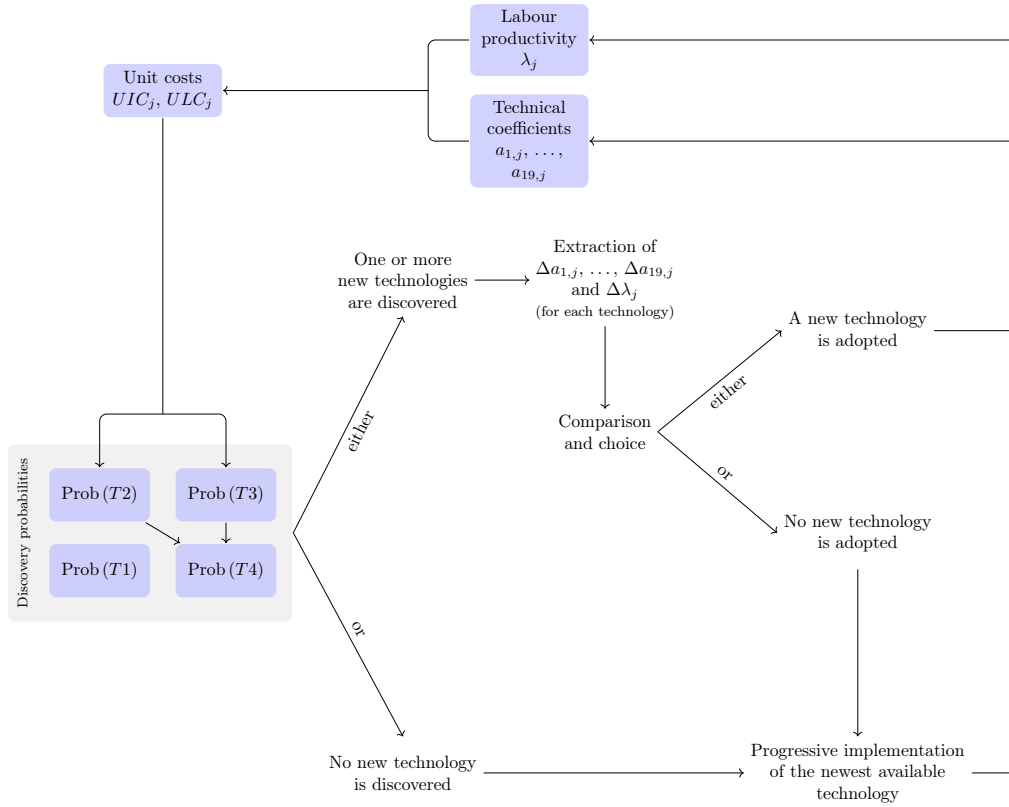


Figure 2.6.4: Conceptual framework of the innovation module

Sections 2.6.1.5 and 2.6.1.11).

Data sources: Innovation

The variations in labour productivity are calculated from EU Klems data from 2010-2015, whereas historical changes in the technical coefficients are based on a structural decomposition analysis, isolating the effect of technological innovation, from WIOD Rev. 1, 1995-2009 data for Italy.

Other parameters and values in this section are calibrated to data from 2010-2019/20.

2.6.1.5 Investment and profits

2.6.1.5.1 Profits

In national accounts, industry value added (VA) is the difference between the value of output and total (domestic plus imported) intermediate costs:

$$VA_{i,t} = y_{i,t} - \hat{z}_{i,t}, \quad (2.18)$$

where $\hat{z}_{i,t} = \sum_j z_{j,i,t} + \sum_j z_{j,i,t}^{imp}$. Gross profits ($\hat{\Pi}$) are defined as the industry's value added minus value added taxes (VAT) and total labour costs, which are given by the sum of the gross wage bill ($GW B$, see Section 2.6.1.7) and social contributions levied on employers (T^{SF}):

$$\hat{\Pi}_{i,t} = VA_{i,t} - VAT_{i,t} - (GW B_{i,t} + T_{i,t}^{SF}). \quad (2.19)$$

The model includes an option to introduce a carbon tax (T^{carb}) on industrial emissions:

$$T_{i,t}^{carb} = \tau_t^{carb} GHG_{i,t}, \quad (2.20)$$

where τ^{carb} is the carbon tax rate. The default setting is carbon tax = ON. Equation (2.19) therefore becomes:

$$\hat{\Pi}_{i,t} = VA_{i,t} - VAT_{i,t} - (GW B_{i,t} + T_{i,t}^{SF}) - T_{i,t}^{carb}. \quad (2.19')$$

Finally, gross profits are subject to corporate income taxation. Thus, an industry's net profits (Π) are equal to:

$$\Pi_{i,t} = (1 - \tau^{CIT}) \hat{\Pi}_{+i,t}, \quad (2.21)$$

where τ^{CIT} is the corporate income tax rate and $\hat{\Pi}_+ \equiv \max \{ \hat{\Pi}, 0 \}$ is the positive part of $\hat{\Pi}$.

2.6.1.5.2 Investment

Firms update their desired productive capacity striving to be able to satisfy their demand at the normal capacity utilisation level. The desired investment (I^*) in industry i is increasing with the degree of 'overutilisation' (with respect to the normal rate of capacity utilisation), and conversely it decreases in case of 'underutilisation'. Formally:

$$I_{i,t}^* = \min \{ I_{i,t-1}^* [1 + \omega_i^I (uc_{i,t-1} - uc^N)], 0 \}, \quad (2.22)$$

where uc and uc^N are the actual and the normal capacity utilisation rates, respectively, and $\omega^I \in [0, 1]$ is a positive parameter reflecting the pace of adjustment.

The capacity utilisation rate is defined as the ratio between current and full capacity output:

$$uc_{i,t} = \frac{y_{i,t}}{y_{i,t}^{FC}}. \quad (2.23)$$

If the capacity constraint on production is activated (see Section 2.6.1.2), then uc can only take values between 0 and 1. Full capacity output is given by:

$$y_{i,t}^{FC} = \epsilon_{i,t}^K K_{i,t}, \quad (2.24)$$

where K is the current stock of physical capital and ϵ^K denotes capital productivity. The latter changes over time following changes in technical coefficients:

$$\epsilon_{i,t}^K = \epsilon_{i,0}^K \frac{\sum_{j=1}^{19} a_{j,i,0}}{\sum_{j=1}^{19} a_{j,i,t}}. \quad (2.25)$$

The initial of capital productivity is given by the ratio between initial real output-capital ratio (y_i^0/K_i^0) and the fixed normal capacity utilisation rates (uc^N).²⁰

$$\epsilon_{i,0}^K = \frac{y_i^0}{K_i^0 \cdot uc^N}. \quad (2.26)$$

Industries' capacity to undertake their desired level of investment is constrained by their capacity to finance part of these investments out of their profits. In particular, the maximum investment an industry can finance (I_i^{max}) is determined by profits after debt repayment and taxes ($\Pi_{i,t}^{free}$), the fixed equity-to-liability ratio (elr), and the rate of leverage (lev).²¹

$$I_{i,t}^{max} = \frac{\max\{\Pi_{i,t}^{free}, 0\}}{p_{t-1}^K elr} \cdot \left(\frac{\min\{elr, (1 - lev_{i,t-1})\}}{elr} \right), \quad (2.27)$$

where $I_{i,t}^{max}$ is deflated by the price of capital (p_t^K), which is a sum of the prices of all industries weighted by their share in GFCF supply (**b**, see also below), since the desired level of investment is defined in real terms. The first term of the equation defines the maximum amount of investments that a sector can afford under the fixed value of the equity-to-liability elr . While the second part of the equation takes into account the idea that if the actual industry specific equity-to-liabilities ratio ($1 - lev_{i,t-1}$) is lower than the fixed value elr , the industry is able to afford a lower level of investments.

Another possible option is to omit the effect of lev_i on financing conditions, implying that everyone is constrained at elr :

$$I_{i,t}^{max} = \frac{\max\{\Pi_{i,t}^{free}, 0\}}{p_{t-1}^K elr}; \quad (2.27')$$

or no constraints on investment at all.

Consequently, the actual gross fixed capital formation in real terms is defined as the minimum between desired and maximum possible investment:

$$I_{i,t} = \min\{I_{i,t}^*, I_{i,t}^{max}\} \quad (2.28)$$

However, we assume that most public sectors do not face the same financing restrictions. Thus, for $i = m = \{\text{education, public, health}\}$ it follows:

$$I_{m,t} = I_{m,t}^* \quad (2.29)$$

The model also incorporates external shocks on $I_{i,t}$ in 2020 and 2021 driven by the Covid-19 pandemic. The capital stock is then a function of investment, the previous period capital stock and capital depreciation.

²⁰Therefore, we are assuming that in the initial period, industry i produced under normal capacity utilisation.

²¹Where elr is defined as a specific ratio of equity over the sum of equity and debt (liabilities), while lev is defined as industry specific debt over total assets (capital stock in nominal terms).

$$K_{i,t} = I_{i,t} + (1 - \delta_i)K_{i,t-1}. \quad (2.30)$$

Note, that the investments above represent the investment demands from each industry and do not correspond to the final demand of gross fixed capital formation faced by each industry ($\hat{f}_{i,t}^I$). We redistribute investment demand by multiplying it with vector $\mathbf{b}$.

$$\hat{f}_{i,t}^I = \mathbf{b} \cdot \sum_i I_{i,t}^{nom} \quad (2.31)$$

where $I_{i,t}^{nom}$ is the nominal level of investment

$$I_{i,t}^{nom} = p_{t-1}^K I_{i,t}, \quad (2.32)$$

while $\mathbf{b}$ allocates total investment demand to final goods demand for all sectors, i.e. its entries are the fractions of investment demand directed at each sector. Subsequently, total investment demand is split between domestic demand ($f_{i,t}^I$) and demand for imports ($f_{i,t}^{I,imp}$) according to:²²

$$f_{i,t}^I = (1 - \gamma_i^I) \cdot \hat{f}_{i,t}^I; \quad f_{i,t}^{I,imp} = \gamma_i^I \cdot \hat{f}_{i,t}^I. \quad (2.33)$$

2.6.1.5.3 Loans and equity

The initial stock of equities (Eq_i^0) in each industry is determined as the difference between the initial stock of loans (H^0) and the nominal value of their capital stock.²³

$$Eq_i^0 = K_i^0 - H_i^0 \quad (2.34)$$

We may now define profits free for investment as a function of net profits, industry specific interest rates (i^{ind}) and debt repayment (dr).

$$\Pi_{i,t}^{free} = \Pi_{i,t} - i_{i,t}^{ind} H_{i,t} - dr_{i,t} \quad (2.35)$$

The interest rate paid by industries is given by a markup (ω^H) over the policy rate (i_t). The mark-up is increasing in the leverage of each industry, with the fixed parameter $\omega^H \in (0, 1)$ representing the sensitivity of an industries' interest rate to its leverage. When an industry's leverage is equal or smaller than or equal to $(1 - elr)$ the additional term is equal to zero.

$$i_{i,t}^{ind} = i_t + \omega^H \cdot \max \left\{ \frac{elr}{1 - lev_{i,t-1}} - 1, 0 \right\}, \quad (2.36)$$

where the leverage ratio, $lev_{i,t-1}$, is equal to:

²²See section 2.6.1.3.

²³See section 2.6.1.6 for more details.

$$lev_{i,t-1} = \frac{H_{i,t-1}}{K_{i,t-1}^{nom}}. \quad (2.37)$$

The repayment of the principal ($dr_{i,t}$) is a fixed proportion of total outstanding debt and the average lifetime of the investments associated with new investments in fixed capital (lt_i), implying that an industry aims to have paid off debt incurred from an investment at the time the investment is fully depreciated:

$$dr_{i,t} = \frac{H_{i,t}}{lt_i}. \quad (2.38)$$

The remaining profits are then used to partially finance the investment of the period and to pay dividends.

The direct expenditure for investments (Inv) corresponds to the share of the investment that the industry finances out of its profit, which in turn depends on the the values of elr and $lev_{i,t-1}$:

$$Inv_{i,t} = I_{i,t}^{nom} elr \cdot \left(\frac{elr}{\min \{elr, (1 - lev_{i,t-1})\}} \right), \quad (2.39)$$

As aforementioned, there are two other possibilities on the restriction of investments and also impacting its equity to loans ratio when financing it, that are considered here. For both other scenarios, (2.27') or no restrictions, investment is financed at the elr :

$$Inv_{i,t} = I_{i,t}^{nom} elr. \quad (2.39')$$

Subsequently, dividends paid to equity owners are computed as a fixed proportion (div) of the

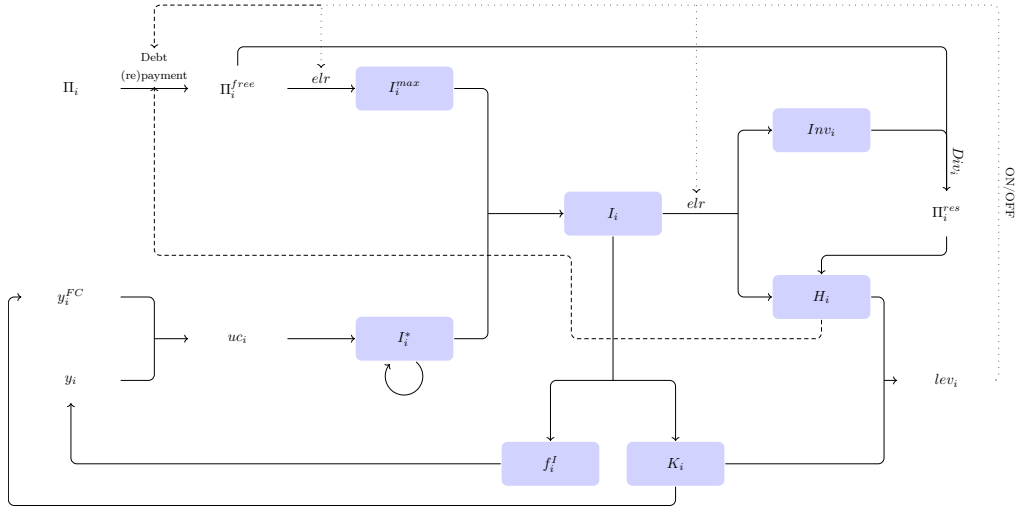


Figure 2.6.5: Conceptual framework of the Investments and profits module

remaining profits after investments:

$$Div_{i,t} = div \left(\Pi_{i,t}^{free} - Inv_{i,t} \right). \quad (2.40)$$

and residual profits ($\Pi_{i,t}^{res}$) are given by:

$$\begin{aligned}\Pi_{i,t}^{res} &= \Pi_{i,t}^{free} - Inv_{i,t} - Div_{i,t} \\ &= (1 - div) \left(\Pi_{i,t}^{free} - Inv_{i,t} \right),\end{aligned}\quad (2.41)$$

Consequently, the dynamics of private debt in each industry depend on the portion of nominal investments financed by loans and debt repayment (also in the form of residual profits):

$$H_{i,t,+} = H_{i,t-1} + I_{i,t}^{nom} - Inv_{i,t} - \Pi_{i,t}^{res} - dr_{i,t}. \quad (2.42)$$

Using an accounting identity, the amount of equity of each sector is calculated as

$$E_{i,t} = K_{i,t} - H_{i,t}. \quad (2.43)$$

Accordingly, equity increases with investment financed out of profits and debt repayments and decreases with depreciation.

Data sources: Investment and Profits

- elr is defined as 1–Book debt to capital ratio for large firms in Western Europe. Available at: <https://pages.stern.nyu.edu/~adamodar/pc/datasets/dbtfundEurope.xls>, 05/01/2021 update.
- uc^N values are approximate to: Setterfield and Avritzer (2020, p. 909)
- $\mathbf{b}$ is based on calculations from Eurostat IO data
- for div we opted for a maximum of 30% which is below the market average of 40.37% for net cash returned over free cash flow to equity (post-debt), which measure how much cash is actually returned to shareholder. It is defined as: $(dividends + buybacks - stockissuances)/(netIncome + depreciation - capitalexpenditure + newdebtraised - debtrepaid)$. However, the data considers only publicly traded companies in Western Europe, thus the economy average is very likely well below the measured 40.37%. Data is available at: <https://pages.stern.nyu.edu/~adamodar/pc/datasets/divfundEurope.xls>, 05/01/2021 update.
- lt_i , δ_i , $K_{i,0}$, $y_{i,0}$ are based on and calculated from EU Klems data for Italy in 2010
- $I_{i,0}$ is derived from OECDstat, table 8A: capital formation by activity. Available at: https://stats.oecd.org/Index.aspx?DataSetCode=SNA_TABLE8A#.

- The covid shock is based on the Bank of Italy's projections for the variations of consumption, investments, exports and imports, available at <https://www.bancaditalia.it/pubblicazioni/proiezioni-macroeconomiche/2020/Proiezioni-Macroeconomiche-Italia-dicembre-2020.pdf>
- Other parameters and values in this section are calibrated to data from 2010-2019/20.

2.6.1.6 Finance and wealth distribution

The finance module is divided into two parts: *i.* the allocation of wealth between individuals and their financial income, and *ii.* the issuance of private loans and the revenue of the financial industry.

2.6.1.6.1 Wealth distribution

Individuals allocate their savings and wealth between three assets: deposits, government bonds and equities. The first necessary step to determine the wealth distribution is to obtain the initial allocation of total financial wealth in these three assets among low, middle and high skill individuals. Due to data limitations we apply some simplifying assumptions to obtain a framework that respects the stock-flow consistency of the model and can be adapted to different countries.

Initial wealth distribution

The distribution of wealth among skills is based on data from the world inequality database. Apart from providing the value of per adult national wealth for a wide variety of countries, the database also contains data on the share of national wealth owned by individuals in each income percentile for a few countries. Hence, we first obtain the value of total national wealth (W^0) multiplying per adult national wealth (w^0) by the number of adults (N^A).

$$W^0 = w^0 N^A. \quad (2.44)$$

Next we map the total national wealth to the three skills using the income percentiles. We attribute the share of wealth of the bottom 0-30th percentile of the population to low-skill individuals (ζ_{low}^W), that between the 30th and 70th percentile to middle-skill individuals (ζ_{middle}^W), high-skill individuals own the wealth corresponding to the 70-99.9th (ζ_{high}^W), and the remaining wealth is assigned to capitalists that constitute 1% of the adult population.²⁴ Once the initial

²⁴That is a good approximation for France where the skill shares of the 20+ population can be expressed so that low-skill individuals represents the lower 0-30.7% interval, the middle-skill ones the 30.7-71.6% interval and the high-skill

wealth is distributed among skills, the second step is to distribute such wealth among the available assets. For this purpose additional assumptions are required. Firstly, generally speaking we assume that the degree of financial sophistication increases with the skill level. Therefore, we assume that low-skill individuals hold only bank deposits that pay no interest, meaning that they receive no financial income.

$$D_{low}^0 = W_{low}^0, \quad (2.45)$$

where D^0 is the amount of deposits in the initial period. Next, we assume that middle-skill individuals' distribute their wealth between deposits and bonds. The initial amount of bonds held by the middle skill group is determined by the initial stock of bonds available (B^0), weighted by the share of middle-skill individuals' initial wealth in proportion to the wealth of all groups holding bonds (middle-skilled, high-skilled and capitalists). The remaining part is allocated to deposits.

$$\begin{cases} B_{middle}^0 &= B^0 \left(\frac{W_{middle}^0}{W_{middle}^0 + W_{high}^0 + W_{cap}^0} \right) \\ D_{middle}^0 &= W_{middle}^0 - B_{middle}^0 \end{cases} \quad (2.46)$$

High-skill individuals and capitalists allocate their wealth between deposits, bonds and equities. For these two groups we also adopt a simplifying assumption to determine the initial distribution of wealth between assets. Their initial amount of bonds is computed in the same way used for the middle-skill individuals', and an analogous approach is used for the initial equity holdings which are a function of the initial stock of these assets (Eq^0) and the initial distribution of total wealth between high skills individuals and capitalists.

Therefore the initial wealth held by high skill individuals is allocated in the following way:

$$\begin{cases} Eq_{high}^0 &= Eq^0 \left(\frac{W_{high}^0}{W_{high}^0 + W_{cap}^0} \right) \\ B_{high}^0 &= B^0 \left(\frac{W_{high}^0}{W_{middle}^0 + W_{high}^0 + W_{cap}^0} \right) \\ D_{high}^0 &= W_{high}^0 - Eq_{high}^0 - B_{high}^0 \end{cases} \quad (2.47)$$

and analogously for capitalists:

$$\begin{cases} Eq_{cap}^0 &= Eq^0 \left(\frac{W_{cap}^0}{W_{high}^0 + W_{cap}^0} \right) \\ B_{cap}^0 &= B^0 \left(\frac{W_{cap}^0}{W_{middle}^0 + W_{high}^0 + W_{cap}^0} \right) \\ D_{cap}^0 &= W_{cap}^0 - Eq_{cap}^0 - B_{cap}^0 \end{cases} \quad (2.48)$$

individuals the 71.6-100% interval. See the supplementary information of D'Alessandro et al. (2020).

In addition, given that, due to lack of data, it is not possible to compute an exact gender-specific initial wealth, we divide the initial wealth of each type of assets for each skill level equally between male and female individuals.

Accumulation of new wealth

We apply the same rules for the accumulation of new wealth from savings (S). However an additional assumption required at this stage is that all new assets issued domestically are purchased by residents and that these don't own any foreign assets. Besides the lack of data, any attempt to model foreign assets ownership would require a multi-national model which is beyond the scope of this model. Therefore, the total flow of new equities and bonds is given by the change in equities over all industries (ΔE_{qt}) and the government's deficit (G_t^D).²⁵

As aforementioned, low skill individuals transfer all of their saving (if any) in deposits:

$$\Delta D_{low,g,t} = S_{low,g,t}. \quad (2.49)$$

Middle skill individuals divide their saving between bonds and deposits, for the amount of deposits we use a modified version of equation (2.46) where the total amount of bonds available (G_t^D) is distributed among the groups (defined by gender and skills) using the total wealth of the previous period as weights:

$$\begin{cases} \Delta B_{middle,g,t} = G_t^D \left(\frac{W_{middle,g,t-1}}{W_{middle,t-1} + W_{high,t-1} + W_{cap,t-1}} \right) \\ \Delta D_{middle,g,t} = S_{middle,g,t} - \Delta B_{middle,g,t} \end{cases} \quad (2.50)$$

Finally high skills and capitalists divide their wealth among all possible assets

$$\begin{cases} \Delta E_{qhigh,g,t} = \Delta E_{qt} \left(\frac{W_{high,g,t-1}}{W_{high,t-1} + W_{cap,t-1}} \right) \\ \Delta B_{high,g,t} = G_t^D \left(\frac{W_{high,g,t-1}}{W_{middle,t-1} + W_{high,t-1} + W_{cap,t-1}} \right) \\ \Delta D_{high,g,t} = S_{high,g,t} - \Delta B_{high,g,t} \end{cases} \quad (2.51)$$

$$\begin{cases} \Delta E_{qcap,t} = \Delta E_{qt} \left(\frac{W_{cap,t-1}}{W_{high,t-1} + W_{cap,t-1}} \right) \\ \Delta B_{cap,t} = G_t^D \left(\frac{W_{cap,t-1}}{W_{middle,t-1} + W_{high,t-1} + W_{cap,t-1}} \right) \\ \Delta D_{cap,t} = S_{cap,t} - \Delta B_{cap,t} \end{cases} \quad (2.52)$$

Please note that savings are not used to finance ΔE_q , but instead - as aforementioned in section 2.6.1.5 - changes in equity are financed by profits, be they used for debt repayments or to finance

²⁵See Sections 2.6.1.5 and 2.6.1.8 for more details.

part of investments, or decrease with depreciation. Hence, changes in equity are determined entirely on the side of the industries and capture changes in the *value* of existing shares, not a change in the amount of shares.

Subsequently, stocks of assets for any group (defined by skill and gender plus capitalists) are given by:

$$\begin{cases} Eq_{s,g,t} &= \Delta Eq_{s,g,t} + Eq_{s,g,t-1} \\ B_{s,g,t} &= \Delta B_{s,g,t} + B_{s,g,t-1} \\ D_{s,g,t} &= \Delta D_{s,g,t} + D_{s,g,t-1} \end{cases} \quad (2.53)$$

The sum of the three assets define the group's total wealth in period t ($W_{s,g,t}$):

$$W_{s,g,t} = Eq_{s,g,t} + B_{s,g,t} + D_{s,g,t}, \quad (2.54)$$

It is worth pointing out that after the initial periods, the total wealth hold by individuals of the same skill level but different gender will not necessarily be equal anymore due to the possible differences in income and hence savings.

Finally we define the gross financial income (GFI) for each group.

$$GFI_{middle,g,t} = i_t B_{middle,g,t-1} \quad (2.55)$$

$$GFI_{high,g,t} = i_t B_{high,g,t-1} + \sum_i Div_{t,i} \frac{Eq_{high,g,t-1}}{Eq_{t-1}} \quad (2.56)$$

$$GFI_{cap,t} = i_t B_{cap,t-1} + \sum_i Div_{t,i} \frac{Eq_{cap,t-1}}{Eq_{t-1}} \quad (2.57)$$

Their respective net financial income is obtained deducting a fixed tax rate levied on financial income (τ^F)

$$NFI_{s,g,t} = (1 - \tau^F) GFI_{s,g,t}. \quad (2.58)$$

2.6.1.6.2 Financial sector

The financial sector holds the stock of deposits (D_s) and of private loans (H_i). The dynamics of the private loans are described in Section 2.6.1.5. However, the interest and principal payments on private loans by all industries to the financial industry must correspond to the revenue received by the financial industry in the input-output intermediate trade. Hence, we may use the initial values for intermediate trade (Z), interest paid by each industry ($i_{i,t}^{ind}$) and the average lifetime of fixed capital by industry (lt_i).

$$\begin{aligned} Z_{i,fin}^0 &= i_i^{ind} H_i^0 + dr_i^0 \\ &= i_i^{ind} H_i^0 + \frac{H_i^0}{lt_i}, \end{aligned} \quad (2.59)$$

As seen in Section 2.6.1.5, the interest rates paid by industries (i_i^{ind}) is also an increasing function of their leverage. However, to recover a closed form expression for the initial stock of loans (and thus, leverage) using the initial values of intermediate trade, basic interest rates and other fixed parameters of the model, we set to zero the term $\left(\omega^H \cdot \max \left\{ \frac{elr}{1-lev_{i,t-1}} - 1, 0 \right\}\right)$.²⁶ It then follows:

$$Z_{i,fin}^0 = \left(i^0 + \frac{1}{lt_i}\right) H_i^0, \quad (2.60)$$

Therefore, the initial stock of loans is equal to:

$$H_i^0 = Z_{i,fin}^0 \left(i^0 + \frac{1}{lt_i}\right)^{-1}. \quad (2.61)$$

An alternative option to using the technical coefficient IO structure, is to use that the repayment of interest and principal over private loans from all industries constitutes the intermediate trade received by the financial sector. According to this equality, debt related payments of industry i must be equal to the final output of the financial industry multiplied by the technical coefficient $a_{i,fin,t}$ of the Leontief matrix:

$$i_{i,t}^{ind} H_{i,t-1} + dr_{i,t-1} = a_{i,fin,t} y_{fin,t}, \quad (2.62)$$

Data sources: Finance and Wealth

The data for the amount and the distribution of wealth are derived from the world inequality database.

The initial interest rate faced by firms is based on ECB data (<https://sdw.ecb.europa.eu/browse.do?node=9691390>).

2.6.1.7 Labour

Employment is determined by industries, which require labour in order to meet their production plans. The population development, but also its skill distribution are crucial factors of labour supply.

2.6.1.7.1 Skill distribution

The population development described in section 2.6.1.1 determines the working age population for males and females. Initially, this population is allocated to the three skill levels – low, medium and high – according to its initial distribution $\sigma_{s,g,0}$. This skill composition for each

²⁶This simplification is equivalent to assuming that all industries have a leverage below or equal to $1 - elr$, which allows us to obtain a closed form expressions for the initial H^0 . This is still better than assuming that all the initial loans are at equilibrium $H_i^0 = K_i^0(1 - elr)$.

age cohort from 15 years onwards changes over time due to differences in skill specific unemployment rates, causing people to transition between skills and labour supply to adjust to labour demand. For the population maturing into the labour force there is an additional exogenous skill specific trend derived from the data, capturing for instance implications from developments in education. Here, the dynamics of the skill developments on the population are illustrated by example of the low skilled population. The high skilled population is derived analogously for each age cohort, whereas the middle skilled population is obtained as the residual. The low skilled population in age cohort II is governed by

$$\begin{aligned} Pop_{low,g,t}^{II} = & (1 + \omega_{low}^u [u_{middle,t-1} - u_{low,t-1}]) [Pop_{low,g,t-1}^{II} - Mat_{low,g,t}^{II}] \\ & + [Mat_{g,t}^I (\sigma_{low,g,t-1}^{II} (1 + \omega_{low}^u [u_{middle,t-1} - u_{low,t-1}]) + g_{low,g}^{entry})] \\ & - \kappa_{II,g}^- Pop_{g,t-1}^{II} \sigma_{low,g,t-1}^{II}, \end{aligned} \quad (2.63)$$

where $Mat_{g,t}^I = \varrho_I Pop_{g,t-1}^I$ is the amount of people maturing into the working age, $\sigma_{low,g,t-1}^{II}$ is the existing share of low skilled people in the age group II at $t-1$, $u_{s,t}$ are the skill specific unemployment rates and ω_{low}^u captures the sensitivity of low skill labour supply of the age cohort n with respect to the difference in skill specific unemployment rates. If $u_{middle,t} > u_{low,t}$, so that unemployment is higher for middle skilled employees, more people will adapt towards a low skill. Here only the neighbouring skill's unemployment rate matters, assuming that a direct transition from low to high skill does not occur. In addition, $g_{low,g}^{entry}$ captures the trend in skill endowment upon entering the working age.

$Mat_{g,t}^{II} = \frac{Pop_{low,g,t-1}^{II}}{t^{II}}$ reflects the amount of low skilled people maturing out of the age cohort, with t^{II} being the amount of years that span the age cohort. Finally, $\kappa_{II,g}^- Pop_{g,t-1}^{II}$ reflects total deaths in age group II, which has to be scaled by the share of low skilled population in that group, $\sigma_{low,g,t-1}^{II}$ in order to capture deaths in the low skilled portion.

For age cohorts $n = III, IV$, the dynamics of the low skilled population are governed by

$$\begin{aligned} Pop_{low,g,t}^n = & (1 + \omega_{low}^u [u_{middle,t-1} - u_{low,t-1}]) [Pop_{low,g,t-1}^n - Mat_{low,g,t}^n] \\ & + Mat_{low,g,t}^{(n-1)} (1 + \omega_{low}^u [u_{middle,t-1} - u_{low,t-1}]) \\ & - \kappa_{n,g}^- Pop_{g,t-1}^n \sigma_{low,g,t-1}^n, \end{aligned} \quad (2.64)$$

where $Mat_{low,g,t}^n = \frac{Pop_{low,g,t-1}^n}{t^n}$, analogous to $Mat_{low,g,t}^{II}$ and $\kappa_{n,g}^- Pop_{g,t-1}^n \sigma_{low,g,t-1}^n$ reflects deaths in age group n . Finally, the dynamics of the low skilled population of age group V are given by

$$\begin{aligned} Pop_{low,g,t}^V = & Pop_{low,g,t-1}^V - \kappa_{V,g}^- Pop_{g,t-1}^V \sigma_{low,g,t-1}^V \\ & + Mat_{low,g,t}^{IV} (1 + \omega_{low}^u [u_{middle,t-1} - u_{low,t-1}]) \end{aligned} \quad (2.65)$$

Here, no skill transition takes place anymore within the age group and no maturation out of it occurs.

2.6.1.7.2 Labour supply

However, the labour supply is not directly given by the working age population, but also determined by the labour force participation rate (LFPR) of that population. The LFPR depends on the expected labour income when being in the labour force compared to the expected labour income when being out of the labour force.

Employed people will leave the labour force if

$$Y_{s,g,t}^{e,LF} = (1 - u_{s,g,t-1})w_{s,g,t-1}\bar{h}_{t-1} + u_{s,g,t-1}ub_{s,g,t-1} - ob_{s,g,t-1} \quad (2.66)$$

decreases, with $\bar{h}$ being the average amount of working hours, $ub_{s,g,t}$ reflecting the unemployment benefits for the respective skill and gender group and $ob_{s,g,t}$ capturing benefits received only when not being part of the labour force, which only applies to low skilled individuals. On the other hand people being out of the labour force or unemployed face

$$Y_{s,g,t}^{e,OLF} = (1 - u_{s,g,t-1})w_{s,g,t-1}\bar{h}_{t-1} + u_{s,g,t-1}ob_{s,g,t-1} - ob_{s,g,t-1}. \quad (2.67)$$

These expressions explicitly assume that unemployment benefits are only received by people who have been employed the previous period. Hence, a person entering the labour force but not finding a job would only receive the same benefits as if she had remained out of the labour force. Subsequently, the gender and skill specific LFPR, $\psi_{s,g}$, moves according to

$$\psi_{s,g,t} = \psi_{s,g,t-1} + (1 - e_{s,g,t-1})\omega_s^\psi \left(\frac{\Delta Y_{s,g,t}^{e,OLF}}{Y_{s,g,t-1}^{e,OLF}} \right) + e_{s,g,t-1}\omega_s^\psi \left(\frac{\Delta Y_{s,g,t}^{e,LF}}{Y_{s,g,t-1}^{e,LF}} \right), \quad (2.68)$$

where ω_s^ψ captures the sensitivity of labour force participation with respect to expected labour income differences. The equation is restricted such that $\psi_{s,g,t} \leq 1$ is guaranteed. Thus, rises in expected labour income relative to out of labour force benefits increases the LFPR and vice versa in the case of decreases. The weights are based on the employment rate, as only currently employed people will receive unemployment benefits, in case of unemployment next period.

2.6.1.7.3 Labour productivity

Labour productivity is defined by industry and changes over time according to technical progress as described in section 2.6.1.4.

2.6.1.7.4 Employment

Employment by industry is determined by how much labour is required to produce the planned

output, and hence depends on labour productivity, as well as hours worked:

$$L_{i,t} = \frac{y_{i,t-1}}{\lambda_{i,t} h_i}, \quad (2.69)$$

where h_i reflects constant annual hours worked, defined by industry.

Employment by industry, by skill is subsequently given by

$$L_{i,s,t} = L_{i,t} \cdot \max\{\rho_{i,s,0} + \sum_t g_{i,s}^\rho c_t, 0\}, \quad (2.70)$$

where $\sum_t g_{i,s}^\rho c_t$ captures the trend development over time. Here, $\rho_{i,s,0}$ reflects the initial skill distribution for an industry and $g_{i,s}^\rho$ are the initial trends in these distributions, which are exogenously derived from data. The conversion rate $c_t = 1 - \sqrt{\frac{t}{40}}$ renders the effect of the initial trend diminishing over time.

Employment by industry, by skill and by gender is subsequently determined by male ($m_{i,s,t}$) and female shares in employment

$$L_{i,s,g=m,t} = L_{i,s,t} m_{i,s,t}; \quad L_{i,s,g=f,t} = L_{i,s,t} (1 - m_{i,s,t}) \quad (2.71)$$

where the share of male employment is given by

$$m_{i,s,t} = m_{i,s,t-1} + \omega_s^m (u_{g=m,s,t-1} - u_{g=f,s,t-1}) + g_{i,s}^m c_{t-1} \quad (2.72)$$

Here, the last summand again captures the development of the initial exogenous trend $g_{i,s}^m$, whose effect is decreasing over time. The second summand reflects the effect of the comparative labour market tightness of male employment, by capturing the development of male to female unemployment, with $\omega_s^m > 0$ being a sensitivity parameter. If the male unemployment exceeds female unemployment, $m_{i,s,t}$ increases, as the male labour market is less tight. The equation is restricted such that $m_{i,s,t} \leq 1$ is guaranteed.

2.6.1.7.5 Wage

Finally, the wage depends on labour productivity growth $g_{i,t}^\lambda$, changes in gender and skill specific employment rates $\Delta e_{s,g,t}$ and inflation r_t (see section 2.6.1.9) according to

$$w_{i,s,g,t} = w_{i,s,g,t-1} (1 + \omega^\lambda g_{i,t}^\lambda + \omega^L \Delta e_{s,g,t} + \omega^p r_t) \quad (2.73)$$

where again $\omega^{\lambda/L} > 0$ capture the sensitivity of wages with respect to labour productivity/employment respectively and $\omega^p > 0$ being the sensitivity of the wage with respect to inflation.

Finally, the gross wage bill is given by

$$\begin{aligned} GWB_{i,s,g,t} &= w_{i,s,g,t} L_{i,s,g,t} h_i \\ \Rightarrow GWB_{i,t} &= \sum_s \sum_g GWB_{i,s,g,t} \end{aligned} \quad (2.74)$$

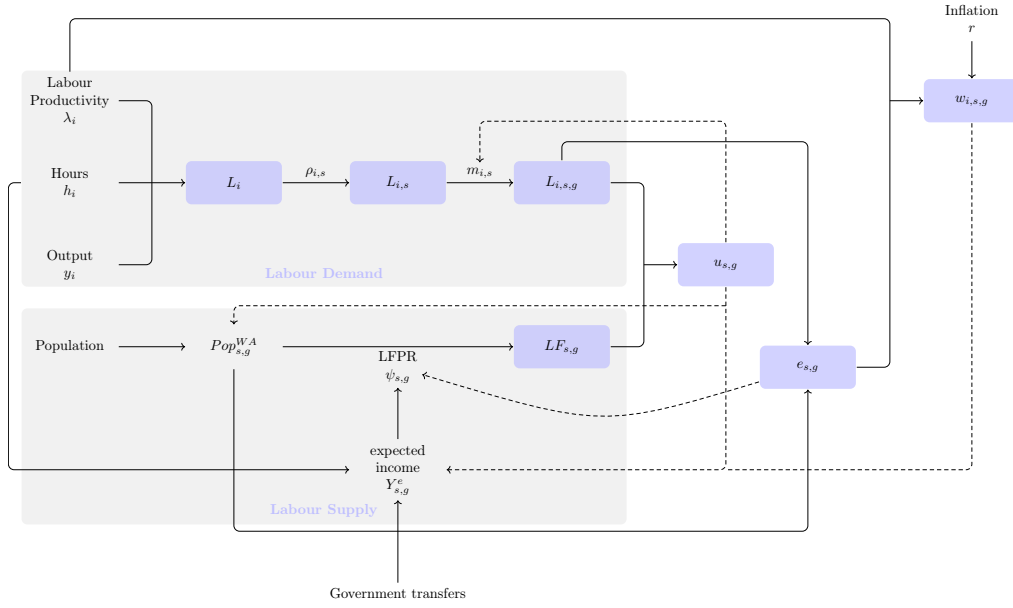


Figure 2.6.6: Conceptual framework of the labour module

Thus, the definition of GWB here includes social security contributions paid by employees, but excludes the ones paid by employers.

Data sources: Labour

- $\sigma_{s,g,0}^n$, $g_{s,g}^{entry}$, $u_{s,g,0}$ are based on and calculated from Eurostat data for Italy in 2010 (and beyond for the trends) and $\psi_{s,g,0}$ is calculated to respect real world data.
- h_i , $\lambda_{i,0}$, $\rho_{i,s}$, $m_{i,s,0}$, $w_{i,s,0}$, $g_{i,s}^\rho$, $g_{i,s}^m$ are based on and calculated from EU Klems data for Italy in 2010 (and beyond for the trends)
- Other parameters and values in this section are calibrated to data from 2010-2019/20.

2.6.1.8 Government

This section mainly focuses on government accounts, with its revenues and expenditures. Due to the abundance of sub- and superscripts present in this section, we will omit the time subscript t for simplicity where reasonable.

2.6.1.8.1 Public revenues

Public revenues are given by taxes G^T which are divided in: tax on household (G^{HH}), corporate

income tax (G^{CIT}), value added tax (G^{VAT}) and social contributions (G^{SC}). In addition, a carbon tax (G^{CARB}) may be activated:

$$G^T = G^{HH} + G^{CIT} + G^{VAT} + G^{SC} [+G^{CARB}] \quad (2.75)$$

In what follows we will discuss each tax in more detail, however in order to simplify the exposition we firstly define V as the gross annual income which is subject to social security contributions and income tax, respectively for employment categories $k \in \{E, U, P\}$. Therefore, the annual taxable income for sector (i), skill (s), gender (g) and employment category (k) is equal to:

$$V_{i,s,g}^K = \begin{cases} w_{i,s,g} \cdot h_{i,s,g} & \text{for employed individuals} \\ GUB_{s,g} & \text{for unemployed individuals} \\ GPB_{s,g} & \text{for retired individuals} \end{cases} \quad (2.76)$$

Notice that in case of employed ($k = E$) we distinguish between sector, skill and gender while for unemployed ($k = U$) and retired ($k = P$) we distinguish between skill and gender only.

Social security contribution

The individual compulsory social security contributions for the employed population are approximated as a fixed percentage of gross wages.

$$t_{s,g}^{S,E} = \tau^{S,E} \cdot V_{i,s,g}^E \quad (2.77)$$

In addition, there is a component of the compulsory social security contributions paid by the employers $t_{s,g}^{S,F}$, which is computed as percentage of the gross wages:

$$t_{i,s,g}^{S,F} = \tau^{S,F} \cdot V_{i,s,g}^E. \quad (2.78)$$

The amount of social contribution paid by the employed and by the employer respectively, by skill and gender are equal to:

$$T_{s,g}^{S,E} = \sum_i t_{i,s,g}^{S,E} \cdot L_{i,s,g}; \quad T_{s,g}^{S,F} = \sum_i t_{i,s,g}^{S,F} \cdot L_{i,s,g} \quad (2.79)$$

Consequently, the total amount of contribution for each category paid by the employed and the employer respectively are given by

$$T^{S,E} = \sum_s \sum_g T_{s,g}^{S,E}; \quad T^{S,F} = \sum_s \sum_g T_{s,g}^{S,F} \quad (2.80)$$

Therefore the total amount of revenues from compulsory social contributions is equal to:

$$G^{SC} = T^{S,E} + T^{S,F} = (\tau^{S,E} + \tau^{S,F}) \cdot GWB \quad (2.81)$$

Income tax

The progressive tax is based on a schedule given in Table 2.6.2. Income and the respective marginal tax rates are split into m brackets ranging $[\underline{b}_f, \bar{b}_f]$ respectively for $f = \{1, \dots, m\}$.

The taxable annual income by category (employed, unemployed and pensioner) is computed

Table 2.6.2: Tax schedule for taxable annual income

Country	Fraction of taxable income	Rate %
	$\underline{b}_f \vdash \bar{b}_f$	τ_f^I
Bracket 1	$0 \vdash \bar{b}_1$	0
Bracket 2	$\bar{b}_1 \vdash \bar{b}_2$	τ_2^I
...		...
Bracket m	$> \bar{b}_{m-1}$	τ_m^I

after the social security contribution is discounted :

$$\tilde{V}_{i,s,g}^K = (1 - \tau^{S,K}) V_{i,s,g}^K, \quad (2.82)$$

with $\tau^{S,U} = \tau^{S,P} = 0$. Subsequently, the individual income tax per category is computed as:

$$t_{i,s,g}^{I,K} = \sum_f^{m-1} \tau_f^I \cdot \min[(\bar{b}_f - \underline{b}_f), \max((\tilde{V}_{i,s,g}^K - \underline{b}_f), 0)] + \tau_m^I \cdot \max[(\tilde{V}_{i,s,g}^K - \bar{b}_{m-1}), 0] \quad (2.83)$$

For each bracket, the equation checks which amount of income lies between the lower and higher border of the bracket and multiplies it with the respective tax rate. For the last bracket all income above $\bar{b}_{m-1}$ is being taxed at τ_m^I .

Total income taxes paid by each employment category, per skill s and gender g is given by:

$$T_{s,g}^{I,K} = \begin{cases} \sum_i t_{i,s,g}^{I,K} \cdot L_{i,s,g} & \text{for employed} \\ t_{s,g}^{I,U} \cdot L_{s,g}^U & \text{for unemployed} \\ t_{s,g}^{I,P} \cdot N_{s,g}^P & \text{for retired} \end{cases} \quad (2.84)$$

Financial income tax

Financial income is subject to a flat tax rate τ^F on the sum of gross financial incomes ($GFI_{s,g}$).

$$T_{s,g}^F = \tau^F \cdot GFI_{s,g}, \quad (2.85)$$

where $GFI_{s,g}$ is the sum of the gross financial income of each category. See section 2.6.1.6 for more details. The same logic applies to the financial income of capitalists.

Therefore, total revenues from the household income taxation is equal to:

$$G^{HH} = \sum_K \sum_s \sum_g T_{s,g}^{I,K} + \sum_s \sum_g T_{s,g}^F + T_{cap}^F \quad (2.86)$$

VAT

The VAT is characterised by an economy wide effective tax rate τ^{VAT}

$$T_i^{VAT} = VA_i \cdot \tau^{VAT}. \quad (2.87)$$

Total revenue from VAT is given by the sum of the VAT collected in every sector:

$$G^{VAT} = \sum_i T_i^{VAT} \quad (2.88)$$

Corporate income tax

Corporate profits are subjected to a flat tax rate τ_{CIT}

$$T_i^{CIT} = \hat{\Pi}_i \cdot \tau^{CIT} \quad (2.89)$$

The total amount of corporate income tax revenue is equal to

$$G^{CIT} = \sum_i T_i^{CIT} \quad (2.90)$$

Carbon tax

In case the carbon tax is active, it provides additional revenue for the government in the amount of

$$T^{CARB} = \sum_i T_i^{carb} = \sum_i GHG_i \cdot \tau_t^{carb} \quad (2.91)$$

See subsection 2.6.1.5 for more details.

2.6.1.8.2 Public expenditure

Government expenditure is defined as G^E and is composed of: investments (G^I), transfers and benefits (G^{Tr}), wages (G^W), interest on bonds (G^B), and final consumption ($\hat{f}^G$).

Hence:

$$G^E = G^I + G^{Tr} + G^W + G^B + \hat{f}^G \quad (2.92)$$

Transfers and benefits

Transfers and Benefits G^{Tr} are mainly composed by unemployment benefits and pensions, but also smaller transfers like child benefits or sickness and disability contributions are considered.

Unemployment benefits

The nominal gross unemployment benefit for skill s and gender g ($GUB_{s,g}$) is a constant fraction (uw) of skill and gender specific average wages ($\frac{GWB_{s,g}}{L_{s,g}}$); please note that we assume that all the recipients of unemployment benefits are completely unemployed.

However, not all unemployed people receive unemployment benefits which is why we apply a coverage ratio (χ) to the unemployed population of skill s and gender g ($L_{s,g}^U$). Hence, gross unemployment benefits are determined by:

$$GUB_{s,g} = uw \cdot \frac{GWB_{s,g}}{L_{s,g}} \cdot \chi \cdot L_{s,g}^U. \quad (2.93)$$

Therefore the total nominal gross benefits for unemployment (GUB) is equal to:

$$GUB = \sum_s \sum_g GUB_{s,g}. \quad (2.94)$$

Pensions

The second main source of expenditure in the transfers is given by pensions paid to retired individuals: the population aged 65 or more (N^P) whose income depends on public pensions and individual benefits that are proportional to average annual income of each skill category. Let pw be the pension-to-wage ratio, assumed to be a fixed parameter calculated as:

$$pw = \frac{D62^{OS}}{\sum_{s,g} (\frac{GWB_{s,g}}{L_{s,g}} \cdot N_{s,g}^P)} \quad (2.95)$$

where $\frac{GWB_{s,g}}{L_{s,g}}$ is the skill specific average annual gross wage, $N_{s,g}^P$ is the population aged 65 or over in skill s and gender g and $D62^{OS}$ is the total cash transfers in the “Old age and Survivors” benefits categories in Eurostat’s COFOG database (see below).

Pension transfers are assumed to vary in proportion to the wages of employed workers of the same skill group. Consequently, the gross pension benefit ($GPB_{s,g}$), by skill and gender group is given by:

$$GPB_{s,g} = pw \cdot \frac{GWB_{s,g}}{L_{s,g}} \cdot N_{s,g}^P \quad (2.96)$$

Other social protection transfers

We also include other social protection transfers to provide a realistic picture of the welfare system. Table 2.6.3 presents a non-comprehensive breakdown of the largest components in social protection expenditure of the public sector ($S13$) by example of Italy in 2010, which are distributed between social benefits other than social transfers in kind ($D62$) and social transfers in

kind ($D632$).

The modelling of social protection expenditures focuses on cash transfers, ignoring in-kind

Table 2.6.3: General government expenditure on social protection. Source: Eurostat COFOG.

Code	COFOG99	Total million €	D62	D632
GF10	Social protection	317 240	298 695	4 962
GF10.1	Sickness and disability	27 664	25 358	1 200
GF10.2	Old age	214 091	207 065	1 357
GF10.3	Survivors	41 506	40 591	0
GF10.4	Family and children	14 885	10 663	1 462
GF10.5	Unemployment	14 375	14 054	0
GF10.6	Housing	534	157	302
GF10.7	Social exclusion n.e.c.	2 932	807	641
GF10.8	R&D Social protection	65	0	0
GF10.9	Social protection n.e.c.	1 218	0	0

Total = Total general government expenditure;

D62 = Social benefits other than social transfers in kind;

D632 = Social transfers in kind - purchased market production;

n. e. c. = not elsewhere classified

provision of goods and services, which instead are aggregated into the final consumption expenditure of the general government, as in the national accounts. For simplicity, we assume that all other social transfers other than unemployment benefits and pensions are non-taxable. GF 10.2, 10.3 and 10.5 have already been addressed in the form of pensions and unemployment benefits. For GF10.1, we assume that the *Sick and Disability* contributions under $D62^{SD}$ are paid in equal amounts to all working-age adults (18-64), independent of skill level and gender, as the number of actual recipients is not modelled. The amount per person (sd) is calculated from COFOG data for the initial period, and is subsequently updated according to inflation.

We divide total cash benefits for *Family and Children* ($D62^{FC}$) by the number of children in the base year which gives a yearly benefit fc per under-age person, which is again updated to inflation. In addition, for simplicity we assume that only the working age population (N^{WA}) has children eligible for child benefits and that they are distributed equally across this population, independent of age and skill. Hence, the number of children (N^F) for each skill and gender group is given by:

$$N_{s,g}^F = N^F \cdot \frac{N_{g,s}^{WA}}{\sum_s \sum_g N_{g,s}^{WA}} \quad (2.97)$$

and consequently the family and children benefits received per group are:

$$FC_{s,g} = fc \cdot N_{s,g}^F \quad (2.98)$$

The associated government expenditure is then equal to:

$$D62^{FC} = \sum_s \sum_g FC_{s,g} \quad (2.99)$$

The *residual benefits* (*RB*) (e.g. housing, but mainly social exclusion n.e.c.) are assumed to be allocated to low skilled individuals who are out of the labour force, as they tend to be the most vulnerable population group. Hence, government expenditure on these amounts to $D62^{RB} = \sum_g rb \cdot N_{low,g,t}^{olf}$, where *rb* are the residual benefits per person, calculated for the base year and updated according to inflation.

Government investment, wages and consumption

Apart from transfers, the government also invests, pays labour compensation to the people it employs and consumes final demand goods.

Investment

Government investment is a constant fraction ζ^I of total investment. That is, for $i \in \{1, \dots, 19\}$:

$$G^I = \zeta^I \sum_i I_i, \quad (2.100)$$

where ζ^I is calculated from input-output and government account data as the ratio between government's and overall GFCF for the initial period.

Wages

Wages paid by the government comprise all wages in the public sector and a constant fraction ζ^w of the gross wage bill in sectors 16 and 17 (education, and health). That is for $m \in \{16, 17\}$: $GW B_{15} + \zeta^w \sum_m GW B_m$. The underlying assumption here is, that the fraction of public servants employed in sectors other than 15, 16 and 17 is negligible. The coefficient ζ^w is calculated from KLEMS and government account data for the initial period. Subsequently, incorporating the social security contribution paid by the employer, labour compensation expenditures by the government amount to:

$$G^W = (1 + \tau^{S,F}) \cdot \left(GW B_{15} + \zeta^w \sum_m GW B_m \right) \quad (2.101)$$

Final Demand

Governments' expenditure for final demand changes over time according to an exogenous trend

(g_i^G) , which is computed from Input-Output data:

$$\hat{f}_{i,t}^G = \hat{f}_{i,0}^G + \sum_t g_i^G c_t \quad (2.102)$$

Consequently, total government expenditure into final demand consumption is

$$G^f = \sum_i \hat{f}_{i,t}.$$

Subsequently, final demand of the government is split between domestic demand ($f_{i,t}^G$) and demand for imports ($f_{i,t}^{G,imp}$) according to:²⁷

$$f_{i,t}^G = (1 - \gamma_i^G) \cdot \hat{f}_{i,t}^G; \quad f_{i,t}^{G,imp} = \gamma_i^G \cdot \hat{f}_{i,t}^G. \quad (2.103)$$

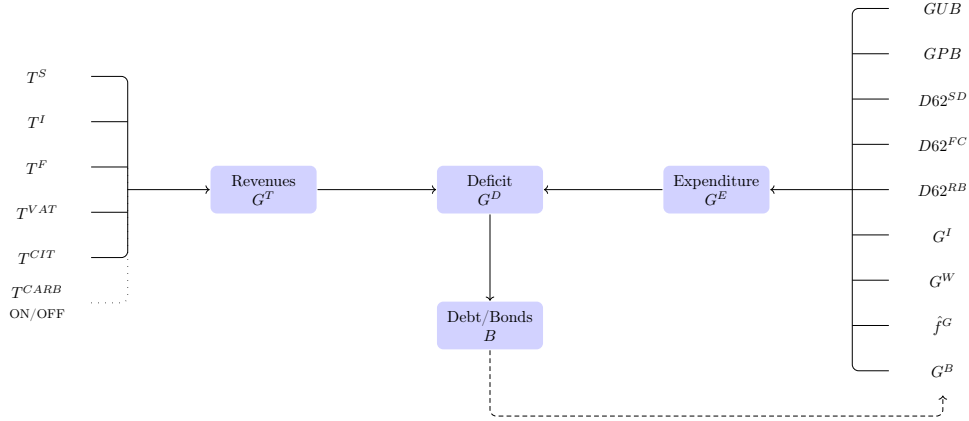


Figure 2.6.7: Conceptual framework of the government module

Public debt

Finally, the government also has to pay back interest on its debt. The difference between revenues and expenditures returns the yearly public deficit:

$$G_t^D = G_t^E - G_t^T. \quad (2.104)$$

To simplify the analysis we assume that all the yearly deficit is transformed into bonds (B). Therefore the stock of public debt at the end of period t is equal to:

$$B_t = B_{t-1} + G_t^D \quad (2.105)$$

²⁷See section 2.6.1.3.

Government bonds are assumed to be perpetual securities that pay an annual coupon whose unit value is equal to the nominal interest rate. Therefore, the yearly debt payment for the government is equal to:

$$G_t^B = i_t B_{t-1} \quad (2.106)$$

2.6.1.8.3 Central bank

The central bank is only interested in an inflation target (r^{ast}), adopting the following rule:

$$i_t = i_{t-1} \cdot \omega^i (r_{t-1} - r^*), \quad (2.107)$$

which is however restricted not to become negative.

Data sources: Government

If not otherwise specified, the data here refers to Italy in 2010.

- The rates for social security contributions ($\tau^{S,E}$ and $\tau^{S,F}$) are derived from EU Klems data.
- Data on personal income tax brackets and tax rates are taken from the Italian Revenue Agency (Agenzia delle Entrate): <https://www.agenziaentrate.gov.it/portale/web/english/personal-income-tax-rates-and-calculation>). The corporate and financial income tax rates for the period 2010-2021 are taken from Italy's Agenzia delle Entrate and Camera dei Deputati, respectively (see <https://www.agenziaentrate.gov.it/portale/web/english/corporate-income-tax-ires> and <https://temi.camera.it/leg18/post/la-tassazione-delle-rendite-finanziarie.html>).
- The VAT rate (τ^{VAT}) approximates the effective economy wide rate from 2010-2019 based on Worldbank data (<https://data.worldbank.org/indicator/GC.TAX.GSRV.VA.ZS?locations=IT>).
- uw is calculated as the percentage of wages paid as unemployment benefits weighted by the average unemployment duration^a. The coverage ratio is subsequently calculated to fit $D62^U$ from the Eurostat COFOG database, based on data on unemployed individuals ($L_{s,g}^U$) derived from Eurostat.
- The average skill and gender specific wage, also used to derive pw , for the initial period is calculated from EU Klems data. The initial number of pensioners ($N_{s,g}^P$) is also calculated from and $D62^{OS}$ based on Eurostat data, to calculate pw .

- Sickness and Disability transfers per working age person are calculated using $D62^{SD}$ and the size of the working age population, both based on Eurostat data. An analogous approach was taken to calculate rb and fc , from the same database.
- As aforementioned, wage (ζ^w) and investment (ζ^I) fractions for the government, as well as initial final demand ($f_{i,0}^G$) and its trend (g_i^G) are calculated using Input-output and government account data as well as EU Klems data. The trend results from available data from 2010-2017.
- Initial outstanding government debt (B_0) is taken from Eurostat and initial interest rate set by the central bank is taken as the rate for main refinancing operations from ECB data (https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html). The inflation target (r^*) is set to 2% according to the ECB's target.

^aUnemployment benefits are 75% of wages in the first 3 months of unemployment and decrease 3% per month afterwards. The maximum duration of unemployment benefits, which are a function of time employed, is of two years.

2.6.1.9 Prices

In the initial period prices are all normalised to one and subsequently they are driven by a markup μ computed over the unit cost in the previous period and the VAT rate:

$$p_t = \max \left\{ \frac{UFC_{i,t-1}(1 + \mu_{i,t})(1 + \tau_i^{VAT})}{p_{i,0}}, p_{i,t-1} \right\} \quad (2.108)$$

where UFC is the unit full cost, which is composed of the unit post-tax labour cost (ULC) and the unit costs of intermediate inputs (UIC):

$$UFC_{i,t-1} = ULC_{i,t-1} + UIC_{i,t-1}. \quad (2.109)$$

In turn, ULC is equal to the gross wage bill ($GW B$) plus the employers' social contributions ($T^{S,F}$) divided by the industry output:²⁸

$$ULC_{i,t-1} = \frac{\sum_s \sum_g GW B_{i,s,g,t-1} + T_{i,s,g,t-1}^{S,F}}{y_{i,t-1}}, \quad (2.110)$$

while UIC is the sum of the demand of intermediate goods of the industry i from other sectors, (either domestically or abroad) and the capital depreciation, divided by output.

$$UIC_{i,t} = \frac{\hat{z}_{i,t-1} + \delta_i K_{i,t-1}}{y_{i,t-1}}, \quad (2.111)$$

²⁸As for the definition of $GW B$ and the rates of $T^{S,F}$ see Section 2.6.1.8.

where $\hat{z}_{i,t-1} = \sum_j z_{j,i,t-1} + \sum_j z_{j,i,t-1}^{imp}$.

The mark up is an increasing function of capacity utilisation

$$\mu_{i,t} = \mu_{i,t-1} (1 + \omega^\mu \max \{uc_{i,t-1} - ucN, 0\}), \quad (2.112)$$

where $\omega^\mu > 0$ is a measure of the sensitivity to capacity over-utilisation.

The maximum function in (2.108) captures price stickiness and guarantees, that technical development, and hence a decrease in UFC does not lead to deflation.

Finally, in case the carbon tax is active, prices are additionally affected by it according to

$$p_t = \max \left\{ \frac{UFC_{i,t-1}(1 + \mu_{i,t})(1 + \tau_i^{VAT})}{p_{i,0}}, p_{i,t-1} \right\} \cdot (1 + \Delta UCC_{i,t}), \quad (2.108')$$

where $UCC_{i,t} = \frac{T_{i,t}^{carb}}{y_{i,t}}$ reflects the unit carbon costs faced by industry i in case the carbon tax is active, leading to an additional price increase.

Finally, the consumer price index is computed as the weighted average of the industry prices:

$$CPI_t = \sum_i \zeta_i^p p_{i,t}, \quad (2.113)$$

where $\zeta_i^p = \frac{C_{i,t}^{real}}{\sum_i C_{i,t}^{real}}$ is the share of real household demand allocated to sector i (see section 2.6.1.10). The same weights are used to calculate inflation (r) as $r_t = \sum_i \zeta_i^p r_{i,t}$, with $r_{i,t} = \frac{p_{i,t} - p_{i,t-1}}{p_{i,t-1}}$, being industry specific inflation.

Data sources: Prices

Initial industry specific markups ($\mu_{i,0}$) are approximated based on Christopoulou and Vermeulen (2012, Appendix C) using correspondence tables from Eurostat (https://ec.europa.eu/eurostat/web/nace-rev2/correspondence_tables).

The value for initial (2010) inflation is taken from OECD data (<https://data.oecd.org/price/inflation-cpi.htm>) Other parameters and values in this section are calibrated to data from 2010-2019/20.

2.6.1.10 Consumption

2.6.1.10.1 Disposable income

Individual income comes from labour, pensions, unemployment benefits and other social security payments, and financial assets. As described in Section 2.6.1.8, let $T_{s,g}^{I,k}$ and $T_{s,g}^{S,k}$ denote the progressive income tax and social security contributions for each $k \in \{E, U, P\}$. The net wage bill (NWB) of the employed population of gender g and skill level s is given by:

$$NWB_{s,g} = GWB_{s,g} - T_{s,g}^{I,E} - T_{s,g}^{S,E}, \quad (2.114)$$

where GWB is the gross wage bill. Similarly, the net unemployment benefits (NUB) of the unemployed population are:

$$NUB_{s,g} = GUB_{s,g} - T_{s,g}^{I,U} \quad (2.115)$$

and the net pension benefits (NPB) of the retired population are:

$$NPB_{s,g} = GPB_{s,g} - T_{s,g}^{I,P}, \quad (2.116)$$

where GUB and GPB are gross unemployment and pension benefits, respectively.

Financial income is disaggregated by skill and gender but not by employment status. The equation for net financial income introduced in Section 2.6.1.6 is $NFI_{s,g} = (1 - \tau^F) GFI_{s,g}$, which also holds for capitalists.

Other group specific social transfers are given by the sickness and disability transfer per capita (sd), family and children benefits per group ($FC_{s,g}$) and residual benefits per capita (rb) for the low skilled OLF population.

A final source of income is mixed income, which “*is the remuneration for the work carried out by the owner (or by members of his family) of an unincorporated enterprise.*”²⁹ Due to lack of data about its distribution, it is allocated as an equal per capita income (mI) to all individuals. Total mixed income is calculated as a fraction of the GWB , calculated for the initial period, and consequently moves proportional to the development of it in order to capture dynamics in the labour market³⁰.

Thus, the individual disposable income of employed individuals (YD^E), unemployed individuals (YD^U), pensioners (YD^P), people out of the labour force (YD^{OLF}) and capitalists (YD^{CAP}) is, respectively:

$$YD_{s,g}^E = \frac{NWB_{s,g}}{L_{s,g}} + \frac{NFI_{s,g}}{N_{s,g}^A} + sd + \frac{FC_{s,g}}{N_{s,g}^{WA}} + mI, \quad (2.117)$$

$$YD_{s,g}^U = \frac{NUB_{s,g}}{L_{s,g}^U} + \frac{NFI_{s,g}}{N_{s,g}^A} + sd + \frac{FC_{s,g}}{N_{s,g}^{WA}} + mI, \quad (2.118)$$

$$YD_{s,g}^P = \frac{NPB_{s,g}}{N_{s,g}^P} + \frac{NFI_{s,g}}{N_{s,g}^A} + mI, \quad (2.119)$$

$$YD_{s,g}^{OLF} = \frac{NFI_{s,g}}{N_{s,g}^A} + sd + \frac{FC_{s,g}}{N_{s,g}^{WA}} + mI + rb, \quad (2.120)$$

$$YD_{CAP} = \frac{NFI_{CAP}}{N_{CAP}} + mI, \quad (2.121)$$

²⁹<https://data.europa.eu/data/datasets/gzn7y7xzogclggm12wea?locale=en>

³⁰Total mixed income is then divided by a value roughly equal to the average propensity to consume (0.75) which results in a total private consumption in the first period that matches the values of households' consumption in the input-output tables. We believe that this reflects the fact that mixed income is mostly directed towards lower income households with larger marginal propensities to consume.

with $rb = 0$ if $s \neq low$. Also, $L_{s,g}$, $L_{s,g}^U$, and $N_{s,g}^P$ are the number of employed, unemployed, and retired individuals of skill level s and gender g , $N_{s,g}^A$ is the overall adult and $N_{S,G}^{WA}$ the working age population of skill level s and gender g , and N_{CAP} is the number of capitalists. Overall, the model distinguishes 25 income groups.

2.6.1.10.2 Consumption expenditure

An individual's total consumption expenditure is assumed to be a fraction (the marginal propensity to consume, α) of expected disposable income (YD^e)

$$C_{k',s,g,t}^{tot} = \alpha_{k',s,g,t} \cdot YD_{k',s,g,t}^e, \quad (2.122)$$

where $k' \in \{E, U, OLF, P, CAP\}$ indicates the occupational status of the individuals.

Following Jackson and Victor (2019a), we assume that the expected disposable income at time t is a linear extrapolation of past income growth:

$$YD_{k',s,g,t}^e = YD_{k',s,g,t-1} \left(1 + \frac{YD_{k',s,g,t-1} - YD_{k',s,g,t-2}}{YD_{k',s,g,t-2}} \right). \quad (2.123)$$

The marginal propensity to consume out of income in turn depends on the level of income itself and over time adjusts to inflation. Hence, $\alpha_{k',s,g,t}$ can be unique for each group identified by skill, gender, and occupational status, depending on their level of income. The initial values of the marginal propensity to consume ($\check{\alpha}$) are differentiated by income, based on income quintile specific values obtained from Banca D'Italia.

$$\check{\alpha}_{k',s,g,t} = f^Y(YD_{k',s,g,t}^e), \quad (2.124)$$

where the function $f^Y(\cdot)$ is derived from the aforementioned data and the income distribution in the model (see datasources). In particular, the marginal propensity to consume tends to decrease in income.

Subsequently, to counteract the effect of inflation on real consumption, we assume that $\alpha_{k',s,g,t}$ adjusts with respect to the inflation faced by each group:

$$\alpha_{k',s,g,t} = \check{\alpha}_{k',s,g,t} (1 + \bar{i}_{k',s,g,t})^{1 - \check{\alpha}_{k',s,g,t}}, \quad (2.125)$$

where the effect of $\check{\alpha}_{t-1}$ captures the characteristic that consumers already consuming a high proportion of their income have less freedom to react to inflation, while $\bar{i}_{k',s,g,t}$ is the average inflation rate experienced by each group and computed as the inflation of the each consumption category i_t^c weighted by the share of consumption expenditure previously dedicated to it by the

specific population group, $\beta_{k',s,g,t-1}^c$.³¹

$$\bar{i}_{k',s,g,t} = \sum_{cc} \beta_{k',s,g,t-1}^c i_t^c, \quad (2.126)$$

where i_t^c is the inflation rate of COICOP category c . A list of the consumption categories is provided in table 2.6.4.

As long as $C_{k',s,g,t}^{tot} \leq YD_{k',s,g,t}$, consumption will be wholly financed out of disposable

Table 2.6.4: List of consumption categories

Category no.	Description	COICOP code
1	Food and non-alcoholic beverages	CP01
2	Alcoholic beverages, tobacco and narcotics	CP02
3	Clothing and footwear	CP03
4	Housing	CP041 – 043
5	Water supply and miscellaneous services relating to the dwelling	CP044
6	Electricity, gas and other fuels	CP045
7	Furnishings, household equipment and routine households maintenance	CP05
8	Health	CP06
9	Private transport	CP071 – 072
10	Transport services	CP073
11	Communications	CP08
12	Recreation and culture	CP09
13	Education	CP10
14	Restaurants and hotels	CP11
15	Personal care	CP121
16	Other	CP122 – 127

income and $C_{k',s,g,t}^{tot} = C_{k',s,g,t}^y$, where $C_{k',s,g,t}^y$ refers to consumption out of income. However, if $C_{k',s,g,t}^{tot} > YD_{k',s,g,t}$, either because $\alpha_{k',s,g} > 1$ or $YD_{k',s,g,t}^e > YD_{k',s,g,t}$, the remaining consumption expenditure is finance out of wealth ($C_{k',s,g,t}^w$):

$$C_{k',s,g,t}^y = \min(\alpha_{k',s,g,t} \cdot YD_{k',s,g,t}^e, YD_{k',s,g,t}) \quad (2.127)$$

$$C_{k',s,g,t}^w = \max(\alpha_{k',s,g,t} \cdot YD_{k',s,g,t}^e - YD_{k',s,g,t}, 0) \quad (2.128)$$

³¹Here, in contrast to $r_{i,t}$, inflation $i_{c,t}$ measures the cumulative inflation with respect to prices in the initial period, which are normalised to 1.

Consequently, savings are defined as $S_{k',s,g,t} = YD_{k',s,g,t} - C_{k',s,g,t}^{tot}$.

After determining the amount of total consumption expenditure and its financing, C^{tot} of each group is distributed among the 16 COICOP consumption categories and subcategories (c) in table 2.6.4, through an income, and therewith group, specific vector β_t , allocating the respective expenditure shares to the respective consumption categories. Again, β_t depends not only on income, but is also adjusted with respect to consumption good specific inflation, capturing price elasticity. The initial values per c are given by

$$\check{\beta}_{k',s,g,t}^c = f^\beta(YD_{k',s,g,t}^e), \quad (2.129)$$

where the function $f^\beta(\cdot)$ is based on calculations based on Istat data, as well as the income distribution in the model. In addition, price changes lead to responses in β_t^c over time as we assume that demand is price elastic. More precisely, we consider the elasticity as the sensitivity to a price increase in a c compared to price changes faced over all consumption commodities. Thus, the crucial determinant of changes in consumption patterns is the relative price change of a c faced by a group. Therefore, let

$$\hat{i}_{k',s,g,t}^c = i_t^c - \bar{i}_{k',s,g,t} \quad (2.130)$$

be the difference between the inflation of the specific c and average inflation faced by the group.

It follows that each c element $\beta_{k',s,g,t}^c$ is given by:

$$\beta_{k',s,g,t}^c = (1 + \epsilon^c \cdot \hat{i}_{k',s,g,t}^c) \check{\beta}_{k',s,g,t}^c + \check{\beta}_{k',s,g,t}^c \left(\sum_{cc} \left(-\epsilon^c \cdot \hat{i}_{k',s,g,t}^c \check{\beta}_{k',s,g,t}^c \right) \right), \quad (2.131)$$

where ϵ^c reflects the elasticity of $\beta_{k',s,g}^c$ with respect to relative price changes, $\hat{i}_{k',s,g}^c$. The last summand implies that the overall changes in all consumption expenditure shares due to price changes, $\sum_c \left(-\epsilon^c \cdot \hat{i}_{k',s,g,t}^c \check{\beta}_{k',s,g,t}^c \right)$ are reallocated to the individual $\beta_{k',s,g,t}^c$ s according to their initial share in order to guarantee $\sum_c \beta_{k',s,g,t}^c = 1$.³² The elasticity, ϵ^c , is initially set to 0 for all COICOP, but subsequent variations in the elasticities warrant its presence of equation

³²As an example to illustrate the function of the last summand assume for simplicity $c \in \{1, 2\}$, one population group and that prices are in the initial state with $p^1 = p^2 = 1$. The expenditure shares are $\beta^1 = \check{\beta}^1 = 0.4$ and $\beta^2 = \check{\beta}^2 = 0.6$. Subsequently, the next period features a price increase of the first good, with now $p^1 = 1.2$ and consequently, $i^1 = 0.2$ and $i^2 = 0$. Hence, $\bar{i} = 0.08$ and accordingly $\hat{i}^1 = 0.12$ and $\hat{i}^2 = -0.08$. Now assuming ad hoc $\epsilon^1 = -0.8$ and $\epsilon^2 = -0.6$ and omitting the last summand in (2.131), the new expenditure shares become $\beta^1 = (1 - 0.8 \cdot 0.12) \cdot 0.4 = 0.3616$ and $\beta^2 = (1 + 0.6 \cdot 0.08) \cdot 0.6 = 0.6288$. This illustrates that the increase in β^2 partly but not fully compensate for the drop in β^1 . (Changing the elasticities, could also lead to overcompensation (if $\epsilon^1 < \epsilon^2$) or exact compensation (if $\epsilon^1 = \epsilon^2$.) Therefore, in this example $\sum_c \beta^c = 0.9904 < 1$ and the remaining $1 - \sum_c \beta^c = \sum_c \left(-\epsilon^c \cdot \hat{i}_{k',s,g,t}^c \check{\beta}_{k',s,g,t}^c \right) = 0.0096$ of consumption expenditure have to be spend on consumption categories. The last summand in (2.131) does exactly that, assuming that this redistribution is proportional to the initial share of expenditure.

(2.131).

Finally, consumption expenditure for each population group on every consumption category ($C_{k',s,g}^c$) is given by

$$C_{k',s,g,t}^c = C_{k',s,g,t}^{tot} \cdot \beta_{k',s,g,t}^c. \quad (2.132)$$

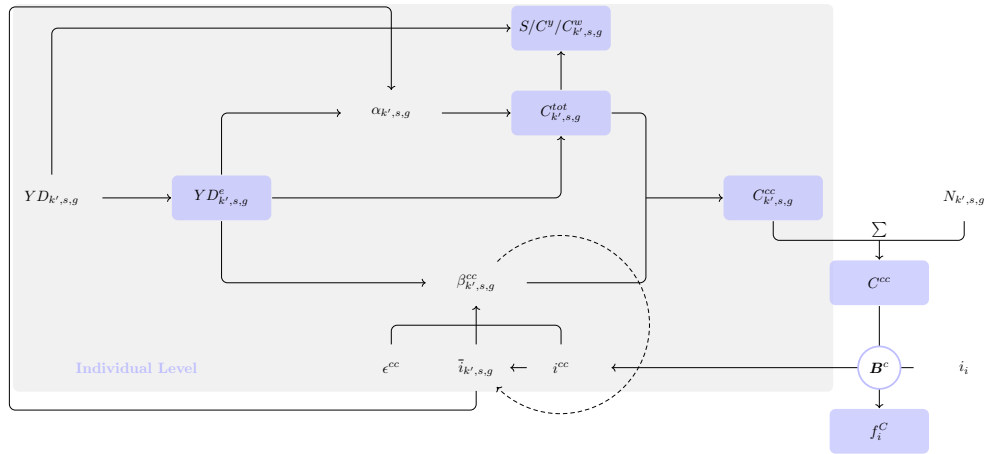


Figure 2.6.8: Conceptual framework of the consumption module

2.6.1.10.3 Bridging COICOP and NACE

As the production side of the economy is defined over NACE categories i , and the consumption side over COICOP c , a bridge matrix between these two classifications is needed in order to translate changes in prices and demand.

The COICOP-NACE bridge matrix B^c assigns to each COICOP category the respective share of every NACE category. For instance, it depicts what percentage of the COICOP category “food and non alcoholic beverages” is produced by the NACE category “agriculture, forestry and fishing”. As an example, accordingly the first column captures the composition the COICOP category “food and non-alcoholic beverages” as shares of the NACE categories.

Inflation for COICOP categories can then be imputed using price changes by industry and the bridge matrix: $i_{(c),t} = \left(B_{(i \times c)}^c\right)^T i_{(i),t}$, where $i_{(c),t}$ is a vector of inflation of consumption categories and $i_{(i),t}$ of sectors.

Moreover, for the input-output production, the total domestic private consumption demand for each industry i , f_i^C is required. Total economic nominal demand per c , C_c , can be determined

as the sum of the products of groups' consumption per c and the size of each group.

$$C_{c,t} = \sum_s \sum_g (C_{E,s,g,t}^c \cdot L_{s,g,t}) + \sum_s \sum_g (C_{U,s,g,t}^c \cdot L_{s,g,t}^U) + \sum_s \sum_g (C_{OLF,s,g,t}^c \cdot N_{s,g,t}^{OLF}) \quad (2.133)$$

$$+ \sum_s \sum_g (C_{P,s,g,t}^c \cdot N_{s,g,t}^P) + C_{CAP,t}^c \cdot N^{CAP} \quad (2.134)$$

$$= \sum_{k'} \sum_s \sum_g (C_{k',s,g,t}^c \cdot N_{k',s,g,t}) \quad (2.135)$$

where $N_{k',s,g,t}$ reflects the amount of individuals in the respective population group.

Real demand per c in turn is given by $C_{c,t}^{real} = \frac{C_{c,t}}{1+i_{c,t}}$. Subsequently, the bridge matrix allows real final demand to be transformed from COICOP to NACE classifications:

$$C_{(i),t}^{real} = B_{(i \times c)}^c C_{(c),t}^{real} \quad (2.136)$$

In addition, an external Covid shock affects $C_{i,t}^{real}$ for 2020 and 2021. Subsequently, nominal demand can simply be computed by multiplying each $C_{i,t}^{real}$ with the prices from the respective industry: $\hat{f}_i^C = C_{i,t}^{real} \cdot p_i$.

Finally, in order to determine domestic production, the demand determined above, has to be divided between domestic and foreign demands, with respective shares being derived from Eurostat Input-Output data. Domestic final demand (f_i^C) is subsequently given by

$$f_{i,t}^C = \hat{f}_i^C (1 - \gamma_i^C), \quad (2.137)$$

with γ_i^C being the share of consumption demand for imports, which are in general assumed to be constant (see section 2.6.1.3).

Data sources: Consumption

$f^Y(\cdot)$ is defined over income quintiles. The respective values of the propensity to consume is calculated from equivalent income and expenditure data provided by Banca d'Italia (https://www.bancaditalia.it/pubblicazioni/indagini-famiglie/bil-fam2010/en_suppl_06_12n.pdf?language_id=1, Appendix B, table B2), and the average income per quintile is approximated by the initial income distribution in the model. $f^\beta(\cdot)$ is calculated from Istat data from 2010 (additionally using data from 2014-2018 to approximate missing values for 2010). The matrix is balanced in order to respect total consumption expenditure for the first year, using the RAS algorithm. Again, average income per quintile is approximated by the initial income distribution in the model. The COICOP NACE bridge matrix (B^c) is based on data provided by Cai and Vandyck

(2020), that are elaborated from Eurostat, and subsequently is balanced with IO data from Eurostat using the RAS algorithm. The step-by-step procedure is, in a nutshell:

- download the matrix A1 of final demand composition in CPA-COICOP terms from Cai and Vandyck (2020);
- create the vector R of total household demand from Eurostat IO data;^a
- compute S1 as the column sum of A1;
- calculate the coefficient $\zeta = \frac{\sum_i R_i}{\sum_i S1_i}$ to correct any difference between the total consumption in COICOP and NACE terms. This step is necessary because RAS requires that tot row equals tot column;
- Calculate the RAS column-constraints $S2 = \zeta \cdot S1$;
- Run RAS departing from A1, give as row (column) constraints R (S2) to obtain the new bridge matrix A2;
- calculate the matrix of production shares (B1) to obtain the sum by COICOP = 1, i.e. with respect to the column sum of A2;

Total initial mixed income is based on Eurostat IO data and the Covid shock on consumption is based on the Bank of Italy's projections for the variations of consumption, investments, exports and imports, available at <https://www.bancaditalia.it/pubblicazioni/proiezioni-macroeconomiche/2020/Proiezioni-Macroeconomiche-Italia-dicembre-2020.pdf.pdf>

^aNote that R will represent the row-constraint of RAS, i.e. the row sum of the new matrix after-RAS must equal R.

2.6.1.11 Energy

2.6.1.11.1 Energy input-output

In order to allow for detailed modelling of energy flows, we require an input-output structure for energy flows linked to the economic system. For this purpose, the Eurostat physical energy flow accounts (PEFA) prove to be useful, presenting data on the physical flows of energy (in TJ) defined over 5 tables:

A Supply: each of the 37 row represents an energy output (natural, products, residuals³³)

³³“Residuals refer to flows of solid, liquid and gaseous materials, and energy, that are discarded, discharged or

and the column indicate the origins (NACE sector, rest of the world, environment);

B Use: each of the 37 row represents an energy input (natural, products, residuals) and the column indicate the demand (NACE sector, rest of the world, households). Table B is further divided into transformation (B.1) and end use (B.2);

C Emission relevant: returns the fraction of energy (from B) that is responsible for emissions.

Note that we want to link the physical energy flows with the economic monetary trade (e) by distinguishing among domestic intermediate trade, imported intermediate goods, exports, domestic household final demand and household final demand for imports.³⁴ This is possible, in principle, because the PEFA follows the same statistical procedure and categorisation applied in building national accounts. However, the correspondence is not straightforward because PEFA only provides total values by sector avoiding any reference to inter-sectoral trade.

Table 2.6.5 reports a simplified version of the PEFA, for Italy in 2010, in which we consider the total amount for the energy products (category P00). Although the total energy flows over all activities must be equal in tables A and B, they differ if one looks at the total values by sector and households. This is because the total sectoral supply and use values does not distinguish between domestic and international trade. Therefore we follow a logical procedure to recover the total sectoral supply and use at the domestic level in order to reconstruct the input-output energy matrix that traces the physical flows associated to the monetary ones (as reported in matrix $\mathbf{Z}$). First, we recover the total imports by households assuming that the energy flow is proportional to the share of monetary imports ($\sim 10.7\%$ corresponding to 257.67 TJ) and subtract them to the total energy imports. This entails that the remaining flow ($8178.41 - 257.67 = 7920.74$) is utterly due to intermediate imports. Knowing that, we can recover the total domestic energy use ($10737.68 - 7920.74 = 2816.94$) that must equal total supply. What remains from total energy supply by NACE ($6517.79 - 2816.94 = 3700.85$) will exactly cover the change in inventories plus domestic household demand and exports ($55.15 + 2156.55 + 1489.15 = 3700.85$).³⁵ This decomposition facilitates the translation of monetary trade into energy flows. In order to obtain industry-specific values, the calculated total values are rescaled with initial industry shares.

emitted to the environment (e.g., emission to air) by establishments and households through processes of production, consumption or accumulation but may also flow within the economy, as is the case when, for example, solid waste is collected as part of a waste collection scheme. "(Eurostat, 2014, p.7). The total residuals from NACE sectors and households is close to the total net inland energy consumption. See Eurostat (2014) for a detailed description of the conceptual foundations of PEFA.

³⁴As natural energy inputs lack the monetary dimension and hence cannot be directly linked with the monetary economic flows, they are excluded here and only the 20 energy products are considered.

³⁵Note that the values related to "Changes in inventories" are not further considered because of irrelevance.

The last step consists in recovering the physical input-output matrix, that associates for each intermediate exchange across domestic sectors the amount of energy required – by applying the RAS bi-partitive algorithm. Indeed, for any initial matrix, this algorithm allows to find a unique final matrix that respects given row- and column- constraints, under the condition that the total sum by row and by column equalise. In our case, we use the initial IO monetary matrix ($\mathbf{Z}$) and we impose the domestic sectoral energy supply (E_i^S) and use (E_i^D) as row and column constraints, respectively. RAS subsequently adjusts $\mathbf{Z}$ in an iterative process so that it respects the energy constraints but maintains its general structure, returning a new matrix $\mathbf{E}^Z$ which row (column) sums will be equal to E_i^S (E_i^D). This allows us to establish a direct linkage between each monetary and physical flow, in the domestic intermediate trade, namely

$$E_{i,j,t}^Z = z_{i,j,t} \cdot \eta_{i,j,t}^Z \quad (2.138)$$

where $\eta_{i,j,t}^Z$ is the cell-specific conversion factor that results from the RAS procedure, which changes over time due to innovation (see below).

In case of imported intermediate goods ($\sum_j z_{j,i,t}^{imp}$), exports ($f_{i,t}^E$), final domestic household demand ($f_{i,t}^C$) and final household demand for imports ($f_{i,t}^{C,imp}$) PEFA provides the respective energy supply (for $\sum_j z_{j,i,t}^{imp}$) and demand (for $f_{i,t}^E$, $f_{i,t}^C$ and $f_{i,t}^{C,imp}$) so that the conversion from monetary to physical units is straightforward.

Analogously, the process results in

$$E_{i,t}^{Z,imp} = \left(\sum_j z_{j,i,t}^{imp} \right) \cdot \eta_i^{Z,imp}; \quad (2.139)$$

$$E_{i,t}^E = f_{i,t}^E \cdot \eta_i^E; \quad (2.140)$$

$$E_{i,t}^C = f_{i,t}^C \cdot \eta_i^C; \quad (2.141)$$

$$E_i^{C,imp} = f_{i,t}^{C,imp} \cdot \eta_i^{C,imp}. \quad (2.142)$$

Note that we assume coefficients to be constant in the case of imports. However, domestic energy efficiency is assumed to change in line with changes in intermediate demand (see section 2.6.1.4) due to innovation according to $\eta_{i,j,t}^Z = \eta_{i,j,0}^Z \cdot \frac{a_{i,t}}{a_{i,0}}$ and $\eta_{i,t}^C = \eta_{i,0}^C \cdot \frac{a_{i,t}}{a_{i,0}}$, where $a_{j,t}$ reflects unit total intermediate demand for industry j : $a_{j,t} = \sum_i a_{i,j,t}$.

2.6.1.11.2 Gross inland consumption

In order to obtain an output comparable to a wide range of data, including energy balances, we calculate the gross inland consumption (GIC or Total primary energy supply TPES).³⁶ For this

³⁶“Total energy consumption is made up of production plus imports, minus exports, minus international marine bunkers plus/minus stock changes. It is also called Total primary energy supply or Gross inland energy consump-

Table 2.6.5: Total supply and use energy flows by sectors and households. Source: PEFA, Italy 2010, in TJ, own elaboration. “RoW” stands for rest of the world, indicating total import (export) in case of table B (A).

ENERGY PRODUCTS (P00)	SUPPLY (A)	RESIDUALS (A)	DEMAND (B)	EMISSION (C)
A	311.95	145.02	146.99	126.48
B	485.07	30.53	31.96	23.89
C	439.791	1836.0832	2196.3921	1176.9637
<i>C19</i>	<i>3768.47</i>	<i>407.20</i>	<i>4175.66</i>	<i>332.59</i>
<i>D</i>	<i>1274.83</i>	<i>1139.13</i>	1956.09	1575.83
<i>E</i>	<i>35.24</i>	<i>173.37</i>	134.32	104.35
<i>F</i>	<i>0.17</i>	<i>157.00</i>	157.05	74.39
G	<i>189.35</i>	<i>285.47</i>	473.93	196.71
H	5.85	842.21	847.31	763.72
I	0.60	135.80	135.82	90.41
J	0.94	20.77	21.60	10.54
K	1.61	25.78	25.81	14.55
L	0.04	16.39	16.39	14.51
M	1.89	62.69	62.94	37.10
N	0.33	42.90	42.97	27.93
O	0.01	73.18	73.18	42.54
P	0.30	38.74	39.01	25.51
Q	0.80	134.11	134.70	102.69
R	0.09	26.76	26.76	19.16
S	0.49	38.55	38.80	21.41
T	0.00	0.00	0.00	0.00
U	0.00	0.00	0.00	0.00
Tootal by NACE	6517.79	5631.67	10737.68	4781.28
Total by households	0.00	2414.23	2414.23	2147.00
Changes in inventories	0.00	110.96	55.15	0.00
Total by RoW	8178.41	0.00	1489.15	0.00
Total	14696.21	8156.86	14696.21	6928.28

purpose, we assume some simplifications based on the relations that emerge from the data.

To derive the GIC, first we need to acquire the net inland consumption (NIC). In PEFA it is given by the sum of the overall residuals (E^R), for both industries and households, reported in Table A. We calculate the constant industry specific fraction of energy residuals to energy demand (η_i^R) given the values for 2010:

$$\eta_i^R = \frac{E_i^R}{E_i^D}. \quad (2.143)$$

tion and represents the quantity of all energy necessary to satisfy inland consumption.”(<https://www.eea.europa.eu/data-and-maps/indicators/total-energy-consumption-outlook-from-iea>).

In the case of households, total energy residuals and energy use equalise. Consequently, the NIC is given by

$$NIC = \sum_i \left(\left(\sum_k E_{k,i,t}^Z + E_{i,t}^{Z,imp} \right) \cdot \eta_i^R + E_{i,t}^{C,imp} + E_{i,t}^C \right) \quad (2.144)$$

Subsequently, following the logic of PEFA, we subtract (add) the consumption made abroad (on the territory) by the residents (foreigners) and the statistical discrepancies from the NIC to obtain the GIC. However, since these are features that are not modelled in this framework, we instead calculate the GIC/NIC ratio (≈ 0.9325) in order to capture their relation based on PEFA data for Italy. Finally, the result is converted from TJ to ktoe (1 TJ = 23.8846 ktoe) to be comparable with energy balances.

2.6.1.11.3 Energy sources

Total energy demand by industry ($E_{i,t}^{\hat{Z}} = \sum_j E_{j,i,t}^Z + E_{i,t}^{Z,imp}$), as well as total household demand ($E_t^{\hat{C}} = \sum_i E_{i,t}^{C,imp} + \sum_i E_{i,t}^C$) are subsequently distributed over five energy sources (ϕ), namely: solid, liquid, gas, biomass and renewable. The share for each energy source changes due to the expansion of renewable energy substituting other sources. This expansion are based on the growth rates for electricity, transportation and all energy demand reported in the MISE-MATTM-MIT (2019). We input the actual mean yearly growth rates between 2010 and 2020 and the projected yearly growth rates between 2020 and 2030 in the Italian National Energy Plan to the remaining simulated periods. The share of renewable energy ($\zeta_{i,t}^{\phi=r}$) is consequently given by

$$\zeta_{i,t}^{\phi=r} = \zeta_{i,t-1}^{\phi=r} + g_{i,t-1}^{\phi}, \quad (2.145)$$

where $g_{i,t}^{\phi}$ is the aforementioned time specific growth rate in percentage points. In terms of substitution of other energy sources, we assume that they decline with respect to $g_{i,t}^{\phi}$ proportional to their initial share in non-renewable energy sources:

$$\zeta_{i,t}^{\phi=x} = \zeta_{i,t-1}^{\phi=x} - \frac{\zeta_{i,0}^{\phi=x}}{1 - \zeta_{i,0}^{\phi=r}} \cdot g_{i,t-1}^{\phi}, \quad (2.146)$$

where $x = s, l, g, b$ captures all energy sources apart from renewable sources.

The PNIEC, provides specific shares ($g_{i,t}^{\phi}$) for the transport and electricity sector. All other sectors are assumed to feature a common economy wide average. The same approach is taken with regard to shares in energy sources for households ($\zeta_t^{C,\phi}$), with a specific growth rate in renewable energy supply ($g_t^{C,\phi}$).

Consequently, energy demand by energy source is given by:

$$E_{i,\phi,t}^F = E_{i,t}^{\hat{Z}} \cdot \zeta_{i,t}^{\phi} \quad (2.147)$$

for industries, and by

$$E_{\phi,t}^H = E_t^{\hat{C}} \cdot \zeta_t^{C,\phi} \quad (2.148)$$

for households.

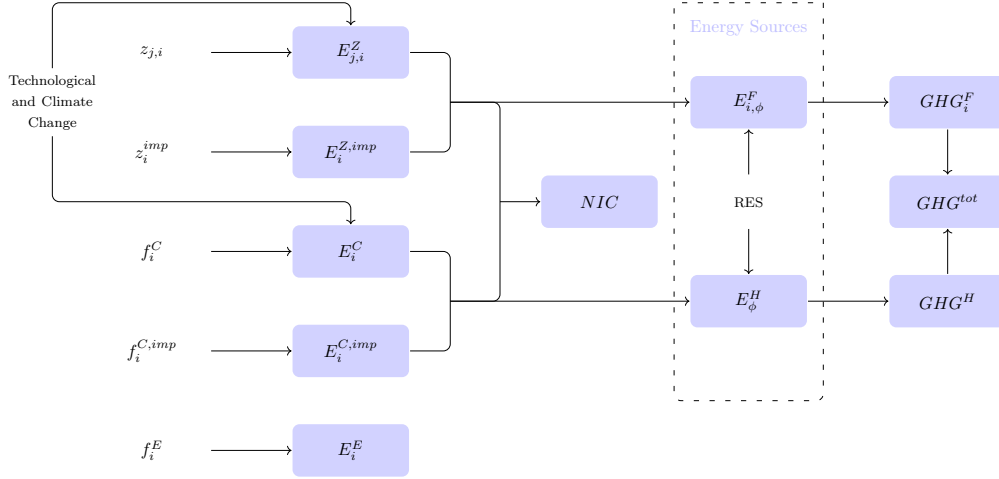


Figure 2.6.9: Conceptual framework of the energy module

2.6.1.11.4 Emissions

First, the computed total energy use is transformed into emission relevant energy use (EE). This is done using a ratio of emission relevant to total energy use ($\gamma_{i,\phi}^{EE}$) respectively for each industry and energy source, which is derived from PEFA (table C), so that emission relevant energy by industry and energy source is given by:

$$EE_{i,\phi,t}^F = E_{i,\phi,t}^F \cdot \gamma_{i,\phi}^{FE} \quad (2.149)$$

Again, the same reasoning applies to households, leading to

$$EE_{\phi,t}^H = E_{\phi,t}^H \cdot \gamma_{\phi}^{HE}, \quad (2.150)$$

where γ_{ϕ}^{HE} are again the ratio of total energy use to emission relevant use for each source.³⁷ The CO2 and GHG emissions by sector and households are provided by Eurostat. However, since it does not report the correspondence with the energy source used, we recover the emission factor of conversion from the Italian Institute for environmental protection and research (ISPRA). In the last reports (ISPRA, 2021b), it provides the ratio of conversion, from TJ to CO2 eq., per each energy source. Subsequently, we calculate the sector (households) source specific coefficients of conversions $v_{i,\phi}^F$ (v_{ϕ}^H) to translate the emission relevant energy use (from table C

³⁷Note that renewable energies are not emitting so that $\gamma_{i,\phi=r}^{FE}$ and $\gamma_{\phi=r}^{HE}$ are zero.

of PEFA) into CO₂ emissions.³⁸ Namely

$$CO2_{i,\phi,t}^F = EE_{i,\phi,t}^F \cdot v_{i,\phi}^F \quad (2.152)$$

$$CO2_{\phi,t}^H = EE_{\phi,t}^H \cdot v_{\phi}^H \quad (2.153)$$

Finally, using existing relations between CO₂ and overall GHG emissions for each industry and households, CO₂ emissions are extrapolated to GHG emissions:

$$GHG_{i,t}^F = \sum_{\phi} CO2_{i,\phi,t}^F \cdot \gamma_i^{Fghg} \quad (2.154)$$

$$GHG_t^H = \sum_{\phi} CO2_{\phi,t}^H \cdot \gamma^{Hghg}, \quad (2.155)$$

where γ_i^{Fghg} and γ^{Hghg} are the fraction of GHG to CO₂ emissions for each industry and for households respectively, based on Eurostat data. Thus, total GHG emissions (GHG^{tot}) in the economy are given by:

$$GHG_t^{tot} = \sum_i GHG_{i,t}^F + GHG_t^H. \quad (2.156)$$

Data sources: Energy and Emissions

Most data – i.e. GIC/NIC ratio (as average from 2014–2018), energy residual ratios (η^R), initial shares of energy sources (ζ^ϕ), share of emission relevant energy use ($\gamma_{\phi}^{FE/HE}$) and GHG to CO₂ fractions (γ^{ghg}) – are based on Eurostat data, mainly PEFA. The construction of the energy IO structure additionally used Eurostat IO together with the PEFA. Also, CO₂ emission coefficients (v) are based not only on the PEFA but also on the total GHG emissions, by sector and households, provided by Eurostat (<https://ec.europa.eu/eurostat/web/environment/air-emissions>) and on the Emission Factors of conversion provided by the last national inventory report drawn up by ISPRA (ISPRA, 2021b, p. 80). Finally, the trend in renewable energy source expansion (g^ϕ) is derived from MISE-MATTM-MIT (2019).

If not otherwise specified, data was retrieved for Italy in 2010.

³⁸ISPRA (see box below) provides the coefficients of CO₂ emissions per TJ of energy respective for each energy source (v_ϕ). Using this we normalise v_ϕ for each source with respect to v_{solid} . Subsequently, applying these normalised weights (ι_ϕ), we calculate the overall sectoral weighted average CO₂-to-energy coefficient for the initial period as

$$v_i = \frac{CO2_i}{\sum_{\phi} \iota_{\phi} EE_{i,\phi}^F}. \quad (2.151)$$

Then, we rescale v_i with respect to ι_ϕ , so that $v_{i,\phi}^F = v_i \iota_\phi$, with $\iota_{solid} = 1$. This procedure is applied to each industry and households respectively and allows to directly translate TJ of emission relevant energy use into CO₂ eq. emissions.

2.6.1.12 Climate damage

Climate change affects the economy directly by multiplication with output. The production function embodied in the input-output model described in Section 2.6.1.2 is:

$$y_{j,t} = \min \left\{ \frac{z_{1,j,t}}{a_{1,j,t}}, \frac{z_{2,j,t}}{a_{2,j,t}}, \dots, \frac{z_{19,j,t}}{a_{19,j,t}} \right\}, \quad (2.157)$$

where y_j is the total output of sector j , $z_{i,j}$ represents interindustry sales by sector i to sector j , and $a_{i,j}$ is the technical coefficient representing the relation between sector j 's output and its input from sector i . Introducing a sectoral climate damage multiplier $(1 - \Lambda_{j,t}) \in [0, 1]$ into equation 2.157 yields:

$$\begin{aligned} y_{j,t} &= (1 - \Lambda_{j,t}) \min \left\{ \frac{z_{1,j,t}}{a_{1,j,t}}, \frac{z_{2,j,t}}{a_{2,j,t}}, \dots, \frac{z_{19,j,t}}{a_{19,j,t}} \right\} \\ &= \min \left\{ \frac{z_{1,j,t}}{(1 - \Lambda_{j,t})^{-1} a_{1,j,t}}, \frac{z_{2,j,t}}{(1 - \Lambda_{j,t})^{-1} a_{2,j,t}}, \dots, \frac{z_{19,j,t}}{(1 - \Lambda_{j,t})^{-1} a_{19,j,t}} \right\}, \end{aligned} \quad (2.158)$$

meaning that climate damage increases the input from sector i per output of sector j . A value of Λ equal to 0 implies no damage, whereas a value equal to 1 implies no production. As the initial observed $a_{i,j,0}$ already incorporate the effect of the contemporary temperature, subsequent technical coefficients are scaled not by the absolute value of $\Lambda_{j,t}$, but by its relative change with respect to the previous period: $a_{i,j,t} = a_{i,j,t}^{bd} \cdot \frac{\tilde{J}_{j,t-1}}{\tilde{J}_{j,t}}$, where $a_{i,j,t}^{bd}$ is the technical coefficient before climate damage is being taken into account and $\tilde{J}_{j,t} = 1 - \Lambda_{j,t}$ is the realised damage multiplier.

Damage is modelled after Desmet and Rossi-Hansberg (2015) and Burzyńska et al. (2019), with modifications to account for extreme climate events. Let T denote annual mean temperature in °C. The average multiplicative effect of climate on production is:

$$1 - \mathbb{E}[\Lambda_{j,t}] \equiv J_{j,t} = \max \{g'_j + g''_j T_t + g'''_j T_t^2; 0\}, \quad (2.159)$$

where (g'_j, g''_j, g'''_j) is a triple of real-valued parameters with $g''_j > 0$ and $g'''_j < 0$, implying that the optimal temperature for economic activity in sector j is $T_j^* = -g''_j / 2g'''_j$. Thus, sectoral production tends to decrease with temperature when $T > T_j^*$ and to increase with temperature when $T < T_j^*$ (this is consistent with studies suggesting that global warming will be associated with an increase in output at certain latitudes; see e.g. Burke et al., 2015 and Orlov et al., 2020). The percentage change in output due to warming from a base period b to a comparison period $t \geq b$ is given by $(J_{j,t} - J_{j,b}) / J_{j,b}$.

In the model a distinction is made between agricultural and non-agricultural activity, that is between sector 1 and other sectors. The estimates by Desmet and Rossi-Hansberg (2015) yield $(g'_1, g''_1, g'''_1) = (-2.24, 0.308, -0.0073)$ for agriculture and $(g'_{j'}, g''_{j'}, g'''_{j'}) = (0.3, 0.08,$

-0.0023) for every $j' \neq 1$. On average, damages on agricultural production are therefore minimised when $T \approx 21.1$ degrees, whereas non-agricultural damages are minimised when $T \approx 17.4$ degrees. This also means that agricultural production is zero when mean temperature falls below 9.34 degrees or above 32.85 degrees, while non-agricultural production is zero when mean temperature is below -3.41 degrees or above 38.20 degrees.

The possibility of extreme events — such as droughts and floods — is accounted for by assuming that for each sector j and period t , $\Lambda_{j,t}$ is a random variable having a Beta distribution:

$$f(\Lambda_{j,t}; \varphi_{j,t}, \nu_{j,t}) = \frac{\Lambda_{j,t}^{\varphi_{j,t}-1} (1 - \Lambda_{j,t})^{\nu_{j,t}-1}}{\int_0^1 \Lambda_{j,t}^{\varphi_{j,t}-1} (1 - \Lambda_{j,t})^{\nu_{j,t}-1} d\Lambda}, \quad (2.160)$$

where

$$\varphi_{j,t} = \bar{\varphi} |T_j^* - T_t|, \quad (2.161)$$

whereas $\nu_{j,t}$ satisfies:

$$J_{j,t} = 1 - \frac{\varphi_{j,t}}{\varphi_{j,t} + \nu_{j,t}},$$

which gives:

$$\nu_{j,t} = \frac{\varphi_{j,t} J_{j,t}}{(1 - J_{j,t})}. \quad (2.162)$$

The parameter $\bar{\varphi}$ is set to 10 to ensure that Λ has a reasonably small variance. In the special (but unlikely) case where $T_j^* = T_t$, a small constant (10^{-9}) is added to the right-hand side of (2.161) and to the denominator of the right-hand side of (2.162) to avoid multiplication and division by zero. Similarly, when T_t is such that $J_{j,t} = 0$, a small constant is added to the numerator of the right-hand side of (2.162). Figure 2.6.10 shows the probability distribution of damages to

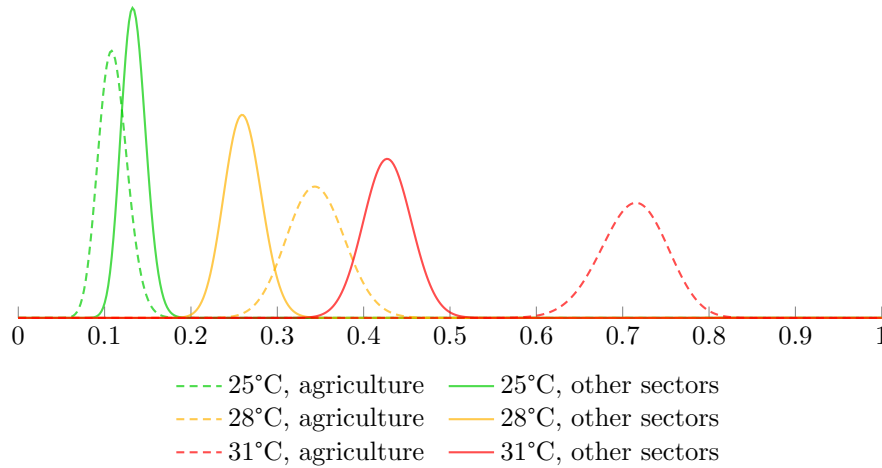


Figure 2.6.10: Probability distribution of Λ at different temperature levels

agriculture and non-agricultural production at different temperature levels. Greater deviations

from T^* result in higher average damages (relative to a world where temperature is at its optimal level) and a greater likelihood of extreme events on the right tail of the distribution.

Data sources: climate damage

The data collection and processing procedure follows Burzyńska et al. (2019). Monthly data on historical and projected mean temperatures are taken from the World Bank's Climate Change Knowledge Portal (henceforth CCKP, see <https://climateknowledgeportal.worldbank.org/download-data>). Projections under four different Representative Concentration Pathways (RCP2.6, RCP4.5, RCP6.0 and RCP8.5) are available at 20-year intervals from 2020 to 2100. Monthly data were averaged to generate annual projections at 20-year intervals, which were then interpolated linearly to obtain year-by-year values. For each Representative Concentration Pathway, the CCKP provides projections made by 16 different climate models. These models were ranked according to their temperature projection for the period 2040-2059 under RCP4.5, and the model corresponding to the median projection was chosen for analysis.

Next, temperature data were weighted by population to account for differences in population size across countries. A scaling parameter was constructed using the time series of monthly population-weighted, country-level data provided by Dell et al. (2012) for the period 1995-2005. Dell et al.'s data were averaged at the year level, and the value for each year was then divided by the corresponding (unweighted) value provided by the CCKP. Finally, the mean scaling parameter for the period 1995-2005 was multiplied by CCKP's projected temperatures to obtain population-weighted projections.

2.6.1.13 Time use

Each individual allocates time to 5 different activities: *physiological overhead* (PO), *study* (St), *paid work* (PW), *unpaid work* (UW) and *leisure* (Le). Here we assume that the allocation of disposable time depends on skill level, gender and employment status. Therefore there are 24 relevant categories.³⁹

The initial allocation of time is based on time use survey data. The driver behind subsequent dynamics in time allocations are changes in paid working time of employed people. These translate into changes in unpaid working time of employed, which in turn prompt adjustments in other population groups.

In every period, the daily working hours for gender and skill specific groups are computed as

³⁹We do not consider the time use of capitalists.

the average hours of work of employed people in the group over all industries:

$$PW_{s,g} = \frac{1}{44 \cdot 5} \cdot \frac{\sum_i h_i \cdot L_{i,s,g}}{\sum_i L_{i,s,g}}. \quad (2.163)$$

where $\frac{1}{44 \cdot 5}$ is the conversion term from annual to daily hours.

Firstly, we assume that the amount of time devoted to physiological overhead and study are fixed over time for every group, i.e. $\Delta PO_t; \Delta St_t = 0 \forall t$, while the time devoted to leisure works as a residual category.

Secondly, we assume that for employed individuals, a change in the time devoted to paid work is associated with an additional proportional – gender specific – variation in the time devoted to unpaid work.

Therefore, the variation of time allocation for the employed is equal to:

$$\begin{cases} \Delta PW_{s,g,t}^E = PW_{s,g,t} - PW_{s,g,t-1} \\ \Delta UW_{s,g,t}^E = \omega_g^{uw} \Delta PW_{s,g,t} \\ \Delta LE_{s,g,t}^E = -(1 + \omega_g^{uw}) \Delta PW_{s,g,t} \end{cases} \quad (2.164)$$

where ω_g^{uw} is the gender specific sensitivity of unpaid work w.r.t. changes in paid work.

Individuals that are not employed (i.e. unemployed, out of the labour force and retired) only adjust their time allocation between unpaid work and leisure. The reaction of not employed people's unpaid working time to changes in employed people's unpaid working time is determined by the assumption that there has to be a specific amount of unpaid working time for the economy to function. Thus, the total amount of unpaid work in the economy is assumed to be fixed. Therefore, every time the employed subjects adjust their own time devoted to unpaid work, they trigger an adjustments throughout the entire economy.

According to this assumption, the total amount of unpaid working time that has to be reallocated to the part of the population that is not employed is given by the total change in UW of the employed population:

$$\Delta UW_t^E = \sum_s \sum_g \Delta UW_{s,g,t}^E \cdot L_{s,g,t}, \quad (2.165)$$

Subsequently, this time is allocated to not employed groups according to their previous share of overall UW in the not employed population. This share is given by

$$\zeta_{s,g,t}^{UW,\tilde{k}} = \frac{UW_{s,g,t-1}^{\tilde{k}} N_{s,g,t-1}^{\tilde{k}}}{\sum_{\tilde{k}} \sum_s UW_{s,g,t-1}^{\tilde{k}} N_{s,g,t-1}^{\tilde{k}}}; \quad \tilde{k} = \{U, O, P\} \quad (2.166)$$

therefore the change in unpaid work of each group in period t is equal to:

$$\Delta UW_{s,g,t}^{\tilde{k}} = \frac{\zeta_{s,g,t}^{UW,\tilde{k}} \cdot \Delta UW_t^E}{N_{s,g,t}^{\tilde{k}}}. \quad (2.167)$$

Finally, the variation in leisure time is given by the residual

$$\Delta LE_{s,g,t}^{\bar{k}} = -\Delta UW_{s,g,t}^{\bar{k}}. \quad (2.168)$$

Data sources: time use

The initial allocation of time use is based on microdata from the 2013 Italian Time Use Survey and the gender specific sensitivity of unpaid work (ω_g^{uw}) results from a regression analysis from aforementioned data.

2.6.2 Stock-flow matrices

This Section presents the balance-sheet and the transaction flow matrices of the model. The variables presented for the different stocks and flows inside each matrix are defined in the previous sections. Given the peculiarities of the financial sector, we present it in separate columns while the stocks and flows of the remaining 18 industries are presented together in one column (ind i). The unique characteristic of the financial sector NACE category is that it includes monetary intermediation which is why its balance sheet captures the issuance of loans and provision of deposits. Please note, that the model is not fully stock-flow consistent since it considers an open economy without modelling the foreign sector. Additionally, mixed income does not have a use of funds counterpart, as we do not explicitly model its origin, however it is necessary to include in the model in order to obtain results consistent with real world data.

Table 2.6.6: Aggregate Balance sheet

	Low-skill	Middle-skill	High-skill	Cap.	Ind. [†] _i	Fin.	Gov.	Σ
Deposits	D_l	D_m	D_h	D_{cap}		$-D$		0
Bonds		B_m	B_h	B_{cap}			$-B$	0
Equity			Eq_h	Eq_{cap}	$-\sum_i Eq_i$	$-Eq_{12}$		0
Loans					$-\sum_i H_i$	H		0
Fixed Capital					$\sum_i K_i$	K_{12}		$\sum_{i=1}^{19} K_i$
Net worth	$-V_l$	$-V_m$	$-V_h$	$-V_{cap}$			$-V_G$	$-V$
Σ	0	0	0	0	0	0	0	0

[†] Represents all industries with the exception of the financial sector which has a specific balance-sheet.

Table 2.6.7: Transaction Flow Matrix

	Low-skill				Middle-skill				High-skill				Capitalists	Industries [†]	Financial		Government [‡]		Σ	
	E	U	OLF	P	E	U	OLF	P	E	U	OLF	P		current	capital	current	capital	current	capital	
Wages	Σ _i GW B _{i,l}				Σ _m GW B _{i,m}				Σ _h GW B _{i,h}					− Σ _i GW B _i		− GW B ₁₂		− (GW B ₁₅ + ζ ^w (GW B ₁₆ + GW B ₁₇))		0
Pensions	GP B _l				GP B _m				GP B _h									− Σ _j GP B _j		0
UB	GU B _l				GU B _m				GU B _h									− Σ _j GU B _j		0
FCB	FC _l	FC _l	FC _l		FC _m	FC _m	FC _m		FC _h	FC _h	FC _h							− Σ _s FC _s		0
SDB	SD _l	SD _l	SD _l		SD _m	SD _m	SD _m		SD _h	SD _h	SD _h							− Σ _j SDB _j		0
RB	RB																	− RB		0
MI	MI _l ^E	MI _l ^U	MI _l ^I	MI _l ^P	MI _m ^E	MI _m ^U	MI _m ^I	MI _m ^P	MI _h ^E	MI _h ^U	MI _h ^I	MI _h ^P	MI _{cap}							Σ _s Σ _h MI _s ^h
SSC	− T _{l1} ^{S,E}				− T _{m1} ^{S,E}				− T _{h1} ^{S,E}									Σ _s T _{s1} ^{S,E}		0
Income tax	− T _{l1} ^{I,E}	− T _{l1} ^{I,U}	− T _{l1} ^{I,P}		− T _{m1} ^{I,E}	− T _{m1} ^{I,U}	− T _{m1} ^{I,P}		− T _{h1} ^{I,E}	− T _{h1} ^{I,U}	− T _{h1} ^{I,P}							Σ _s Σ _h T _{s1} ^{I,h}		0
Financial income tax*						− T _m ^F				− T _h ^F			− T _{cap} ^F					T ^F		0
SSC employer														− Σ _i T _{i12} ^{S,F}		− DT ₁₂ ^{S,F}		Σ _i T _{i1} ^{S,F} + T ₁₂ ^{S,F}		0
CIT														− Σ _i T _{i12} ^{C,IT}		− T ₁₂ ^{C,IT}		Σ _i T _{i1} ^{C,IT} + T ₁₂ ^{C,IT}		0
Carbon tax														− Σ _i T _{i12} ^{carb}		− T ₁₂ ^{carb}		Σ _i T _{i1} ^{carb} + T ₁₂ ^{carb}		0
VAT														− Σ _i T _{i12} ^{V,AT}		− T ₁₂ ^{V,AT}		Σ _i T _{i1} ^{V,AT} + T ₁₂ ^{V,AT}		0
Profits*									$\frac{E_{gh}}{w_{gh} + E_{hcap}} (\sum_i Div_i + Div_{i2})$	$\frac{E_{gh}}{w_{gh} + E_{hcap}} (\sum_i Div_i + Div_{i2})$				− Σ _i Π _i	Σ _i (In v _i + π _i ^{res} + dr _i)	− Π ₁₂	(In v ₁₂ + π ₁₂ ^{res} + dr ₁₂)		0	
Intermediate goods														Σ _i (Σ _m Z _{m,i} − Σ _s Z _{m,s} − Σ _s Z _{m,i} ^{imp})		Σ _i Z _{12,i} − Σ _m Z _{m,12} − Σ _m Z _{m,12} ^{imp}				
Consumption ^{*,1}	− Σ _i C _{i,l}				− Σ _i C _{i,m}				− Σ _i C _{i,h}				− Σ _i C _{cap}	Σ _i (f _i ^C + f _i ^G + f _i ^E)		f ₁₂ ^C + f ₁₂ ^G + f ₁₂ ^E		− Σ _i f _i ^C		Σ _i (f _i ^E − f _i ^{C,imp} − f _i ^{G,imp})
Investments														Σ _i f _i ^I	− Σ _i I _i ^{nom}	f ₁₂ ^I	− I ₁₂ ^{nom}		− G ^I	− Σ _i f _i ^{I,imp}
Interest on bonds*					i B _m				i B _h				i B _{cap}					− i B		0
Σ*	Sav _l				Sav _m				Sav _h				Sav _{cap}	Σ _i Π _i ^{res} + In v _i + dr _i − Σ _i I _i		Σ _i Π ₁₂ ^{res} + In v ₁₂ + dr ₁₂ − Σ _i I ₁₂		− G ^D		MI + f ^E − M − G ^I − G ^W
Δ Deposits*	− Δ D _l				− Δ D _m				− Δ D _h				− Δ D _{cap}			Δ D				0
Δ Bonds*					− Δ B _m				− Δ B _h				− Δ B _{cap}					Δ B		0
Δ Loans*														Σ _i Δ H _i		− Σ _i Δ H _i				0
Σ	0				0				0				0	0		0		0		MI + f ^E − M − G ^I − G ^W
Δ Equities*									Δ E _h				Δ E _{cap}		− Δ E _l		− Δ E ₁₂			0

[†] Represents all industries given by subscript i except financial services (12).

[‡] Note that investment and wage costs covered in government funded sectors are represented twice. For consistency, we model their payment in the industry column itself, as for all other production, but the public sector differs from other sectors as it produces a range of non-market output.(For instance see <https://ec.europa.eu/eurostat/documents/3859598/5902521/KS-RA-07-015-EN.PDF>) Thus, payments to facilitate these activities (i.e. wages and investment) have to be funded directly by the government. However, note that the inclusion of these costs in the government accounts are purely for accounting purposes, in order to determine the government deficit. Households income and taxes are split between employed (E), unemployed (U), out of labour force (OLF) and retired (P) to indicate taxes and benefits specific to each of these occupational statuses. As there is no differentiation of income sources or uses, for simplicity we depict each group aggregated over both genders.

* R presents flows that are aggregated by skill indicated by the subscript $s=l$ low, m middle and h high-skill.

1 includes consumption out of wealth.

UB = unemployment benefits, FCB = family and children benefits, SDB = sickness and disability benefits, RB = residual benefits, MI = mixed income, SSC (employer) = social security contribution employee (employer), CIT = corporate income tax, VAT = value added tax.

2.6.3 List of variables and parameters

2.6.3.1 Subscripts and Superscripts

Table 2.6.8: Subscripts and Superscripts

Subscript/Superscript & Range	Description
$c \in \{1, 2, \dots, 18\}$	CIOCOP goods categories
cap	capitalists
$g \in \{f, m\}$	gender
$i(j) \in \{1, 2, \dots, 19\}$	industries
$k \in \{E, U, P\}$	employment status excluding <i>OLF</i> and <i>cap</i>
$k' \in \{E, U, OLF, P, cap\}$	employment status including <i>OLF</i> and <i>cap</i>
$\tilde{k} \in \{U, OLF, P\}$	employment status of not employed population
$n \in \{I, II, \dots, V\}$	age cohorts
$s \in \{low, middle, high\}$	skills
t	time in years
$\phi \in \{solid, oil, gas, biomass, renewables\}$	fuel sources
$real/nom$	variable in real/nominal terms

2.6.3.2 Variables and Parameters

Table 2.6.9: Variables and Parameters

Name	Description	Defined in section
κ^+	fertility rate	Population
$\kappa_{n,g}^-$	mortality rate	
RL	reproductive years	
ϱ_n	maturity rate	
$Pop_{n,g}$	number of individuals in age (n) and gender (g) group	
y_i	output of sector i	Input-Output
$z_{i,j}$	domestic intermediate trade from industry i to industry j	
$f_i^{(x)}$	domestic final demand (of $x = C, I, G, E$)	
$a_{i,j}$	technical coefficient	
$\mathbf{L}$	Leontief inverse matrix	
$l_{i,j}$	ij th element of $\mathbf{L}$	
y_i^{FC}	output under full capacity utilisation	
GDP	gross domestic product	
p_i	price of goods of sector i	
$\tilde{p}_i$	commodity price index for deflation	
$\gamma_{i(j)}^x$	import share of $x = \{z, C, G, I\}$ of industry $i(j)$	
$z_{i,j}^{imp}$	imports of sector i goods in sector j	

$f_i^{x,imp}$	demand for imports in demand category $x = \{C, G, I\}$ for sector i	
$\hat{f}_i^x$	overall demand of category $x = \{C, G, I\}$ for industry i	
f_i^x	domestic demand of category $x = \{C, G, I\}$ for industry i	
$M_{i,t}$	total nominal total imports by industry	
f_i^E	demand for exports for industry i	
g_i^E	exogenous growth rate in exports	
ε^E	elasticity of exports to domestic price variations	
λ_i	labour productivity	
$\tilde{a}_j$	mean intermediate input requirement per unit output	
$T1 - T4$	type of technology	
$\text{Prob}_{j,t}(Tz)$	probability of technology x ($z \in \{2, 3, 4\}$) emerging	Innovation
$\dot{\theta}_i$	rate of change of relative unit costs, $\frac{UIC}{ULC}$	
ϑ	parameter scaling effect of $\dot{\theta}_i$ on $\text{Prob}_j(Tz)$	
$\Delta^{\lambda/a}$	changes from labour/resource saving technology	
$\lambda_j^{\max}, a_{i,j}^{\max}$	upper bound for λ_j and $a_{i,j}$ in case of diffusion	
VA_i	value added in sector i	
$\hat{\Pi}_i$	gross profits;	
Π_i	post tax profits;	
Π_i^{free}	profits free for investment;	
Π_i^{res}	residual profits	
$\hat{z}_i$	intermediate demand of industry i , domestically and imports	

VAT_i	value added tax paid by industry i	Investment and Profits
I_i^*	desired investment	
I_i^{max}	maximum investment industry i can finance	
I_i	actual investment	
K_i	stock of physical capital	
$uc_i^{(N)}$	(normal) rate of capacity utilisation	
ω_i^I	sensitivity of desired investment to over-/underutilisation of capacity	
ϵ_i^K	productivity of capital	
elr	equity to liabilities ratio	
lev	leverage ratio (debt to assets)	
p^K	the price of capital	
δ_i	rate of capital depreciation	
$\mathbf{b}$	vector allocating investment into final demand	
Eq_i	stock of equity	
H_i	stock of debt	
i_i^{ind}	industry specific interest rates	
ω^H	sensitivity of mark up on interest rate w.r.t. lev	
dr_i	repayment of debt principal	
lt_i	average lifetime of new investments	
Inv_i	direct expenditure for investment paid by sector i	
div_i	rate of dividends paid to equity owners	

Div_i	total dividends paid to equity owners by industry i	
$(w) W$	(per adult) national wealth	
N^A	amount of individuals in adult population	
ζ_s^W	share of total wealth belonging to skill s	
$W_{s,g}$	amount of wealth held by group	Finance and Wealth
$D_{s,g}$	deposits held by group	
$Eq_{s,g}$	equity held by group	
$B_{s,g}$	bonds held by group	
$GFI_{s,g}$	gross financial income	
$NFI_{s,g}$	net financial income	
$Pop_{s,g}^n$	number of individuals in age(n), gender (g) and skill(s) group	
$u_{s,g}$	unemployment rate	
$e_{s,g}$	employment rate	
$\sigma_{n,s,g}$	skill distribution in population group	
ω_s^u	sensitivity of skill distribution w.r.t unemployment rates	
$Mat_{s,g}^n$	number of individuals of group maturing out of cohort n	
$g_{s,g}^{entry}$	exogenous trend in skill endowment upon entering working age	
$Y_{s,g}^{e,LF/OLF}$	expected relative labour income when joining/staying in the labour force	
$w_{i,s,g}$	hourly wage	
$(\bar{h})h_i$	(average) annual working hours	
$ub_{s,g}$	unemployment benefits received by individual	

Labour

$ob_{s,g}$	benefits received by individual when being out of labour force
$\psi_{s,g}$	labour force participation rate
ω_s^ψ	sensitivity of ψ w.r.t. changes in expected labour income
$L_{i,s,g}$	number of individuals from population group employed in i
$\rho_{i,s}$	skill distribution of employment per industry
$g_{i,s}^\rho$	exogenous trend in ρ
c	conversion rate
$m_{i,s}$	share of male employment
$g_{i,s}^m$	exogenous trend in $m_{i,s}$
ω_s^m	sensitivity of $m_{i,s}$ w.r.t. rate of change of relative gender specific u_s
$\omega_{i,s}^\lambda$	sensitivity of $w_{i,s,g}$ w.r.t. g_i^λ
g_i^λ	rate of change of λ_i
ω^p	sensitivity of $w_{i,s,g}$ w.r.t. inflation
$\omega_{s,g}^L$	sensitivity of $w_{i,s,g}$ w.r.t. $e_{s,g}$
$GW B_{i,s,g}$	gross wage bill
$G^{T/HH}$	public revenue from: taxes/taxes on households/
$G^{CIT/VAT/SC}$	corporate income tax/value added tax/social contributions
G^{CARB}	carbon tax
$V_{i,s,g}^K$	annual income subject to social security contributions
$\tilde{V}_{i,s,g}^K$	taxable annual income
$GU B_{s,g}$	gross unemployment benefits

$GPB_{s,g}$	gross pension benefits	
$\tau_{s,g}^{S,E}$	rate of social security contributions paid by employee	
$\tau_{s,g}^{S,F}$	rate of social security contributions paid by employer	
τ_f^I	marginal income tax rate of income bracket f	
τ^F	tax rate on financial income	
τ^{VAT}	value added tax rate	
τ^{CIP}	corporate income tax rate	
τ^{carb}	carbon tax rate	
$t_{s,g}^{S,E}$	amount of employee's social security contributions	
$t_{s,g}^{S,F}$	amount of employee's social security contributions paid by employer	
$t_{i,s,g}^{I,K}$	amount of individual's income tax	
$T_{s,g}^{S,E/F}$	total amount of social security contributions for specific group paid by employees/employers	
$T_{s,g}^{I,K}$	total amount of income taxes paid by population group	
$T_{s,g}^F$	total financial income tax paid by population group	
T_i^{VAT}	total value added tax paid by sector	
T_i^{CIT}	total corporate income tax paid by sector	
T_i^{carb}	total carbon tax paid by sector	Government
$\underline{b}_f/\bar{b}_f$	lower/upper bound of income tax bracket	
G^E	government overall expenditure	
$G^{I/Tr/w/B}$	expenditure on: investments/transfers and benefits/wages/bond interests	
$\hat{f}^G$	final government consumption expenditure	

$GUB_{s,g}$	gross unemployment benefits for population group
$GPB_{s,g}$	gross pension benefits for population group
ϖ	GUB fraction of population group's average wages
χ	coverage ratio of unemployed population
pw	pension-to-wage ratio
$D62^x$	public transfers in category $x = \{OS, FC, SD, RB\}$
sd	sickness and disability transfers received per person
fc	yearly benefit per under-age person
$FC_{s,g}$	family and children benefits received by population group
rb	residual benefits per person
$N_{s,g}^F$	number of children for population group
$N_{s,g}^{WA}$	number of people in working age population in population group
ζ^I	fraction of government investment with respect to total investment
ζ^w	fraction of wages paid by government in sectors education and health
g_i^G	exogenous trend in government final consumption
G^D	government deficit
i	policy rate of interest
ω^i	sensitivity of i w.r.t. divergence of inflation from target
r^*	inflation target set by central bank
UFC_i	unit full cost
ULC_i	unit labour cost

UIC_i	unit costs of intermediate inputs
μ_i	mark up
ω^μ	sensitivity of μ_i w.r.t. divergence of capacity utilisation from uc^N
UCC_i	unit carbon costs faced by industry i
CPI	consumer price index
ζ_i^p	share of real household demand allocated to sector i
$r_{(i)}$	(sector specific) inflation
$NWB_{s,g}$	net wage bill
$NUB_{s,g}$	net unemployment benefits
$NPB_{s,g}$	net pension benefits
mI	mixed income per capita
$L_{s,g}^U$	number of unemployed in population group
$N_{s,g}^P = Pop_{s,g}^5$	number of retired in population group
N_{CAP}	number of capitalists
$YD_{k',s,g}^{(e)}$	(expected) disposable income
$C_{k',s,g}^{(tot/y/w)}$	consumption expenditure (total/out of income/out of wealth)
$\alpha_{k',s,g}$	propensity to consume out of income
$\check{\alpha}_{k',s,g}$	propensity to consume out of income before being adjusted for inflation
$f^Y(\cdot)$	function translating $YD_{k',s,g}^{(e)}$ into $\check{\alpha}_{k',s,g}$
i^c	cumulative inflation for consumption category c
$\bar{i}_{k',s,g}$	average cumulative inflation experienced by population group

Consumption

$\hat{i}_{k',s,g}^c$	difference between i^c and $\bar{i}_{k',s,g}$
$S_{k',s,g}$	savings of population group
$\beta_{k',s,g}^c$	fraction of consumption expenditure of population group allocated to each c
$\check{\beta}_{k',s,g}^c$	$\beta_{k',s,g}^c$ before being adjusted for inflation
$f^\beta(\cdot)$	function translating $Y D_{k',s,g}^{(e)}$ into $\check{\beta}_{k',s,g}^c$
ϵ^c	elasticity of $\beta_{k',s,g}^c$ w.r.t. $\hat{i}_{k',s,g}^c$
$C_{(k',s,g)}^c$	total consumption expenditure (by each group) spend on c
$C_{cc/i}$	total consumption expenditure in the economy spend on c on sector i
$\mathbf{B}^c$	COICOP-NACE bridge matrix
E_i^S / E_i^D	energy supply/demand of industry i
$\mathbf{E}^Z_{i,j}$	matrix of physical energy flows between domestic industries
$\eta_{i,j}^Z$	cell specific conversion factor of monetary to energy flow for domestic intermediate trade
$E_i^{Z,imp}$	demand for imported energy from domestic industry i
$E_i^{\hat{z}}$	total energy demand by industry i
E_i^E	exported energy supply
E_i^C	energy demand for households' domestic consumption
$E_i^{C,imp}$	energy demand for households' imported consumption
$E^{\hat{C}}$	total household energy demand
GIC	gross inland consumption
NIC	net inland consumption
η_i^R	ratio of energy residuals to energy product demand

Energy

E_i^R	energy residuals	
$\zeta_i^\phi(\zeta^{C,\phi})$	share of energy demand of industry i (of households) covered by energy source ϕ	
$g_i^\phi(g^{C,\phi})$	growth rate of shares of energy demand of industry i (of households) covered by <i>renewable</i> energy	
$E_{i,\phi}^F(E_\phi^H)$	energy demand of industry i (of households) by energy source	
$\gamma_{i,\phi}^{EE}(\gamma_\phi^{HE})$	source specific ratio of emission relevant to total energy use for industry i (for households)	
$EE_{i,\phi}^F(EE_\phi^H)$	source specific emission relevant energy use for industry i (for households)	
$v_{i,\phi}^F(v_\phi^H)$	source specific coefficient translating emission relevant energy use into CO ₂ emissions for industry i (for households)	
$CO2_{i,\phi}^F(CO2_\phi^H)$	source specific CO ₂ emissions by industry i (by households)	
$\gamma_i^{Fghg}(\gamma^{Hghg})$	fraction of GHG to CO ₂ emissions for industry i (for households)	
$GHG_i^F(GHG^H)$	GHG emissions by industry i (by households)	
GHG^{tot}	total GHG emissions	
Λ_i	climate damage to production	
$J_i = 1 - \mathbb{E}[\Lambda_i]$	average multiplicative effect of climate on production	
$\tilde{J}_i$	realised multiplicative effect of climate on production	
$a_{i,j}^{bd}$	technical coefficient net of climate damage	Climate Damage
$T^{(*)}$	(optimal) mean temperature	
g, g', g''	parameters translating T into J_i	
φ_i, ν_i	parameters of the Beta distribution of climate damage	
$PW_{s,g}^E$	average daily paid working hours for group	

$PO_{s,g}^{E/\bar{k}}$	average daily hours devoted to physiological overhead for group
$St_{s,g}^{E/\bar{k}}$	average daily hours devoted to studying for group
$UW_{s,g}^{E/\bar{k}}$	average daily hours devoted to unpaid work for group
$LE_{s,g}^{E/\bar{k}}$	average daily hours devoted to leisure for group
ω_g^{uw}	gender specific sensitivity of unpaid work w.r.t. changes in paid working time
$\zeta_{s,g}^{UW,\bar{k}}$	share of overall unpaid work in the not employed population handled by group

2.6.3.3 Calibration Parameters

Table 2.6.10: Calibration Parameters. The parameters were calibrated using data between 2010 and 2019/20. The letters *s* (skill), *i* (industries) and *c* (consumption goods) indicate parameters that have different values for subscripts of a single variable.

Parameter	Baseline Value	Equation
Tech. probability sens.	11.93	2.13-14
Initial prob. T2	0.67	2.13
Initial prob. T3	0.47	2.14
Skill transition sens. (<i>s</i>)	[0.69,0.75]	2.63-65
Labour force participation sens. (<i>s</i>)	0.75	2.68
Gender employment subst. sens. (<i>s</i>)	[0.03,0.08]	2.72
ω employment	0.45	2.73
ω lab. productivity	0.99	2.73
ω price	1	2.73
Investment sens. (<i>i</i>)	[0,0.225]	2.22
Dividends rate	0.3	2.40
Interest on loans sens.	0.12	2.36
Price-elasticity of exports	-0.5	2.11
Price-elasticity of consumption (<i>c</i>)	0	2.131
Mark-up sens. (<i>i</i>)	[0,0.067]	2.112
Interest rate sens.	1	2.107
Scaling Δ labour productivity	0.65	
Seed	1	

Chapter 3

Emissions, inequality and employment – The role of the energy transition in a just low-carbon transition

This chapter corresponds to: Heydenreich, T., Cano Ortiz, D., Morlin, G. S., D'Alessandro, S., Samsó, R., and Demaria, F. (*In Preparation*). The role of the energy transition in a just low-carbon transition.[†]

Abstract

Providing a good life for all within planetary boundaries requires a fundamental transformation of the socioeconomic system. The energy transition and related socioeconomic outcomes, including employment, play an important role. This work analyses the socioeconomic implications of the energy transition within the context of three socioeconomic paths, featuring for instance different degrees redistribution, for Italy. By integrating economic effects of electrification and a renewable transition in the electricity sector into the Eurogreen model, this paper establishes a particularly suitable framework for this task, allowing for a thorough and integrated analysis of socioeconomic implications. We find that, while redistribution policies alone tend to increase economic activity and reduce inequality, they do so at the expense of higher energy use and emissions. **In contrast, the energy transition can provide a “triple” dividend in all three socioeconomic paths: Increasing economic activity and employment while decreasing inequality, all by a few percent, it simultaneously substantially reduces energy use and emissions.** Structural changes in employment are marginal on the macroeconomic scale, given the relatively

[†]Given the extensive and complex character of the Eurogreen model, this work is a highly collaborative effort. I am the lead author of this project, having conceptualised it and integrated the novel energy and electricity modules into the model. In addition, I led the investigation and the writing process, as well as conducted the formal analysis. More details on individual contributions are provided in the authorship statement of this Chapter.

small share of overall employment in energy supplying sectors. In addition, our results reveal that a redistribution can have consequential implications for the innovation process, shifting it towards improvements in labour productivity at the expense of production efficiency in terms of other inputs. Consequently, substantial redistribution measures not only directly increase emissions via economic activity but also lead to a higher ecological impact *ceteris paribus* through impacts on innovation dynamics. This highlights the need for ambitious energy transition efforts and the potentially crucial role of structural reductions in demand and economic activity in the context of a just transition.

3.1 Introduction

It is well established that addressing the climate crisis will require a rapid and extensive structural transition of current socioeconomic system (e.g. Fanning et al., 2022; IPCC, 2023), within which the transition of the energy system is crucial. Here, addressing socioeconomic implications and in particular inequality is crucial as evident in the just transition¹ literature (e.g. Wang and Lo 2021). Apart from considerations of justice, fairness and the disproportional impact of the affluent (Chancel, 2022; Wiedmann et al., 2020), it is important for political acceptance that perceived losses from the transition are addressed (Groh and Möllendorff, 2020).

Jobs play a crucial role for a fair transition. Firstly, apart from providing income, jobs are closely tied to a sense of identity and culture (Carley and Konisky, 2020). Thus, losing or switching a job has far deeper implications than those reflected in income. In addition, even switching jobs itself can be challenging because of gaps in skills or other complications like geographical distance (Carley and Konisky, 2020). Accordingly, policies for the ecological transition should take into account the implied structural changes in employment in order to guarantee that 'no one is left behind' and facilitate public support.

To analyse potential transition pathways and their socioeconomic implications, the field of Ecological Macroeconomics is particularly suited. It addresses several shortcomings of frameworks that are dominated by neoclassical economic theory like many Integrated Assessment Models (IAMs)². For instance, the aggregated representative agent framework dominant in IAMs prevents an integrated analysis of inequality (Emmerling and Tavoni, 2021; Rao et al., 2017). Many shortcomings are particularly relevant in the context of an extensive structural transformation (Hafner et al., 2020) (e.g. optimisation and smooth adjustment paths (Rezai et al., 2013) or full utilisation of resources precluding involuntary unemployment (Rezai and Stagl, 2016)) and can create biased or misleading policy recommendations (Hafner et al., 2020; Mercure et al., 2019; Stern et al., 2022).

However, Ecological Macroeconomics has paid relatively little attention to inequality and its implications (Hardt and O'Neill, 2017) and in particular a thorough analysis of inequality and employment challenges within comprehensive socioeconomic and energy transition pathways is lacking. On the one hand, prominent models in the ecological macroeconomic literature like LowGrow (Jackson and Victor, 2019a) and MEDEAS (Capellán-Pérez et al., 2020) lack a structure that allows for an integrated analysis of inequality and disaggregated employment effects.

¹The IPCC defines a just transition as "A set of principles, processes and practices that aim to ensure that no people, workers, places, sectors, countries or regions are left behind in the transition from a high-carbon to a low-carbon economy" (IPCC, 2022a, p.1806).

²IAMs are defined as models that incorporate insights from several fields of knowledge (IPCC, 2022a).

On the other hand, the Eurogreen model (D'Alessandro et al., 2020), while featuring a suitable disaggregated structure to capture socioeconomic outcomes thoroughly, so far lacks a consistent integration of the economic feedback from the energy transition. Generally, analyses of the just transition as part of the energy transition tend to focus on volumes of net employment and not be embedded into overall economic developments and socioeconomic transition pathways, in particular post-growth scenarios (García-García et al., 2020).

In this vein, this paper contributes the literature in two major ways: 1) It improves the conceptualisation of the energy transition in the Eurogreen model, which enables to capture direct feedback effects from two main elements of the energy transition to the economy, namely electrification and the renewable transition in the electricity sector. 2) It adds crucial results to the literature analysing transition pathways, in particular the just transition literature, by thoroughly evaluating the socioeconomic implications of the energy transition within broader socioeconomic paths, including post-growth scenarios. Here a particular focus is laid on disaggregated effects on employment. Note that throughout this paper we distinguish between socioeconomic paths and scenarios, since socioeconomic paths only describe certain aspects and have to be combined with energy transition assumptions to form full scenarios.

To achieve these objectives, we modify the energy module and introduce a novel electricity module into the Eurogreen model. Subsequently, we apply the revised model to Italy and assess the socioeconomic impacts of a low-carbon energy transition within three overall socioeconomic transition paths inspired by D'Alessandro et al. (2020), namely a baseline, a redistribution and a post-growth path. While the baseline features no additional policies, the redistribution path introduces new redistribution policies and the post-growth path exhibits more extensive redistribution and measures like a reduction in consumption.

Specifically, we set out to address the following research questions from a novel perspective, adding new and detailed insights with crucial policy implications: “What are the implications of a low-carbon energy transition within socioeconomic transition paths?” and in particular “What are the implications for employment structure and dynamics?”

The paper proceeds as follows: Section 3.2 briefly introduces the Eurogreen model and its application to Italy as well as more extensively describes the novel energy and electricity modules. The quantitative results are presented in Section 2.4 and policy implications are discussed in Section 3.4. Section 2.5 summarizes the main findings.

3.2 Methods

3.2.1 The main structure of the Eurogreen model

The Eurogreen model is a system dynamics ecological macroeconomic model for the national economy following post-Keynesian theory and stock-flow consistent principles, with past applications to France (Cieplinski et al., 2021b; D'Alessandro et al., 2020) and Italy (Cieplinski et al., 2021a; Cieplinski et al., 2023; Distefano and D'Alessandro, 2023).

In the Eurogreen model, economic production is driven by aggregate demand, i.e. household and government consumption, investment and exports, while also being constrained by the availability of capital and labor. Production follows an open input-output model, with economic activity segmented into 21 industries. Prices are determined according to a refined version of the Leontief price model (Miller and Blair, 2009), with industry-specific mark up applied over production costs.

Individuals are differentiated by skill, gender and employment status and pooled into households by region and income quintile. Household consumption depends on group-specific marginal propensity to consume and composition of the consumption bundle, and evolves with real disposable income and changes in relative prices. Employment in each industry is driven by industry output, labour productivity and working hours, with developments in skill shares being driven by changes in labour productivity. Labour productivity, along with inflation and (un)employment rates lead to wage adjustments, which in turn affect government transfers like pensions. Investment is chiefly motivated by bringing the capacity utilisation towards the normal rate given expected output. The government collects taxes, provides transfers and creates direct final demand.

The model features endogenous data-driven technological change. Here, the development in respective costs incentivises innovation to improve labour productivity on the one hand or technical coefficients on the other, and the impact of technological developments is driven by investment. Energy demand is determined from monetary flows using interindustrial trade and household consumption specific energy intensities.

Overall, the model used here mainly follows the structure described in Campigotto et al. (2024) with main aspects of divergence described in the supplementary materials. As the focus of this paper is on the energy transition, we subsequently outline the novel aspects relating to energy and the energy transition.

3.2.2 Novel energy-economy links in Eurogreen

The newly introduced interaction between energy and economic developments essentially consists of two elements: 1) We directly link monetary and physical energy flows, allowing to translate changes in energy flows, especially electrification, into changes in the economic structure. 2) We introduce an electricity module that is able to capture the economic impacts of a renewable energy transition, in particular associated investment and implications for the production structure of the electricity sector.

The remainder of this section introduces the major features of these two novel elements, with more detailed information provided in the supplementary material.

3.2.2.1 Energy and production structure

We revise the energy module of the Eurogreen model in order to facilitate the modelling of changes in the production structure driven by electrification. Using a range of datasets and assumptions, we obtain an input-output (IO) production (A^E), as well as final demand (FD^E) structure that reflect only monetary flows related to energy as well as the associated energy flows. The aggregate energy products we consider are: Coal and coal products, Oil and petroleum products, Natural gas, Biofuels, Electricity and Heat.

3.2.2.1.1 Electrification process By translating physical into monetary flows, the novel energy extension is able to capture the impact of electrification on the economic production and consumption structure. Figure 3.1 illustrates the implementation of the electrification process in Eurogreen. Electrification develops through two channels, namely the composition of final demand and the input-output production structure of the economy.

For final demand, in a first step, the monetary final demand structure before electrification (FD^{be}) is split into an energy (FD^E) and a non-energy (FD^{NE}) related part, using demand- and sector-specific shares ($em\ share^{FD}$). Subsequently, FD^E is translated into physical units of energy and the scenario specific electrification is applied to these physical units. For instance, 1TJ of fossil fuel demand from different sectors is replaced by 1TJ of energy from the electricity sector, depending on efficiency gains assumptions due to electrification. This new structure of final demand in physical units is subsequently converted back into monetary terms ($FD^{E,S}$). In addition, the demand for sectors that primarily distribute but do not produce certain energy products (e.g. natural gas) is scaled in proportion to the change in the demand for the respective energy product ($Distribution^{FD}$). The combination of the changes due to the electrification in $FD^{E,S}$ and $Distribution^{FD}$ and the unaffected FD^{NE} subsequently yields a new final demand structure in monetary terms (FD) with its respective shares reflecting energy flows

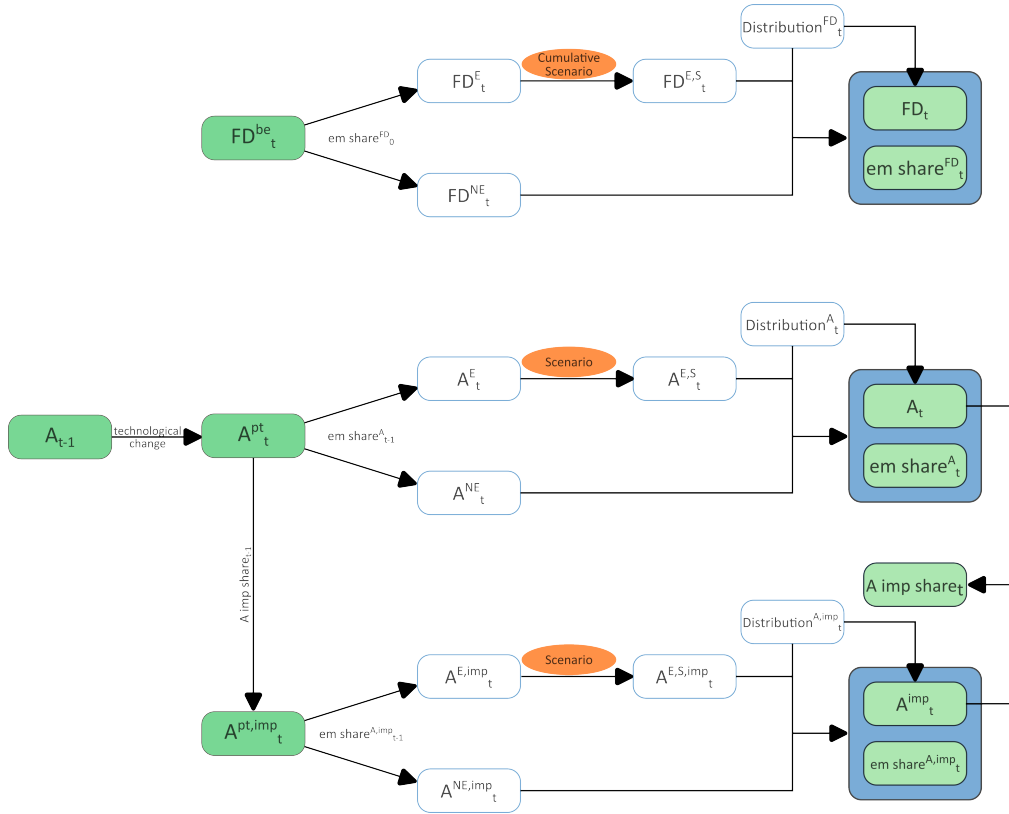


Figure 3.1: Illustration of the electrification process. Implementation of the electrification process for final demand (FD) and production structure (A). The green boxes on the left are an output of the overall Eurogreen model, which are subsequently modified by scenario specific electrification dynamics (orange), before the green boxes on the right feed back into the overall Eurogreen model. The superscripts mean before electrification (be), post technology (pt), energy (E), non-energy (NE) and energy after scenario (E, S). Moreover, $em\ share$ refer to the fraction of monetary flows that result from energy flows and $imp\ share$ to import shares.

($em\ share^{FD}$).

While the general approach is the same for the modelling of electrification of the IO structure, it follows a slightly different procedure than the one of final demand. Whereas final demand, in each period, is calculated in the absence of electrification and subsequently electrification scenarios are applied in a cumulative fashion, changes in the IO structure occur iteratively period by period. The reason for this approach is that changes in the IO structure interact dynamically with technological change. In turn, the final demand structure before electrification is determined anew each period, given changes in the income structure.

We assume that electrification leads imported fossil fuels to be replaced by energy from the domestic electricity sector. In addition, note that the import shares of the IO structure ($A\ imp\ share$) develop dynamically in line with electrification given the separate process of domestic and im-

ported structures.³

Reflecting the increasing difficulty of electrifying procedures, in each period a constant fraction of the remaining fossil fuels is replaced, leading to an exponential decay in the effective rate of electrification with respect to its initial values. Moreover, the proposed modelling of electrification allows for efficiency gains (ρ) as a result of the electrification process, i.e. x TJ of fossil fuels are replaced by $(1 - \rho)x$ TJ from the electricity sector.

3.2.2.2 Electricity module

In addition to the electrification process, we develop a new integrated electricity module as illustrated in Figure 3.2.

Electricity demand (E demand) derived from the energy module is met by a combination of

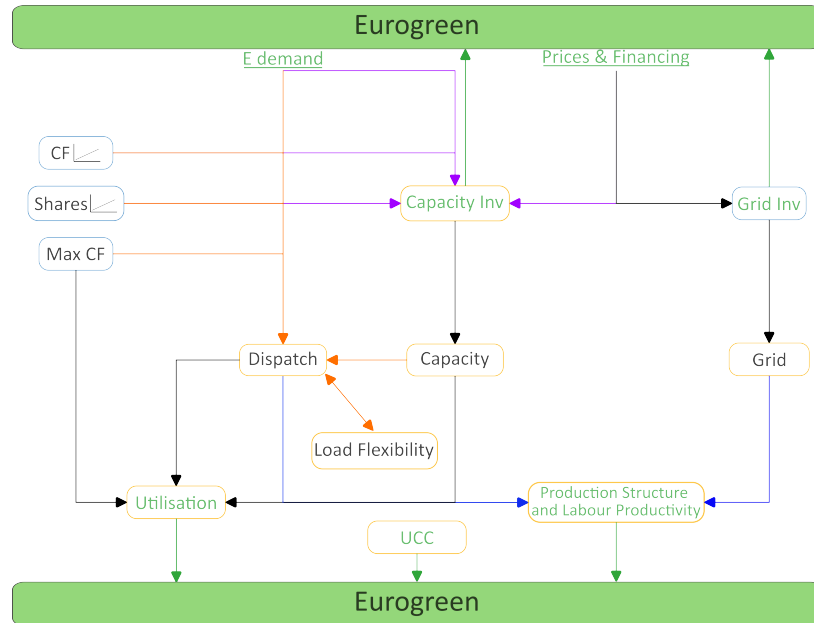


Figure 3.2: Simplified structure of the electricity module. The blue blocks are exogenous inputs, whereas the orange blocks reflect endogenous variables. Green underlined text without box represents inputs from other modules into the electricity module and green text within boxes reflects variables that feed back into other modules. The graph symbol indicates exogenous dynamic developments. CF stands for capacity factor, E demand for electricity demand and UCC refers to unit capital costs.

gas (incl. natural gas and gas to power), solar pv, wind (onshore), hydro (incl. dam, run of river and PHS), coal, petroleum, nec (incl. geothermal and biomass) and batteries which are here

³The import shares before and after electrification also differ for final demand, but since each period the process initiates with import shares before any electrification process, we do not explicitly consider their dynamic development here.

assumed to be lithium-ion (Schmidt et al., 2019).

The module relies on an exogenous pathway for a fully decarbonised power system in 2050 in Italy based on the open source power system cost optimisation model FLEXIES (Maeder et al., 2021) to model a shift in capacity and production towards renewables taking into account challenges of a renewable transition like flexibility. We consult values for the limited transmission scenario. Using implicit capacity factors (CF) and production shares ($Shares$) by technology for 2050, calculated from (Maeder et al., 2021), these values are linearly interpolated between the last year with available data and 2050, considering also the coal phaseout in 2025 in Italy. In a counterfactual no-energy transition scenario, CF and $Shares$ are kept constant at their last historic value, except for the coal phaseout.

Desired capacity by technology is subsequently driven by trend based expectations of electricity demand, $Shares$ and CF , and desired investment aims to bring capacity to the desired level. In line with the learning curve literature (e.g. Rubin et al., 2015), investment costs are affected by installed capacity, except for batteries which instead are based on costs projections from Schmidt et al. (2019). Desired grid investment is highly simplified, extrapolating yearly investment costs from MISE-MATTM-MIT (2019) in the case of an energy transition, and aiming to keep the initial grid to electricity demand ratio constant in the counterfactual.

Subsequently, technologies are dispatched to meet their respective demand, determined by E demand and $Shares$. If demand cannot be met by all technologies (i.e. demand exceeds $Capacity$ scaled with CF by technology), technologies other than wind, solar and hydro can ramp up production up to the capacity factor $max\ CF$ based on Jaxa-Rozen et al. (2022). For this residual production, flexibility considerations play a role for both demand and supply. With respect to demand this corresponds to the general load flexibility with flexibility requirements increasing with variable renewables, i.e. wind and solar, being directly fed into the grid following Carrara and Marangoni (2017) and Sullivan et al. (2013). For supply, flexible production potential is differentiated by technology following Carrara and Marangoni (2017) and Sullivan et al. (2013). Residual production technologies are dispatched according to priorities that minimise emissions.

The production structure and labour productivity of the overall electricity sector is subsequently determined by the average of the individual technologies weighted by their share in production. Given the link of skill shares to labour productivity dynamics (see subsection 3.2.1) developments in the electricity module affect the skill share demanded by the electricity sector.

In addition, other aggregate electricity-sector wide variables are calculated as inputs for the overall Eurogreen structure, notable the rate of capacity utilisation and unit capital costs (UCC), e.g. for price determination.

3.2.3 Scenarios

Applying this extended model to Italy, we run six different scenarios, based on three distinct socioeconomic paths inspired by D'Alessandro et al. (2020), namely a reference, a redistribution and a post-growth path.

While the reference path does not introduce any policy changes, the redistribution path intro-

Table 3.1: Policies, parameters and type of implementation for the socioeconomic paths. Phased in linearly reflects a linear interpolation so that the indicated value is reached in 2050, whereas immediate implies a jump to the indicated value in the period where the policy is introduced.

Socioeconomic path	Policy	Parameter Values	Implementation
Reference	–	–	–
Redistribution	Progressive Income Tax	[0.1,0.25,0.55,0.85,0.9]	Immediate
	Focalised Benefits	3500 €	Immediate
	Min and Max Wage	10 and 40 € per hour	Immediate
Post-growth	Basic Income	5000 €	Immediate
	Min and Max Wage	10 and 40 € per hour	Immediate
	Working Time Reduction	20% reduction	Phased in linearly
	Consumption Reduction	25% reduction	Phased in linearly
	Job Transition	20% of unemployed	Phased in linearly
	Wage Indexation	0.7	Immediate

duces policies aimed at reducing income inequality, like a reform of the income tax. The post-growth path features more profound changes and redistribution, by introducing for instance a basic income in combination with structural reductions in working time and consumption. Note that potential fiscal costs of these policies are offset by increases in the tax on financial income and corporate profits. The scenario policies and parameters are outlined in Table 3.1 with more information provided in Appendix 3.A.1. Note that, since we aim to assess the implications of a successful energy transition and not how to enable it, we consider no financial or scale constraints on investments in the scenarios.

Each of these socioeconomic paths is simulated with and without an energy transition, leading to the six scenarios displayed in Table 3.2.

3.2.3.1 The energy transition

Combining measures specifically aimed at reducing greenhouse gas emissions, the energy transition consists of four main developments: 1) an increase in carbon costs; 2) the phaseout of fossil fuel subsidies; 3) electrification of fossil fuel use, and 4) a renewable energy transition in the electricity sector.

The carbon tax increases linearly from 50 € per tCO₂eq at the moment of its implementation to 200 € in 2050. Similarly, fossil fuel subsidies are phased out linearly until 2050. In all scenarios, the proceeds from the carbon costs⁴ and the reduction in fossil fuel subsidies are distributed to households as an equal per head payment (subsequently referred to as climate income).

Electrification rates are calculated to be in line with an end use share of electricity of approximately 49% in 2050 following IEA (2021). See Appendix 3.A.2 for the calculation and exact values. In addition, efficiency gains of approximately 40% from electrification are assumed (Eyre, 2021). These rates are applied equally across all industries and household consumption. To analyse the impact of these assumptions on the outcomes, we additionally run scenarios a) without efficiency gains from electrification, and b) with a final end use share of electricity of approximately 71%, based on the ‘Radical Action’ scenario in Eurelectric (2023).⁵

Table 3.2: The six scenario combinations.

Socioeconomic path	Energy Transition	Scenario Name
Reference	NO	Fossil Baseline
Reference	YES	Green Baseline
Redistribution	NO	Fossil Redistribution
Redistribution	YES	Green Redistribution
Post-growth	NO	Fossil Post-growth
Post-growth	YES	Green Post-growth

3.2.3.2 Sensitivity to stochastic processes

In order to increase the robustness of the scenario analysis, we run a Monte-Carlo sensitivity analysis with 200 runs for each scenario, drawing the seed from a random uniform distribu-

⁴Carbon costs also include EU ETS costs. We assume that, in the case of a carbon tax implementation, the ETS price will follow the carbon tax trajectory.

⁵While Eurelectric (2023) is provided by a sector association, we run its projections as a robustness check since they are EU-specific and hence more tailored than the global IEA scenario. To our knowledge other specific projections for this parameter are not available in the literature.

tion. Thus, the sensitivity analysis is restricted to technological innovation which is the only stochastic process in the model.

3.3 Results

3.3.1 Main dynamics

The results are presented as median values obtained from the Monte-Carlo simulations, generally with an interval of one median absolute deviation above and below the median. Figure 3.3 displays how selected macroeconomic indicators evolve in the six scenarios. Economic activity as measured by GDP (Figure 3.3 a) initially increases due to redistribution policies increasing demand and output (see Figure 3.C.1) in both redistribution and post-growth scenarios. Over time, GDP in the post-growth scenarios decreases due to the gradual reduction in consumption and exports. In addition, the energy transition boosts final demand, output and GDP in all three socioeconomic paths (see also Figure 3.C.1). Both, redistribution policies as well as the energy transition show a permanent level effect on economic activity.

Total employment (Figure 3.3 b) is highest in the post-growth scenarios mainly owing to the

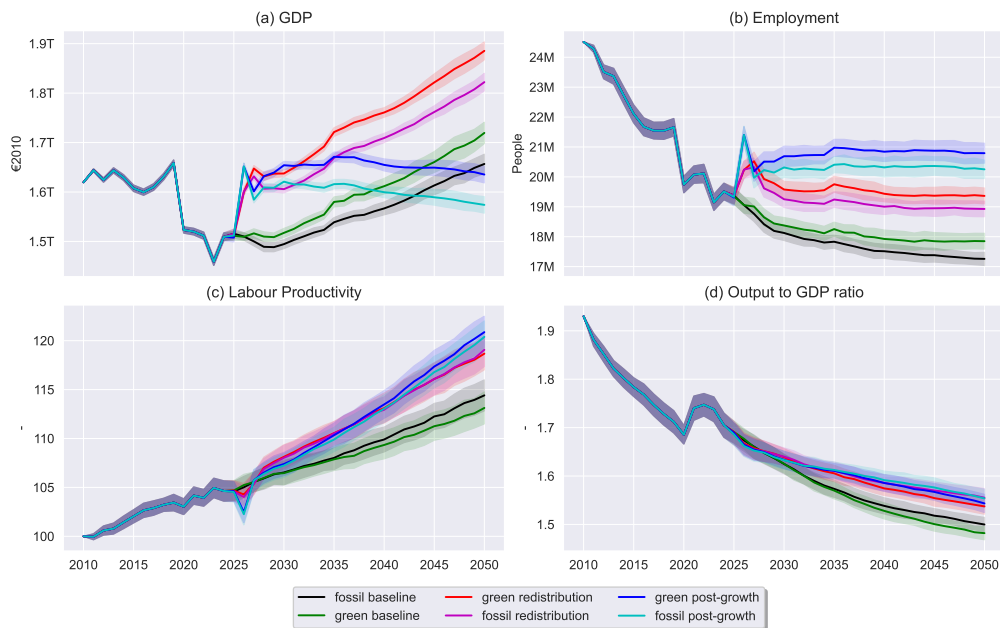


Figure 3.3: Main macroeconomic dynamics. Dynamics of a) GDP, b) total employment, c) labour productivity and d) output to GDP ratio. The median values plotted over time are obtained from the Monte-Carlo simulations. The dispersion of outcomes is indicated by an interval of $\pm$ one median absolute deviation around the median.

working time reduction policy. The higher level of economic activity explains the second highest

level of employment appearing in the redistribution scenario. The energy transition increases economic activity and hence employment in all cases. Dynamics of employment are mainly driven by the scale of economic output (and the working time reduction policy in post-growth scenarios), but also affected by technological developments.

Production efficiency, here measured as output per GDP, is mainly driven by technological innovation, which features a general trade-off between improvements in production efficiency in terms of intermediate inputs and capital, and labour productivity. Thus, the large increases in labour productivity (Figure 3.3 c) in the post-growth and to a lesser extent in the redistribution scenarios are accompanied by less efficient production than the reference paths (Figure 3.3 d). In addition, the energy transition is associated with more efficient production (e.g. due to energy efficiency gains), but does not exhibit a clear impact on labour productivity. While the composition of final demand and output can potentially also play a role between socioeconomic paths, it remains relatively stable within them, i.e. it is not substantially affected by the energy transition (Figure 3.C.5).

These technological developments have crucial implications for environmental indicators like energy use and emissions. For the scenarios without an energy transition, total energy use and emissions largely follow the same order as economic activity (Figure 3.4 a–b). However, the less efficient production, in particular in the post-growth scenarios, leads to comparatively higher energy use and emissions even in periods of lower economic activity in terms of GDP. The energy transition dramatically reduces energy use, due to efficiency gains, and also emissions in all scenarios (by approximately 20–23 and 40–43% respectively by 2050). In this way, the transition reduces environmental impacts even in the context of higher economic activity, reaching a comparative absolute decoupling. Still it is noteworthy that none of the analysed transitions achieves emission values compatible with net zero in 2050.⁶

Redistribution policies in the post-growth scenarios, and to a lesser extent in the redistribution scenarios, persistently reduce inequality, as measured by Gini coefficient (Figure 3.4 c). The energy transition also clearly decreases inequality in all socioeconomic scenarios, potentially because of the positive impact on employment.

Functional inequality in terms of the labour share is also initially improved by both redistribution and post-growth scenarios (Figure 3.4 d). However, only the more extensive post-growth policies, permanently relocating more power to labour, lead to a shift towards and increasing trajectory of the labour share. Note that, while a relatively lower GDP in the post-growth scenarios narrows this gap for labour compensation, in particular between post-growth and redistribution

⁶According to MATTM-MISE-MIT-MPAF (2021), the decarbonisation scenario features 65–85 million tCO₂eq and LULUCF potential (i.e. not relying on carbon capture and storage (CCS)) is 45 million tCO₂eq.

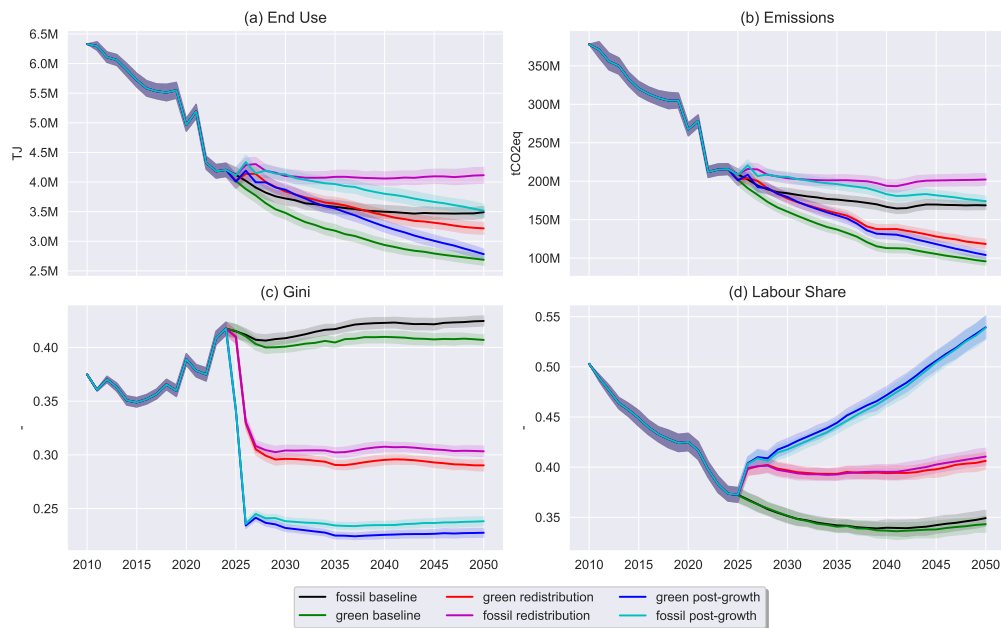


Figure 3.4: Main ecological and inequality dynamics. Dynamics of a) Energy end use, b) greenhouse gas emissions, c) the Gini coefficient and d) the labour share. The median values plotted over time are obtained from the Monte-Carlo simulations. The dispersion of outcomes is indicated by an interval of $\pm$ one median absolute deviation around the median.

paths, the ranking of paths remains the same (see Figure 3.C.2). The energy transition does not appear to have a significant impact on the development of the labour share.

In terms of the magnitude of the effects, socioeconomic policies tend to have a more significant impact on outcomes such as inequality, economic activity, and employment than the energy transition itself. For instance while the energy transition increases employment by approximately 2.5–3.5% by 2050, the difference between the green post-growth and the fossil baseline is 20%. Yet, the level effect of the energy transition on economic activity, employment and inequality is still unambiguous and substantial. For the indicators energy use and emissions, clearly the effect of the energy transition far outweighs socioeconomic policies.

Notably, even within the energy transition socioeconomic mechanisms are crucial. The impact of the energy transition on economic activity, employment and inequality is not only due to higher levels of investment but significantly driven by the redistribution of the carbon cost revenue to households (Figure 3.C.3). Without this climate income, the effects of the energy transition are markedly smaller and more temporary, with inequality even increasing in the final years with respect to the fossil baseline. A carbon tax without climate income does not increase but decrease demand *ceteris paribus* counteracting the increase in investment. The structural change of the economy due to the energy transition appears to lead to higher levels of inequality

in the face of similar GDP and employment values in 2050. Thus, depending on the policy context the transition can increase or decrease inequality. In this paper we focus on the version of the transition with a climate income for the households as already implemented for instance in Austria or planned in Germany.

Varying the energy transition parameters adds additional insights to the results, as illustrated in Figure 3.C.4. Higher levels of investment in the face of more extensive electrification efforts are accompanied by higher levels of economic activity and employment, as well as slightly lower inequality. While no efficiency gains evidently lead to larger amounts of energy use and emissions, the electrification scenario based on Eurelectric (2023) achieves emissions in line with net zero targets in 2050, albeit within the range relying on CCS.

After analysing socioeconomic implications on the aggregate macroeconomic level, we subsequently take a more disaggregated perspective assessing impacts on the employment structure.

3.3.2 Employment structure

The employment structure in the three main energy supplying sectors in 2050 changes significantly with the energy transition (Figure 3.5). Whereas mining and petroleum lose employment, the increases in the electricity industry more than compensate for the losses in the other sectors. Thus, overall the energy transition increases employment in these energy supplying sectors, within all socioeconomic policy conditions.

However, an analysis of the overall employment structure by industry in 2050 (Figure 3.6a) reveals that the energy industries only account for a marginal fraction of employment. Accordingly, the energy transition does not have a major effect on the macroeconomic employment structure. Socioeconomic paths have a slight impact with for instance redistribution and post-growth scenarios featuring higher shares of employment in services, possibly due to higher household demand.

The overall picture looks similar for the macroeconomic skill shares of employment in 2050 (Figure 3.6b). Again, the energy transition appears to have a negligible effect on the skill shares, whereas socioeconomic paths seem to affect skill shares more, albeit also not substantially. Notably, the redistribution and in particular the post-growth scenarios tend to feature a higher degree of job polarization. This development is probably driven by the rapid increases in labour productivity in these scenarios, but can also be affected by changes in output and employment structure.

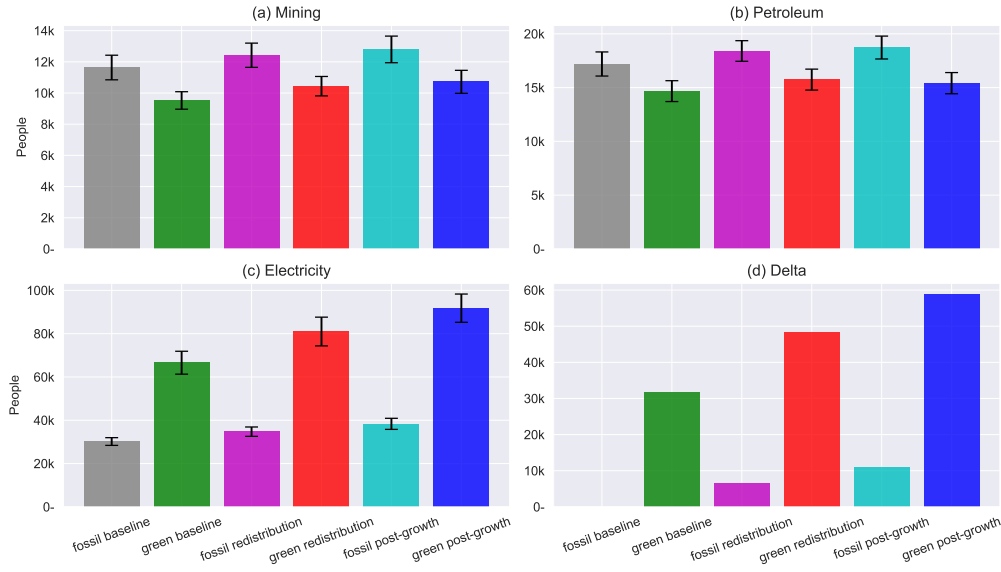


Figure 3.5: Employment in the main energy supplying sectors. Total employment in the a) mining, b) petroleum and c) electricity sector as well as the overall change of employment of these three sectors compared to the fossil baseline, all in the year 2050. The median values plotted are obtained from the Monte-Carlo simulations. The dispersion of outcomes is indicated by an interval of $\pm$ one median absolute deviation around the median.

3.4 Discussion

Assessing the socioeconomic impacts of the energy transition within socioeconomic paths, we find a positive net effect of the energy transition on employment. In all socioeconomic paths we observe higher employment in energy supply and larger employment shares in the construction and the energy sectors. These findings are generally in line with the literature (García-García et al., 2020) and the higher capital intensity of renewable electricity systems (Best, 2017; Timmons et al., 2014). Increases in macroeconomic employment are generally driven by higher levels of economic activity. Here the positive effect of the energy transition derives from the post-Keynesian model closure (Mercure et al., 2019), as the climate income as well as increased investment drive up aggregate demand.

Still, these overall positive effects on employment are accompanied by losses of certain types of jobs, in particular in the fossil fuel supplying sectors. While in the macroeconomic context the employment in these sectors is only marginal, still thousands of people are affected. Here in particular the macroeconomic perspective might not adequately capture pronounced local impacts of the energy transition (Carley and Konisky, 2020).

Redistribution policies have a clear level effect on economic activity with less of a persistent impact on growth rates, broadly in line with the Sraffian Supermultiplier growth models (Freitas

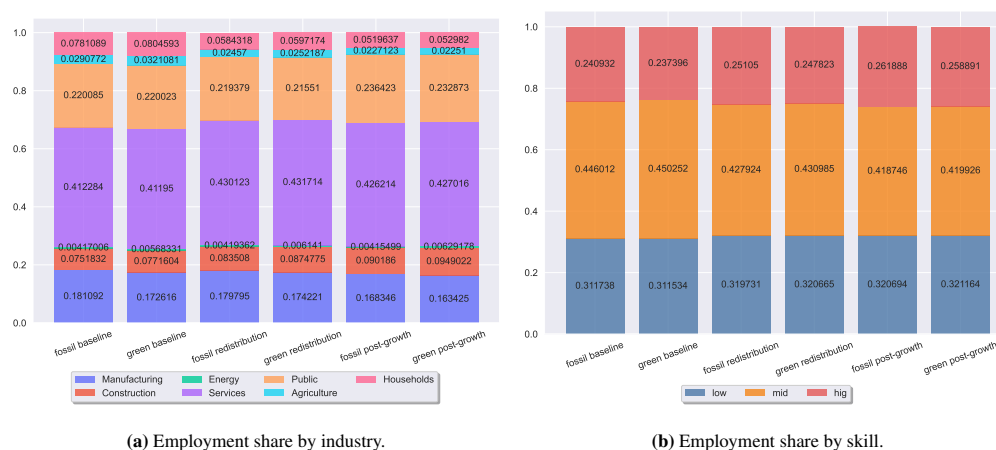


Figure 3.6: Employment share by industry and skill. The median values plotted for the year 2050 are obtained from the Monte-Carlo simulations. Industries are aggregated according to Table 3.B.1.

and Serrano, 2015; Hein, 2023). Since investment is induced in Eurogreen, the long-run growth dynamics are dominated by the autonomous components of demand as in the Sraffian Super-multiplier macroeconomic model.

Overall, on the macroeconomic level the implications of socioeconomic paths seem to thwart the impact of the energy transition on socioeconomic outcomes, including inequality measures, employment and the employment structure by skill.⁷ Still, the energy transition has significant socioeconomic ramifications. For instance, not only redistribution policies but also the energy transition can trigger an increase in economic activity and employment while reducing inequality.

We show that on the macroeconomic level comparatively few people are adversely affected by the energy transition while there are significant overall socioeconomic and ecological benefits. Socioeconomic issues are not predetermined by a low-carbon energy transition but can be more than offset by a suitable mix of socioeconomic policies. In fact we show that the energy transition can overcome the emission-inequality trade-off and provide a “triple dividend” by increasing economic activity and employment, while reducing inequality, energy use and emissions. An important factor here is the climate income for households.

Still, the fact that none of the scenarios achieves emissions in line with net zero underlines that the modelled energy transition is not ambitious enough, in particular when taking into account that the model only accounts for emissions from energy use, so that overall net-zero emissions require even higher levels of reduction. [A more extensive energy transition can certainly be an](#)

⁷The implications in particular on outcome and employment structure would be more pronounced in a more elaborate post-growth scenario where particular, targeted sectors are scaled down, whereas here consumption is reduced generically.

answer to this issue: The higher levels of electrification based on Eurelectric (2023) reach the bracket of net-zero when accounting for CCS. Still, the fact here only emissions from energy use are included highlights that even this transition likely requires additional measures⁸. In addition, this level of emissions still relies on problematic negative emission technologies (e.g. Anderson and Peters, 2016; Smith et al., 2016). In combination with considerations that an extensive energy transition can be confronted with institutional and technical challenges (Davis et al., 2018), these findings potentially allow for an important role of targeted reductions and shifts in consumption and production patterns in order to meet emission targets.

When scrutinising the mechanisms behind the socioeconomic and ecological outcomes of socioeconomic paths, the implications of certain redistribution policies for technological innovation become particularly noteworthy. The discovery of innovations is driven by the development of the relative costs of labour compared to other inputs (Campigotto et al., 2024). The policies in the redistribution and to a larger extent the post-growth scenarios that shift more income and power to labour increase the relative costs of labour and thus lead to more labour saving technological progress. This occurs at the expense of more efficient production in terms of other inputs. This mechanism counteracts the output contracting effect of the consumption reduction and leads to higher end use and emissions in the green post-growth scenario compared to green baseline. Moreover, the shift towards labour-saving technological development can worsen inequality *ceteris paribus* through its impact on labour demand (D'Alessandro et al., 2020) and its composition. This indicates that the current profit-driven investment incentive structure can be problematic for a transition that centers around the reduction of inequalities, not only rendering the transition *ceteris paribus* less ecological but also increasing potential frictions and barriers through the employment structure. This is another layer to the emission-inequality trade-off found in the model via changes in aggregate demand (Campigotto et al., 2024). Thus, redistribution policies towards labour should be accompanied by alternative innovation structures like more publicly driven innovation with specific ecological and socioeconomic goals in mind. Hence, this study underlines the continued importance to study post-growth scenarios (as called for by Hickel et al., 2021; Keyßer and Lenzen, 2021) also to uncover and address potential unintended outcomes adequately. While outperforming other scenarios in terms of employment level and inequality, this innovation process constitutes a major obstacle in terms of energy use and emission reduction in the post-growth scenario.

Taken together, extensive redistribution does not only directly but additionally through this innovation mechanism lead to comparatively higher emissions *ceteris paribus*. Without substantially more extensive energy transition efforts, this renders the structural reduction in demand and eco-

⁸For instance alternative decarbonisation of energy sources as also outlined in Eurelectric (2023) itself.

conomic activity in the post-growth scenario particularly crucial in the context of a just transition.

Lastly, this paper features several limitations: Firstly, the renewable transition in the electricity module and the electrification process are exogenous and therefore not directly affected by the dynamics of variables like prices, carbon taxes or investment costs. Also note that the model in its current form is not particularly suitable to capture potential energy price effects along the transition pathway. Neither does it differentiate the underlying mechanism for price setting (see section 3.2.1) for sector specific characteristics (e.g. electricity), nor does it explicitly model prices for energy products which feature specific characteristics and are particularly challenging to predict (Zhu et al., 2024). Furthermore, final calibration and validation issues remain, particularly with respect to the innovation process in the initial years. Moreover, the impact of the energy transition is systematically underestimated since investment required for the electrification process (e.g. electric cars or heatpumps) are omitted due to a lack of data and the current aggregated input-output production structure in the model, which is not able to capture the nuances between for instance electric or combustion engine cars. In addition, we do not consider biophysical constraints like materials or energy (e.g. Nieto et al., 2020) for any path analysed and set a generic rate of electrification for all industries as well as households. Still, the contributions of the work remain significant: By integrating energy-economic feedback of the energy transition into an ecological macroeconomic model that is particularly suited to evaluate socioeconomic outcomes, it narrows the gap towards a holistic analysis of the energy transition and its socioeconomic and ecological implications within overall transition pathways and provides a basis for further research.

3.5 Conclusion

In this work we explicitly integrate an electricity module and economic effects of electrification dynamics into the Eurogreen model. This renders the model particularly suitable to represent socioeconomic implications of the energy transition. We analyse these within broader socioeconomic transition paths inspired by D'Alessandro et al. (2020), namely a reference, a redistribution and a post-growth policy setting.

On the socioeconomic implications of a low-carbon transition within socioeconomic paths, we find that the energy transition and redistribution both increase economic activity and employment while reducing inequality. However, whereas redistribution policies alone lead to an increase in energy use and emissions, the energy transition can lead to a “triple dividend”, also decreasing energy use and emissions and hence manages to overcome the general emissions-inequality trade-off in Eurogreen (Campigotto et al., 2024). In the settings of this study, it

increases GDP and employment by approximately 3.5–4 and 2.5–3.5 %, while reducing emissions by around 40–43% by 2050. A crucial feature of the energy transition for this “triple dividend” is the introduction of a climate income.

The dynamics and structure of employment on the macroeconomic level are characterized by higher total employment levels and little structural shifts in employment structure, neither in terms of industries nor skills, as a result of the energy transition. Still, while overall employment in the energy sector increases, there are parts of the population that will lose from the energy transition, in particular employees in the mining and petroleum industries. These losses can have pronounced local effects (Carley and Konisky, 2020).

In addition, we reveal that a redistribution towards labour can have unintended consequences by incentivising labour saving technological innovation at the expense of production efficiency in terms of other inputs. This leads to less efficient production in the redistribution and particularly post-growth scenarios, leading to green baseline performing best in terms of energy use and emissions, while post-growth features the highest levels of employment and lowest inequality. While redistribution policies are essential to reduce inequality, this mechanism must be addressed for instance by more publicly driven innovation. **The combination of redistribution increasing economic activity and emissions, as well as inducing this shift in technological development, highlight the importance of structural reductions in economic activity for a just transition in the absence of a substantially accelerated energy transition.**

While this analysis has contributed to a more integrated analysis of the energy transition within socioeconomic paths, the need for further research on this procedure remains. In particular the endogenisation of the energy transition as well as the inclusion of investment requirements for electrification will potentially provide new insights and enrich the understanding of different transition pathways. By revealing unintended consequences, our work also highlights the importance for more evaluation of post-growth pathways in order to address potential challenges head on.

Authorship statement

TH: conceptualisation, methodology, investigation, formal analysis, visualisation, writing (original draft). *DCO*: methodology, investigation, writing (original draft). *GSM*: methodology, investigation, writing (original draft). *SD*: supervision, methodology, investigation. *RSC*: investigation, writing (original draft). *FD*: writing (original draft).

Appendix

3.A Additional details on scenarios

3.A.1 Socioeconomic paths

In this section, we provide more details on the individual policies outlined in Table 3.1:

- *Progressive Income Tax*: The values presented in Table 3.1 depict the values of the marginal tax rates per income bracket after the implementation of the policy.
- *Focalised Benefits*: A government transfer targeted at the most vulnerable, here assumed to be the out of labour force low-skilled population.
- *Min and Max Wage*: An economy wide minimum wage of 10 € and maximum wage of 40 € per hour is implemented.
- *Basic Income*: A government transfer received by every individual in the population.
- *Working Time Reduction*: The working time reduction as accompanied by a wage compensation to maintain labour income, of which 50% is financed by the government.
- *Consumption Reduction*: The propensity to consume of households is reduced⁹ as are exports. Both are reduced ceteris paribus by 25% in 2050, but the dynamics differ slightly. While the process is cumulative and cumulative percentage reduction follows a linear interpolation for household consumption, for exports the relative reduction follows an exponential decay given its iterative interaction with its delayed value. The implementation of the latter is analogous to the implementation of the electrification scenarios (see 3.A.2).
- *Job Transition*: The Job Transition policy offers unemployed people a one year paid further training and education program. The fraction of unemployed population that is recruited linearly increases from 0 at the start of the policy to 20% at the end of the simulation. Unemployed are recruited if the skill specific unemployment rate exceeds 5%. The wage that people in the Job Transition receive is 10€. Also note that while all other policies are implemented in 2025, the job guarantee starts 3 years later in order to avoid labour shortages when demand and output rise rapidly due to the implementation of other socioeconomic policies.
- *Wage indexation*: While not directly a policy, we assume that a post-growth scenario is accompanied by a shift in power towards labour due to policies like the working time

⁹Note that the reduction in household consumption also affects investments in real estate.

reduction with wage compensation, but in particular the Basic Income and Job Transition programmes, making employees less dependent on employed work. This increased labour power is reflected in the improved wage indexation to inflation.

Government transfers (i.e. focalised benefits and basic income) are adjusted to inflation, with the job transition wage adjustment featuring the same level of price indexation as wages in the rest of the economy.

Note that additional expenditure or revenue losses for the government arising from socio-economic policies are offset by increases in the tax on financial income and corporate profits. The financial income tax is supposed to cover 70% of these increases in costs, but can only rise to maximum 90%. The other 30% and shortfalls of the financial income tax revenue are covered by corporate tax increases. The costs are revenue losses due to the progressive income tax and costs from focalised benefit payments in the redistribution paths; and costs arising from the basic income, job guarantee and working time reduction policies for the post-growth ones.

3.A.2 Quantitative electrification scenario

Due to the exponential decay of effective electrification over time, cumulative electrification is given by: $\tau_t^c = \sum_{t_1}^t \tau_{t_1} = \bar{\tau} \sum_{t_0}^t (1 - \bar{\tau})^{(t-t_0)}$. This geometric series converges to $\tau_t^c = \bar{\tau} \left(\frac{1 - (1 - \bar{\tau})^{(t+1-t_0)}}{1 - (1 - \bar{\tau})} \right) = 1 - (1 - \bar{\tau})^{(t+1-t_0)}$. Here, t_0 is the first year of the electrification process and $\bar{\tau}$ is the constant yearly rate of electrification of fossil fuel use. Thus, $\bar{\tau}$ to meet a certain rate of cumulative electrification in a given year is determined by: $\bar{\tau} = 1 - (1 - \tau_t^c)^{\frac{1}{t+1-t_0}}$.

Including efficiency gains (ρ) to this framework requires higher rates of electrification in order to meet the same end use share of electricity. Assuming only fossil fuels are electrified, the share of electricity (sh^e) in t is given by: $sh_t^e = \frac{sh_{t_0}^e + (1 - \rho)\tau_t^c sh_{t_0}^{ff}}{1 - \rho\tau_t^c sh_{t_0}^{ff}}$, where $sh_{t_0}^{ff}$ and $sh_{t_0}^e$ are the fossil fuel and electricity shares in end use at the beginning of the electrification process. Thus, to meet a certain end use share, the cumulative rate of electrification is given by $\tau_t^c = \frac{sh_t^e - sh_{t_0}^e}{sh_{t_0}^{ff}(1 - \rho + sh_{t_0}^e \rho)}$. Note that since we do not distinguish for which processes (industry, transport and buildings) energy products are used and electrify overall fossil fuel demand, for the final share of electricity from IEA, 2021 and for efficiency gains from Eyre, 2021 we use total and not process specific values. The calculated rate of electrification is applied across household consumption and all sectors. Please note that the final end use shares might not exactly meet the input values, since other developments in the model can also affect the end use share of electricity.

3.B Aggregate industries

Table 3.B.1: Aggregate Industry Structure.

Aggregate Industry	Codes of Included Industries
Manufacturing	C, C29
Construction	F
Energy	B, C19, D1, D2, D3
Services	E, G, H, I, J, K, L, M, N, R, S, T, U
Public	O, P, Q
Agriculture	A
Households	T

3.C Additional figures

3.C.1 Dynamics of totals

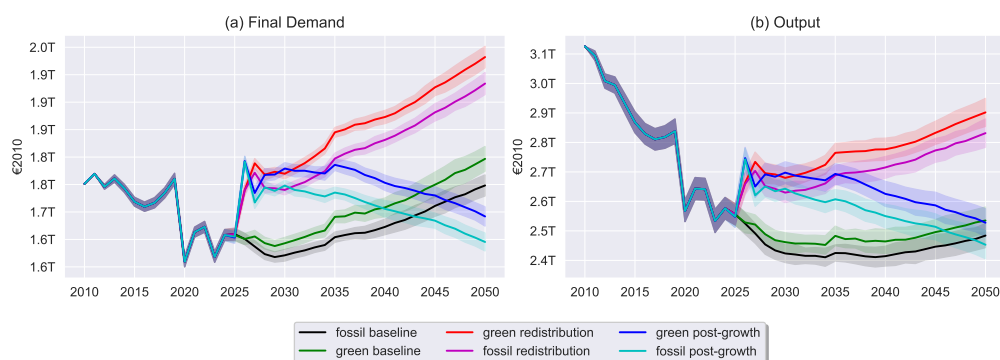


Figure 3.C.1: Dynamics of total final demand and output. The median values plotted over time are obtained from the Monte-Carlo simulations. The dispersion of outcomes is indicated by an interval of $\pm$ one median absolute deviation around the median.

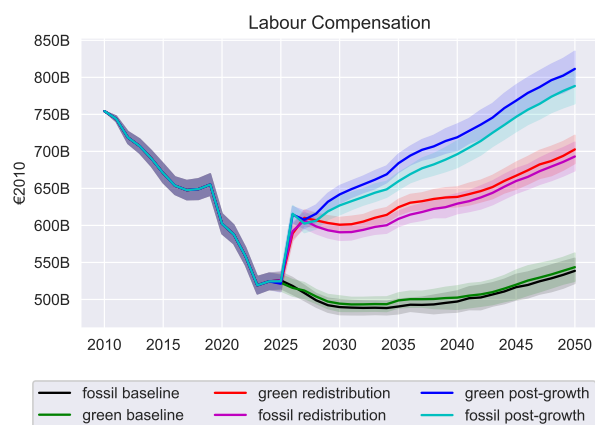


Figure 3.C.2: Dynamics of labour compensation. The median values plotted over time are obtained from the Monte-Carlo simulations. The dispersion of outcomes is indicated by an interval of $\pm$ one median absolute deviation around the median. Note that the labour compensation here is distinct from the compensation of employees in national accounts, as self-employed are also included.

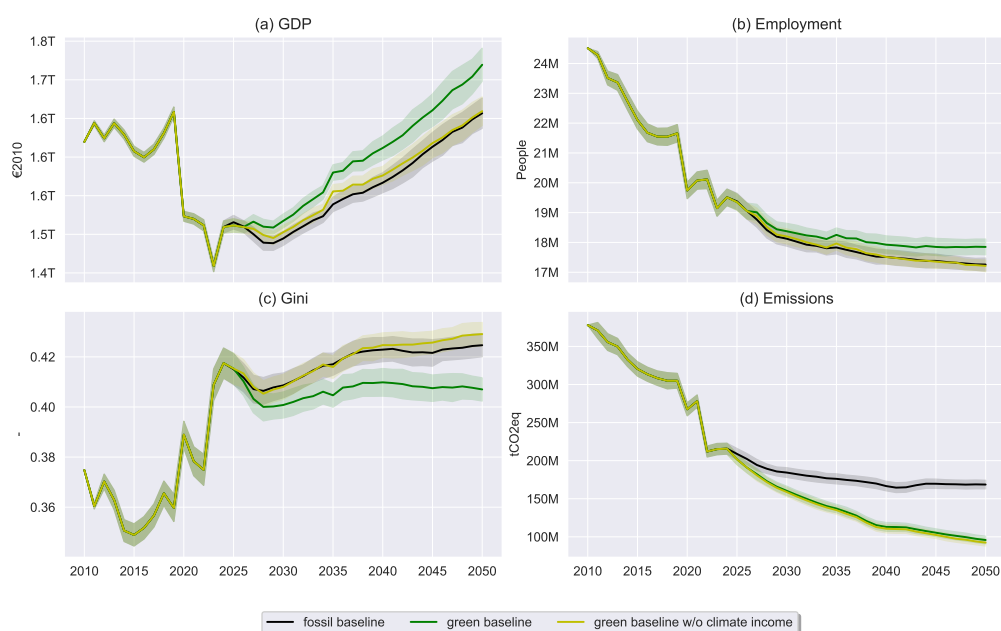


Figure 3.C.3: Main macroeconomic, ecological and inequality dynamics with and without climate income. The median values plotted over time are obtained from the Monte-Carlo simulations. The dispersion of outcomes is indicated by an interval of $\pm$ one median absolute deviation around the median.

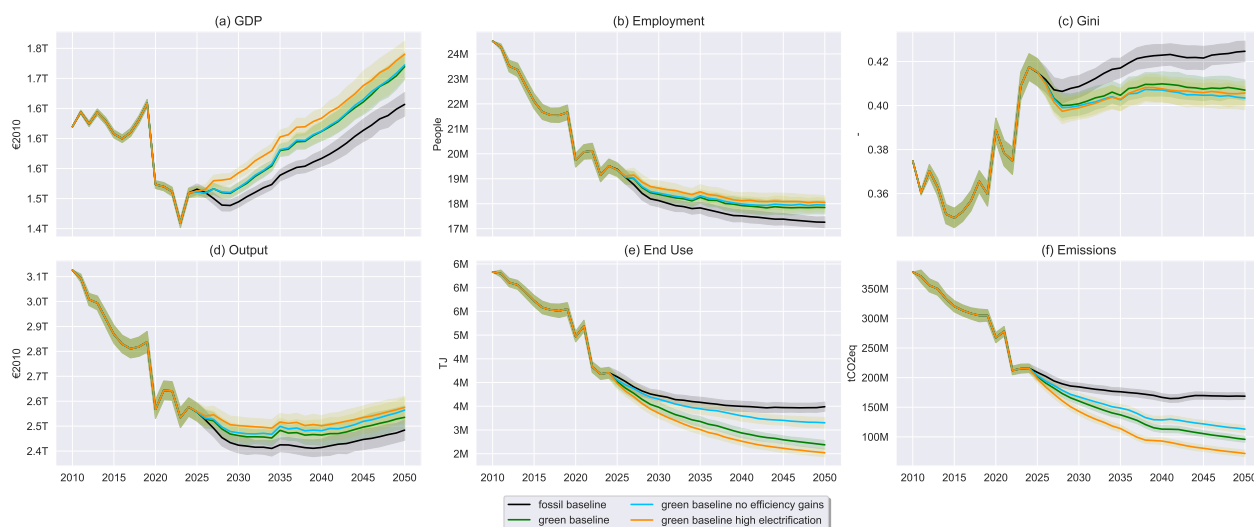


Figure 3.C.4: Main macroeconomic, ecological and inequality dynamics with different energy transition parameters. In the ‘no efficiency gains’ scenario efficiency gains from electrification are set to zero and in the ‘high electrification’ scenario electrification rates are set to target an end use share of electricity of 71% in 2050. The median values plotted over time are obtained from the Monte-Carlo simulations. The dispersion of outcomes is indicated by an interval of $\pm$ one median absolute deviation around the median.

3.C.2 Final period structures

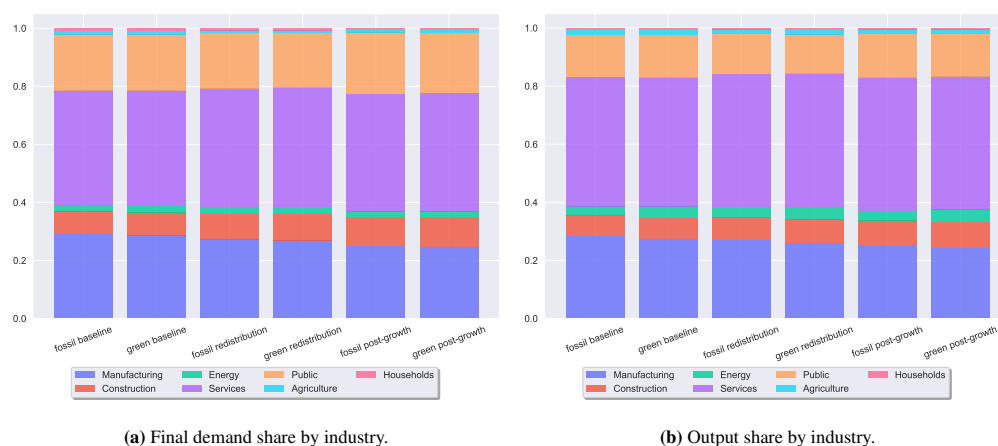


Figure 3.C.5: Final demand and output structure. The median values plotted for the year 2050 are obtained from the Monte-Carlo simulations. Industries are aggregated according to Table 3.B.1.

3.6 Supplementary Material

This Supplementary Material outlines the major changes of the Eurogreen model used in this paper compared to the version presented in Campigotto et al. (2024) module by module.

3.6.1 Demography

3.6.1.1 Migration

Net migration flows based on historic data (Eurostat, 2023, migr_emi and migr_imm8) and projections (Eurostat, 2023, proj_23nanmig) are added to population dynamics.

3.6.2 Input-Output

3.6.2.1 Sector classification

Firstly, since the WIOD (2016 release, Timmer et al., 2015) does not split sectors D35 (Electricity, gas, steam and air conditioning supply), we use additional data from the H2020 LOCOMOTION project (LOCOMOTION, 2021), henceforth referred to as LOCOMOTION data, to split it into D1 (Electricity), D2 (Gas) and D3 (Heat). As initial model data refers to the year 2010 we use WIOD as base data to calculate the matrix of technical coefficients, $\mathbf{A}$, output and final demand. Subsequently, in order to split the sector D35, we weight D sector values by the relative weights of each sector according to LOCOMOTION data (year 2015). For instance $Z_{D1,B} = Z_{D,B} \cdot \frac{Z_{D1,B}}{\sum_{i=1}^3 Z_{Di,B}}$, where $Z_{D,B}$ refers to intermediate trade derived from WIOD data and $Z_{Di,B}$ from LOCOMOTION. In order to calculate the new $\mathbf{A}$ matrix, we first calculate they hybrid versions of $\mathbf{Z}$ and output $\mathbf{x}$ and subsequently re-calculate $\mathbf{A}$.¹⁰ Please note that the merging of theses datasets from different time periods causes an issue for the $\mathbf{A}_{i,D3}$ column, which sum exceeds 1. Since this is illogical from an economic point of view and far away from the column sum value in the LOCOMOTION dataset, we adjust each cell value proportionally to meet the column sum of the LOCOMOTION dataset.¹¹ We subsequently re-calculate the $\mathbf{Z}$ matrix and total output by industry given final demand and the adjusted $\mathbf{A}$ matrix.

In order to facilitate unambiguous concordance between LOCOMOTION and WIOD sector

¹⁰Note that disaggregating $\mathbf{x}$ directly or $\mathbf{Z}$ and final demand ($\mathbf{y}$) separately yields different results. As it is more precise in terms of disaggregation weighting, we chose the second option and re-calculate output as $\mathbf{x} = \mathbf{Z}\mathbf{i} + \mathbf{y}$, where $\mathbf{i}$ is an all ones vector.

¹¹Note that the column sums of $D2$ are quite close between the datasets (0.73 in our dataset and 0.72 in LOCOMOTION). The reason $D3$ is problematic could be due to its by far smallest size making its $\mathbf{A}$ matrix particularly sensitive to changes in output.

classifications for all other sectors, we adjust the overall sector classification in Eurogreen to Table 3.6.1.

Table 3.6.1: Sector classification.

Code	Description
A	Agriculture, forestry and fishing
B	Mining and quarrying
C	Manufacturing
C19	Fossil fuel manufacturing
C29	Manufacture of motor vehicles, trailers and semi-trailers
D1	Electricity
D2	Gas
D3	Heat
E	Water supply; sewerage, waste management and remediation activities
F	Construction
G	Wholesale and retail trade; repair of motor vehicles and motorcycles
H49–51	Transport and storage (Land, Water and Air)
	Transportation and storage (Warehousing and supporting activities; Postal and courier activities) ;
H52,H53,M,N,U	Professional, scientific and technical activities; Administrative and support service activities; Activities of extraterritorial organisations and bodies
I	Accommodation and food service activities
J,R,S	Information and communication; Arts, entertainment and recreation; Other service activities
K	Financial and insurance activities
L	Real estate activities
O	Public administration and defence; compulsory social security
P	Education
Q	Human health and social work activities
T	Activities of households as employers

3.6.2.2 Output constraints

Output is driven by demand, but constrained by the availability of capital and labor in the economy. We therefore explicitly model this constraints to avoid that output associated with final demand exceeds the production capacity of the economy. We do this by computing the maximum output in each industry, given the availability of production capacity in the same period,

and the availability of labor. We thus obtain the industry-specific vector of output under Full capacity, and the industry-specific vector of output under Full employment. The constraints come into play when the output determined by unconstrained final demand exceeds the maximum output. The unconstrained output refers to the vector of output of each industry resulting from the interaction between the input-output structure and the unconstrained final demand, that is, the final demand resulting from the endogenous evolution of each component of aggregate demand without considering the constraints. The constraint operates through the change in the proportion of domestic and imported component of final demand. Implicitly, we adopt the realistic assumption that an increase in demand that cannot be met by domestic production leads to an increase in imports. Therefore, if, unconstrained output is greater than the maximum output for a specific industry, then the import share of final demand for the same industry increases, until the level of domestic final demand is the one compatible with the maximum output level.

With respect to the labour constraint additional clarifications are needed. Since labor demand is split across three different skill levels (low, middle, high), we assume that the labor constraint is determined by the most scarce type of labor (almost always the high skilled). Labor supply and demand are confronted for each skill level, and the smallest fraction of supply over demand is the one that determines the constraint. We thus assume no short-term substitutability among workers of different skill types. Moreover, if the constraint is binding, industries increase the wages of the most scarce skill level in order to attract more workers.

3.6.3 Innovation

3.6.3.1 Capital weighted technological development

In order to capture the impact of capital replacement on the diffusion of technological innovation, we add investment as a driver of technological development.

From KLEMS data (Bontadini et al., 2023), we obtain values for a normal a distribution of $\psi^\lambda = \frac{g_t^\lambda}{\frac{I_{t-1}}{K_t}}$, with all parameters in chain linked volume terms, where g is the growth rate, I and K refer to investment and the capital stock respectively. Here, we split between a distribution of labour productivity improving innovations and productivity deteriorating ones with respective distribution parameters. Subsequently, after drawing a value for ψ^λ we update labour productivity using $\lambda_t = \lambda_{t-1} \cdot \left(1 + \psi_{t-1}^\lambda \frac{I_{t-1}}{K_t}\right)$.¹²

We use the same procedure for innovations of technological progress, based on WIOD 2016 Release (Timmer et al., 2015) and KLEMS data. Splitting between improving and deteri-

¹²Please note that sector T in an exception here, since it does not use any capital for production. We keep its labour productivity constant.

orating innovations is not as straightforward as each industry has several input coefficients. Hence, we assume a technology is input saving if the sum of technical coefficients $\sum_i a_{i,j}$ in terms of chain linked volume falls and vice versa when it rises and split the distributions accordingly. Subsequently, we again normalise changes with the share of new capital, i.e. $\psi^A = \frac{g_{i,j,t}^A}{\frac{I_{j,t-1}}{j, K_t}}$ and after drawing from the distribution update technological coefficients using $a_{i,j,t} = a_{i,j,t-1} \cdot \left(1 + \psi_{i,j,t-1}^A \frac{I_{j,t-1}}{K_{j,t}}\right)$. In case of no innovation, the variables remain constant.

3.6.4 Investment

3.6.4.1 Residential investment

The Gross Fixed Capital Formation of the Real Estate industry is considered to be residential investment, which does not follow the typical business investment that evolves according to the use of capacity. The GFCF of this sector evolves in line with the evolution of household disposable income, with a coefficient estimated for Italy based on data for the period 2000-22.

3.6.4.2 Investment matrix

Supply and demand of capital goods are split across the 21 industries using an investment matrix. This matrix records transactions of capital goods across sectors as shares in the total investment of the demanding sector. The coefficients are computed using data on capital investment by activity from the OECD, which splits each sectors' investment into different categories of capital goods. These categories are assigned to different NACE sectors in the Eurogreen model to obtain an initial matrix of capital goods transactions. The final investment matrix is subsequently derived by applying the RAS algorithm to the initial one, while using totals from the WIOD and OECD data as constraints.

3.6.5 Labour

3.6.5.1 Skill-shares in labour demand

To endogenise developments in demanded skills, we model changes in skill shares as a function of labour productivity dynamics. We use historic data for the overall change in labour productivity by industry (λ_i) in chain linked volumes and historic data for the skill shares (both from KLEMS (Bontadini et al., 2023) spanning the period 2010 – 2019) to calibrate the impact of innovation on skill shares by industry ($\gamma_{s,i}$). We subsequently model the endogenous skill share (s) using $s_{i,s,t+1} = s_{i,s,t} \cdot \left(1 + \left(\frac{\lambda_{i,t}}{\lambda_{i,t-1}} - 1\right) \cdot \gamma_{s,i}\right)$.

3.6.5.2 Job Transition

We implement a new policy, called Job Transition, where unemployed individuals, if selected receive a certain wage by the government in order to adjust their skill level. The policy only becomes relevant when activated and when the unemployment rate is above a certain threshold, which is when a certain fraction of unemployed is recruited.

Skill transition parameters are affected by the scarcity of a skill level (determined as $\phi = \frac{\text{labour demand}}{\text{labour supply}}$). Transition towards the most scarce skill increases by $inc = \phi \cdot sp$, with sp being a sensitivity parameter, whereas the transition towards the other two skills together decreases by the same amount and weighted by their current transition as well as their relative scarcities: $dec_a = cf_a \cdot \frac{\phi_a}{\sum_a \phi_a} \frac{inc}{\sum_a (cf_a \cdot \frac{\phi_a}{\sum_a \phi_a})}$, where cf reflects the current transition factor of that skill and a includes all skills that counteract the increasing skill. Thus, the transition factors are given by: $cf_{t+1} = cf_t + inc$ for the scarcest skill and by $cf_{a,t+1} = cf_{a,t} - dec_a$ for all a . In order for the transition factors never to exceed 1, we also create an endogenous maximum sp value determined by $sp_{max} = \frac{1-cf}{s}$, where all values refer to the skill that is most scarce, i.e. for which the skill transition factor increases. The skill-specific outflow of the Job Transition is subsequently determined by the delayed inflow scaled by the skill-specific transition factor.

3.6.6 Government

We introduce many modifications in the modelling of government final consumption in the updated version of Eurogreen, improving the consistency between components of government spending, government consumption as part of final demand, and the public sector accounts. The components of government final consumption are detailed in the model, in line with the definitions of National Accounts and data on government accounts. Therefore, wages, intermediate costs, and capital costs incurred by the public sector in the production of industries Public, Health, and Education, are treated as components of government final consumption. We find a residual between the production costs in these industries and the value of government final consumption from data, which we keep constant, for consistency, being an additional part of government contribution to final demand. The government sector is also in charge of the Gross Fixed Capital Formation in the industries Public, Health, and Education. The spending on GFCF on these industries thus is considered in the government accounts as part of public spending. Finally, we have also added the spread paid by the Italian government on public debt bonds. The spread in the model reproduces the time series from data for the period 2010-2021. After 2021 the spread is kept constant at its most recent level.

3.6.7 Households

3.6.7.1 Households and consumption

The Households module pools income from individuals into 100 representative households, differentiated by region and income quintile, and derives consumption at the household level. Income allocation across households is based on the observed distribution types of individuals¹³ across region and quintile. To this end, we use the Survey on Household Income and Wealth (SHIW) by the Bank of Italy for 2010, with detailed information for households on their individual members' characteristics, different income sources and regional disaggregation. As income variable we use disposable income. This allows us to pool individual incomes into household income and we obtain total disposable income of the 100 representative households. Finally, using population shares across regions, the number of households and household members per region and quintile are calibrated to match adjusted aggregate household income data from SHIW.

Using total household expenditure per region from National Accounts and demographic data from Istat, we derive average household expenditure per region-quintile to compute households' propensities to consume, using the adjusted average household disposable income. Each period, propensities to consume are applied to expected real disposable income to derive total consumption per representative household. Subsequently, total consumption is split into 16 different consumption goods categories, based on the COICOP classification. Using data from the Household Budget Survey (HBS), we compute the initial expenditure shares of these 16 COICOP categories for the 100 representative households as the average for the period 2014-2017, adjusted to match total expenditure per region and COICOP category in the National Accounts for 2010. Each period, these shares are updated with respect to changes in real income and prices. With respect to income, we use the initial expenditure shares per region across quintiles to derive Engel curves. For prices, we estimate price elasticities¹⁴ using expenditure data from the HBS and price indices per region and COICOP category for 2014-2017.

Expenditure shares per consumption goods are converted into expenditure shares per industry using a COICOP-NACE bridge matrix. We use the 47 COICOP $\times$ 64 NACE benchmark matrix from Ignacio Cazcarro and Kratena (2022), and transform it into a 16 $\times$ 21 matrix for Eurogreen applying the RAS algorithm, using WIOD totals adjusted with purchaser price-basic price conversion factors from Ignacio Cazcarro and Kratena (2022) as constraints.

¹³There are 26 types of individuals in Eurogreen: children, capitalists and a remaining adult population that is split into 24 categories, $2 \times 3 \times 4$ combinations of gender, skill and occupational categories.

¹⁴These refer to own-price elasticities, but using the ratio of the own price to the general price level of the corresponding region.

3.6.7.2 Transformation of purchaser prices to basic prices

Firstly, we assume that taxes are paid as a proportion of *nominal* values, whereas the transport margins refer to *real* values. In addition, purchaser prices (pp) *without taxes* are one in the initial period. This is due our assumption that real values reflect initial ones, i.e. basic prices (bp) are all one in the initial period.

From Ignacio Cazcarro and Kratena (2022), we obtain the following data:

- m : the margin rate applying to basic prices
- t : the tax rate applied to basic price + margins values
- s : the margin share, i.e. the distribution of margins across sectors

Thus, $pp = bp(1 + m)(1 + t)$. As aforementioned, we assume that m and s reflect real shares, whereas t should be applied to nominal values. Thus, in order to allow everything to be applied to nominal data, we adjust the m and s using basic prices. Note that in the following the subscript 0 indicates initial values gathered from Ignacio Cazcarro and Kratena (2022), i.e. reflecting real shares, whereas t indicated nominal values in that period.

$$m_{i,t} = m_{i,0} \frac{\sum_j p_{j,t} s_{j,0}}{p_{i,t}} \quad (3.1)$$

$$s_{i,t} = s_{i,0} \frac{p_{i,t}}{\sum_j s_{j,0} p_{j,t}} \quad (3.2)$$

$$pp_{i,t} = \frac{p_{i,t} + m_{i,0} \sum_j s_{j,0} p_{j,t}}{1 + m_{i,0}} (1 + t_{i,t}) \quad (3.3)$$

, where p_i is the basic price and pp_i the purchaser price. As we can see from (3.3), initial pp (i.e. all bp are equal to one) will be given by $pp_{i,0} = 1 + t_{i,0}$.

Using these expressions, we can transform nominal consumption expenditure in pp (c_{pp}) into nominal expenditure in bp (c_{bp}) and nominal taxes less subsidies paid (tls) with the following steps:

1. removing taxes: $c_{pp,i,t}^{nt} = \frac{c_{pp,i,t}}{1+t_{i,t}} \Rightarrow tls_{i,t} = \frac{t_{i,t} c_{pp,i,t}}{1+t_{i,t}} = c_{pp,i,t} - c_{pp,i,t}^{nt}$
2. calculating margins: $tm_{i,t} = c_{pp,i,t}^{nt} \frac{m_{i,t}}{1+m_{i,t}}$
3. allocating margins $tma_{i,t} = s_{i,t} \sum_j tm_{j,t}$
4. calculating nominal consumption in basic prices: $c_{bp,i,t} = c_{pp,i,t}^{nt} - tm_{i,t} + tma_{i,t}$

This leads to the following overall equation:

$$c_{bp,i,t} = \frac{c_{pp,i,t}}{(1 + t_{i,t})(1 + m_{i,t})} + s_{i,t} \sum_j \left(\frac{c_{pp,j,t} m_{j,t}}{(1 + t_{j,t})(1 + m_{j,t})} \right) \quad (3.4)$$

Finally, basic prices can be used to deflate $c_{bp,i,t}$.

Conversely, nominal basic price consumption can be translated into nominal purchaser price consumption using the following:

$$c_{pp,i,t} = \left((1 + m_{i,t})c_{bp,i,t} - s_{i,t} \sum_j (m_{j,t}c_{bp,j,t}) \right) (1 + t_{i,t}) \quad (3.5)$$

3.6.8 Prices

3.6.8.1 Input-output price model

Eurogreen has an input-output price model inspired in the work of Leontief (Miller and Blair, 2009). Therefore, the intermediate demand expressed in the input-output matrix contributes to the production costs of the final good of each industry, together with unit labor costs (w) and unit capital costs (δ). The explicit introduction of the unit capital costs represents an improvement with respect to the traditional Leontief price-model. Another important refinement is that we treat nominal profits as proportional to production costs, according to an industry-specific mark-up (μ) over average production costs. The equation below describes the input-output price model of Eurogreen.

$$p = [I - A' \circ (1 + \mu)]^{-1} [(w + \delta) \circ (1 + \mu)] \quad (3.6)$$

Here p , w , and δ are column vectors of the basic price, unit wage costs, unit capital costs, μ is the diagonal matrix of industry-specific mark-ups and A is the matrix of technical coefficients and I the identity matrix.

3.6.8.2 Consumer Price Index and inflation

The Consumer Price Index (CPI) is computed as a weighted sum of the purchaser prices of each industry. We use the standard method of using updated weights, that is, each period the weights are changed to reflect the composition of the consumption bundle in the period immediately before. We also compute specific CPIs per each region-quintile representative household, using their specific consumption bundles to compute the updated weights, and these are the CPI used to derive the effect of prices on consumption per household. The inflation rate is simply computed as the rate of change in CPI.

3.6.9 Carbon costs

3.6.9.1 ETS

Carbon costs derived from ETS policies follow the characteristics of the EU Emissions Trading Scheme (ISPRA, 2021a). That scheme targets certain sectors that in the Eurogreen model correspond mainly to Mining (B), Manufacturing (C), Refining of petroleum (C19), Electricity (D1), Heat (D3), Transport (H) and Water (E). However, not all producers in these sectors are required to participate in the trading scheme. We use the ratio of verified emissions in ETS to total emissions to compute the shares of each sector participating in the scheme. Applying these shares, we split total sector emissions between those subject to carbon tax costs and those subject to ETS costs. The ETS system works in the following way: each period there is a total number of emission allowances, which sets the cap of overall emissions for these industries as a whole, and a subset of these are freely allocated to each sector. At the end of the period firms must surrender a number of allowances at least equal to their emissions. Sectors in deficit can buy allowances from the government in auctions or must pay a penalty if they fail to surrender the required amount of allowances.

The data for modelling the ETS system in Eurogreen is taken from the European Environment Agency (EEA). The number of emission allowances, both total and freely allocated, are taken from historic EEA data. Free allowances are set to arrive at 0 in 2030, total allowances reduce at an annual rate of 2.2%, auction prices are taken from historic EEA data and extrapolated onwards, and ETS penalties are set at 100€ per ton of CO₂. These parameters for the baseline are set accordingly with current EU regulation.

3.6.9.2 Carbon costs and prices

Since our model is in basic prices, we allocate changes in purchaser prices due to developments in levied taxes to the *taxes less subsidies (tls)* element of prices. Here we assume that all taxes for final consumption but also all taxes embedded in the chain of production are fully passed on to the final consumer.

Embedded carbon costs *Carbon tax:* To obtain a per unit production carbon tax rate, we calculate emissions per industry and unit output (ϑ_i) in line with the process described in section 3.6.10. Since the input-output production structure is linear, this value is independent of the volume of production. Multiplying ϑ_i with the carbon tax rate yields unit carbon costs from the carbon tax (ucc_i^{tax})

ETS: Due to the allocation of free pollution permits, the ETS costs per unit production cannot

be determined by the industries in advance since they depend on total output. Thus, we assume that industries calculate expected unit ETS costs (ucc_i^{ets}) using expected output based on trend based expectations.

Thus, total unit carbon costs per unit output passed on to the consumer are $ucc_i = ucc_i^{tax} + ucc_i^{ets}$. Since initial carbon costs only apply to ETS and are relatively small, we simplify and assume that all carbon costs will be added on top of initial values of tls . Thus, we dynamically adjust tls using $ucc_{i,t}$. This might cause marginal level effect distortions but should not affect the dynamic effects of the policy. Since we assume that all carbon costs are passed on to the final consumer, final embedded ucc per unit of basic price goods ($fcc_{i,t}^{bp,real,e}$) is given by $fcc_t^{bp,real,e} = ucc_t^T L_t + ucc_t^T (\gamma_t^Z \circ L_t)$, where $\gamma_{i,j}^Z$ refers to the import shares of intermediate inputs and L is the Leontief inverse matrix. Please note that second summand arises due to our assumption that other trade partners also implement an equivalent carbon price, which arises from the fact that Italy's main trading partners are other EU countries. Note that this expression is used for the change in tls , whereas the following is applied to discern the amount of carbon tax paid by consumers to the *domestic* government, since the domestic government only levies the taxes on domestic production: $ucc_t^T L_t$.

Subsequently, we divide this expression by the basic price of the respective industry and transform it to purchaser prices using $\frac{fcc_{i,t}^{bp,nom,e} + m_{i,t} \sum_j s_{j,t} fcc_{j,t}^{bp,nom,e}}{1 + m_{i,t}}$. Again, due to trade there is a discrepancy between tls paid by the consumers and tls paid by the consumers to the *domestic* government. Thus, to capture only carbon costs revenue for the domestic government the purchaser price conversion becomes: $\frac{(1 - \gamma_{i,t}^C) fcc_{i,t}^{bp,nom,e} + m_{i,t} \sum_j s_{j,t} (1 - \gamma_{j,t}^C) fcc_{j,t}^{bp,nom,e}}{1 + m_{i,t}}$, where $\gamma_{i,t}^C$ captures the import share by basic price good of household consumption.

Fossil fuel subsidies Using for simplicity initial period output values, we split total government expenditure on fossil fuel subsidies derived from MASE (2022) into subsidies received by the industries B (Mining), C19 (Petroleum), D1 (Electricity) and H (Transport) and calculate subsidies received per unit output for each of these industries ($usff_i$). Again, assuming that a reduction in these subsidies will be passed on to the final consumer, we calculate final embedded reduced subsidies ($fsff$) as $fsff_t^{bp,real,e} = usff_t^T L_t + usff_t^T (\gamma_t^Z \circ L_t)$ for changes in tls and $usff_t^T L_t$ for domestically paid government taxes, again due to our aforementioned assumptions that these types of developments occur equivalently in trading partner regions. Subsequently we again transform this expression to terms of nominal purchaser prices as described above for carbon costs.

Direct carbon costs To calculate direct carbon costs, i.e. costs that households pay directly on their produced emissions, require some intermediate steps due to the way we model the electrification process. Since for household consumption we model it as a cumulative process (see section 3.6.10.2), each period we first calculate the consumption in the absence of electrification and then adjust it according to scenario settings. However, since electrification affects per unit consumption emissions and thus per unit consumption carbon costs and the purchaser price, we have to introduce an additional tls and pp (purchaser price) variable so that we have one tls/pp that incorporates electrification and one that does not. The variables without electrification (be) are applied to the household module, which determines consumption in the absence of electrification. Subsequently, the variables after electrification (ae) are introduced after the electrification process and used to calculate actual tls paid and CPI.

We calculate a per unit consumption direct carbon costs ($hhucc_{i,t}$) using emissions per unit household consumption determined according to the process described in section 3.6.10. As aforementioned, we conduct this process twice, once assuming the initial consumption-energy link (e.g. in the absence of electrification) and once considering the dynamic impact of the electrification process. These per real unit basic prices costs are subsequently transformed into costs per nominal unit of purchaser prices according to the approach described above. Note however, that here there is no issue of trade that has to be taken into account as the tax applies to the final consumption which occurs within the country.

Impact on price and profits All of the calculated carbon costs per nominal unit of purchaser price are subsequently added to the respective tls :

$$\begin{aligned} tls_{i,t}^{be} &= tls_{i,0} + fcc_{i,t}^e + fsff_{i,t}^e + hhucc_{i,t}^{be} \\ tls_{i,t}^{ae} &= tls_{i,0} + fcc_{i,t}^e + fsff_{i,t}^e + hhucc_{i,t}^{ae} \end{aligned}$$

also leading to respective purchaser prices (see section 3.6.7.2).

Finally, for the carbon tax and fossil fuel subsidies industries can price in the taxes perfectly but as aforementioned ETS costs can only be estimated a priori. Thus, if industries price in too much or too little ETS costs, this will affect their profits by the difference of $ucc_i^{ets} \cdot x_i^{real}$, where x_i^{real} reflects real output, and actual ETS costs.

3.6.10 Energy

3.6.10.1 Energy input-output structure

The main steps towards and features of the novel energy input-output structure are illustrated in Figure 3.6.1 and are subsequently outlined in detail. The main underlying database for

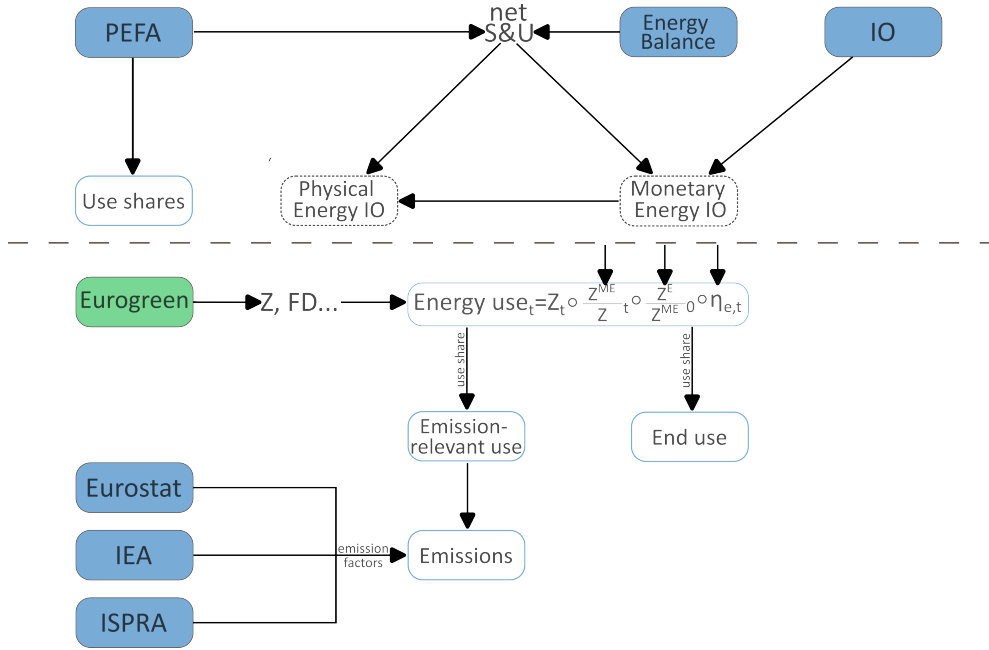


Figure 3.6.1: Illustration of energy input-output structure. In a first step PEFA (Physical Energy Flow accounts; Eurostat, 2023, env_ac_pefasu) and Energy Balance (Eurostat, 2023, nrg_bal_c) data are combined and transformed into net supply and use (net S&U) form. Subsequently, consulting input-output (IO) data (WIOD; Timmer et al., 2015 and LOCOMOTION; LOCOMOTION, 2021) an input output structure in monetary terms only reflecting energy flows (monetary energy IO) and a corresponding physical one (physical energy IO) are established. These outputs are used to calculate the share of monetary intermediate trade (Z) that reflects energy ($\frac{Z^{ME}}{Z}$), as well as the respective physical amount of energy associated with a unit monetary flow ($\frac{Z^E}{Z^{ME}}$). The Hadamard product of these matrices is finally allocated to specific energy products using product specific shares of the energy flows (η). This process is also applied to final demand (FD) and other energy relevant flows. Finally, energy use is translated into emissions, collecting emission factors by combining several datasources (Eurostat, IEA and ISPRA).

this procedure are the Physical Energy Flow accounts from Eurostat (PEFA) (Eurostat, 2023, env_ac_pefasu), but they do not distinguish between use of imported and domestic energy products, nor between energy supply directed at intermediate, household or export demand. Thus, we have utilize various assumptions in order to be able to transform the supply and use tables into an input-output structure.

Own use and net supply As a first step, in order to capture autoproduction, we assume that each sector first uses heat and electricity they produce themselves before procuring them from other industries. Thus, each sector's (i) own use by energy product ($e \in \{elec, heat\}$) is given by

$$own\ use_{i,e} = \min(use_{i,e}, supply_{i,e}). \quad (3.7)$$

Net supply and use are subsequently given by use and supply after accounting for own use, where own use for products other than electricity and heat is 0:

$$net\ supply_{i,e} = supply_{i,e} - own\ use_{i,e} \quad (3.8)$$

$$net\ use_{i,e} = use_{i,e} - own\ use_{i,e}. \quad (3.9)$$

Split D sector As with the WIOD, PEFA only features an aggregate D35 sector, which we split into D1, D2 and D3 sectors (see Table 3.6.1).

To allocate energy use, we use Eurostat energy balance data (Eurostat, 2023, nrg_bal_c) as approximation.¹⁵ To allocate energy use to D1 and D3, we use transformation input values for electricity and heat main producers¹⁶ However, these databases are not comparable one to one due to different approaches and classifications.¹⁷ Thus, we apply the following approximation rules:

- If the energy balance value is higher than the PEFA use of a product, we allocate the PEFA value to sector D1 and D3 in their proportion given by the energy balance
- If the energy balance value is lower than the PEFA value, we allocate the residual use to all three sectors using the monetary output as weights.
- For heat production, we add geothermal heat used for electricity production to heat production in the Energy balance table, and subsequently allocate its supply and use to the electricity sector D1. The residual heat use in PEFA is allocated to all three sectors according to their monetary output, assuming it is also linked to distribution and not only energy production processes.
- For electricity use, we allocate electricity use from pumped storage to D1 and electricity use from electricity and heat generation to D1 and D3, weighted by their monetary output. The residual electricity is allocated to the three sectors according to their monetary output, assuming it is linked to transmission and distribution as well as D2 production.

¹⁵For unambiguous concordance, we have to merge the disaggregate PEFA products “Transport diesel (without bio)” and “Heating and other gasoil (without bio)” to “Gas oil and diesel oil”.

¹⁶Electricity only and CHP are allocated to D1 and heat only is allocated to D3.

¹⁷For instance, the balance entries only consider production, not distribution and transmission of electricity and heat and do not consider D2 at all. Another example is that PEFA is set up according to the residence principle and the D sector excludes production of electricity through waste. Also, PEFA solid biofuels does not include animal waste, and biogas does not include biogas from thermal processes, whereas the energy balance does include both. Basic oxygen steel furnace gas not explicitly included in the energy balance, but part of the PEFA energy product derived gases.

- For other energy products like Motor spirit, that occur in PEFA but have 0 entries in the Energy balance, we simply allocate between the three industries using their monetary output.

To allocate the supply, we assume that D2 supplies all gases produced by the D sector and D1 supplies all electricity. For heat, as aforementioned in a first step we allocate the geothermal heat used for electricity production to D1. We split the remaining heat supply between D1 and D3 according to their respective heat supply share given in the energy balance.¹⁸ Note however, that in the end we do not model D2 and D3 as energy supplying sectors due to their role as distributors (see 3.6.10.1.3).

Trade To distinguish between imported and domestically supplied demand for energy products as well as demand by category and origin (i.e. intermediate or household, as well as export and inventories), we calculate an import share of net supply by source: $\vartheta = \frac{\text{imports}}{\text{total supply} - \text{own use}}$. We subsequently assume that this share holds for all types of demand. Thus, to obtain domestic use of domestic supply, we scale every type of use by the same product specific import share:

$$\text{domestic net use} = \text{net use} \cdot (1 - \vartheta) \quad (3.10)$$

Note that for final demand categories use and net use are equivalent and for imports, supply and net supply.

This process is conducted for each type of energy product (e) respectively.

3.6.10.1.1 Monetary energy IO

In order to link energy and monetary flows in both directions, we first establish monetary Z matrices that only reflect trade in energy products. This link is also the reason why we focus only on energy products in the PEFA and omit natural inputs as well as residuals as they cannot be linked to monetary flows straightforwardly. We use three different approaches for three different types of energy supplying sectors to discern how much of their monetary trade is linked to energy.

Full energy sectors: The first category of sectors are sectors where it is reasonable to assume that their products are only energy related. These sectors are C19 (Petroleum) and D1 (Electricity). For them, we set the monetary flows related to energy equal to their monetary flows.

Mining: The mining sector (B) clearly supplies a range of goods other than energy. In order to

¹⁸PEFAs total electricity and heat (minus geothermal electricity) supply are reasonably close to the energy balance supply values (97 and 103% respectively).

discern the energy related flows, we utilise the aforementioned LOCOMOTION dataset, which divides the mining sectors into several subsectors. This allows us to calculate cell specific shares of the Z matrix B row that reflect energy flows. We apply the same logic to final demand of the mining sector.

Biofuels: For sectors providing biofuels (A), we cannot utilise one of the previous two approaches as they clearly provide a range of products and LOCOMOTION does not provide the required disaggregation. Since this sector however only net supplies one energy product we assume that its monetary flows are proportional to the use of this product by other industries.¹⁹ In a first step, we determine, which interindustrial flow from the respective sector (i) features the highest energy intensity of the respective energy product (e), i.e. $\frac{net\ use_{j,e}}{Z_{i,j}}$.²⁰ We subsequently assume this flow to *only* reflect energy related trade and scale its value by the respective relative use of other sectors and final demand categories.

Own consumption: Finally for all sectors apart from the full energy sectors and mining, we set the diagonal entry of the Z matrix reflecting monetary energy related flows (Z^{ME}) equal to the diagonal entry of the monetary Z matrix, in order to be able to represent own consumption. This should not be interpreted such that these cells only reflect energy flows but that an increase in output will proportionally increase this cell and consequently proportionally scale up production of energy products for own consumption.

Imports: To generate Z^{ME} for imports, we first assume the same supply structure by industry and product as represented in domestic data, since we do not have information about the sectoral supply of imports. Import demand is given by $imp\ net\ use = net\ use \cdot \vartheta$.

Subsequently we follow in general the same structure as described above with the following notable distinctions:

- For the mining sector we calculate import specific shares reflecting energy/subsectors.
- Imports that are subsequently exported as is are not modelled in Eurogreen. Thus, we ignore imports to directly meet export demand.

3.6.10.1.2 Physical energy IO

To link physical energy supply and use to an physical energy input-output structure (Z^E), we adhere to the following sequential procedure for each energy product:

¹⁹Here we assume that each product has the same sale structure, i.e. is spread across all demand categories the same way no matter its sector of origin. This means that the supply from each industry by good is determined by its net supply share and the same applies to all demand categories.

²⁰Note that if an industry i is not the only net supplier of that product, $net\ use_{j,e}$ is scaled by industry i 's net supply share of that product, again reflecting the fixed product sale assumption.

1. If there is only one sector supplying or only one sector demanding that energy product the allocation is straightforward.
2. If (1.) is not applicable, but when considering only the energy sectors determined here (i.e. A, B, C19 and D1), we obtain only one supplying industry, again the allocation is straightforward with demand being scaled by the net supply share of the supplying sector.
3. If neither (1.) nor (2.) are applicable, we resort to the RAS algorithm to balance our matrices. Here the column sums are given by net use by industry and the row sums by net supply. The underlying matrix is Z^{ME} . Note that we remove diagonal entries in Z^{ME} before applying the RAS, since this step only allocates net supply and use.

As all these procedures focus on net supply and demand only, in a last step we add own use by product and industry in the diagonal of the matrices. Subsequently, the A matrix of physical energy flows (A^E) is given by

$$A^E = Z^E \hat{x}^{-1} \quad (3.11)$$

Dividing total by source specific energy flows, gives cell specific shares by energy product e ($\eta_{e,i,j}$). These steps allow us to generate physical flows from monetary ones:

$$Z_e^E = Z \circ \frac{Z^{EM}}{Z} \circ \frac{Z^E}{Z^{EM}} \circ \eta_e, \quad (3.12)$$

Final demand: We apply the same procedure to obtain energy flow for final demand. Note however, that we do not consider GFCF or government as they are not linked to direct energy consumption, but their energy demand is captured in intermediate consumption. In a first step, we again use the fixed product sales assumption, implying that households obtain an energy product by industries according to their net supply shares of that product.

Following the same procedure as for interindustrial trade, we obtain specific monetary energy shares, for household final demand and exports respectively and by industry. The same goes for shares of energy products in total energy supply and the total energy supply over monetary energy flows fraction.

Imports: Imports follow the same general logic with only one distinction. As aforementioned we assume the same supply structure as for domestic products. However, 'Brown coal and peat' is only imported and not supplied by domestic industries. We allocate it to be supplied by the importing mining sector.

Types of use: With this procedure so far we have obtained an energy input-output structure tracing supply and use. However, so far we considered total use, which will lead to an issue of double counting. Thus, in the next step, we use PEFA accounts to calculate product and industry specific shares of total use that reflect 1.) end use; 2.) emission relevant use. We do the same for

final demand by households. To split sector D, here we assume that all subsectors feature the same emission relevant and end use shares. The only difference we ascribe to the product heat, since only D1 uses heat for transformation purposes. Thus, we set the end use share of heat for D2 and D3 equal to one and re-calculate the end use share for D1.

Energy product aggregation Throughout the whole analysis, maintained energy product classification represented in the Eurostat PEFA accounts in order to maintain the highest level of detail. However, subsequently and in order to translate energy use in emissions, we aggregate energy products according to Table 3.6.2.

Table 3.6.2: Energy product aggregation.

Aggregation	PEFA products
Coal and coal products	Hard Coal
	Brown coal and peat
	Secondary coal products (coke, coal tar, patent fuel, BKB and peat products)
	Derived gases (=manufactured gases excl. biogas)
Natural gas	Natural gas (without bio)
	Crude oil, NGL, and other hydrocarbons incl. oil shale/sands (excl. bio)
	Motor spirit (without bio)
Oil and petroleum products	Kerosenes & jet fuels (without bio)
	Naphtha
	Transport diesel (without bio)
	Heating and other gasoil (without bio)
	Residual Fuel Oil
	Refinery gas, ethane and LPG
	Other petroleum products incl. additives/oxygenates and refinery feedstocks
Nuclear fuel	Nuclear fuel
	Wood, wood waste & other solid biomass, charcoal
Biofuels	Liquid biofuels
	Biogas
Electricity	Electrical energy
Heat	Heat

3.6.10.1.3 Distribution

For energy and monetary exchange regarding the D subsectors, we find discrepancies, i.e. mon-

etary flows that represent energy but no energy flows.

A major reason for this is that we only model initial supply and final/transformation use of energy products, not the transmission of these flows. This is particularly relevant for sectors who distribute energy products supplied by other sectors, i.e. D2 - distributing but not supplying much of the natural gas –, for sector D1 imports – distributing but not supplying electricity and for sector D3 – distributing heat produced elsewhere. The monetary flows of these distribution services are clearly linked to energy, but no energy is reflected in some of these flows. In order to mitigate this issue, we conceptualise sectors D2 and D3 not as energy supplying sectors, but purely as distribution sectors. Since their contribution to energy supply in using our assumptions is rather minor,²¹ we assume that their role as distributors is more relevant. Accordingly, changes in their demand are not linked to energy supplied by these industries, but to energy demanded of the respective energy product (natural gas for D2 and heat for D3). We assume the production structure of the industries to stay constant except for changes due to their final energy use (i.e. for discrepancy flows, particularly from B, we set the energy intensity to 0). Note that for imported distribution demand we only consider imported energy products, whereas for domestic distribution we consider domestic and imported energy, since imported products are also distributed using domestic infrastructure.

Note that the approach to scale distribution sectors separately is an approximation and might slightly overestimate the change in these distribution sectors, since some energy is directly traded, without running through the distribution sectors. The respective monetary flows are scaled separately, first the direct energy flows without distribution sectors involved, due to changes in energy supply and subsequently the flows from energy supplier to distribution and from distribution to point of use via the distribution mechanism. Note that we only have one net supplier of gas (B) and heat (D1), but for heat (as for electricity) most sectors supply to themselves via autoproduction, which does not require distribution infrastructure and services. Thus, to scale the heat distribution sector we calculate the percentage change of heat use excluding autoproduction.

Finally, D1D1 imports features monetary but no energy flows, again likely due to the transmission and distribution function of the D1 sector. However, again we keep the value in the

²¹Considering the supply of all sectors, i.e. not only the ones we model as energy supplying sectors, D2 supplies 1.8% of biogas, 0.5% of derived gases and 6.9% of natural gas, whereas D3 supplies 3% of heat. Even with removing the energy supply of these sectors we still model > 90% of energy related to respectively intermediate domestic and imported trade and final demand domestic and imported. Also note that particularly D2 (0.79), but also D3 (6.01) have significantly less energy intensive outputs (in terms of net energy supplied) compared to energy sectors like C19 (50.90) or B (42.97). Due to the higher price of electricity and its inclusion of transmission and distribution, D1 features a lower than B and C19, but still significantly higher intensity of 11.19.

A matrix constant and since we assume that all electricity from electrification is supplied by domestic electricity production, there is not reason to assume this value should change due to the electrification process.

Note that our energy use and emissions differ from PEFA and other accounts as we only model energy provided by our determined energy producing sectors, which overall accounts for $> 90\%$ of the energy, however these values vary strongly among energy products (e.g. liquid biofuels are not modelled at all). Another difference to emission accounts is that we only consider emissions from energy use and adjust these to the residence principle.

3.6.10.2 Energy module scenario implementation

Figure 3.6.2 depicts the implementation of scenarios in the Energy Module, which are subsequently explained in more detail.

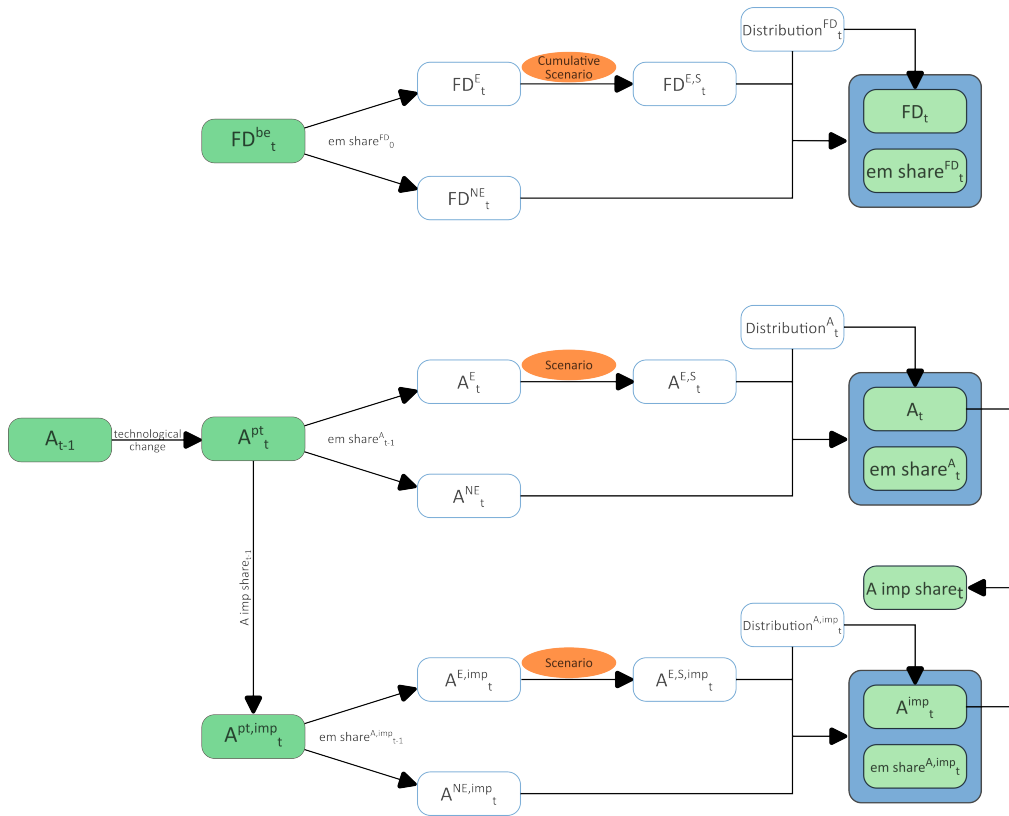


Figure 3.6.2: Illustration of the electrification process. Implementation of the electrification process for final demand (FD) and production structure (A). The green boxes on the left are an output of the overall Eurogreen model, which are subsequently modified by scenario specific electrification dynamics (orange), before the green boxes on the right feed back into the overall Eurogreen model. The superscripts mean before electrification (be), post technology (pt), energy (E), non-energy (NE) and energy after scenario (E, S). Moreover, $em\ share$ refer to the fraction of monetary flows that result from energy flows and $imp\ share$ to import shares.

Final demand For each period, we obtain the vectors of final demand from the consumption and international trade modules. We use the *initial* share of final demand expenditure that relates to energy to split the final demand into energy and non energy demand. The reason to use the initial value is that the consumption behaviour in the rest of the model does not react to changes in energy use, hence we assume that the initial relationship between monetary flows and monetary flows reflecting energy holds over time, i.e. we assume the bundles of goods per category do not change.

This change is subsequently introduced in the next step, where we adjust the energy flows according to exogenous demand scenario assumptions, where we use the calculated energy intensities of energy related monetary flows to translate changes in energy use into monetary changes. Note that the demand scenario implementation here reflects *cumulative* changes in demand patterns, as the consumption inputs up to this point reflect current consumption patterns absent any change on energy consumption patterns. We assume the consumption that is not related to energy does not change in this step. However, overall expenditure can change through the electrification process. To keep overall expenditure independent of the electrification process, we calculate a novel β parameter (representing the share of expenditure spend on each good) after electrification. Applying this novel β to the total expenditure gives us new final expenditure. Note that we conduct this process individually for each type of household allowing for differentiation in electrification developments along income or regional lines. This results in new consumption vectors which feed back into final demand in the rest of the model and generates updated shares of final demand expenditure that relates to energy to translate monetary consumption into energy use.

A Matrix As the A matrix not only interacts with the IO production function but also with dynamics in technological development, we cannot use the same approach as for final demand, conducting a cumulative post-multiplication of otherwise unchanged values. For the A matrix we require a year-by-year interaction with the technological change module, which is why we apply a slightly different approach here:

We assume that the change in energy use occurs each period *after* technological change has taken place and that technological change affects both non energy and energy related flows equally. Thus, the A matrix after technological change is split into energy and non energy related flows according to their respective shares in the *previous* period, as technological change does not alter these shares. Again, we simulate changes in energy supply and use patterns, using the energy intensities of energy related monetary flows to translate changes in energy use into monetary changes. Note that the scenario implementation in this case not *not cumulative*, as

we adjust the A matrix period-by-period. Again, we obtain a new A matrix and $em\ share^{A22}$ values, which feed back into the IO and the technology modules.

We apply the same procedure for imported intermediate goods, with the only addition, that the import share now becomes endogenous and dynamic due to changes in energy flow patterns. Again, since technological change does not affect the import shares, we obtain the post-technology but pre energy change A imports matrix using the *previous* period value of import shares. Also note that we determine imports before the electrification process and hence explicitly electrify domestic and imported energy separately, i.e. we assume domestic electrification has no direct impacts on imported intermediate goods required per unit output. Thus the A matrix we apply the previous period import shares to, is the domestic A matrix before electrification (A_t^{pt}).

3.6.10.2.1 Electrification process

Final demand As aforementioned, for final demand we conduct a *cumulative* electrification scenario implementation, changing demand obtained from the rest of the model each period by $\sum_0^t \tau_t$, where τ_t is the effective yearly rate of electrification w.r.t. the initial period of electrification.²³ Note that we assume a constant rate of electrification of present fossil fuel use each year, which implies an exponential decay of fossil fuel use due to electrification over time. Thus, the effective yearly rate of electrification w.r.t. the initial period of electrification is given by $\tau_t = (1 - \sum_0^{t-1} \tau_t) \cdot \bar{\tau}$, where $\bar{\tau}$ is the constant yearly rate of electrification. Electrified energy in TJ is given by $EE_t = ff_t \cdot \sum_0^t \tau_t$, where ff_t refers to fossil fuel energy use in the absence of any past or present electrification process, i.e. demand from the rest of the model transformed to energy using initial values. For each industry energy supply is reduced by $EE_t \cdot \eta_{i,t}^f$, with $\eta_{i,t}^f$ referring to the share of fossil fuels supplied by industry i , again in the absence of any electrification process.²⁴ In contrast, the energy supply of D1 is increased by $(1 - \rho)EE_t$, where ρ represents the fractional efficiency gain from electrification.²⁵ This yields a new demand vector in energy terms which feeds back into the energy module.

Distribution: As mentioned above, we have to consider changes in distribution sectors associated with energy supply and use. To address this issue, we adjust final demand for the D2/D3 sector in the absence of electrification process by the relative change in gas/heat demanded in

²²Note that $em\ share^A$ is equivalent to $\frac{A^{EM}}{A}$ from the previous subsection. We assume that the fraction $\frac{A^E}{A^{EM}}$ remains constant.

²³Note that here we consider electrification of fossil fuel use only, i.e. τ_t refers to the proportion of fossil fuel energy that is being electrified.

²⁴Note that with this formulation there is no hierarchy in fossil fuel phase down, but they are all affected by the same rate τ_t .

²⁵Note that this also implies an increase in heat supply.

t before and after the electrification process. Note that we apply the same procedure to all three types of relevant final demand (domestic household demand, imported household demand and exports), with the only difference in the scaling of D2/D3 demand: For the domestic case we scale it with imported and domestic gas demand, whereas for imports and exports we only consider the respectively traded gas demand.

A matrix As described above, the A matrix is not adjusted cumulatively, but on a year by year basis, given its interaction with technological developments. We assume that in period t , there is an initial A matrix, which is first altered by technological innovation leading to A_t^{pt} . This A matrix is subsequently altered by the electrification process leading to a final A_t , which feeds back into the production process in t and consequently becomes the basis for technological developments in $t + 1$.

Thus in t , we obtain A_t^{pt} translating it into energy following 3.6.10.2. For the A matrix, we assume that only end use is being electrified, as transformation use can be used for autoproduction or transformation from one energy product into another (e.g. in C19), which cannot necessarily be replaced by electricity. Thus, here we only model electrification of industry end use. Using the end use shares from the previous period (es_{t-1}), we generate end use by industry and product before the electrification process. Subsequently, we obtain electrified end use ($EE_{j,t} = \sum_{ff} \overline{\tau}_j \left(\sum_i a_{i,j,e,t}^{E,pt} es_{j,e,t-1} \right)$, where ff includes coal and coal products, natural gas and Oil and petroleum products) by multiplying the end use of fossil fuel products with the electrification share. For column j of the energy matrix $A_t^{E,pt}$ supply of fossil fuels by i is adjusted according to $a_{i,j,t}^E = a_{i,j,t}^{E,pt} - EE_{j,t} \eta_{i,j,t}^{A,f}$, where $\eta_{i,j,t}^{A,f}$ is the share of fossil fuels demanded by j that is supplied by i before the process of electrification. The electrified energy is subsequently added to the D1 sector supply: $a_{D1,j,t}^E = a_{D1,j,t}^{E,pt} + (1 - \rho) EE_{j,t}$, where again ρ represents the fractional efficiency gain from electrification. Following 3.6.10.2, we re-translate the new A matrix into monetary terms and feed it back into production in period t .

As we only affect end use with the electrification process, but keep transformation use constant, the end use share of fossil fuels is dynamic and develops according to: $es_{j,e,t+1} = \frac{1}{1 + \frac{1 - es_{j,e,t}}{(1 - \overline{\tau}) es_{j,e,t}}}$ for $e \in ff$. In addition, the end use share of heat in D1 is dynamic as well. For all other industries the end use share of heat and electricity is 1, so that electrification does not affect it. However, D1 uses a large share of heat for transformation to electricity which is why the end use share of heat will increase with the electrification process (i.e. adding end use of heat) according to: $es_{D1,Heat,t+1} = \frac{1}{1 + \frac{1 - es_{D1,Heat,t}}{es_{D1,Heat,t} + \frac{\Delta end\ use_{t+1}}{use_t}}}$, where $\Delta end\ use$ captures the total addition in end use of heat in D1 due to electrification in TJ and use reflects total use of heat in D1 before the electrification process.

Also note, that using a constant rate of electrification of fossil fuels each year leads the effective rate of electrification of fossil fuels to follow an exponential decay .

As aforementioned, electrification of domestic energy supply is separately modelled from electrification of imported energy. Thus, in an additional step, we explicitly electrify intermediate energy imports replacing them with *domestic* electricity production. This procedure follows the same approach as described in the previous paragraphs for the domestic A matrix. We assume that the same rate of electrification by industry applies to domestic as well as imported fossil fuel use. Consequently, es does not differ between imported and domestically supplied products.

Distribution: For the distribution sector, we apply a similar procedure as for final demand, i.e. D2 is driven by developments in gas demand and D3 by heat. We assume that the production structure (i.e. the columns of the A matrix) of the distribution sectors is not affected by electrification, apart from end use they substitute. On the other hand, the production structure of other industries w.r.t. to the distribution sectors will change with electrification, e.g. if a sector requires less gas, it will also demand less gas distribution services. Thus, we scale the rows of D2 and D3 in the A matrix in the following way:

Flows from D2 (D3) are scaled by the respective change in demand for gas (heat) of the receiving sector. Note that for imports we only consider imported energy demand, whereas for domestic distribution we consider total (domestic and imported) demand, since both have to be transported using domestic infrastructure.

Note that imported electricity also has to run through domestic infrastructure, however we abstract from this aspect here, assuming that imported electricity does not affect domestic electricity sector inputs. Note that the import flow D1D1 is purely linked to distribution, i.e. there is no related energy flow in our framework. It is the only remaining discrepancy from 3.6.10.1 that we do not address explicitly, which could potential cause issues if electrification would also be driven by imported electricity.²⁶ However, as we assume the electricity for electrification is purely produced domestically, we expect this issue to not significantly affect our results.

Dynamic source shares: As we address electrification in terms of final use only, and final and transformation use feature distinct patterns, the source shares of different energy products also become dynamic for each interindustrial flow. For a given interindustrial flow i, j electrification of final use in j should imply that the share of primarily transformation fossil fuel products from i to j should increase and the share of primarily final use fossil fuel products should decrease.

²⁶In the current model structure, electrification driven also by imported electricity would replace energy from other sectors imports with flows from imported D1. However, since imported D1D1 does not represent any energy, overall energy use of the D1 sector will decline.

These dynamic shares also imply that the amount of autoproduction per unit output remains constant, since transformation use does not change.

For each interindustrial i, j flow, the share after electrification by product e is given by $\frac{tu_e + eu_e + \Delta eu_e}{\sum_e (tu_e + eu_e + \Delta eu_e)}$, where tu is transformation use, eu end use before the electrification process in this period and Δeu reflects the change in end use by product in i, j due to the electrification process, i.e. $\frac{a_{i,j,e,t}^{E,pt} + \Delta eu_{e,t+1}}{\sum_e (a_{i,j,e,t}^{E,pt} + \Delta eu_{e,t+1})}$. The last is given by $\Delta eu_{e,t+1} = -EE_{j,e,t} \eta_{i,j,e,t}^{A,f}$ for each fossil fuel product e .²⁷

For simplicity, we keep the energy intensities constant.

3.6.10.3 Emission factors

The main datasources for emission factors are energy emission data from the IEA by fuel (IEA, 2023b).²⁸ We adjust the IEA dataset in the following way:

1. To adjust the emission data to the residence principle of national accounts, we use the ‘Air emissions accounts totals bridging to emission inventory totals’ data from Eurostat (Eurostat, 2023, env_ac_aibrid_r2). This allows us to subtract emissions by non-residence on the territory and add emissions by residence abroad to the IEA data. Since these adjustments mainly relate to transport (Eurostat, 2009), we assume they are all energy related and specifically refer to the use of ‘Oil and petroleum products’.
2. As IEA reports biofuels and waste combined, but we only consider biofuels, we calculate emissions from biofuels using emission factors from ISPRA (2021b). In line with the IPCC guidelines, we do not include CO2 emissions from biofuels.

3.6.11 Electricity

In order to improve the conceptualisation of the energy transition and its feedback effects on the economy, we created a simplified electricity system model. Here, electricity demand from the energy module is addressed by the following production and storage technologies: Gas (incl. natural gas and gas to power), solar pv, wind (onshore), hydro (incl. Dam, run of river and PHES), coal, petroleum, nec (incl. geothermal and biomass) and batteries which we assume to be lithium-ion (Schmidt et al., 2019). In addition, the grid is expanded in order to facilitate a transition towards renewables.

²⁷Note that we do not consider end use changes of electricity and heat here, since the electrification process is only provided by sector D1, which in turn does not provide any fossil fuel products and exhibits constant source supply shares. Thus, we only consider end use changes of fossil fuels due to electrification here.

²⁸The only discrepancy between IEA and our product aggregation is oil shale (see IEA, 2023a). However, emission relevant use of the respective category is zero so there is no impact in terms of emission intensities.

Figure 3.6.3 represents a simplified illustration of the main structure of the electricity module subsequently explained in more detail.

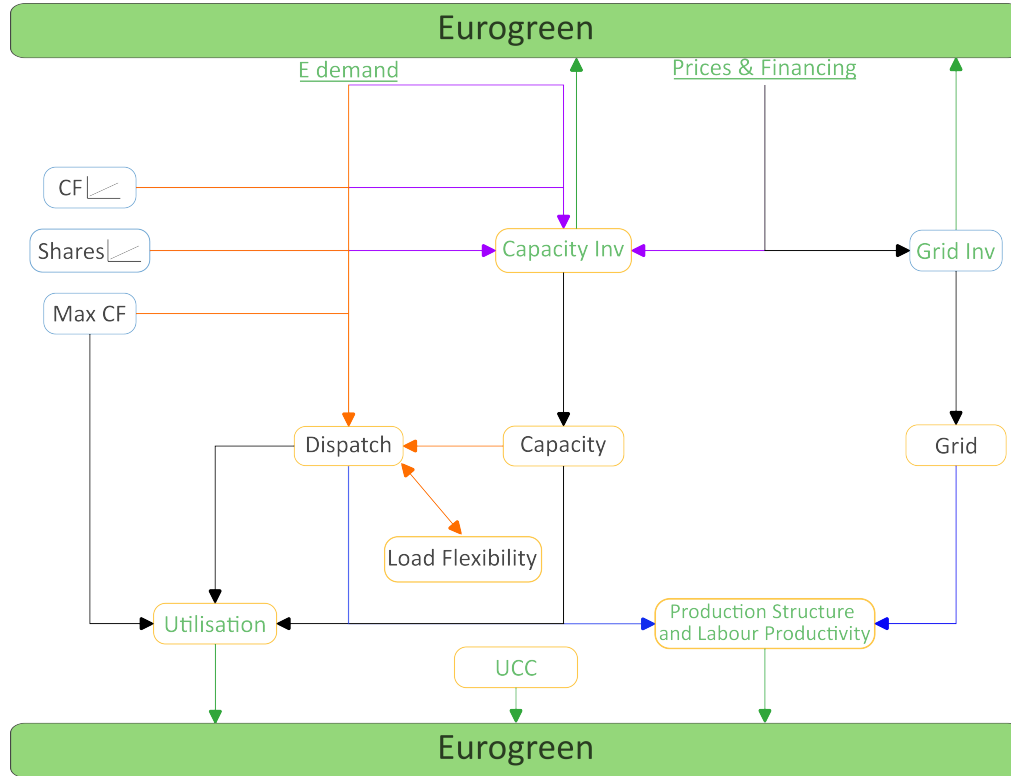


Figure 3.6.3: Simplified structure of the electricity module. The blue blocks are exogenous inputs, whereas the orange blocks reflect endogenous variables. Green underlined text without box represents inputs from other modules into the electricity module and green text within boxes reflects variables that feed back into other modules. The graph symbol indicates exogenous dynamic developments. CF stands for capacity factor, E demand for electricity demand and UCC refers to unit capital costs.

3.6.11.1 Exogenous pathway

The yearly time resolution of the Eurogreen model does not allow to model significant challenges of the transition to renewables like flexibility, curtailment and capacity values explicitly. Thus, here we apply an exogenous pathway for Italy for fully decarbonised electricity production in 2050 based on the open source power system cost optimisation model FLEXIES as presented in Maeder et al. (2021).²⁹ The pathway provides capacity and production by technology in Italy in 2050. Using this data, we calculate production shares by technology, as well as implicit capacity factors for the year 2050. We use historic data where available and subsequently,

²⁹Note that the more realistic limited transmission expansion scenario is applied here.

apply linear interpolation between last available historic and 2050 values for these variables.³⁰ For a counterfactual no renewable transition setting, these variables are kept constant at their last historic value. We also consider the coal phaseout until 2025, which is replaced by renewables and gas. Note that the sum of all shares exceeds one and increases increasing decarbonisation as production for storage is counted twice (the original production of the electricity as well as the storage discharge) as done in Maeder et al. (2021).

For the grid expansion, we simply extrapolate yearly investment costs allocated to transmission and distribution in the MISE-MATTM-MIT (2019) over the whole modelling timeframe. In the counterfactual no transition setting, we assume that grid investment aims at expanding proportional to overall expected electricity demand in order to maintain the initial grid to electricity demand ratio.

3.6.11.2 Capacity investment

Using expected electricity demand (trend based expectations) in combination with the pathway based shares by technology and capacity factors, the industry determines desired capacity for all production and storage technologies.³¹ Subsequently, investment is conducted in order to adjust the current capacity stock towards the desired one.³²

However, it is potentially constrained by a maximum scale expansion by technology and by financial conditions as in the Investments modules. If binding, the financial constraint applies proportionally to all investments from the electricity sector (i.e. production and storage capacities as well as the grid).

Following the standard learning curve literature (e.g. Rubin et al., 2015), investment costs change according to capacity and technology specific learning rates. Note that this is not the case for the grid and for batteries, since there is virtually no initial capacity stock for batteries to base learning effects on. Hence, battery costs follow cost projections presented in Schmidt et al. (2019) and grid costs remain constant.

Depreciation and expected depreciation follow a pipeline delay, i.e. capital is fully productive over its lifetime and then fully depreciates once the lifetime is over. This approximation better reflects actual depreciation behaviour, which tends to be concave, than geometric forms of

³⁰Note that this is a very strong simplification, allowing for not feedbacks between costs and investment for instance. Thus, we do not obtain logistic growth patterns as expected for renewables like wind and solar.

³¹Please note that the historic and projected values of shares by technology and capacity factor here are ahead by one period, since they characterise planning for the future period. For instance in 2010, the values for 2011 are fed into investment consideration, whereas 2010 (i.e. the delayed values) are applied in the dispatch rule.

³²Note however that whereas the planning period is one year ahead, the construction time of most technologies is significantly more.

depreciation (Pietzcker et al., 2017). Note that for the grid and for overall industrial monetary capital we stick to geometric depreciation, since we consider capital in financial terms.

Investment costs and demand We assume a linear expense schedule of investment costs over the time of construction. Investment demand is subsequently transformed in investment supply using GFCF shares from LOCOMOTION. For batteries all investment is allocated to C27: manufacture of electrical equipment.³³

3.6.11.3 Dispatch and production structure

Electricity dispatch occurs in two phases: First, each technology produces according to the minimum of scenario demand – given by electricity demand and pathway technology shares – and supply – given by capacity and pathway capacity factor.

In the case of a deficit, in the second phase underutilised capacity is ramped up. Here, we assume that the pathway capacity factor for renewables (wind, solar and hydro) is equal to the maximum capacity factor (i.e. we assume priority of dispatch). For other technologies, we determine additional production potential using the difference of the maximum capacity factor (data from Jaxa-Rozen et al. (2022)) and the pathway capacity factor. In order to broadly reflect flexibility characteristics, we split deficit load as well as additional production potential by technology into flexible load and baseload. For simplicity we assume that the excess capacity as well as excess demand have the same flexibility features as overall capacity and demand. Thus, flexible additional production potential is determined by overall additional production potential and flexibility parameters from Carrara and Marangoni (2017) and Sullivan et al. (2013). Flexible excess demand is given by the flexibility needs of the load (calculated based on load curves from Jaxa-Rozen et al. (2022)) and increases with VRES (i.e. wind and solar) that is directly fed into the grid, i.e. not used to charge storage³⁴ in line with the respective flexibility parameters. Excess baseload demand and production potential are calculated as residual. Capacities for excess baseload and flexible load are dispatched according to exogenous priorities minimising emissions.³⁵

³³This is most likely an overestimation as part of the costs will go to other sectors for instance for installment. However, we are not aware of any data on this distribution of these costs for batteries.

³⁴Production that is used to charge storage is assumed to be the same for all production technologies.

³⁵The highest priorities are allocated to renewables (hydro, wind and solar), followed by nuclear and then nec (which includes biomass and geothermal). Lastly fossil fuels are ranked from least dirty to dirtiest (i.e. gas, petroleum, coal).

Production structure To determine the electricity production structure of the electricity sector, the grid as well as production and storage technologies have to be considered. However, since the modelling of the grid is quite exogenous and ad hoc, there remain large uncertainties regarding its role in the production structure. In order to take into account of this uncertainty, we allow for two different rules for the development of the production share of the grid within the electricity sector: Either, it is determined by the development of grid capacity with respect to electricity production, i.e. we implicitly assume that grid production increases proportional with grid capacity. On the one hand this assumption potentially allows us to capture an increasing role of the grid within electricity production when shifting to renewables. However, it also reflects a very strong assumption, enabling a quite uncertain and more or less exogenous variable to have a large impact. Thus, an alternative rule is to simply keep the production share of the grid within the electricity sector constant at its initial value.

Subsequently, the remaining production share is allocated to production and storage technologies (excl. batteries)³⁶. The overall electricity sector production structure is subsequently calculated as a weighted average of technology specific production and import structure data from LOCOMOTION and production shares as weights.

Furthermore we apply the same weights when calculating the sector's labour productivity as a weighed average based on labour productivity values from the Exiobase 3 EEMRIO dataset (Stadler et al., 2021), described in Stadler et al. (2018). Skill shares develop in line with developments in labour productivity as in the Labour module. As the shift to renewables affects labour productivity it consequently also influences the skill share of the electricity sector.

3.6.11.4 Integration in the Eurogreen structure

Added variables In order to link the module with the general Eurogreen structure, addition electricity sector aggregated variables are required:

Due to the aforementioned uncertainty regarding the role of the grid, we calculate the capacity utilisation solely as the fraction of electricity production to potential production given by production and storage capacities and maximum capacity factors. Since this value however changes significantly in a transition towards renewables, particularly given the scenario assumptions used here, the normal rate of capacity utilisation for the electricity sectors is not fixed but determined as an exponentially-weighted moving average.

In order to determine per unit capital costs, we first calculate depreciation in real monetary

³⁶Battery dispatch does not affect the electricity production structure due to lack of available data on the production structure of a battery sector according to our knowledge.

terms. Monetary capital stocks for each production technology increase with real monetary payments for investments and depreciate with a rate reflecting the inverse of their lifetime. Combined with the price of capital (given by the capital shares of technologies in overall electricity demand, their respective investment structures, industries' prices and tfs) we obtain capital costs. These are subsequently distributed over potential production in terms of real monetary output (i.e. full capacity output) scaled by the normal rate of capacity utilisation.³⁷

Note that this module introduces a differentiation between capital prices and investment prices, which are calculated analogously (using shares of technologies, their investment structures and industries' prices and tfs) but are based on different distributions of technologies (capital prices consider the composition of the current capital stock, whereas investment prices consider the composition of new investment). The investment price is applied to investment costs and tfs paid, whereas the price of capital is used to translate capital values into monetary values (i.e. for capital costs and equity and leverage). Note that learning effects reduce the price of capital and hence reduce the value of the capital stock *ceteris paribus*.

Finally, the lifetime of the aggregate electricity sector is determined as the inverse of the depreciation rate calculated from the aggregate monetary stock and depreciation.³⁸

Investment constraints The investment constraint compares desired new investment spending (considering potential scale constraints) with potential investment spending as determined in the Investment module net of ongoing financial obligations from past investment derived from the linear expense schedule over the time of construction. We assume that new investment are constraint but that ongoing financial obligations from past investment are priorities and are al-

³⁷For the expected full capacity output in the Technology module, we scale expected capacity with current maximum capacity factors. In addition, in order to avoid circularity with the output constraint, we define full capacity output in real terms using the by one TIME STEP delayed electricity to output ratio. The reason to use the delay is that electricity per output depends on final demand and output composition. Thus, calculating a pre-constraint full capacity output (which depends on electricity per output) is problematic as this would depend on potential constraints in all other sectors, i.e. there is not a direct link between capacity and full capacity output but it is mediated via demand and output composition due to the link via electricity per unit output. Hence, for instance if another sector is constrained, this would affect electricity per output and hence full capacity output for the electricity sector, which is why for the electricity sector it cannot be considered independently before and after the output constraint. Thus, for simplicity, and considering that it is quite constant over time, we use the delayed electricity per output to calculate full capacity output for the output constraint.

³⁸Note that the alternative would have been to calculate the lifetime as the weighted average of the lifetimes of types of capital. The only technology where these two values differ is for the grid, where we found depreciation rates differing from the inverse of lifetime. Since the rest of the model structure calculates lifetime as the inverse of the depreciation rate, we chose to apply the same approach here to maintain consistency.

ways financed and serviced, even if the potential maximum investment is below their value. If binding (i.e. the fraction of maximum investment spending net of ongoing financial obligations over desired investment spending ≥ 1), the financial constraint fraction is applied equally to new investments of all technologies (i.e. production technologies, batteries and grid).

Labour productivity and production structure Since the initial values for labour productivity and the production structure are from different years and not necessarily compatible between the datasets used for the overall model (KLEMS (Bontadini et al., 2023) and WIOD) and the electricity module (LOCOMOTION, Exiobase), initial values stay in line with the overall model (i.e. KLEMS and WIOD) in maintain aggregate coherence (in the D sector), but are dynamically affected through developments in the electricity sector via the growth rates of the variables therein.

Subsequently, the effect of the electricity module on labour productivity and the input structure (incl. import structure) are modelled temporally *after* the technological progress (i.e. they affect the post technology variables) but for the input and import structure *before* the electrification process. As with the electrification process, the changes from the electricity module for the next period are not foreseen when choosing a technology in the Technology module, but impact technological progress by altering the base variable for innovation for the next period.³⁹

3.6.12 Climate damage

Climate damage is turned off in this work.

³⁹Note that in order to avoid circularity, the impact of the production and import structure is delayed by the minimum amount of time (i.e. TIME STEP). The circularity arises since electricity demand determines the production structure, but the production structure drives the A matrix and hence output and electricity demand. The import structure for instance affects tIs through carbon pricing and hence the price, final demand and output. The impact of labour productivity is also delayed by TIME STEP in order to avoid circularity with the output constraint, also allowing for coherence among the implementation of the production structure into the overall model structure.

Chapter 4

Trade and decoupling of fossil fuel use embedded in EU consumption

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Abstract

The EU is frequently recognized as a frontrunner in tackling climate change; however, this assessment primarily relies on production-based evaluations, overlooking the significant role of imports. We conduct a detailed consumption-based analysis of how EU consumption drives global fossil fuel use, combining input-output with structural decomposition analysis and the subsystem approach. We find that the embedded EU fossil fuel footprint has experienced a notable decline between 2000 and 2014, but at rates incompatible with 1.5 °C. We identify trade patterns to be an important upward driver of the EU footprint through intermediate production, also thwarting the impact of the energy transition with effects from changes outside the EU lagging within EU developments. Addressing these outsourcing patterns to more fossil fuel intense production could reduce the EU footprint by almost 20%. We find that more than 50% of fossil fuels embedded in imports are linked to indirect imports. Thus, we argue for the EU Carbon Border Adjustment Mechanism to include indirect imports, particularly of electricity. Yet, given the problematic role of growth, even energy transition efforts along the global supply chain will likely need to be complemented by demand side measures, potentially entailing post-growth pathways.

4.1 Introduction

Fossil fuels are the dominant driver of anthropogenic CO₂ emissions (Friedlingstein et al., 2022) and in order to keep global warming below 1.5°C, fossil fuel use has to decline immediately and rapidly (SEI et al., 2020; SEI et al., 2021) and a substantial amount cannot be extracted (IPCC, 2022b; Welsby et al., 2021). Hence, addressing climate change means to phase out fossil fuels. However, assessing this issue solely in terms of emissions overlooks lock-in effects and uncertainty in leakage (Shearer et al., 2020). It can thus paint an overoptimistic picture, with natural gas playing an increasingly important role in the global energy system (Jackson et al., 2019). Even though a switch from coal to gas reduces emissions temporarily, it still constitutes an emitting fossil fuel required to be phased out in the absence of negative emission technologies (Peters et al., 2020).

The EU provides an insightful case study since on the one hand it is one of the major fossil CO₂ emitters, but on the other hand has made most progress as a region to reduce the fossil fuel dependence of its energy system (Jackson et al., 2019) and continues to have relatively ambitious goals (European Commission, 2021c). However, many climate mitigation analyses (Eskander and Fankhauser, 2020; Peters et al., 2017) and policies take a production-based approach, focusing on national/regional production structures, and hence do not fully incorporate the responsibility of EU consumption in driving global fossil fuel use. This is a particularly crucial shortcoming for richer regions like the EU, which typically net import environmental impacts (Wiedmann and Lenzen, 2018), as shown for emissions often increasingly from lower income countries (Wang et al., 2024). Hence, carbon leakage (we refer to weak carbon, or here fossil fuel, leakage, i.e. increases in emissions abroad driven by demand, regardless of climate policies (Peters and Hertwich, 2008a)) plays an important role with implications for the effectiveness of current climate policies. This is not only due to geographical re-allocation of production structures and hence policy coverage, but also has been shown directly increase overall emissions (Hoekstra et al., 2016).

To tackle the concern of trade and embedded emissions, the European Commission has proposed a Carbon Border Adjustment Mechanism (CBAM), planned to commence in 2026 following a transitional phase. It will focus on direct emissions and in the first phase on the cement, iron and steel, aluminium, fertilisers and electricity sectors. Importers will have to buy certificates at the price of the EU emissions trading system (ETS), minus carbon prices that were due at the source of production (European Commission, 2021a; European Commission, 2021b). A CBAM addresses leakage by targeting distortions in competitiveness due to differing carbon prices and tilts the policy focus towards consumption-based emissions (Böhringer et al., 2022).

Thus, in this work, we analyse the dynamics and decoupling of fossil fuel dependence embedded in EU consumption, taking into account leakage by allocating responsibility of environmental impacts along the supply chain to the final consumers (Davis and Caldeira, 2010; Peters and Hertwich, 2008b). In order to trace the drivers behind decoupling or the lack thereof in detail, we combine an environmentally extended multiregional Input-output (EEMRIO) framework with the subsystem approach and structural decomposition analysis (SDA). In addition, we evaluate the potential of the proposed CBAM to address the fossil fuel footprint by quantifying its coverage using the subsystem approach and by simulating an artificial no-trade scenario.

We provide the first IO-SDA study of EU consumption focusing explicitly on fossil fuels complementing existing emission and energy studies (see Fernández-Amador et al., 2023; Karstensen et al., 2018; Wood et al., 2020b for the EU). As aforementioned, emission-based studies can provide a distorted picture whereas total energy use omit the role of the energy mix and hence does not adequately reflect its environmental impact. In addition, we add crucial level of detail to existing studies by singling out the energy transition and its regional developments and combining the subsystem approach with SDA. This allows us to trace drivers and policy implications more rigorously. Finally, addressing the CBAM, we introduce a novel application of subsystem approach allowing us to provide the first economy wide consumption-based quantification of directly and indirectly imported fossil fuel use into the EU.

The paper is organized as follows: Section 4.2 introduces the application of the EEMRIO-SDA and the subsystem framework. The quantitative results are presented in Section 4.3 and policy implications and methodological issues are discussed in Section 4.4. Finally, Section 4.5 summarizes the main findings.

4.2 Material and Methods

Figure 4.1 illustrates the application and outcomes of the individual methodologies used in this paper. We use the Subsystem Approach (production structure) to answer questions about the magnitude of the total embedded fossil fuel footprint (FP), how it is structured and what their dynamics are. Subsequently, we apply a Structural Decomposition Analysis to FP and its components to quantify their respective drivers. The simulation of EU production standards provides an estimate of the magnitude of the overall impact of outsourcing and hence potential gains in tackling this issue. Finally, the novel application of the Subsystem Approach (production & trade structure) is applied to quantify direct and indirect embedded imports.

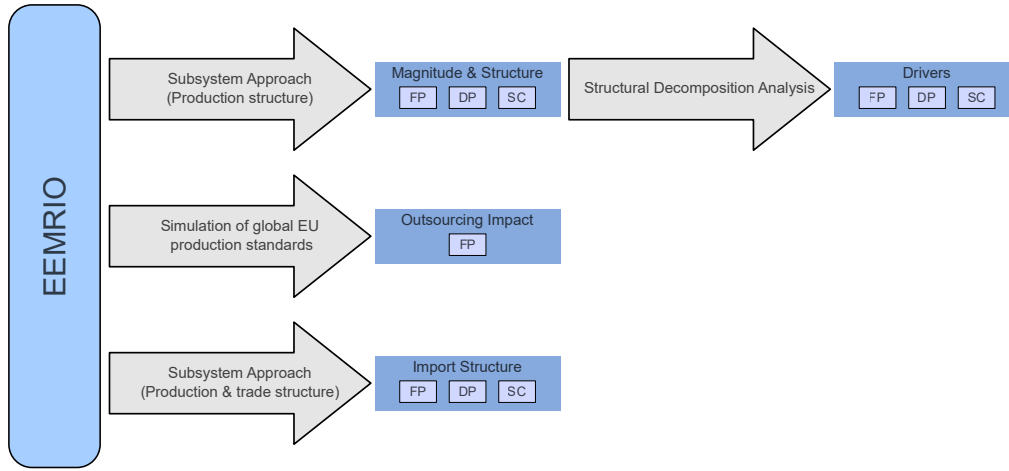


Figure 4.1: Flowchart linking the methodologies used and the results obtained. FP (fossil fuel footprint), DP (fossil fuels used in direct final production) and SC (fossil fuel use embedded as intermediate inputs) indicate which components the methodology is applied to and/or provides results for.

4.2.1 Footprints

We first apply the standard Leontief EEMRIO approach in order to calculate consumption-based fossil fuel use. Total output x is given by $Ax + y = x$, where A is the matrix of technical coefficients, capturing direct input requirements per unit output and y is final demand. Isolating x produces $(I - A)^{-1}y = Ly = x$, with L being the so called Leontief inverse, where element L_{ij}^{rs} reflects total inputs (direct and indirect) required by sector j in region s from sector i in region r per unit output.

In order to trace environmental impacts, the environmental extension vector (e) assigns the impact (here TJ of fossil fuel use) per unit output to each sector in each region. Subsequently, L is pre-multiplied with the diagonalised vector $\hat{e}$.

In addition, we modify the final demand vector y to a matrix $\hat{Y}$, where each region to region demand vector is diagonalised. The final matrix Q produces total environmental impact by sector and region of final consumption if summed over rows (i.e. embedded environmental footprint), and region and sector of origin if summed over columns. It is given by

$$Q = \hat{e}L\hat{Y}. \quad (4.1)$$

The final demand matrix aggregates final consumption from the EU27 countries, yet note that due to data issues the Netherlands is excluded from our analysis. We include consumption of households, non-profit organisations serving households and the government as they can all be linked directly to lifestyles.

For more details, the Supplementary Material provides more data on the analysis and Miller and

Blair (2009) present a thorough explanation of IO analysis.

4.2.2 Subsystem Approach

To distinguish different components in the chain of production within the environmental footprint, we apply the subsystem approach. The general approach is described in detail in Alcántara and Padilla (2009) and features a range of applications tracing environmental impacts such as Ma et al. (2019). In its typical application it reveals the production and emission structure of sectors, rendering it particularly useful for more focused policy recommendations (Alcántara et al., 2017).

Essentially, the framework distinguishes between intermediate inputs required on the one hand and final production on the other. It does so by using from the standard IO approach that $x = Ly$ and inserting this expression for x into the basic equation $Ax + y = x$, resulting in $ALy + y = x$ (Alcántara and Padilla, 2009).

On the one hand, we use this approach to distinguish the FP between fossil fuels used in direct final production (DP) and embedded as intermediate inputs through the rest of the supply chain (SC):

$$Q = \underbrace{\hat{e}AL\hat{Y}}_{\text{SC}} + \underbrace{\hat{e}\hat{Y}}_{\text{DP}} = Q_{SC} + Q_{DP}. \quad (4.2)$$

This is the production structure subsystem approach in Figure 4.1. To obtain respective consumption-based intensities of final demand, we simply omit $\hat{Y}$ in equations (4.1),(4.2), i.e. we divide FP/SC/DP by chain linked final demand.

On the other hand, to address the potential coverage of the EU CBAM, we rewrite (4.2) in regional block matrices

$$\begin{bmatrix} \widehat{e^C} \\ \widehat{e^N} \end{bmatrix} \begin{bmatrix} A^{CC} & A^{CN} \\ A^{NC} & A^{NN} \end{bmatrix} \begin{bmatrix} L^{CC} & L^{CN} \\ L^{NC} & L^{NN} \end{bmatrix} \begin{bmatrix} \hat{Y}^C \\ \hat{Y}^N \end{bmatrix} + \begin{bmatrix} \widehat{e^C} \\ \widehat{e^N} \end{bmatrix} \begin{bmatrix} \hat{Y}^C \\ \hat{Y}^N \end{bmatrix} = \begin{bmatrix} Q^C \\ Q^N \end{bmatrix}, \quad (4.3)$$

where the superscript C indicates all countries that are covered by or linked to the EU ETS directly (EU27, Norway and Switzerland) and N denotes all other countries who are targeted by the CBAM. Consequently, the matrix Q^C (Q^N) captures the environmental impact produced in C (N) to meet EU demand. This application is what Figure 4.1 refers to as the production and trade structure subsystem approach.

To focus on the coverage of the EU carbon price system (ETS and CBAM), we solve the SC

part of (4.3), which yields

$$\begin{aligned} Q_{SC}^C &= \hat{e}^C (A^{CC} L^{CC} + A^{CN} L^{NC}) \hat{Y}^C + \hat{e}^C (A^{CC} L^{CN} + A^{CN} L^{NN}) \hat{Y}^N \\ Q_{SC}^N &= \hat{e}^N (\underbrace{A^{NC} L^{CC}}_{\text{direct}} + \underbrace{A^{NN} L^{NC}}_{\text{indirect}}) \hat{Y}^C + \hat{e}^N (\underbrace{A^{NC} L^{CN}}_{\text{direct}} + \underbrace{A^{NN} L^{NN}}_{\text{indirect}}) \hat{Y}^N \end{aligned} \quad (4.4)$$

Here, Q_{SC}^C derives from production occurring in the EU ETS area, which can be completely covered by the EU carbon price. On the other hand, we distinguish Q_{SC}^N between direct and indirect embedded imports with the latter being systematically out of scope of the EU carbon price system according to current proposals.

4.2.3 Simulation

To establish an estimate of the scope of CBAM potential in terms of magnitude of FP reduction, we simulate global EU production standards along the whole production chain (SC and DP). We create an average EU A_{EU} – including all imports of intermediate inputs, but assumed to be produced in the EU – and e_{EU} – reflecting average fossil fuel intensity for EU production by sectors – from WIOD data for the year 2014 and simulate that all EU demand is met with this production structure: $Q_{EU} = \hat{e}_{EU} L_{EU} \hat{Y}_{EU}$, where $L_{EU} = (I - A_{EU})^{-1}$ and Y_{EU} reflects all EU demand (imports and domestic) now assumed to be only directed at EU producers.

4.2.4 Structural Decomposition Analysis

As a final step in our analysis, we assess the dynamic drivers of the environmental footprint and its components. We mainly follow Dietzenbacher et al., 2020 and decompose the FP into the following multiplicative components which drive its development:

- Fossil fuel intensity: total energy intensity, energy mix
- Leontief: total sectoral input requirements, regional origin of inputs
- Final demand: population, total demand per capita, distribution of consumption among sectors ($\times 2$), regional origin of production.

We approximate the decomposition form using the average of the two polar forms as illustrated in Dietzenbacher and Los (1998). Finally, we combine data in previous year prices and current prices and chain the results (Arto and Dietzenbacher, 2014). This means that to analyse the changes between year t and $t - 1$, we use data of year $t - 1$ in current prices and of year t in previous year prices, so that only real changes are considered. Subsequently, we chain the

results over the period of analysis.

For the full decomposition and more details see the Supplementary Material.

4.2.5 Data

Our main data source is the MRIO database from the WIOD 2016 release (Timmer et al., 2015), which covers 27 EU countries, 16 other major countries and a rest of the world (ROW) region with 56×56 industry resolution from 2000 to 2014. Please note, that the aforementioned exclusion of the Netherlands in this study does not only apply to final demand, but the whole input-output structure. To avoid double accounting we use the official WIOD emission relevant energy use extensions (Corsatea et al., 2019), removing electricity, heat use and losses (Usubiaga-Liaño et al., 2021). Population data is obtained from Eurostat.

To address the issue that the WIOD ends in 2014 and its last years coincide with the Great Recession and the ensuing European debt crisis (subsequently referred to as EU Crises), we complement our analysis with more recent data: Friedlingstein et al. (2022) (updated from Peters et al., 2011) and the Exiobase 3 EEMRIO dataset (Stadler et al., 2021), described in Stadler et al. (2018).

It is important to note, that for these more recent analysis, we study GHG emissions, mainly due to issues of data uncertainty and policy relevance. To maintain comparability and coherence with the WIOD analysis, when analysing Exiobase, we only include GHG emissions from combustion, remove the Netherlands, consider consumption in final demand and use the industry-by-industry format. For a discussion on the role of growth, we additionally collect GDP and consumption data from Eurostat. The analysis is coded in Python, using the pymrio tool (Stadler, 2021) to parse Exiobase data.

4.3 Results

4.3.1 Fossil Fuel Footprints

Figure 4.2 illustrates that while the global FP (d) from consumption increased relatively continuously throughout the analysed period of time, the EU FP (a) decreased since the onset of the EU Crises. Consequently, EU consumption drives a decreasing share of global FP, dropping to below 11% in 2014. Despite its improvements, EU consumption remains a significant driver of global fossil fuel use with less than 7% of the global population driving more than 10% of fossil fuel use.

SC (c) remains the dominant part throughout the period of analysis. DP (b) has been decreasing

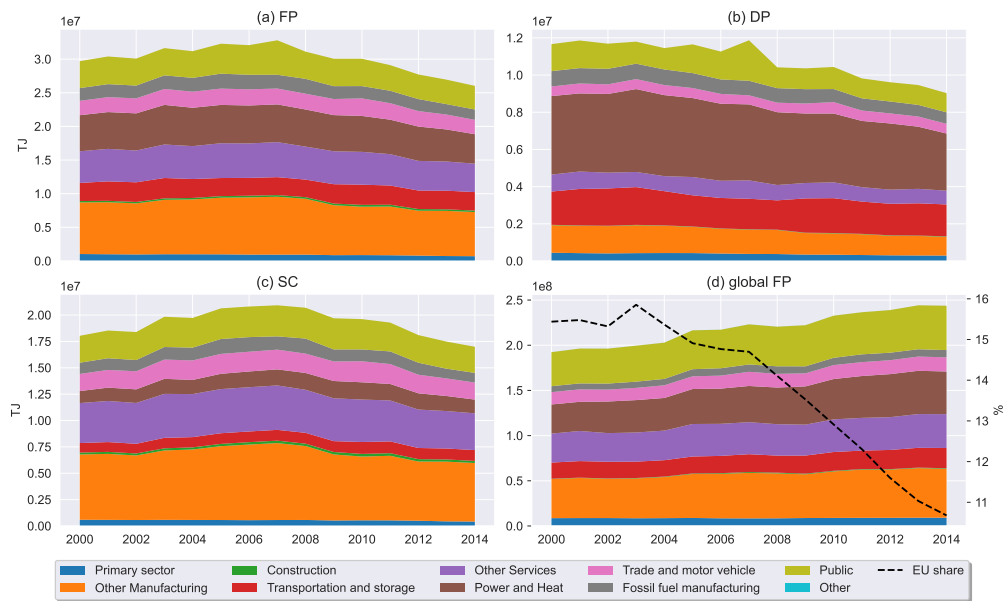


Figure 4.2: The evolution of the EU (a) and global (d) fossil fuel footprint between 2000 and 2014 in TJ. The EU footprint is additionally divided into fossil fuels used in final production (b) and fossil fuel use embedded in intermediate production (c). The dashed line represents the share of EU with respect to global fossil fuel footprint.

steadily, while, after initial increases, SC only started to decline after 2008 with the onset of the EU Crises. Over the whole period, DP fell by almost 25%, whereas SC only fell by around 5%, implying an increasing relative importance of SC (Figure 4.3 (d)).

The overlap of the acceleration in FP and particularly SC decline with the EU crises however casts doubts on the actual decoupling occurring. Consulting consumption-based intensities (Figure 4.A.1; references with the prefix “A” refer to the Appendix A), we still observe continuous declines over the whole period and hence relative decoupling. Yet, given that the average yearly rate of decline is below 2% for FP, with SC featuring a rate of only 1.3% and clearly lagging behind, a continuous decline in the face of growing demand remains questionable.

In order to scrutinise which developments could facilitate and which impede decoupling of FP and its components, we subsequently analyse their underlying drivers.

4.3.2 Drivers

As depicted in Figure 4.3 (a–c) the quantity of final demand is a major upwards driver of fossil fuel use until the EU crises, opposed primarily by efficiency improvements. This applies to FP and each of its components.

The second most important driver reducing FP is the composition of final demand among different sectors. Its effect is less volatile and more pronounced in SC, than in DP. This illustrates,

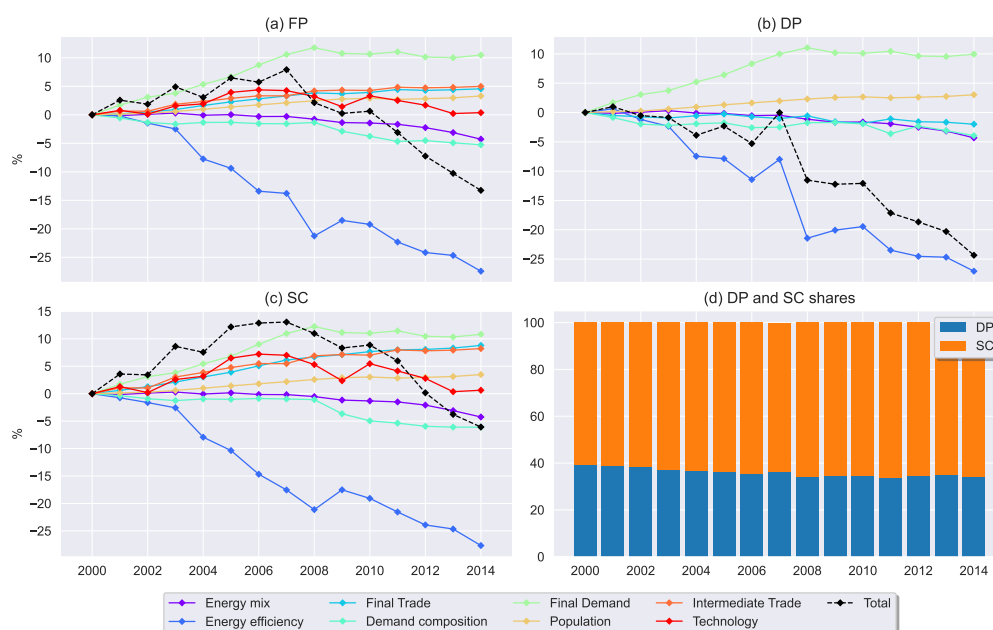


Figure 4.3: The relative cumulative impact of each driver alongside total relative cumulative change respectively for the fossil fuel footprint (a) and its components: fossil fuels used in final (b) and in intermediate (c) production. The barplot (d) illustrates the dynamics of the share of final and intermediate production in the fossil fuel footprint.

that the shift in demand, particularly away from the primary sector, manufacturing and construction has intuitively reduced DP, but more crucially, particularly the shift away from construction and manufacturing has reduced SC fossil fuel use, as it relies on a highly fossil fuel intense supply chain (see Figure 4.A.2).

Interestingly, the energy mix, here defined as share of fossil fuel energy over total energy required per unit production, reduced FP and its components by less than 5% until 2014. However, note that this transition is accelerating rapidly since 2007 and continues after 2014 (BP, 2022; Eurostat, 2022), suggesting a more extensive role in recent years. Also notable is that the energy mix effect mainly coincides with the period of economic crises in the EU, contributing to a reduction in absolute fossil fuel use.

Technology (here referring to the input-output structure, describing the inputs required per unit production independent of sourcing) dynamics suggest that input reduction or substitution to less fossil fuel intense inputs has not materialised yet when considering the whole supply chain. A growing population in the EU also has lead to increases in demand and hence FP.

4.3.2.1 Trade

Trade plays a more complex role in the dynamics of FP, and prompts the fundamental difference between DP and SC dynamics. On the one hand, there is little outsourcing of DP (less than

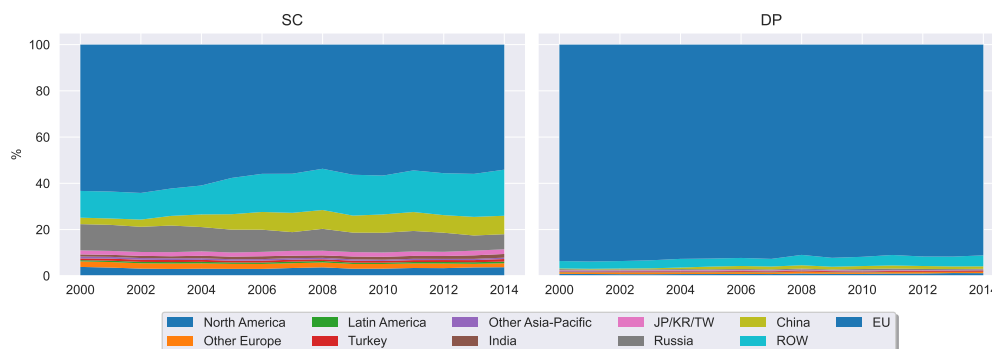


Figure 4.4: The contribution to intermediate (SC) and final (DP) production of the EU fossil fuel footprint by region of origin.

10%, Figure 4.4) and the shift in trade patterns seem to lead to slightly less fossil fuel intense DP processes. In contrast, trade is the most dominant upward driver of SC, with intermediate and final trade having impacts of similar magnitude. This illustrates that on the one side, input production is directly shifted to regions with more fossil fuel intense production. On the other hand it shows that even though the trade pattern of final demand does not tend to lead to higher DP fossil fuel use, it typically induces significantly more fossil fuel intensive supply chains, rendering the outsourcing of fossil fuel use even more opaque. SC is significantly more outsourced than DP, with almost half of SC fossil fuel use occurring outside the EU (Figure 4.4). This is not only due to a fall in fossil fuel use for SC production within the EU, but also driven by an absolute increase outside EU borders (Figure 4.A.3).

Moreover, we observe that the energy transition has been significantly more impactful in the EU than for production abroad (Figure 4.5). Even though almost 50% of SC occurs outside EU borders, shifts in EU energy efficiency and particularly in the energy mix far outpace improvements in the rest of the supply chain. This indicates that outsourcing production has an increasingly detrimental impact on FP, shifting production to relatively more environmental harmful production structures. It also underlines that the shift to renewables alone within the EU as only reduced FP by little over 4% *ceteris paribus* and that there is significant potential to reduce FP through developments outside the EU or shifts in sourcing patterns, with trade being the dominant upwards driver of FP next to demand.

One measure to address outsourcing and developments outside EU borders is the EU CBAM. Thus, we subsequently estimate the potential impact of this policy on the EU FP.

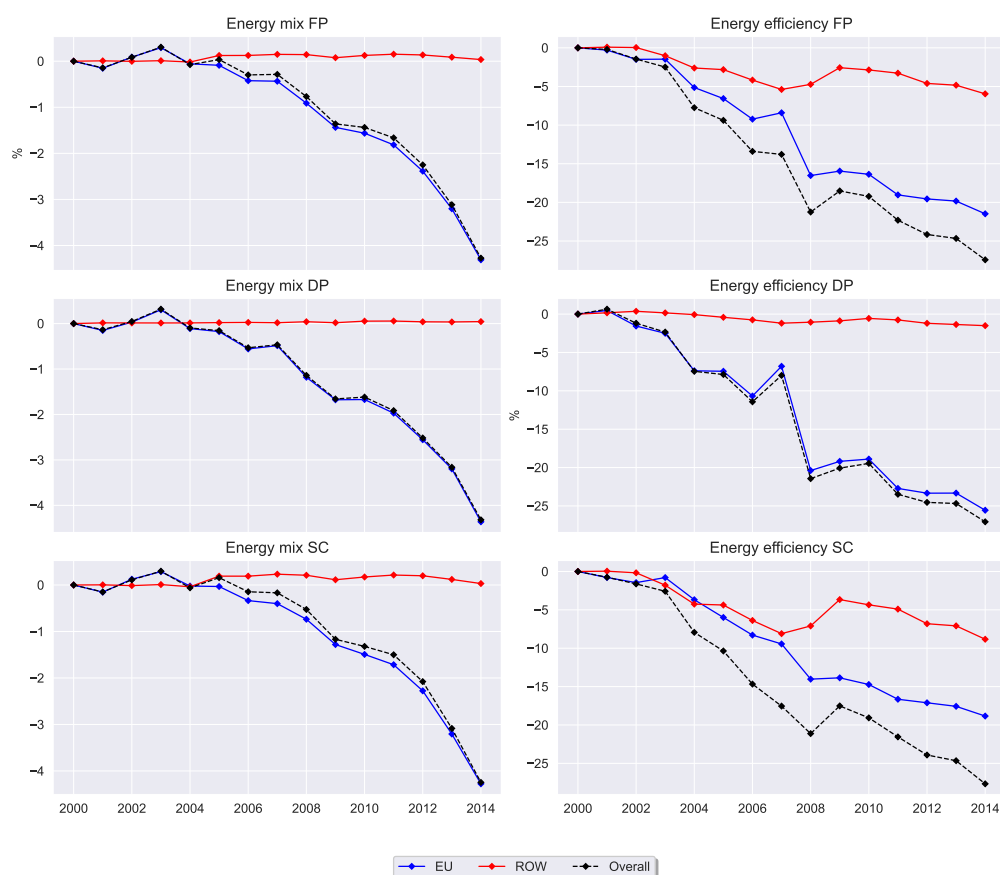


Figure 4.5: The relative cumulative impact of energy efficiency and energy mix split into the impact due to the respective development within and outside the EU (ROW) respectively for the fossil fuel footprint (a) and its components: fossil fuels used in final (b) and in intermediate (c) production.

4.3.3 The Carbon Border Adjustment Mechanism

Simulating an EU production structure meeting all EU demand, we find that, in 2014, moving the whole supply chain to average EU production standards would reduce FP by around 19%, illustrating the potential of addressing outsourcing to more fossil fuel intense production structures.

However, given the highly fragmented global supply chain, the EU CBAM faces potential issues of coverage, as it systematically excludes indirect imports. Production within the EU still makes up the vast majority of fossil fuel use (Figure 4.6). However, whereas this component is falling, fossil fuel use driven by imports is increasing due to a rise in indirect imports of fossil fuel use. This increasing importance of developments outside EU borders for the EU FP again underlines the relevance of the CBAM. However, we find that the EU carbon pricing mechanism systematically omits an increasing share of FP, almost 1/4 in 2014, and specifically the CBAM, even if

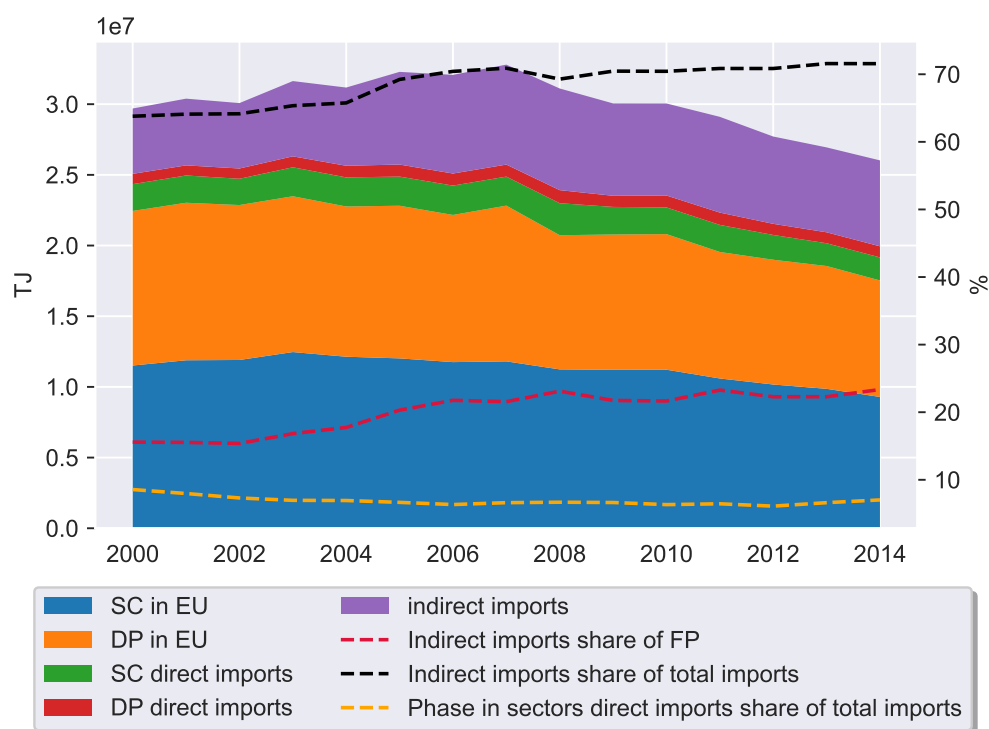


Figure 4.6: The EU fossil fuel footprint disaggregated according to origin (within or outside EU) and element in the production chain (final or intermediate production). Additionally, imports are split into direct and indirect imports. The dashed lines represent the dynamics of the labelled shares.

applied to all sectors' direct imports, would not address more than 70% of imported fossil fuel use. The sectors covered in the first phase will account for less than 10% of imported fossil fuel use, which is a clear overestimation due to the coarse sectoral detail in the database.

In order to validate and update the CBAM analysis to the year 2019, also with more policy relevant environmental extensions, we repeat the same analysis with Exiobase data (Figure 4.A.7). In general, we confirm the relative magnitudes of the different components, albeit direct imports play a relatively bigger role. Indirect imports make up a smaller and relatively constant share of total imports that is however still above 50%. The share of indirect imports relative to the whole FP is stagnating since the end of the 2000s between 17 and 19% and actually seems to have fallen after 2014. Finally, Exiobase allows for a more targeted analysis of the phase-in sectors due to its higher sectoral detail, revealing that only around 2.5% of all imported emissions would be addressed in 2019. Overall Exiobase confirms and extends our qualitative results for emissions to 2019, underlining the importance of indirect imports.

4.4 Discussion

In line with the energy and emission literature on the EU, we find that consumption-based impacts increase until the EU crises and fall subsequently (Karstensen et al., 2018; Wood et al., 2020b), with intensity improvements and consumption levels being dominant downwards and upwards drivers until the EU crises (Fernández-Amador et al., 2023; Wood et al., 2020b). Overall, the EU is making some progress towards reducing its fossil fuel dependence, but the pace is far from the 6% annually called for between 2020 and 2030 to limit warming to 1.5 degrees (SEI et al., 2020). Required levels of reduction are not even reached in the face of economic stagnation (see also Le Quéré et al., 2019 and Lamb et al., 2021 for emissions).

We do not only find a clear lack of the degree of decoupling but query whether meaningful continuous absolute decoupling of the EU FP has occurred, given the apparent reliance of reductions on economic underperformance with the FP intensity falling by less than 2% per year on average. We identify SC as a significant drag on decoupling developments.

Consulting more recent consumption-based *emission* data seems to confirm the reliance on stagnating economic growth for the accelerated reductions of the footprint between 2008 and 2014. An update of Peters et al. (2011) in Friedlingstein et al. (2022) shows that consumption-based fossil CO₂ emission reductions of the EU level off around the mid 2010s, coinciding with the pick up of stable economic growth and only fall again with the onset of the Covid-19 pandemic (Figure 4.A.5). We validate these developments using the Exiobase dataset, as it allows better comparability with our specific analysis (see Section 4.2.5), also confirming the stagnation of reductions in FP (Figure 4.A.6). We find this stagnation for SC as well as for DP but note that this recent data exhibits notable uncertainty (Stadler et al., 2021). A structural shift that facilitates continuous absolute decoupling, particularly at a pace in line with limiting warming to 1.5 °C, has clearly not occurred yet.

This lack of decoupling of consumption-based measures is confirmed for EU emissions and energy use (Kan et al., 2019; Papież et al., 2022), with Kan et al. (2019) finding a mixed picture when it comes to the decoupling of different fossil fuels.

We find that trade is an important driver counteracting decoupling and only slightly outpaced by the scale of final demand. Its negative impact is channeled entirely through SC. This increasing relative importance of SC is in line with Hertwich and Wood (2018).

The emission literature confirms that a large part (almost 1/3) of the EU footprint is outsourced (Wood et al., 2020b) and predicts an increasingly important relative role of imports Wood et al. (2020a) which we find for fossil fuels. In addition, Wu and Chen (2017)'s finding that the vast majority of energy embodied in trade is associated to intermediate trade supports our results

that trade is mainly driving SC. The literature also supports trade being an upwards driver of the EU footprint for emissions (Wood et al., 2020b) and energy (Fernández-Amador et al., 2023). However, the effect of trade in our analysis is significantly larger and more in line with Hoekstra et al. (2016)'s global findings for emissions. Reasons to explain these variations range from differences in the environmental extension to distinct scopes footprints and final demand and diverse datasets. In addition, Zhang et al. (2021)'s findings underline the importance of trade. They find the emission intensity of the global value chain to be higher than of the EU and identify regional sourcing patterns as the main upwards driver of global value chains. Overall the literature generally tends to qualitatively confirm the upwards driver of trade on EU footprints. Thus, the role of trade and developments abroad are crucial for decoupling dynamics, and focusing only on a domestic energy transition is a profoundly narrow perspective, with shifts in the energy mix in the EU only contributing to little over 4% decline in FP. Here, the CBAM constitutes a promising tool, but in its currently proposed form will not cover the majority of embedded fossil fuel imports, which constitute indirect imports.

4.4.1 Implications for Decoupling

Evaluating individual drivers of FP and its components in detail, we aim to establish underlying reasons for why FP fell during the EU crises years but a structural shift towards continuous absolute decoupling has not materialized yet.

On the one hand, some of the drivers might only exhibit temporal changes with a return to pre-crisis conditions, endangering the continuous reduction of SC and FP. Certainly, rising levels of consumption will have an impact on the scale of economic activity and fossil fuel use. Another potentially temporal driver is the change in demand composition, due to cyclical demand behaviour. For instance Berger and Vavra (2014) show cyclically distinct behaviour of durable goods. This interpretation is in line with Eskander and Fankhauser (2020)'s findings of a procyclical movement of emission intensities. Also Song et al. (2022) find a significant role of consumption composition for consumption-based emissions for the U.S. and remark on the role of the economic recession on different types of products.

On the other hand, the overall impacts of developments in the energy transition interact with other drivers. Firstly, rising levels of consumption might again continue to counteract efficiency improvements. Efficiency improvements have not led to significant decoupling in SC until the economic crises, potentially also due to rebound effects which can be very significant ($\geq 50\%$) (Brockway et al., 2021). In line with Le Quéré et al. (2019), they only achieved substantial reductions in energy and fossil fuel use during phases of economic underperformance. The role of the energy mix will also be affected by the scale of demand and economic activity. The im-

pact of the energy mix has gained traction around the time of the financial crisis, but we cannot ascertain in how far the pace and displacement of fossil fuel use due to the energy mix was fostered by the EU Crises itself (Le Quéré et al., 2019). A rising energy demand clearly hampers the improvement of the energy mix and replacement of fossil fuels *ceteris paribus*. This notion is also supported by data in BP (2022) and Eurostat (2022) which agree that there is a temporary (2015 – 2017) increase in the fossil fuel share in Europe coinciding with the end of the EU Crises. However, the share continues to decrease after 2017, illustrating that the energy mix can also move away from fossil fuels in the face economic growth. That growth is a potential hindrance, rendering decarbonization more challenging is in line with a range of studies analysing the historic decarbonization of regions (e.g. Le Quéré et al., 2019) mainly from a territorial perspective.

Furthermore, trade patterns interact with the impact of the energy transition on the EU FP. Whereas relatively ambitious climate and energy measures in the EU have improved production structures within its borders, the impact of the transition in regions exporting into the EU is lagging behind (Figure 4.5), thwarting the impact of the energy transition on the EU FP. Our period of analysis spans the introduction of important EU policies including the EU Emission Trading System (ETS) in 2005 and the EU Energy Service Directive in 2006, which however, in line with our findings, are found to have limited impact on energy and emission footprint measures, consistent with leakage (Fernández-Amador et al., 2023; Papież et al., 2022). Here, Wang et al. (2024) find that for many EU countries an increasing share of embedded emissions originate in lower income countries, which tend to feature higher emission intensities. To enable a structural shift towards sufficient decoupling and reductions in FP, these interactions with trade and growth have to be taken into account carefully, in particular with respect to SC.

4.4.2 Policy Implications

Our study underlines the issue of policy coverage in global supply chains. Energy transition advancements within the EU have increasingly limited leverage on the EU FP, primarily due to developments in SC. One policy option to address this issue and promote an energy transition along the global supply chain is the CBAM. It has enormous potential to reduce the EU FP as indicated in subsection 4.3.3, but our analysis in subsection 4.3.3 underlines the importance of indirect imports in the EU FP, as also noted in Böhringer et al. (2022). We find that direct and indirect imports feature significantly different structures in terms of country as well as sector of origin (Figure 4.A.4). Thus, only addressing direct impacts will not necessarily have effective ripple effects on indirect imports. However, including electricity used to produce imported goods, as indicated as a future option by the CBAM proposal, could already be a significant

improvement, covering a major part of indirectly imported fossil fuel use (Figure 4.A.4). Still, substantial practical, legal and political challenges around the CBAM remain (Böhringer et al., 2022).

Yet, even if covering indirect emissions, the CBAM should not remain the only tool addressing emissions abroad, as by itself it is associated with serious concerns about equity (Grubb et al., 2022). Complementary measures could help to phase out fossil fuels along international supply chains, while alleviating some concerns about equity. These can be directly in the form of clean technology transfers or support through climate finance (Wang et al., 2024), for instance using the revenue generated from the CBAM (Böhringer et al., 2022).

Still, energy efficiency measures face rebound issues and eventual thermodynamic limits. Moreover, a shift in energy mix faces technical and institutional difficulties (Davis et al., 2018). Crucially, as aforementioned economic growth can be a significant hindrance to the energy transition and the displacement of fossil fuels.

Given the urgency and scale of the challenge, this underlines the importance of complementing energy transition efforts with demand side measures (Creutzig et al., 2018; Grubb et al., 2020). Concrete policy proposals include infrastructure improvements, charges and taxes and recycling or product lifespan requirements (Grubb et al., 2020). Applied in the EU, they can facilitate the energy transition and the phasing out of fossil fuels globally by reducing the pressure of energy demand along the entire supply chain. Furthermore, they can mitigate rebound effects from energy efficiency improvements. Note that we find the FP intensity to fall by less than 2% per year on average illustrating that stagnating demand is not sufficient, but demand would need to change more fundamentally while the energy transition would need to be expanded rapidly. This might imply post-growth pathways (Hickel and Kallis, 2020; Hickel et al., 2021), even for climate ambitious regions like the EU.

4.4.3 Methods and Limitations

Typical for IO analysis, this analysis is prone to issues due to aggregation bias and linearity assumptions. Aggregating EU countries into one EU region certainly results in the loss of country specific characteristics, but has apparent political relevance. In addition, results for large economies like the EU tend to be relatively robust (Moran and Wood, 2014; Wood et al., 2020b).

Furthermore, note that the *ceteris paribus* interpretation of the SDA has to be scrutinised with caution. The drivers cannot be assumed to be independent. For instance energy efficiency improvements can affect economic growth or sector specific demand via the rebound effect, or the energy mix can alter energy efficiency.

Moreover, we systematically underestimate the EU FP by excluding the Netherlands, direct final use of fossil fuels and gross fixed capital formation (GFCF). This is because of data issues, the specific scope of the study and in order to avoid the introduction of an additional bias respectively. GFCF can distort consumption-based accounting, as investment demand is partly driven by production for exports. Hence, GFCF should not be allocated to the country where it is demanded, but to the country where it is producing for (Södersten et al., 2018).

The impact assessment of the EU CBAM is mainly for illustrative purposes, not exactly quantifying its potential impact. On the one hand the simulation of global EU production structures is only a thought experiment, since many production activities like mining occur in certain locations outside the EU due to the location of natural resources (Grubb et al., 2020), which is why the EU mining industry and its production structure is not representative to a mining industry that would produce all products required by EU consumption. Also note that industries outside the EU can produce less fossil fuel intensively than within the EU. Thus, the resulting change does not constitute an upper limit on the possible emission reduction due to sourcing patterns. On the other hand, the coverage of the EU carbon price system crucially depends not only on the type of imports but on the sectoral coverage of the policies, which we neglect given the coarse sectoral detail of the WIOD.

Finally, we chose the WIOD even though the timeseries ends in 2014 as it provides current year price and previous year price data, allowing to remove price effects in a structural decomposition analysis. When updating the fossil fuel based results derived from this data with emission data from other databases we expect a certain comparability of the results given the dominance of fossil fuels as driver of emissions (Friedlingstein et al., 2022). Also note, that a reason fossil fuels were used until 2014 was that emission reductions can paint a distorted positive picture. However, here we use the data to confirm the absence of significant reductions, which in all likelihood implies an absence of significant fossil fuel reductions.

4.5 Conclusion

Applying the IO subsystem approach in combination with SDA to analyse the embedded fossil fuel footprint of EU consumption, we find that the EU FP has declined steadily, but at an insufficient rate to limit global warming to 1.5 °C.

We identify trade as one important driver counteracting decoupling and pushing up FP, with its adverse impact entirely driven by intermediate production. In addition, trade interacts with the effects of the energy transition on FP, with the impact of the energy transition in regions exporting into the EU lagging behind EU developments. The shift in energy mix in the EU only

contributed to little over 4% decline in FP. This underlines the importance of extending a territorial policy focus, promoting an energy transition along the global supply chain, in particular to address SC. We find that targeting outsourcing patterns to more fossil fuel intense production could reduce the EU FP by almost 20%. The EU CBAM has the potential to provide a suitable tool to address this issue. However, to effectively reduce the fossil fuel intensity of EU consumption, the EU CBAM has to incorporate indirect imports, which constitute the majority of embedded fossil fuel imports. Here, including indirect imports of electricity would already pose a significant improvement. Yet, given the urgency and scale of the challenge, in combination with the problematic role of economic growth, our results underline the importance to complement the supply side transition with demand side measures, potentially involving post-growth pathways.

Future research can provide more targeted policy recommendations by reducing the geographical scale or focus on specific sectors or households, also when evaluating the CBAM coverage and impact. More recent data will allow to trace drivers, in particular the impact of the energy transition, after the EU Crises, providing crucial implications about interactions with economic growth and potential to achieve sufficient decoupling.

CRedit authorship contribution statement

Till Heydenreich: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The code to replicate the results can be found in the Supplementary Material.

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Appendix

4.A Additional Graphs

4.A.1 Fossil fuel intensities

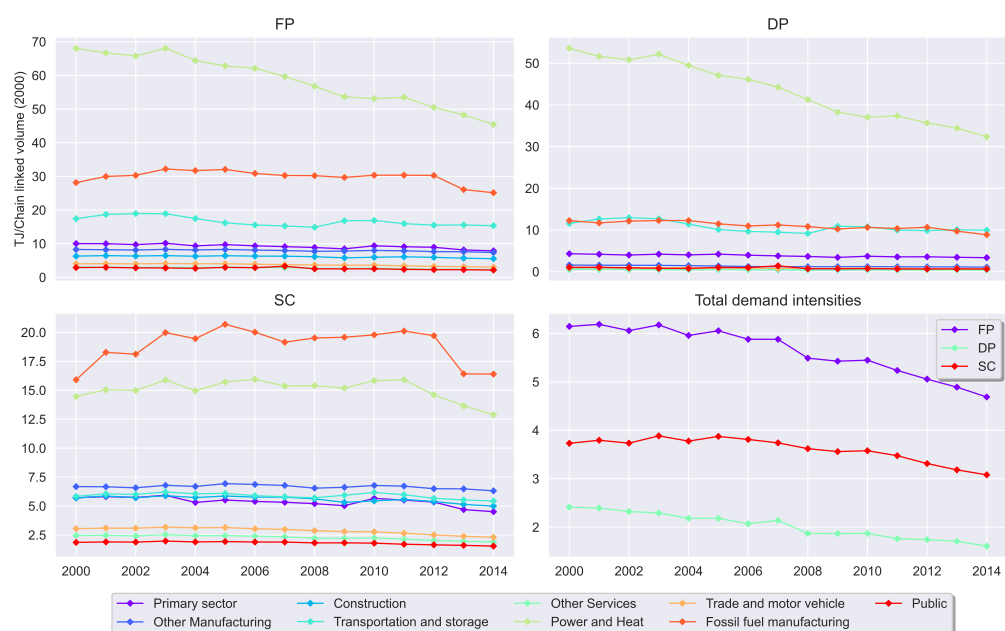
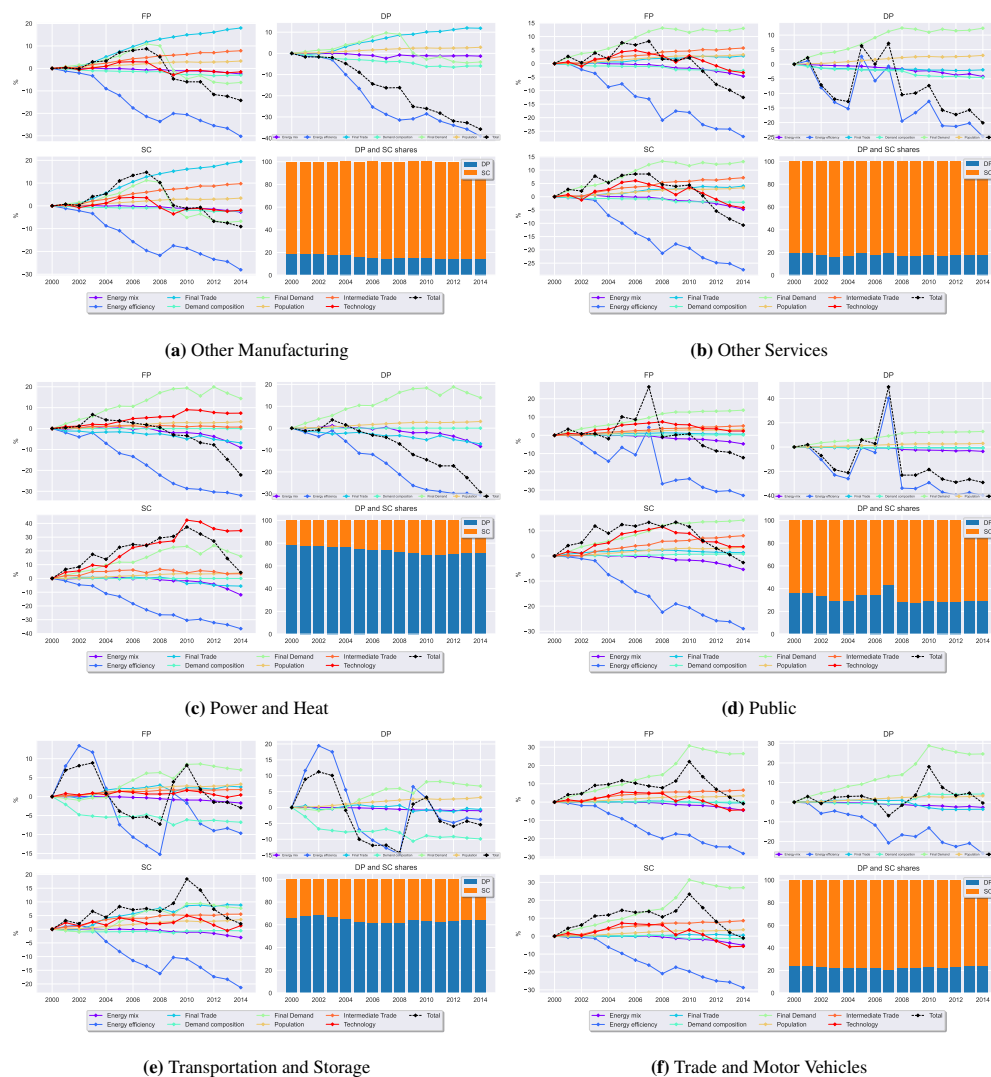


Figure 4.A.1: The dynamics of fossil fuel intensities from a consumption-based accounting framework per chain linked volume of final consumption. The total intensities for each industry category (a) are split into intensities linked to final (b) and intermediate (c) production, as are intensities for average overall consumption (d).

4.A.2 Structural decomposition analysis results by good



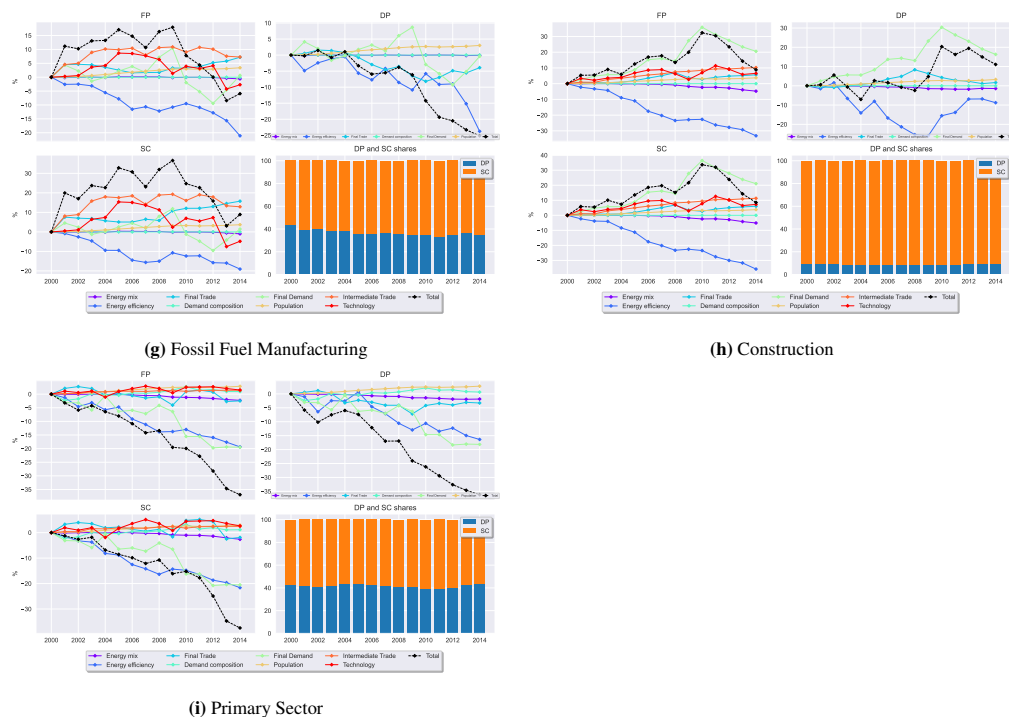


Figure 4.A.2: The relative cumulative impact of each driver alongside total relative cumulative change for sector specific fossil fuel footprint and its components, fossil fuels used in final and in intermediate production. The barplot illustrates the dynamics of the share of final and intermediate production in the fossil fuel footprint of sector specific demand.

4.A.3 Input structure

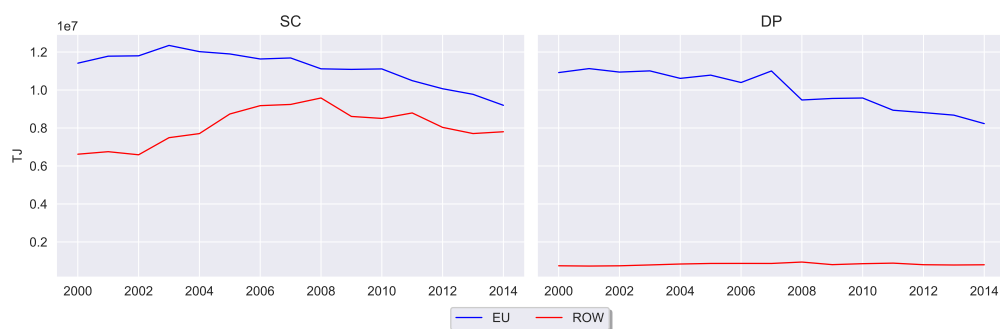


Figure 4.A.3: Absolute levels of fossil fuels embedded in final and intermediate production of the EU fossil fuel footprint by EU and non-EU (ROW) origin.

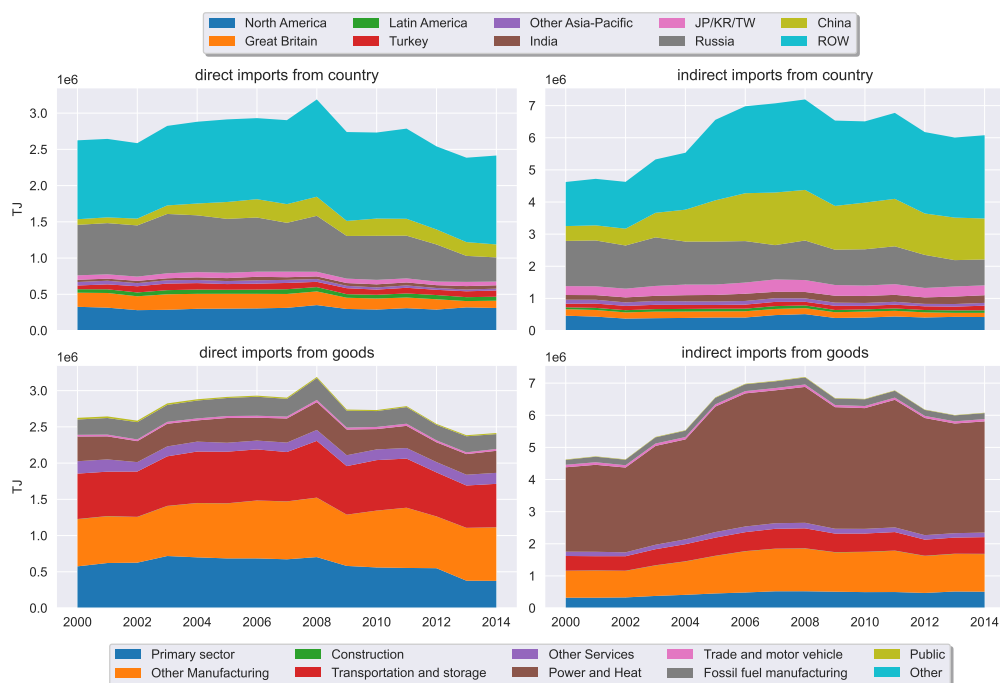


Figure 4.A.4: Fossil fuels embedded in direct and indirect imports into the EU, structured by region and sector of origin respectively.

4.A.4 Emissions

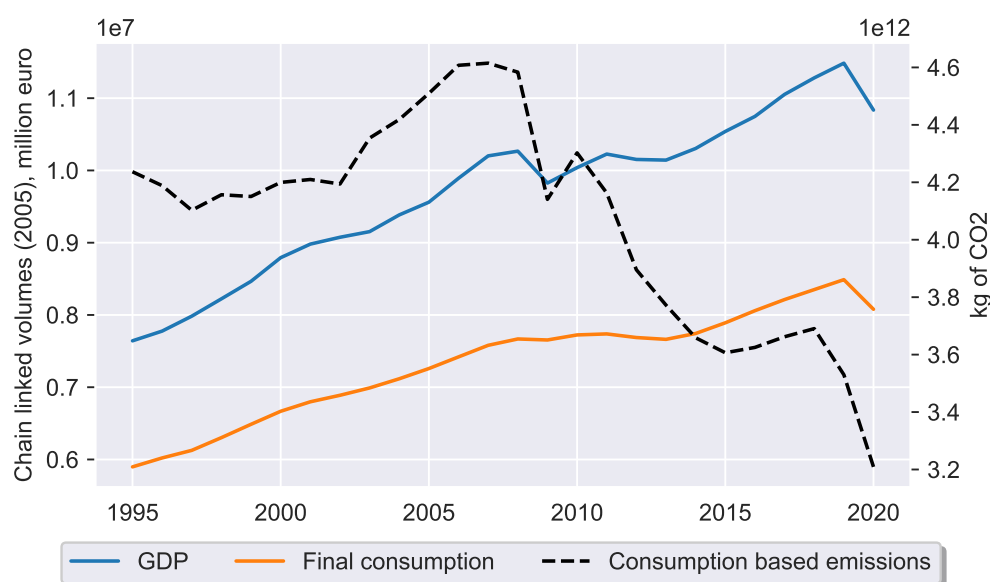


Figure 4.A.5: EU 27 GDP, consumption and fossil CO₂ emission footprint. Data sources: Eurostat (GDP and consumption) and update of Peters et al. (2011) in Friedlingstein et al. (2022) (emission footprint).

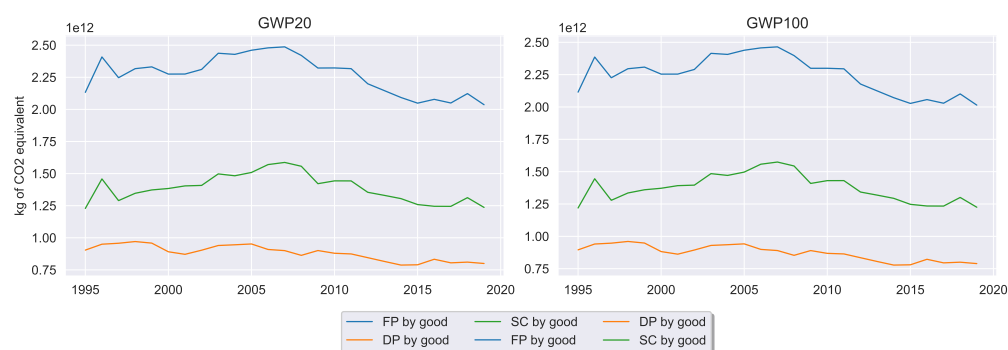


Figure 4.A.6: The evolution of the EU emission footprint, using GWP20 and GWP100 data, decomposed into fossil fuels embedded in intermediate and final production. Based on Exiobase data.

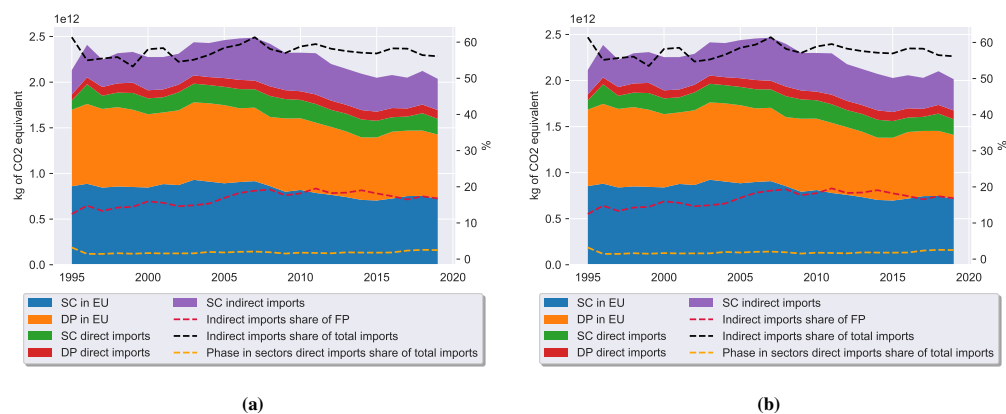


Figure 4.A.7: The EU emission footprint, using (a) GWP20 and (b) GWP100, is disaggregated according to origin (within or outside EU) and part of the production chain (final or intermediate production). Additionally, imports are split into direct and indirect imports. The dashed lines represent the dynamics of the labelled shares. Based on Exiobase data.

4.6 Supplementary Material

4.6.1 Footprint analysis

Aggregate sector	NACE sectors
Primary Sector	A01, A02, A03, B
Other Manufacturing	C10-C12, C13-C15, C16, C17, C18, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31-C32, C33
Fossil fuel Manufacturing	C19
Power and Heat	D35
Construction	F
Trade and motor vehicle	G45, G46, G47
Transportation and storage	H49, H50, H51, H52, H53
Other services	K64, K65, K66, L68, M69-M70, M71, M72, M73, M74-M75, N, R-S, I, J58, J59-J60, J61, J62-J63, E36, E37-E39
Public	O84, P85, Q
Other	T, U

Table 4.6.1: Sectoral aggregation and their NACE Rev. 2 equivalent.

Aggregate product	WIOD products
Fossil fuels	COAL_COKE_CRUDE, DIESEL, FUEL_OIL, GASOLINE, JETFUEL, NATGAS, OTHGAS, OTHPETRO
Other energy products	LIQUID_GASEOUS_BIOFUELS, OTHSOURC, RENEWABLES_NUCLEAR, WASTE

Table 4.6.2: Energy product aggregation and their equivalents in Corsatea et al. (2019).

4.6.2 Structural decomposition analysis

4.6.2.1 Components

Conducting the structural decomposition analysis (SDA) on $Q_{SC} = \hat{e}AL\hat{Y}$ is problematic due to the overlap of A and L in the analysis. Thus, for the SDA use that $Q_{SC} = \hat{e}L\hat{Y} - \hat{e}\hat{Y} = Q - Q_{DP}$.

4.6.2.2 Full decomposition

The individual drivers quantified here are:

- Final demand
 - population
 - total consumption per capita (household, NPISH and government)
 - share of consumption spend on a specific aggregate sector
 - share of aggregate sector consumption spend on a specific individual sector within aggregate sector
 - the import share for each specific final demand sector and its region of origin
- Leontief and A matrices
 - the input requirements independent of country of origin (technology)
 - trade shares of intermediate trade
- Energy use (following and adjusting Dietzenbacher et al. (2020))
 - changes in overall energy use (per unit produced)
 - changes in the energy mix (specifically the fossil fuel share)

As an additional element of disaggregation, we will distinguish between the contribution of changes within EU production and production that occurs outside of the EU. This will help to quantify the contribution of the energy transition in Europe within a globalised supply chain.

We first disaggregate static overall fossil fuel consumption into its aforementioned components and subsequently allocate the contribution of changes in individual components to the overall change in fossil fuel requirements. Here, we mainly follow and adapt the approach presented in Dietzenbacher et al. (2020).

4.6.2.2.1 Final Demand

Final demand is given by

$$y_t = (T_t \circ \tilde{D}_t) \hat{c}_t p_t,$$

where $\circ$ represents the Hadamard product. For m regions and s sectors, the shares of origin of

demand by region is given by $T = \begin{bmatrix} t^{11} & t^{12} & \dots & t^{1m} \\ t^{21} & t^{22} & \dots & t^{2m} \\ & & \dots & \\ t^{m1} & t^{m2} & \dots & t^{mm} \end{bmatrix}$, where each t^{ur} is a $(s \times 1)$

vector, with entry i depicting the share of final demand for good i in r , that is produced in u .

Hence, T is a $ms \times m$ matrix.

In addition, $\tilde{D} = \begin{bmatrix} \tilde{d}^1 & \tilde{d}^2 & \dots & \tilde{d}^m \\ \tilde{d}^1 & \tilde{d}^2 & \dots & \tilde{d}^m \\ & & \dots & \\ \tilde{d}^1 & \tilde{d}^2 & \dots & \tilde{d}^m \end{bmatrix}$ depicts the final consumption shares spend on different

goods for each country, irrespective of origin. Thus, each $\tilde{d}^u$ is a $(s \times 1)$ vector, with entry i depicting the share of final demand in region u , that is spend on good i , irrespective of its origin.

As T , $\tilde{D}$ is a $ms \times m$ matrix. $\hat{c} = \begin{bmatrix} c^1 & 0 & \dots & 0 \\ 0 & c^2 & \dots & 0 \\ & & \dots & \\ 0 & 0 & \dots & c^m \end{bmatrix}$ is a $m \times m$ matrix with each entry on

the diagonal reflecting the total consumption per capita for the specific region.¹ Finally, p is a $m \times 1$ vector depicting the population for each region/country.

The resulting y is then a $ms \times 1$ vector, with its entries given by

$$y_i^u = \sum_r t_i^{ur} \tilde{d}_i^r c^r p^r = \sum_r \frac{f_i^{ur}}{\sum_u f_i^{ur}} \frac{\sum_u f_i^{ur}}{\sum_i \sum_u f_i^{ur}} \frac{\sum_i \sum_u f_i^{ur}}{p^r} p^r,$$

depicting country (u) and sector (i) specific demand. In our analysis only the final demand from EU countries is relevant (r), whereas goods can be produced in all regions (u).

Aggregate sector decomposition

Additionally, to allow analysis of aggregate sector categories individually, we newly add an

intermediate final consumption share matrix $B = \begin{bmatrix} b^1 & b^2 & \dots & b^m \\ b^1 & b^2 & \dots & b^m \\ & & \dots & \\ b^1 & b^2 & \dots & b^m \end{bmatrix}$, where each b^u is a

$(s \times 1)$ vector with entries $b_{i,j}^u$ reflecting the consumption shares spend on the aggregate sector category j , to which sector i belongs.

Then, D (previously $\tilde{D}$) is re-defined, so that its entries do not reflect the total consumption shares of individual sectors, but the consumption shares of sectors, within their aggregate product group. For instance if in total 10% is being spend on sector i and 20% on its aggregate product group, $b_{i,j}^u = 0.2$ and $d_i^u = 0.1/0.2 = 0.5$ (and not $d_i^u = 0.1$ as before).

Thus,

$$y_t = (T_t \circ D_t \circ B_t) \hat{c}_t p_t, \quad (4.5)$$

¹Note that this is the general m country case. In our analysis we only look at the demand of one aggregate region, the EU, so that $\hat{c} = c$, both simply being a 1×1 vector.

or in terms of individual entries

$$y_i^u = \sum_r t_i^{ur} d_i^r b_{i,j}^r c^r p^r = \sum_r \frac{f_i^{ur}}{\sum_u f_i^{ur}} \frac{\sum_u f_i^{ur}}{\sum_j \sum_u f_{i,j}^{ur}} \frac{\sum_j \sum_u f_{i,j}^{ur}}{\sum_i \sum_u f_i^{ur}} \frac{\sum_i \sum_u f_i^{ur}}{p^r} p^r \quad \forall i \in j,$$

where $j = r, s, \dots, t$ reflects the aggregate product group. Again, y_i^u depicts country (u) and sector (i) specific demand. This is the formulation used throughout the whole aggregate and disaggregate analysis.

4.6.2.2.2 Intermediate Trade

In order to distinguish between the role of changes in technology and changes in intermediate trade, A can be rewritten as

$$A = H \circ U, \quad (4.6)$$

with H being the matrix of trade coefficients of intermediate trade $H = \begin{bmatrix} H^{11} & H^{12} & \dots & H^{1m} \\ H^{21} & H^{22} & \dots & H^{2m} \\ & & \dots & \\ H^{m1} & H^{m2} & \dots & H^{mm} \end{bmatrix},$

where each H^{ur} is a $s \times s$ matrix with element h_{ij}^{ur} depicting the share of the intermediate trade element ij demanded in r that originates in u . U on the other hand depicts technology and is

given by $U = \begin{bmatrix} U^1 & U^2 & \dots & U^m \\ U^1 & U^2 & \dots & U^m \\ & & \dots & \\ U^1 & U^2 & \dots & U^m \end{bmatrix},$ where each U^u is a region specific $s \times s$ matrix con-

sisting of the intermediate trade requirements per unit output, irrespective of its region of origin.

The individual elements are given by $a_{ij}^{ur} = h_{ij}^{ur} u_{ij}^r = \frac{a_{ij}^{ur}}{\sum_u a_{ij}^{ur}} \sum_u a_{ij}^{ur}.$

Energy Coefficients

Thirdly, the fossil fuel intensity diagonal matrix $\hat{e}$ can be decomposed into a total energy intensity diagonal ($\hat{q}$) and a diagonal depicting the share of total energy that is constituted by fossil fuels ($\hat{s}$). Thus,²

$$\hat{e}_t = \hat{s}_t \circ \hat{q}_t \quad (4.7)$$

4.6.2.2.3 Full Static Decomposition

Thus, the decomposed FP, SC and DP can be calculated from:

$$Q = \hat{e} L \hat{Y} = \hat{s} \circ \hat{q} L (\hat{T} \circ \hat{D} \circ \hat{B} \circ \hat{C} \circ \hat{P}) \quad (4.8)$$

$$Q_{DP} = \hat{e} \hat{Y} = \hat{s} \circ \hat{q} (\hat{T} \circ \hat{D} \circ \hat{B} \circ \hat{C} \circ \hat{P}) \quad (4.9)$$

$$Q_{SC} = Q - Q_{DP} \quad (4.10)$$

²Note that here we use the Hadamard product, however, the same results are obtained using matrix multiplication.

Note, that we use a diagonalised version of (4.5) in order to allow for sector specific analysis:

The shares of origin of demand by region is given by $\hat{T} = \begin{bmatrix} \hat{t}^{11} & \hat{t}^{12} & \dots & \hat{t}^{1m} \\ \hat{t}^{21} & \hat{t}^{22} & \dots & \hat{t}^{2m} \\ & & \dots & \\ \hat{t}^{m1} & \hat{t}^{m2} & \dots & \hat{t}^{mm} \end{bmatrix}$, where

each $\hat{t}^{ur}$ is a $(s \times s)$ diagonal matrix, with entry i on the diagonal depicting the share of final demand for good i in r , that is produced in u . Hence, $\hat{T}$ is a $ms \times ms$ matrix.

Also, $\hat{D} = \begin{bmatrix} \hat{d}^1 & \hat{d}^2 & \dots & \hat{d}^m \\ \hat{d}^1 & \hat{d}^2 & \dots & \hat{d}^m \\ & & \dots & \\ \hat{d}^1 & \hat{d}^2 & \dots & \hat{d}^m \end{bmatrix}$ depicts the final consumption shares of aggregate sectors

spend on specific sectors for each country, irrespective of origin. Thus, each $\hat{d}^u$ is a $(s \times s)$ diagonal matrix, with diagonal entry i depicting the share of aggregate sector final demand of region u spend on specific sector i , irrespective of its origin. The Matrix $\hat{B}$ has the same

format. As $\hat{T}, \hat{D}$ is a $ms \times ms$ matrix. $\hat{C} = \begin{bmatrix} \hat{c}^1 & \hat{c}^2 & \dots & \hat{c}^m \\ \hat{c}^1 & \hat{c}^2 & \dots & \hat{c}^m \\ & & \dots & \\ \hat{c}^1 & \hat{c}^2 & \dots & \hat{c}^m \end{bmatrix}$ also features a $ms \times ms$

format, where $\hat{c}^u$ is a $s \times s$ diagonal matrix with each entry on the diagonal reflecting the total consumption per capita for region u . Finally, $\hat{P}$ is a $ms \times ms$ matrix, with the same outline as $\hat{C}$, however with entries depicting the population for each region. However, as we only consider aggregate EU demand, here $\hat{T}, \hat{D}, \hat{B}, \hat{C}$ and $\hat{P}$ all feature a $ms \times s$ format

In order to obtain the total or sector specific metric from (4.8)-(4.10), we need to pre- and post-multiply with specific respective summation vectors i : i_{pre} is a $1 \times ms$ row vector and i_{post} is a $s \times 1$ column vector. Thus, to obtain total values of FP, SC and DP, we pre- and post-multiply the respective Q-matrix with these vectors (e.g. $FP = i_{pre} Q i_{post}$). For aggregate sector specific values, individual $s \times 1$ column vectors i_{post}^j are required, where j is the respective aggregate sector. The vector i_{post}^j contains a 1 wherever the specific sector is part of j and 0s otherwise. Accordingly, for instance FP for j is given by $FP^j = i_{pre} Q i_{post}^j$.

4.6.2.2.4 Dynamic Decomposition

As a next step, we decompose changes of FP, SC and DP into changes of their components. To illustrate the basic approach behind the SDA to decompose the changes into its elements we follow Miller and Blair (2009): For illustrative purposes we only focus on output as function of the Leontief and final demand: $x^1 = L^1 y^1$ and $x^0 = L^0 y^0$.³ Then Δx can be decomposed

³Note, that the superscript in this illustrative example captures the time period.

into ΔL and Δy for example as $\Delta x = L^1(y^0 + (\Delta y)) - (L^1 - (\Delta L))y^0 = L^1(\Delta y) + (\Delta L)y^0$ or as $\Delta x = (L^0 + (\Delta L))y^1 - L^0(y^1 - (\Delta y)) = (\Delta L)y^1 + L^0(\Delta y)$, which are both mathematically correct. Given n terms, there are $n!$ equivalent decomposition forms (Dietzenbacher and Los, 1998). Since our additive decomposition is a two-stage model (Su and Ang, 2012) and features $n = 10$ components, we follow the recommendations in the literature (e.g. Su and Ang, 2012; Wang et al., 2017) and apply the decomposition method proposed in Dietzenbacher and Los, 1998. Given the large number of components, we chose to apply the approximate form as has been commonly done in the literature (Arto and Dietzenbacher, 2014; Dietzenbacher et al., 2020; Duarte et al., 2021; Luo et al., 2020; Mi et al., 2017). Approximating the average of these $n!$ forms with the average of the polar forms⁴ leads to $\Delta x = \frac{1}{2}[L^1(\Delta y) + (\Delta L)y^0] + \frac{1}{2}[(\Delta L)y^1 + L^0(\Delta y)] = \frac{1}{2}[\Delta L(y^0 + y^1) + \Delta y(L^0 + L^1)]$ or for n terms, with $x^t = a_1^t a_2^t \cdots a_n^t$:

$$\Delta x = \frac{1}{2}[(\Delta a_1)(a_2^0 \cdots a_n^0 + a_2^1 \cdots a_n^1) + a_1^0(\Delta a_2)a_3^1 \cdots a_n^1 + a_1^1(\Delta a_2)a_3^0 \cdots a_n^0 + \cdots + a_1^0 \cdots (\Delta a_{n-1})a_n^1 + a_1^1 \cdots (\Delta a_{n-1})a_n^0 + (a_1^0 \cdots a_{n-1}^0 + a_1^1 \cdots a_{n-1}^1)(\Delta a_n)].$$

In addition, it is crucial to translate changes in the Leontief into the underlying changes in A . It can be shown that⁵

$$\Delta L = L^1(\Delta A)L^0 = L^0(\Delta A)L^1 \quad (4.11)$$

As (4.11) always only features one term, ΔL can be expressed in either form and there is no need to take an average of the polar forms (Miller and Blair, 2009). Subsequently, combining (4.6) and (4.11) yields:

$$\begin{aligned} \Delta L &= L^0(\Delta(H \circ U))L^1 = L^1(\Delta(H \circ U))L^0 \\ &= L^0\left(\frac{1}{2}\Delta H \circ (U^0 + U^1) + \frac{1}{2}(H^0 + H^1) \circ \Delta U\right)L^1 \\ &= \frac{1}{2}L^0(\Delta H \circ (U^0 + U^1))L^1 + \frac{1}{2}L^0((H^0 + H^1) \circ \Delta U)L^1 \end{aligned} \quad (4.12)$$

$$= \frac{1}{2}L^1(\Delta H \circ (U^0 + U^1))L^0 + \frac{1}{2}L^1((H^0 + H^1) \circ \Delta U)L^0 \quad (4.13)$$

⁴The two polar forms "work through the original ordering (1, . . . , n) from left to right and from right to left." (Dietzenbacher and Los, 1998, p.310), so that year 1 (0) values are on the left, and year 0 (1) values on the right of the Δ term (Miller and Blair, 2009).

⁵See for instance Miller and Blair (2009).

Final Decomposition

For the total FP matrix Q , the polar decomposition forms are given by⁶

$$\begin{aligned}
c_a = & (\Delta \hat{s}) \circ \hat{q}^0 L^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) \\
& + \hat{s}^1 \circ (\Delta \hat{q}) L^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) \\
& + \frac{1}{2} \hat{s}^1 \circ \hat{q}^1 L^1 ((\Delta H) \circ (U^0 + U^1)) L^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) \\
& + \frac{1}{2} \hat{s}^1 \circ \hat{q}^1 L^1 ((H^0 + H^1) \circ (\Delta U)) L^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) \\
& + \hat{s}^1 \circ \hat{q}^1 L^1 ((\Delta \hat{T}) \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) \\
& + \hat{s}^1 \circ \hat{q}^1 L^1 (\hat{T}^1 \circ (\Delta \hat{D}) \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) \\
& + \hat{s}^1 \circ \hat{q}^1 L^1 (\hat{T}^1 \circ \hat{D}^1 \circ (\Delta \hat{B}) \circ \hat{C}^0 \circ \hat{P}^0) \\
& + \hat{s}^1 \circ \hat{q}^1 L^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ (\Delta \hat{C}) \circ \hat{P}^0) \\
& + \hat{s}^1 \circ \hat{q}^1 L^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ (\Delta \hat{P}))
\end{aligned}$$

and

$$\begin{aligned}
c_b = & (\Delta \hat{s}) \circ \hat{q}^1 L^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \\
& + \hat{s}^0 \circ (\Delta \hat{q}) L^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \\
& + \frac{1}{2} \hat{s}^0 \circ \hat{q}^0 L^0 ((\Delta H) \circ (U^0 + U^1)) L^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \\
& + \frac{1}{2} \hat{s}^0 \circ \hat{q}^0 L^0 ((H^0 + H^1) \circ (\Delta U)) L^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \\
& + \hat{s}^0 \circ \hat{q}^0 L^0 ((\Delta \hat{T}) \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \\
& + \hat{s}^0 \circ \hat{q}^0 L^0 (\hat{T}^0 \circ (\Delta \hat{D}) \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \\
& + \hat{s}^0 \circ \hat{q}^0 L^0 (\hat{T}^0 \circ \hat{D}^0 \circ (\Delta \hat{B}) \circ \hat{C}^1 \circ \hat{P}^1) \\
& + \hat{s}^0 \circ \hat{q}^0 L^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ (\Delta \hat{C}) \circ \hat{P}^1) \\
& + \hat{s}^0 \circ \hat{q}^0 L^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ (\Delta \hat{P}))
\end{aligned}$$

⁶Note that we cannot use $L^0(\hat{T}^0 \circ \hat{D}^0 \circ \hat{C}^0 \circ \hat{P}^0) = L^0 \hat{Y}^0 = \hat{X}^0$ here as output considers all demand, whereas we only focus on EU demand.

Thus, the decomposition of ΔQ via the average of its polar decomposition forms is given by:

$$\begin{aligned}
\Delta Q &= \frac{1}{2}c_a + \frac{1}{2}c_b \\
&= \frac{1}{2} \left[(\Delta \hat{s}) \circ \hat{q}^0 L^0(\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) + (\Delta \hat{s}) \circ \hat{q}^1 L^1(\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
&+ \frac{1}{2} \left[\hat{s}^1 \circ (\Delta \hat{q}) L^0(\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) + \hat{s}^0 \circ (\Delta \hat{q}) L^1(\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
&+ \frac{1}{4} \left[\hat{s}^1 \circ \hat{q}^1 L^1((\Delta H) \circ (U^0 + U^1)) L^0(\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) \right. \\
&\quad \left. + \hat{s}^0 \circ \hat{q}^0 L^0((\Delta H) \circ (U^0 + U^1)) L^1(\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
&+ \frac{1}{4} \left[\hat{s}^1 \circ \hat{q}^1 L^1((H^0 + H^1) \circ (\Delta U)) L^0(\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) \right. \\
&\quad \left. + \hat{s}^0 \circ \hat{q}^0 L^0((H^0 + H^1) \circ (\Delta U)) L^1(\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
&+ \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 L^1((\Delta \hat{T}) \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) + \hat{s}^0 \circ \hat{q}^0 L^0((\Delta \hat{T}) \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
&+ \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 L^1(\hat{T}^1 \circ (\Delta \hat{D}) \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) + \hat{s}^0 \circ \hat{q}^0 L^0(\hat{T}^0 \circ (\Delta \hat{D}) \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
&+ \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 L^1(\hat{T}^1 \circ \hat{D}^1 \circ (\Delta \hat{B}) \circ \hat{C}^0 \circ \hat{P}^0) + \hat{s}^0 \circ \hat{q}^0 L^0(\hat{T}^0 \circ \hat{D}^0 \circ (\Delta \hat{B}) \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
&+ \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 L^1(\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ (\Delta \hat{C}) \circ \hat{P}^0) + \hat{s}^0 \circ \hat{q}^0 L^0(\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ (\Delta \hat{C}) \circ \hat{P}^1) \right] \\
&+ \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 L^1(\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ (\Delta \hat{P})) + \hat{s}^0 \circ \hat{q}^0 L^0(\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ (\Delta \hat{P})) \right]
\end{aligned} \tag{4.14}$$

Again, obtaining the respective drivers of total or aggregate sector specific FP, requires pre- and post-multiplication (e.g. $\Delta FP = i_{pre} \Delta Q i_{post}$ and the same pre- and post-multiplication is subsequently conducted with each summand/driver in (4.14)).

On the other hand, the final decomposition of Q_{DP} is given by

$$\begin{aligned}
 \Delta Q_{DP} = & \frac{1}{2} \left[(\Delta \hat{s}) \circ \hat{q}^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) + (\Delta \hat{s}) \circ \hat{q}^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
 & + \frac{1}{2} \left[\hat{s}^1 \circ (\Delta \hat{q}) (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) + \hat{s}^0 \circ (\Delta \hat{q}) (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
 & + \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 ((\Delta \hat{T}) \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) + \hat{s}^0 \circ \hat{q}^0 ((\Delta \hat{T}) \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
 & + \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 (\hat{T}^1 \circ (\Delta \hat{D}) \circ \hat{B}^0 \circ \hat{C}^0 \circ \hat{P}^0) + \hat{s}^0 \circ \hat{q}^0 (\hat{T}^0 \circ (\Delta \hat{D}) \circ \hat{B}^1 \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
 & + \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 (\hat{T}^1 \circ \hat{D}^1 \circ (\Delta \hat{B}) \circ \hat{C}^0 \circ \hat{P}^0) + \hat{s}^0 \circ \hat{q}^0 (\hat{T}^0 \circ \hat{D}^0 \circ (\Delta \hat{B}) \circ \hat{C}^1 \circ \hat{P}^1) \right] \\
 & + \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ (\Delta \hat{C}) \circ \hat{P}^0) + \hat{s}^0 \circ \hat{q}^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ (\Delta \hat{C}) \circ \hat{P}^1) \right] \\
 & + \frac{1}{2} \left[\hat{s}^1 \circ \hat{q}^1 (\hat{T}^1 \circ \hat{D}^1 \circ \hat{B}^1 \circ \hat{C}^1 \circ (\Delta \hat{P})) + \hat{s}^0 \circ \hat{q}^0 (\hat{T}^0 \circ \hat{D}^0 \circ \hat{B}^0 \circ \hat{C}^0 \circ (\Delta \hat{P})) \right],
 \end{aligned} \tag{4.15}$$

with the same pre- and post-multiplication of summation vectors required to get the respective changes and drivers.

Finally, using (4.10), (4.14) and (4.15), we can simply calculate the change in SC and the contribution of its drivers by subtracting DP from FP and subtracting their individual drivers. For instance, to quantify the contribution of final demand per capita to changes in SC, we subtract the contribution of final demand per capita to changes in DP from its impact on FP.

4.6.2.2.5 Regional differences

To add final details to the decomposition, we distinguish between effects of the energy mix and efficiency developments within the EU and the impact these developments in other regions have on the EU fossil fuel consumption requirements.

Following the methodology of Miller and Blair (2009), we decompose $\Delta \hat{s}$ and $\Delta \hat{q}$ into a EU and a non-EU part:

$$\Delta \hat{s} = \Delta \hat{s}_{EU} + \Delta \hat{s}_{ROW} \quad (4.16)$$

$$\Delta \hat{q} = \Delta \hat{q}_{EU} + \Delta \hat{q}_{ROW} \quad (4.17)$$

and apply these equations to (4.14) and (4.15). Here, for instance $\Delta \hat{s}_{EU}$ features the respective diagonal elements of EU sector rows and zeros in all other rows, and vice versa for $\Delta \hat{s}_{ROW}$.

4.6.2.3 Chaining

Following Arto and Dietzenbacher (2014) and Dietzenbacher et al. (2020), we describe changes in real terms by using data for period $t - 1$ in current year prices (cyp), while t is described in previous year prices (pyp). Thus, every Δ and hence the SDA between t and $t - 1$ reflect changes in real terms, as the base year of prices is the same ($t - 1$). Since we can chain the changes in footprints and their components (e.g. $\Delta FP_{t-1,t+1} = \Delta FP_{t-1,t} + \Delta FP_{t,t+1} = (FP_{t+1}^{pyp} - FP_t^{cyp}) + (FP_t^{pyp} - FP_{t-1}^{cyp})$), we can also simply chain the drivers over the years. However, note that this implies that we explain differences between $t - 1$ in cyp and t in pyp, not the timeseries of cyp data. If footprints are the same, no matter if cyp or pyp data are used, this would be the same as quantifying drivers to the cyp timeseries. Yet, cyp and pyp results slightly differ (see Figure 4.6.1).



Figure 4.6.1: The EU fossil fuel footprint based on current year price data compared to results using previous year price data.

Note that we observe only very small (maximum low single digit percentage points) and non-systematic deviations, which is why we assume that the results of the SDA approximate changes in the FP and its components well enough.

Chapter 5

Discussion & conclusion

This Chapter builds on the insights obtained from focusing on the specific aspects of the research question in the respective Chapters to address the overall research question of how European countries can achieve a low-carbon transition of the socioeconomic system that is socially just. In this work, this is conceptualised as a transition that substantially reduces emissions while tackling inequality.

While the research question explicitly addresses EU countries, only Chapter 4 specifically conducts the analysis at the EU level. Chapter 2 and 3 apply the Eurogreen model to Italy. Still, the results obtained in these Chapters can be seen as exemplary for many EU countries, given the overall similar economic, political and institutional context, including overarching EU institutions, policies and goals. While the sectoral structure might differ, applied policy or scenario settings are not sector specific¹. Consequently, macroeconomic impacts and socioeconomic behaviour of the system are expected to be qualitatively similar for many EU countries, as evident in past applications of the Eurogreen to France (Cieplinski et al., 2021b; D'Alessandro et al., 2020) yielding similar structural behaviour. Thus, while some Chapters only explicitly analyse results for Italy, this Chapter extends their insights to EU countries in general.

Having clarified the scope and approach towards addressing the research question, the Chapter first discusses two central aspects of consideration pertaining to a just low-carbon transition in EU countries, namely:

1. The role of economic growth in a just transition (Section 5.1): The interaction of the level

¹A notable exception to this is the energy transition, in particular the transition in the electricity sector. Here, country specific characteristics can lead to significantly different electricity generation mixes (see for instance Maeder et al. 2021), implying different impacts of the energy transition. Yet, qualitative outcomes are still comparable, since the results show the energy transition to have limited impact on the structure of the economy or employment, and a positive impact on the level of output, employment etc. within the post-Keynesian model closure will materialize as long as the energy transition triggers an increase in investment.

of economic activity with inequality on the one hand and energy use and emissions on the other, found in Chapters 2-4, underlines that the approach to economic growth has crucial implications for the feasibility of a just low-carbon transition.

2. Inequality and emissions (Section 5.2): Addressing inequality and emissions simultaneously is the central goal of the just low-carbon transition. Given their tight link and the potential trade-off between them as identified in Chapters 2 and 3, evaluating whether both aims can be achieved simultaneously is fundamental to assess whether a just low-carbon transition is possible.

After discussing these general aspects and characteristics, section 5.3 identifies concrete policies and policy combinations that Chapters 2-4 found to be conducive towards a just low carbon transition. Finally, section 5.4 presents limitations and prospects for future research and section 5.5 summarizes the presented findings.

5.1 The role of economic growth in a just transition

The role economic growth should play in a transition of the socioeconomic system is important and highly contentious (see green growth and degrowth debate, section 1.1). In the context of a just low-carbon transition, specifically its links to emission and inequality reduction are crucial. As Chapter 3 illustrates, economic growth tends to be associated with higher levels of emissions and render a just transition more challenging. This is not to say that absolute decoupling (i.e. falling emissions in the face of economic growth) never occurs, but that policies aimed at reducing emissions must be more ambitious *ceteris paribus* in the face of higher economic growth (see also Slameršak et al. 2024). The finding that pursuing GDP growth narrows the path towards a low-inequality low-emissions outcome also underscores the wasteful and inefficient character of growth as a means towards reducing emissions and inequality.

Chapter 3 adds additional aspects and nuances with respect to the energy transition when considering the role of economic growth as pertaining to a just transition. The energy transition allows to rapidly increase rates of decoupling. While comparatively increasing economic activity it simultaneously leads to a reduction in energy use and emissions. Yet, more economic output is still associated with higher emissions, with absolute emission levels in 2050 **unlikely to be compatible with net zero**, and other ecological implications and challenges regarding the energy transition like materials (Valero et al., 2018; Watari et al., 2019) or land-use (Capellán-Pérez et al., 2017) are not considered. While there is no question that the energy transition must be part of the overall system transition, it does not represent a panacea by itself.

The finding that economic growth is generally associated with higher levels of ecological damage and renders sufficient reductions in ecological impact more challenging is generally supported by the literature. On the one hand, some countries feature longer term declining trajectories of consumption as well as production-based emissions while experiencing economic growth (see for instance Lamb et al., 2021). Yet, considerations about the pace of this process, developments on the global scale and other environmental indicators like materials (Haberl et al., 2020; Hickel and Kallis, 2020) call into question the sustainability of retaining a growth-driven overall socioeconomic system. Moreover, Vogel et al. (2021) underline the inefficiency and wastefulness of economic growth as a provisioning factor mediating energy use and the fulfillment of human needs.

While economic growth is not a goal in itself, but, at best, a means to an end, its socioeconomic implications are not clear-cut. On the one hand, Chapter 3 finds that high levels of growth are generally linked to lower levels of inequality. However, this relationship is mediated by the political context. Moreover, evidently both affect each other mutually, i.e. not only can higher growth lead to less inequality – mainly via higher employment and associated effects – but lower inequality can also cause higher levels of demand and economic growth due to income specific consumption behaviour, in particular a decreasing marginal propensity to consume with rising income. In this context, pursuing growth can potentially reduce inequality, but it does so less efficient, also in terms of environmental implications, than policies directly targeting inequality. In particular, Chapter 3 highlights that the decoupling of inequality and economic growth is clearly possible and that a working time reduction can successfully maintain a high level of employment in the face of decreasing GDP.

Consulting the literature on the significance of economic growth socioeconomic outcomes, it is clear that it is not economic growth itself that by default improves the quality of life or inequality, but that the political and institutional context play a crucial role (Acemoglu and Robinson, 2002; Easterlin and Angelescu, 2012). Alvaredo et al. (2018a) show that rates of income growth at the regional level tend to increase with the level of income. Thus, the richer appropriate a disproportional share of economic growth, facilitating increases in inequality and concentration of wealth within regions. As Sullivan and Hickel (2023) argue, social gains (in their case poverty alleviation and human health) are not the result of capitalist growth but of social movements and politics. Conversely, low growth conditions will not necessarily lead to rising inequality, with implications being highly dependent on the institutional context (Jackson, 2019; Jackson and Victor, 2021). Jackson (2019) even argues that it is rather the policies aimed at stimulating economic growth while facing underlying challenges that exacerbate inequality.

Widening the perspective to a consumption-based analysis, Chapter 4 reveals a historic lack of

decoupling of final consumption² and the embedded fossil fuel footprint. In particular, developments in trade were a crucial element why the fossil fuel footprint remains relatively high and coupled to consumption, also due to a regionally differentiated statuses of energy transition. This underlines that the outcomes of Chapter 2 and 3 omit significant ramifications from developments in trade and outside the territory. So far the energy transition has not achieved much in terms of decoupling EU consumption from fossil fuel use. Here not only trade patterns play an important role but also economic growth renders it more challenging to displace fossil fuels. The fact that consumption based metrics typically exhibit worse decoupling statistics than production based ones (Kan et al., 2019; Liobikienė and Dagiliūtė, 2016; Papież et al., 2021; Papież et al., 2022) additionally illustrates that pursuing economic growth as a goal becomes even more problematic when taking trade into account.

Overall, [the results discussed here maintain](#) that economic growth in its current form and context tends to be rather detrimental towards a just low-carbon transition. Striving for growth in this context is at best unnecessary (in affluent regions like the EU) in terms of inequality and other social thresholds, and highly wasteful in terms of ecological ones. This work highlights evidence that high growth is not required to address inequality or even employment. Both can be addressed more effectively and at less environmental costs with targeted redistribution or working time reduction policies. Thus, the subsequent section focuses on the outset goals of inequality and emission reduction directly.

5.2 Inequality and emissions

Inequality and environmental damage and impact are tightly linked. Thus, addressing one element will have impacts on the other with crucial implications for tackling both simultaneously. Consequently, it is important to understand whether tackling emissions affects inequality on the one hand and whether reducing inequality will necessarily lead to an increase in emissions on the other. These mechanisms have crucial implications for the feasibility and the mechanisms of a just low-carbon transition.

[Results in Chapter 2 suggest that there](#) tends to be a trade-off between inequality and emissions, but this is not ubiquitous with certain policy and institutional contexts leading to low-inequality, low-emission outcomes. The general observed trade-off can be the result of both mechanisms: environmentally beneficial policies having an adverse effect on inequality as well as reductions in inequality leading to increases in environmental impact.

Regarding the former, findings in Chapter 3 indicate that, depending on the exact characteristics

²For the period of analysis, consumption and GDP follow similar dynamics (see Figure 4.A.5).

of the energy transition, it has the potential to increase or reduce inequality³ while simultaneously reducing emissions. The latter effect can be a combination of a marginal propensity to consume that decreases with income or of income specific consumption structures in the Euro-green model, i.e. resulting from a carbon intensity of income that decreases with income (see section 1.1). This mechanism is also found in Chapter 3, with redistribution policies themselves increasing demand, output and emissions.

While there seems to be a general emission-inequality trade-off, the energy transition has the potential to overcome it by reducing inequality and emissions in conjuncture (Chapter 3). Thus, while some environmental policies like the carbon tax can exacerbate inequalities, the findings presented here show that given the right configuration, the energy transition can contribute to lower inequality. A crucial feature of this energy transition is the redistribution of carbon cost revenue to households as lump sum payments. In terms of employment, Chapter 3 illustrates that the energy transition does not pose substantial macroeconomic employment challenges, but rather increases overall employment while triggering little structural changes in the employment composition.

Overall, these findings are in line with the literature, which tends to find a general initial trade-off between inequality and emission reductions, but highlights that their relation is crucially mediated by political and institutional context. Several works also question the significance of the scale of this trade-off: In general, the literature agrees that, the initial impact of environmental policies can exacerbate social issues like inequality and poverty, but that these impacts can be offset or even reversed if policies are designed appropriately (e.g. carbon tax rebate) or additional redistribution measures are introduced (D'Alessandro et al., 2020; Fremstad and Paul, 2019; Soergel et al., 2021). In addition, many studies find a positive employment effect of the energy transition (García-García et al., 2020), which can also have important implications for inequality.

Moreover, in line with the mechanism that redistribution can lead to an increase in emissions, Sommer and Kratena (2017) find carbon footprint income elasticities to decrease with income and Grunewald et al. (2017) establish that in low income countries higher per capita emissions are associated with lower inequality. However, Rao and Min (2018) question the relevance of this inequality-emission-trade-off in terms of magnitude and Oswald et al. (2021) demonstrate that decreases in income inequality do not lead to excessive increases in energy use, but that *“the ‘energy costs’ of equity are small”* (p.1). In a similar vein, the literature establishes that a decent life for all in terms of final energy is possible even at less than contemporary energy

³Note that inequality is measured here using nominal disposable income. Hence, price impacts or the distributional impact of the carbon tax burden are not directly included.

use and hence can be achieved in conjunction with effective climate mitigation (Kikstra et al., 2021; Millward-Hopkins et al., 2020). Bruckner et al. (2022) also find that tackling poverty in line with the Sustainable Development Goals would only lead to increases in emissions by low single digit percentages. Furthermore, Grunewald et al. (2017) find that for richer countries, the relationship between emissions and inequality is reversed, i.e. per capita emissions increase with inequality. Thus, the authors argue that a political economy mechanism, relating to social consensus and the relative power of beneficiaries of pollution, might be dominant in these countries.

Note that Chapter 2 and 3 likely overestimate the degree of an emission-inequality trade-off, since political economy mechanisms as mentioned in Grunewald et al. (2017) counteracting the initial trade-off are not captured. On the other hand, the impact of the energy transition, in particular with respect to global impacts, might potentially be less pronounced, as the results of Chapter 4 underline.

Moreover, Chapter 3 reveals an additional mechanism of the emission inequality trade-off in terms of functional income distribution: Here, a redistribution towards labour can nudge the innovation process towards more labour-saving technologies as opposed to other inputs. This less efficient production implies higher levels of energy use and emissions as well as adverse impacts on inequality through employment channels *ceteris paribus*. This underlines that the profit driven innovation process can play an important role in the emissions inequality trade-off and be a potential obstacle towards a just transition.

Chapter 4 does not explicitly assess the emission-inequality trade-off but adds an international and global inequality perspective to the analysis. The EU makes up a disproportional share of the global embedded fossil fuel footprint in relation to its population and its footprint is falling but at rates incompatible with 1.5 °C. This underlines the unequal distribution of environmental impacts, with the increasing outsourcing patterns of the fossil fuel footprint also indicating significance for ecologically unequal exchange (e.g. Hornborg 1998), i.e. net appropriation of resources and outsourcing of harmful production by high income countries (Dorninger et al., 2021; Wiedmann and Lenzen, 2018). The inequality implications in terms of impact and mitigation become even more pronounced when taking into account the concept of common but differentiated responsibilities depending on each countries' context (Anderson et al., 2020).

In general, the presented findings underline that even though the initial impact of environmental policies can have an adverse effect on inequality, this does not exclude the possibility of addressing both goals simultaneously. Still, this initial effect has to be taken into account, calling for compensatory policy design or *ceteris paribus* higher levels of redistribution or climate mitigation policies. In the EU context, international inequality considerations additionally underline

the importance for more ambitious decarbonization measures.

Taken together, this work finds that achieving a just low-carbon transition, while challenging, is possible. The goals of reducing inequality and emissions should be addressed directly, while limiting unnecessary overall economic growth. In addition, not only territorial, but consumption-based implications along the global supply chain should be taken into account to effectively tackle and not simply outsource energy use and emissions. This calls for a careful combination of a number of socioeconomic and environmental policy measures, which are outlined in the subsequent section.

5.3 Policy implications

As outlined in Sections 5.1 and 5.2, realizing a just low-carbon transition is not straightforward, given the interaction between key elements. It requires a range of simultaneous measures, taking into account trade-offs and synergies between measures and goals. Notably, this Chapter does not explicitly discuss the role of public debt and deficit, which remains a contentious issue among economists (Jackson et al., 2022), with for instance Nersisyan and Wray (2021) noting that not financial but real resources matter in terms of costs of the transition.

The section first presents individual measures that this work shows to be conducive towards a just low-carbon transition and subsequently assesses concrete degrees and combinations of these measures and resulting pathways within this context.

5.3.1 Individual policies

A specific goal of the just transition is to reduce inequality, which can be effectively done by pursuing direct redistribution. As presented in Chapter 2, redistribution policies are crucial to achieve low-inequality, low-emission outcomes. Here, policies that reduce inequality by increasing the income at the bottom of the distribution (e.g. direct transfers or tax reforms) are associated with higher demand, economic activity and emissions. Consequently, they should be accompanied by measures that target the income at the top of the distribution (e.g. maximum income or tax reforms), which can counteract and limit the rise in emissions, while also rendering the income distribution more equal. Thus, these results call for a genuine redistribution from the top to the bottom of the income distribution. The call to specifically target the environmental impact of the richest (Otto et al., 2019), also to allow for more environmental leeway for the poor (Bruckner et al., 2022), is increasingly voiced in the literature.

Another crucial measure towards a just low-carbon transition is evidently a rapid and extensive energy transition. As confirmed by Chapters 2 and 3, in the applied framework, the energy

transition is the most important leverage point for low emissions and hence vital for a just low-carbon transition. In particular, it has the potential to overcome the emission-inequality trade-off. Despite its overall positive macroeconomic effects on socioeconomic variables like employment or inequality when featuring a rebate of carbon costs, compensation measures for people and regions negatively affected might still be required.

Yet, from an empirical consumption-based perspective (Chapter 4), the energy transition has so far failed to dramatically reduce embedded fossil fuel use. Since one of the reasons behind this are the increasing outsourcing and trade patterns, diminishing the impact of the domestic transition, measures that address these developments are important. Here, the presented results suggest that a proposed Carbon Border Adjustment Mechanism can be helpful, but should incorporate indirect embedded imports and be complemented by measures addressing associated equity concerns⁴.

Still given lack of decoupling of embedded consumption-based fossil fuel use and the circumstance that economic growth renders the displacement of fossil fuels more challenging (see Chapter 4), in combination with technical and institutional issues (Davis et al., 2018) and other impacts associated with the the energy transition like materials (Valero et al., 2018; Watari et al., 2019) or land-use (Capellán-Pérez et al., 2017), calls into question the feasibility of an energy transition that has sufficiently substantial and rapid impact. Thus, the results presented here underline the importance of a systematic reduction in consumption and economic activity, in particular in the context of a just transition given the potential additional growth implied by redistribution policies. The positive impact of consumption reduction on emissions is confirmed in Chapters 2 and 3.

In this context, a working time reduction can help to maintain a high level of employment in the face of reductions in economic activity. While these measures all have the potential to contribute towards a just transition, none will achieve it in isolation. Thus, the subsequent section evaluates concrete combinations of measures.

5.3.2 Policy combinations and transition pathways

After having discussed potential policy measures individually, the subsequent paragraphs analyse different combinations of these and the resulting transition pathways. Thus, this section focuses particularly on Chapter 3.

The results underline that redistribution policies directly increase economic activity and emissions. The more extensive redistribution in the post-growth scenarios initially leads to the high-

⁴Note that due to the national focus of the Eurogreen model (see also section 5.4) this measure is not included in the discussion in section 5.3.2.

est level of economic activity and emissions, but also reduces inequality to the highest degree as well as constitutes the only socioeconomic scenarios that permanently alter the trajectory of the functional income distribution. Notably, the assumption that these policies lead to a fundamental power shift between capital and labour, evident in wage negotiations (i.e. the higher level of wage indexation) is crucial in this context.

Despite its more extensive redistribution, the combination with reductions in demand allows the post-growth scenario to outperform the moderate redistribution scenario in terms of energy use and emissions in the long run. Due to the working time reduction policy the post-growth transition also outperforms other scenarios regarding the level of employment, despite its comparative reductions in demand. Combining these socioeconomic conditions with energy transition measures that include a rebate of carbon costs, lead to even higher levels of employment accompanied by lower levels of inequality, energy use and emissions.

Still, relying solely on the energy transition as in the green baseline scenario is insufficient. While leading to the lowest levels of energy use and emissions, it reduces inequality relatively little. Note that a crucial element in the higher emissions in the post-growth scenarios is the innovation process, which post-growth scholars aim to direct towards more resource efficient production (Kallis et al., 2018), but which is instead shifted towards higher labour productivity driven by the structural redistribution.

Finally, the energy transition is assumed to progress at the same pace in all scenarios (i.e. the same share of renewable energy supply). Given the historically limited impact of the transition as indicated in Chapter 4, as well as the fact that higher economic activity renders a given state of the transition *ceteris paribus* more challenging to achieve, additionally underlines the potential role of post-growth paths in order to limit reliance on speculative negative emission technologies (Hickel et al., 2021; Slameršak et al., 2024).

Thus, overall the results of this work suggest that the most promising path towards a just transition is a post-growth one, with extensive redistribution and policies like basic income and working time reduction as well as the associated structural reductions in consumption and production. Certainly, the global energy transition will also have to play a prominent role, as will public intervention into the process of innovation.

5.4 Limitations and future research

This section outlines main points of limitation and, if applicable, corresponding potential future research.

Geographical Limitations

A specific limitation of the Eurogreen model [in the EU context of this thesis](#) is its national focus. It certainly has advantages like allowing for the detailed modelling of national policies and their implications, [and qualitative results can plausibly be inferred to other EU countries, as argued above](#). Yet, [the national analysis](#) neglects developments at a smaller geographical scale as well as important interactions with trade and developments outside national borders. This thesis partly addresses the latter limitation by applying a consumption-based analysis to the global supply chain (Chapter 4), but properly integrating these issues within the dynamic simulation model would enrich the analysis substantially. Thus, integrating developments abroad and trade, for instance by embedding the national analysis within a global one (as done in MEDEAS; Capellán-Pérez et al., 2020), would not only improve the modelling of socioeconomic and ecological outcomes directly, but also allow for important feedback between economic developments and emissions on the hand and dynamics in climate change and climate damage on the other.

Ecological Limitations

In addition to its geographical blind spots, the analysis here has ecological ones. The focus on energy use and emissions omits vital ecological aspects like material requirements, land use or biodiversity. In addition, this work does not take into account supply limits in terms of energy or materials. While including these aspects will certainly provide new and interesting insights into the analysis, they will likely only strengthen the case for a post-growth transition, in particular with regards to the scale of economic activity. Thus, they are unlikely to qualitatively affect the results.

Modelling

This thesis mainly relies on modelling as a tool which brings about its own limitations. Models often remain black boxes for many readers, with quantitative outcomes potentially providing a false sense of precision and scientific authority in the face of high levels of structural and parametric uncertainty. In addition, model results are often driven by implicit underlying assumptions (e.g. Mercure et al. 2019). As Sgouridis et al. (2022) phrase it by adding to the famous “*all models are wrong, but some are useful*”: “*all models are subjective, but few acknowledge it*” (p.2). Thus, the results of this thesis should be carefully considered in this context. Again, the results should not be interpreted as predicting the future but the work constitutes a useful tool to compare scenarios and reveal underlying mechanisms as well as possible futures in a complex system environment.

Post-growth

Furthermore, gaps and room for improvement remain in terms of scenario settings and analysis, specifically with respect to post-growth scenarios and parameters. The reduction in economic activity is driven by a generic exogenous reduction in consumption, equal across all sectors.

On the one hand post-growth, [particularly degrowth](#), calls for targeted reduction of destructive and polluting types of activities (Hickel et al., [2021](#); Kallis et al., [2018](#)). Thus, the scenario and parameter analysis in Chapters 2 and 3 might underestimate the positive impact for a given degree of consumption reduction. While the sectoral structure of the Eurogreen model in principle allows for sector specific targeting, the coarse sectoral aggregation renders it challenging in many instances. For example, in order to model a shift in the structure of diets or modes of transport, the respective sectors would have to be further disaggregated.

On the other hand, replacing the exogenous change in the marginal propensity to consume with direct policy and/or behavioural change that leads to the reduction of certain economic activities, would endogenise these dynamics and further enrich the literature on post-growth policies specifically in terms of quantitative results.

An additional major drawback regarding a post-growth transition is that the Eurogreen model is only able to capture the structure and behaviour of the current socioeconomic system. While this allows model calibration and validation as well as analysing policies that might be conducive to a post-growth transition in the current system, it prevents the evaluation of more fundamental changes in the causal relations within the socioeconomic system. Thus, while being helpful towards evaluating the transition towards a post-growth society, this work is not able to analyse a fundamentally different socioeconomic system that might characterize an eventual post-growth society (Kallis et al., [2018](#)).

While indicated as the most promising path towards a just transition in this work, it is important to underline that a substantial range of future research on post-growth is still required. This includes understanding and addressing growth dependencies in the current socioeconomic system, exploring forms of public financing and assessing novel policies like working time reductions (Hickel et al., [2022](#)). In particular, the potential of working time reduction to maintain high levels of employment hinges on its impact on labour productivity (Lavoie, [2014](#)). A crucial knowledge gap the work presented in this thesis and Ecological Macroeconomics more generally can help to address in the future, is related to the point on geographical limitations; namely setting the post-growth transition into a global context, incorporating global capital and currency flows into the analysis (Hickel et al., [2022](#)). In addition, agent-based models could provide new insights into bottom-up dynamics facilitating a post-growth transition.

In this context it is also important to note that discussing the political feasibility of any of the

modelled scenarios is out of the scope of this thesis. Given the nature of its proposed changes, the post-growth transition is confronted with a range of entrenched barriers (Slameršak et al., 2024; Strunz and Schindler, 2018), calling for more research on how to overcome them (Hickel et al., 2022).

5.5 Concluding remarks

The current socioeconomic system requires a rapid and extensive transition in order to address the fundamental ecological and socioeconomic challenges it faces simultaneously. In this context, this thesis analysis implications for a just low-carbon transition in EU countries within the framework of Ecological Macroeconomics, in particular the Eurogreen model, and complements the applied national simulation-based approach with an empirical consumption-based perspective.

The results of this work highlight that there is no single policy measure that can induce a just transition. Instead, addressing intricate interlinkages between key elements, in particular the emissions-inequality trade-off, requires a range of coordinated complementary measures.

Substantially reducing inequality requires genuine redistribution from the top to the bottom of the income distribution. However, this work shows that this reduction of inequality will also be accompanied by a comparative increase in economic activity, energy use and emissions. While the energy transition, if designed appropriately (i.e. including a mechanism for the rebate of carbon costs), has the potential to reduce emissions and inequality simultaneously, its reduction of inequality is modest. Thus, a just low-carbon transition requires effective genuine redistribution policies in combination with an extensive and rapid energy transition.

However, the historic lack of decoupling, in particular the limited impact of the energy transition to reduce global fossil fuel use driven by EU demand, underlines that relying solely on the regional energy transition to reduce emissions sufficiently is highly risky. This is particularly significant in the context of a just transition, where redistribution induces higher levels of economic activity which further impede energy transition efforts.

Thus, in order to achieve a considerable decline in inequality in combination with a sufficient fall in emissions, structural reductions of certain types of consumption and production might be inevitable. The resulting contraction and shift in economic activity needs to be accompanied by compensatory measures to mitigate potential adverse impacts on socioeconomic outcomes, for instance a working time reduction to maintain high levels of employment.

Consequently, this work indicates that the path most conducive towards a just low-carbon tran-

sition is a post-growth one, featuring extensive redistribution measures in combination with rapid global energy transition efforts and a structural reduction in certain types of consumption and production. Still, a wide range of future research into this fundamental transition of the socioeconomic system is required to fully understand its implications and feasibility concerns.

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